RTR(A) MASTERY

The Complete Examination Arsenal

FIRST EDITION

CAPT. PANKAJ PAHIL

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DGCA · Radio Telephony (Aeronautical)

RTR(A) MASTERY

The Complete Guide to the DGCA Radio Telephone Operator (Restricted) Examination — Written, Practical & Oral

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FIRST EDITION

Capt. Pankaj Pahil Pilot · Flight Instructor · Ground Instructor · 17 Years of Aviation Experience

RTR(A) Mastery — The Complete Guide to the DGCA Radiotelephony Examination.
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Aligned to the DGCA RTR(A) syllabus — CAR Section 7, Series G, Part VI; the Radio Telephone Operator (Restricted) Rules, 2025; and the Bharatiya Vayuyan Adhiniyam, 2024. This book is a study and revision aid prepared in good faith; while every effort has been made to ensure accuracy against current DGCA and ICAO provisions, regulations change. Always cross-check current frequencies, procedures and regulations against the latest official DGCA, AIP and ICAO documentation. This book does not replace official publications and is not endorsed by the DGCA.

Dedication

To every student who looks up at the sky and decides that is where they belong —
and to the instructors who help them get there.

This book is given freely, in service of the next generation of pilots. May your readback always be correct, and your skies always clear.

Foreword

The radio is the thread that ties every flight together. Long before a student touches the controls with any real confidence, they must learn to speak — clearly, briefly, and correctly — to the unseen people who keep the sky orderly. A pilot who is fluent on the radio is calm, precise and safe; one who is not becomes a hazard the moment the frequency gets busy.

For too long, candidates for the RTR(A) have prepared from scattered notes and second-hand summaries — memorising numbers without ever understanding the physics and procedure beneath them. They might pass, but they are not yet *safe* on the air. This book sets out to change that, by building every topic from first principles and carrying it all the way to the exact figures and phraseology the examiner demands.

Clear skies,
Capt. Pankaj Pahil

Preface — why this book exists

Understanding first, memory second.

Every chapter takes a single topic from the DGCA syllabus and builds it from the ground up: plain language first, then the *why* behind every rule, then the exact figures and phraseology the examiner demands, and finally a question bank so you can prove you have it. You will find the same features throughout — colour-coded boxes for definitions, limits, procedures, worked examples and exam traps; mnemonics for the facts that refuse to stick; deep-dive panels for the curious; figures and diagrams to make the abstract concrete; and a full question bank closing every chapter, building toward complete mock papers at the end.

I want to be honest about my intention. This is a work of service, not of profit. My aim is simply to help every student — whatever their background or means — walk into that examination prepared, and walk out a more capable aviator. If a single reader is calmer and clearer on a busy frequency because of these pages, the work will have been worth it.

Work the chapters honestly. Aim for eight out of ten on every self-test before you move on, speak the phraseology out loud daily, and revisit the "Numbers to Memorise" tables until they are second nature. Learn it well — not just to pass, but because one day, on a bad day, the words in this book may be the ones that keep you and your passengers safe.

Clear skies and good luck,
Capt. Pankaj Pahil
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About the Author

Capt. Pankaj Pahil
Pilot · Flight Instructor · Ground Instructor

Capt. Pankaj Pahil has spent seventeen years in aviation — in the cockpit as a pilot and on the ground as a flight and ground instructor, training the next generation of aviators. Through Ghost Aviator (www.GhostAviator.com) he teaches the theory and discipline of aeronautical radiotelephony to students preparing for the DGCA RTR(A) examination. This book distils that classroom experience into a single, structured guide — built from the ground up, in plain language, for anyone who wishes to earn their wings.

Acknowledgements

With gratitude to the students whose questions shaped these pages, to the fellow instructors and controllers who checked the procedures, and to every reader who trusts this book to help them toward the licence. The standards and procedures described here rest on the work of the DGCA, ICAO and the ITU; any errors that remain are the author's own.

How to Use This Book

Every chapter is built the same way, so once you learn the pattern you can move quickly. Each concept is taught in **three passes** — **In plain terms**, then **Why it works**, then the **exact figure or rule** — and the margin boxes below each do one job:

Definition

The exact wording to recite. Blue — "say it like this."

Numbers / Limits

Hard figures — frequencies, distances, percentages. Teal — "memorise this."

Procedure

A step-by-step sequence done in order. Green — "do it in this order."

Worked Example

A problem solved end-to-end. Indigo — "follow the working."

Exam Trap

A common mistake or examiner favourite. Orange/Red — "don't fall for this."

Mnemonic

A memory hook for sticky facts. Purple/Amber — "remember it this way."

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> **Cockpit Reality**
>
> How the rule plays out in flight. Pink — "why it matters in the air."

Collapsible **Deep dives** are optional enrichment. Every chapter ends with a **question bank** (MCQs with explained answers, viva questions, and numerical problems where relevant), a **Numbers to Memorise** table, and a **revision card**. A suggested 8-week study plan appears in Chapter 1. Aim for 8/10 on each self-test, and practise the phraseology aloud — the practical fails more candidates than the paper.

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Glossary of Key Abbreviations & Terms

A quick-reference list of the abbreviations and terms used throughout this book and in the RTR(A) examination. Each topic is explained in full in its chapter.

A

ACARS

Aircraft Communications Addressing & Reporting System — digital datalink for short messages.

ACAS / TCAS

Airborne / Traffic Collision Avoidance System — issues traffic (TA) and resolution (RA) advisories.

ADF

Automatic Direction Finder — airborne receiver pointing to an NDB.

ADS

Automatic Dependent Surveillance — aircraft automatically reports its position (ADS-C / ADS-B).

AFTN

Aeronautical Fixed Telecommunication Network — ground network carrying flight plans, NOTAMs, messages.

AIC

Aeronautical Information Circular — advisory information not meeting NOTAM/AIP criteria.

AIP

Aeronautical Information Publication — the permanent State document (parts GEN/ENR/AD).

AIRAC

Aeronautical Information Regulation & Control — common effective dates every 28 days.

AIS

Aeronautical Information Service.

AMS / AMS(R)

Aeronautical Mobile Service / (Route) — for safety & regularity of flight on civil routes.

AM

Amplitude Modulation — used for aviation VHF voice.

ATC / ATS

Air Traffic Control / Air Traffic Services.

ATIS

Automatic Terminal Information Service — continuous one-aerodrome broadcast.

AVC

Automatic Volume Control — holds received audio level steady.

C – E

CAR

Civil Aviation Requirement — Section 7, Series G, Part VI carries the RTR syllabus.

CAVOK

"Ceiling And Visibility OK" — vis ≥10 km, no cloud <5000 ft / no CB·TCU, no significant weather.

CTA / CTR / TMA

Control Area / Control Zone (surface up) / Terminal Control Area.

DGCA

Directorate General of Civil Aviation — India's regulator.

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DME

Distance Measuring Equipment — gives slant range (UHF).

DSB / SSB

Double / Single Side Band — SSB used for HF voice.

ELT

Emergency Locator Transmitter — 406 MHz (satellite) / 121.5 MHz (homing).

F – I

FIR / UIR

Flight Information Region / Upper Information Region — FIS & Alerting provided.

FIS

Flight Information Service.

FL

Flight Level — constant-pressure surface referenced to 1013.2 hPa.

FM

Frequency Modulation — broadcast use; not aviation voice (capture effect).

GAGAN

GPS Aided GEO Augmented Navigation — India's SBAS.

GNSS / GPS

Global Navigation Satellite System / Global Positioning System.

GPWS

Ground Proximity Warning System (EGPWS adds terrain database).

HF / VHF / UHF

3–30 MHz (sky wave) / 30–300 MHz (line-of-sight ATC voice) / 300 MHz–3 GHz.

IAIP

Integrated Aeronautical Information Package (AIP · Amendment · Supplement · AIC · NOTAM).

ICAO

International Civil Aviation Organization — UN agency, Annex 10 = telecom.

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IFR / VFR

Instrument / Visual Flight Rules.

ILS

Instrument Landing System — localizer + glide path + markers.

INS / IRS

Inertial Navigation / Reference System — self-contained.

ITU

International Telecommunication Union — allocates spectrum (Radio Regulations).

L – Q

LUF / MUF

Lowest / Maximum Usable Frequency — HF sky-wave limits.

MAYDAY

Spoken distress signal, said three times.

MEL / MMEL

Minimum Equipment List / Master MEL — dispatch with items inoperative.

METAR / SPECI

Routine / special aerodrome weather report.

NDB

Non-Directional Beacon (LF/MF, ground wave).

NFAP

National Frequency Allocation Plan (India, via WPC).

NOTAM

Notice to Airmen (NOTAMN new / R replace / C cancel; SNOWTAM, ASHTAM).

PAN-PAN

Spoken urgency signal, said three times.

PAR / SRA

Precision / Surveillance Radar Approach.

PIB

Pre-flight Information Bulletin (route NOTAM digest).

PTT

Press-To-Talk.

QDM/QDR/QFE/QNH/QNE

Magnetic heading to / bearing from station · field pressure · MSL pressure · standard 1013.2 hPa.

R – W

RNAV / RNP

Area Navigation / Required Navigation Performance (RNP adds monitoring & alerting).

RVR

Runway Visual Range.

SACFA

Standing Advisory Committee on Frequency Allocation (transmitter siting).

SAR

Search and Rescue.

SELCAL

Selective Calling — coded four-tone alert.

SIGMET / AIRMET

Warnings to all aircraft / to low-level light aircraft.

SSR

Secondary Surveillance Radar — interrogate 1030 / reply 1090 MHz; Modes A/C/S.

TAF

Terminal Aerodrome Forecast.

UTC

Coordinated Universal Time ("Zulu").

VOLMET

En-route met broadcast for several aerodromes.

VOR

VHF Omnidirectional Range — magnetic radials.

WILCO / ROGER

"Will comply" / "Received all your last transmission."

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WPC

Wireless Planning & Coordination Wing — India's radio regulator.

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The RTR(A) Exam & How to Use This Book

Before a single watt of radio theory, you must understand the licence itself — what it permits, who issues it, exactly how the examination is built and scored, the law that governs it, and the study system that will carry you through every chapter that follows. This is the map for the whole journey.

SYLLABUS MAP

`Part I (vi) Licensing of radio apparatus · examination framework · key definitions (ICAO Annex 10)`

LEARNING OBJECTIVES — BY THE END OF THIS CHAPTER YOU WILL BE ABLE TO…

- Define the RTR(A) licence, the privileges it grants, and the aeronautical mobile service it authorises you to operate in.
- Define aeronautical station, aircraft station and radiotelephony precisely, as the examiner expects.
- Describe the structure of the examination — the written/oral theory paper and the practical transmission test — and how each is scored.
- State the validity, attempts, exemptions and renewal rules (figures to be entered from the current CAR).
- Explain the layered legal framework — Act, Rules, CAR, ICAO Annex 10, ITU Radio Regulations — and how they interlock.
- Use the study system in this book and follow a structured study plan.

1. 1.1 What the RTR(A) is — privileges & the service
1.2 The key definitions you must own
1.3 How the examination is built & scored
1.4 Validity, attempts, exemptions & renewal
1.5 The legal framework
1.6 Eligibility & the application process
1.7 What the examiner is really testing
1.8 How to use this book — the study system
1.9 A structured study plan
1.10 Exam-day strategy & common failure causes

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1.1 What the RTR(A) is — privileges & the service

IN PLAIN TERMS

RTR(A) stands for **Radio Telephone Operator (Restricted) — Aeronautical**. It is the certificate that legally permits a person to **operate an aircraft radio station** and communicate by radiotelephony with air traffic services. The aircraft owns the radio; the RTR is what authorises *you*, the human, to press the transmit button on a live frequency and speak. Without it, a pilot may not exercise the radio privileges of a flight crew licence.

WHY A SEPARATE RADIO LICENCE EXISTS

Radio spectrum is a shared, finite, safety-critical resource governed by international treaty. One untrained operator who blocks a distress call, transmits on the wrong frequency, or causes a runway incursion through a misunderstood clearance endangers everyone on that frequency. The State therefore requires every operator to demonstrate a defined minimum standard of **regulatory knowledge, radio theory and disciplined phraseology** before being licensed. The word **"Restricted"** signals that the privilege is confined to the **aeronautical mobile service** — you are licensed as an aircraft radio *operator*, not as a radio engineer or a general (e.g. maritime) operator.

A student pilot operating within the aeronautical mobile service, communicating with Air Traffic Control via radio link.

Privileges & limitations of the RTR(A)

Privileges: to operate radiotelephone equipment in the aeronautical mobile service — transmitting and receiving voice communications with aeronautical (ground) stations and other aircraft, including handling distress, urgency and safety traffic.

Limitations: restricted to the aeronautical mobile service (not maritime/general); the holder operates equipment but is not certified to install or maintain it; operation is subject to the aircraft radio station being separately licensed (Chapter 3).

1.2 The key definitions you must own

WHY DEFINITIONS FIRST

The examiner frequently opens with a definition — "What is the aeronautical mobile service?" — because precise language is the foundation of safe radio work. Learn these word-for-word; they recur throughout the book.

The core ICAO definitions

Aeronautical Mobile Service (AMS): a mobile radiocommunication service between **aircraft stations and aeronautical stations**, or between aircraft stations, in which **survival-craft and emergency position-indicating beacon** stations may also participate.

Aeronautical Station: a **land station** in the aeronautical mobile service (e.g. an ATC or ground station). In certain cases it may be on board a ship or platform.

Aircraft Station: a **mobile station** in the aeronautical mobile service, located on board an aircraft (other than a survival-craft station).

Radiotelephony: a form of radiocommunication intended primarily for the **exchange of information in the form of speech**.

AMS(R) vs AMS(OR)

The AMS is sub-divided: **AMS(R) — "Route"** is reserved for communications relating to **safety and regularity of flight** along national or international civil air routes; **AMS(OR) — "Off-Route"** covers communications outside those routes. As an RTR(A) operator you work primarily in **AMS(R)**.

Mnemonic

Land station = Aeronautical; Aircraft station = Airborne. "Aeronautical is on the earth." Both live inside the Aeronautical **Mobile** Service because the aircraft end moves.

1.3 How the examination is built & scored

> **IN PLAIN TERMS**
>
> The RTR(A) is tested in two complementary halves. A **theory examination** (written and/or oral) checks your knowledge of regulations, radio principles and procedure. A **practical test** checks that you can actually *operate* — read a passage aloud, transmit and receive messages in correct phraseology, and answer the examiner's spoken questions on the spot.

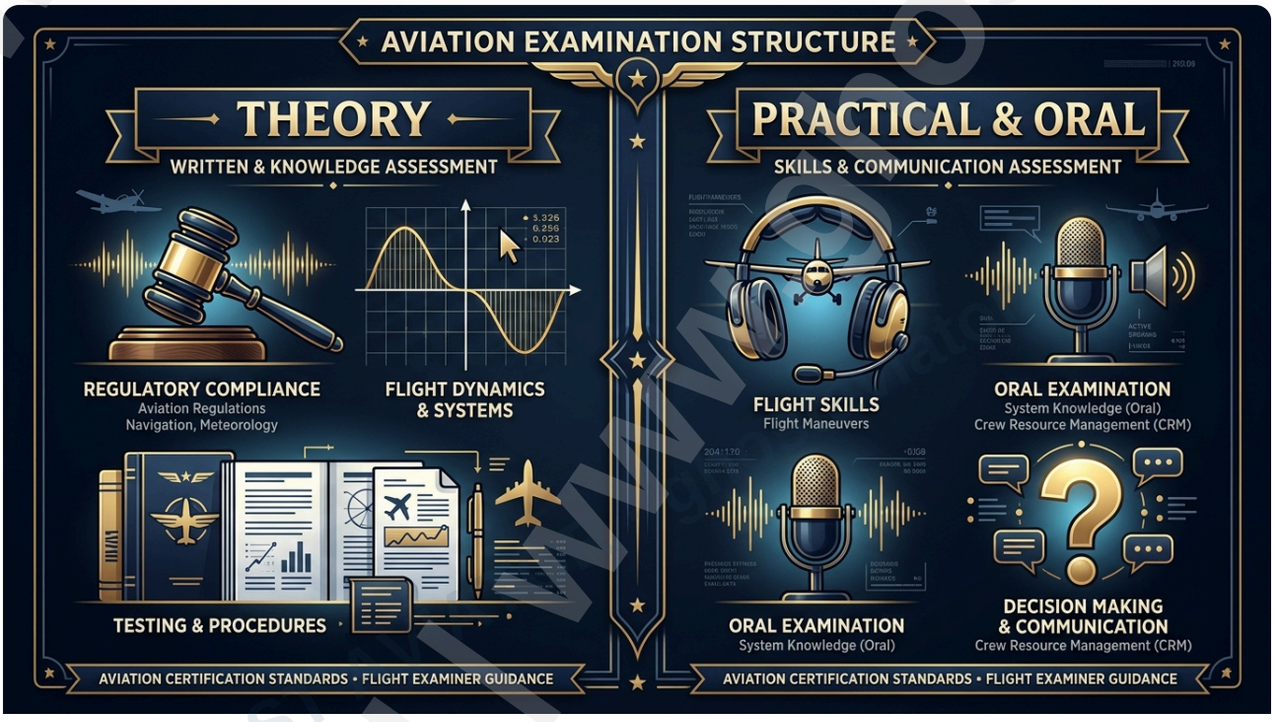


Figure 1.1 — The two halves of the RTR(A) examination: Theory (Written/Oral) and Practical & Oral.

1.4 Validity, attempts, exemptions & renewal

> **IN PLAIN TERMS**
>
> **The theory paper — what it covers**
>
> The theory follows the CAR Section 7 syllabus and breaks into three knowledge blocks: **(I) Aeronautical Telecommunication & Regulations; (II) Radio Principles & Practice;** and **(III) Radiotelephony procedures & phraseology**. Questions are typically objective/short-answer; some centres also conduct an oral on the same material.
>
> **The practical & oral test — what it covers**
>
> Reading a printed passage clearly at a controlled pace; transmitting and receiving a set of messages using correct **phonetic alphabet, numbers, phraseology and read-back**; demonstrating **distress/urgency** handling; and answering spoken questions. This tests *fluency under pressure*, not just recall.

Marks, weighting & pass standard

Reported section weighting on the theory paper is approximately:
Regulations & Procedure ≈ 30%, Radio Principles & Practice ≈ 40%, Radiotelephony ≈ 20%+. **Exact total marks, the pass mark, and time allowed must be quoted from the current CAR** — enter them below and in the table at 1.4.

Exam trap — the practical is where people fail

Candidates over-invest in theory and under-prepare the oral/practical. Examiners fail more people on the spoken test, where nerves wreck phraseology and readability. Budget at least half your preparation for live transmission practice (Module C of this book).

Passing is governed by exact rules on how long a part-pass lasts, how many attempts you get, who is exempt, and how the licence is kept current. These are direct viva favourites and they govern your own paperwork — know the numbers cold.

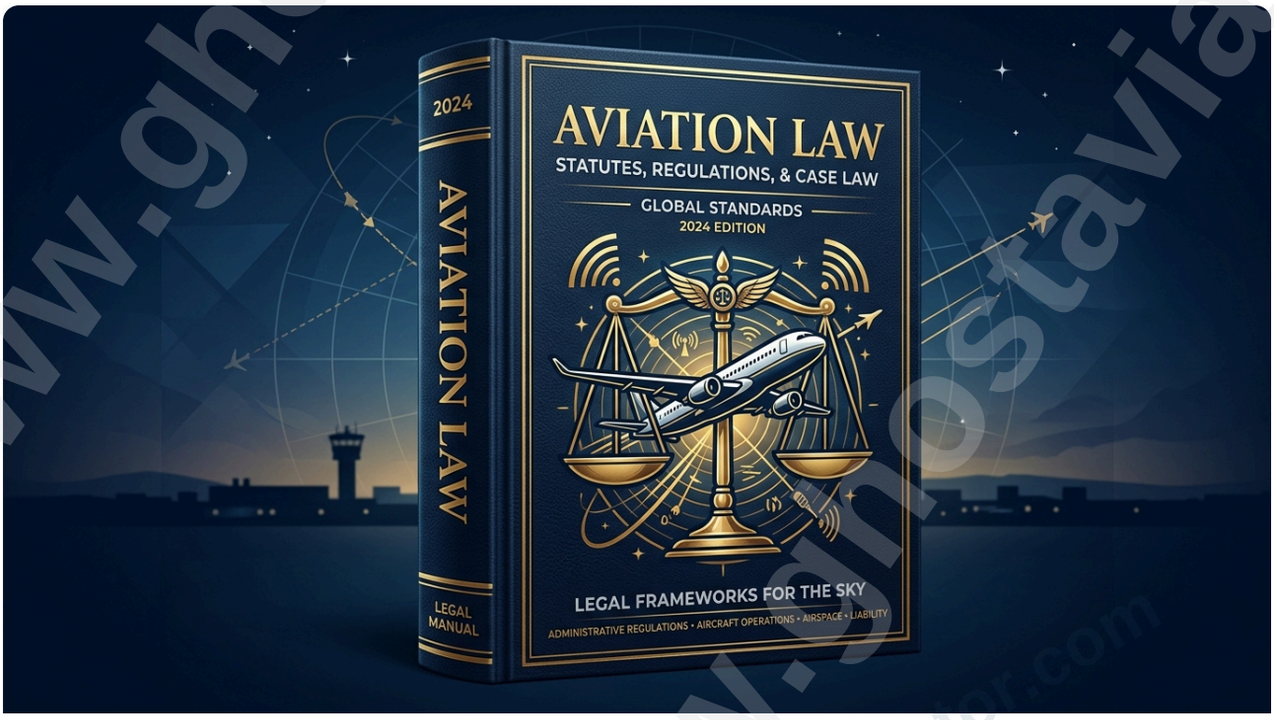
RTR(A) Exam Rules & Validity

Item	Rule (enter exact figure from current CAR / Rules 2025)
Validity of a part-pass	A cleared part is valid for a fixed window within which the remaining part must be passed — *insert exact period.*
Number of attempts / re-sit interval	*Insert permitted attempts and minimum gap between attempts.*
Pass mark (each part)	*Insert minimum percentage.*
Minimum age	*Insert minimum age to hold the licence.*
Exemptions	*Insert categories exempt from part of the exam (e.g. holders of certain higher RTF qualifications or flight crew licences).*
Validity & renewal of the licence	*Insert the licence validity period and any renewal/currency requirement.*

1.5 The legal framework

FIRST PRINCIPLES — THREE LAYERS OF LAW, PLUS TWO INTERNATIONAL FOUNDATIONS

Authority over your radio licence flows downward through nested layers. The **Act** is the top-level national statute; the **Rules** are the detailed mechanics made under it; the **CAR** is the regulator's operating requirement that turns the rules into a syllabus and procedure. Above the national layer sit two international foundations: **ICAO Annex 10** (the technical standards for aeronautical telecommunications) and the **ITU Radio Regulations** (the global allocation of spectrum). India's national law gives domestic legal force to these international standards.



The three-tiered structure of Indian aviation law governing radiotelephony licensing and airwaves coordination.

The Nested Legal Framework

Layer	Instrument	What it does
International (spectrum)	ITU Radio Regulations	Allocates the radio-frequency bands aviation may use
International (aviation)	ICAO Annex 10	Standards & Recommended Practices for aeronautical telecommunications
National Act	Bharatiya Vayuyan Adhiniyam, 2024	India's principal civil aviation statute
National Rules	RTR (Restricted) Rules, 2025	Create & govern the RTR licence and examination
Requirement	CAR Section 7, Series G, Part VI	The examinable syllabus & procedure

> **Mnemonic — "Act → Rules → CAR"**
>
> Remember **"A Real Carrier"**: Act (Vayuyan Adhiniyam) → Rules (RTR Restricted 2025) → CAR (Section 7). Broad to specific, top to bottom — just like a transmitter narrowing a big idea down to one exact frequency.

▶ 🔍 Deep dive — why a new Act and Rules appeared in 2024–25

1.6 Eligibility & the application process

Typical path to the licence

1. Meet the **eligibility** (minimum age; any prescribed nationality/identity requirements).
2. **Apply** to the examining authority with the required form, fee, photographs and identity/age proof.
3. Sit the **theory** examination (and oral where conducted).
4. Sit the **practical & oral** transmission test.
5. On passing both within the validity window, the **licence is issued**.
6. Maintain currency/renewal as required.

1.7 What the examiner is really testing

THE FIVE COMPETENCIES

Behind every question sits one of five competencies. Frame your study around them, not just around topics:

The Five Competencies

Competency	What it looks like in the exam	Where in this book
Regulatory knowledge	Bodies, licences, NOTAM/AIP, definitions	Module A (Ch 1–6)
Radio theory	Electrical, bands, propagation, modulation, equipment, nav aids	Module B (Ch 7–12)
Phraseology fluency	Phonetics, numbers, call signs, ATC scripts	Module C (Ch 13–22)
Readability & delivery	Clear, paced, correct pronunciation on the practical	Module C + practice
Emergency handling	Distress/urgency, comms failure, the right word at the right time	Ch 20

The examiner evaluates phraseology fluency and emergency handling during the live practical testing.

1.8 How to use this book — the study system

IN PLAIN TERMS

Every chapter is built the same way, so once you learn the pattern here you move fast everywhere. Watch for these coloured boxes — each has a job:

Definition

The exact wording to recite. Blue = "say it like this."

Numbers / Limits

Hard figures — frequencies, distances, percentages. Teal = "memorise this."

Procedure

A step-by-step sequence done in order. Green = "do it in this order."

Worked example

A problem solved end-to-end. Indigo = "follow the working."

Exam Trap

A common mistake or examiner favourite. Orange = "don't fall for this."

Mnemonic

A memory hook for sticky facts. Purple = "remember it this way."

Cockpit Reality

How the rule plays out in flight. Pink = "why it matters in the air."

Inside the text you'll also meet the **three-pass method** on each concept — **In plain terms**, then **Why it works**, then the **exact figure or rule**. Collapsible 🔍 **Deep dives** are optional enrichment. Every chapter ends with a **question bank** (MCQs + viva, and numerical problems where relevant) and a **revision card**.

A structured approach combining theory readings with regular phraseology drills yields the best results.

1.9 A structured study plan

AN 8-WEEK FRAMEWORK (COMPRESS OR EXPAND TO SUIT)

Treat theory and practical in parallel — never leave the practical to the last week.

Suggested 8-Week Study Plan

Week	Theory focus	Practical drill (in parallel)
1	Ch 1–2 — exam, bodies, framework	Phonetic alphabet & numbers daily
2	Ch 3–4 — licensing, Q-codes	Time, standard words, radio checks
3	Ch 5–6 — FIR/AIS/NOTAM, met codes	Decode METAR/TAF aloud; ATIS read-back
4	Ch 7–8 — electrical, frequency bands	Call signs & initial contact
5	Ch 9–10 — propagation, modulation	Taxi/clearance phraseology
6	Ch 11–12 — equipment, nav aids	Departure/circuit/approach scripts
7	Ch 13–18 — phraseology & control	Full message exchanges; distress drills
8	Ch 19–24 — revision & mock papers	Mock practical & oral under timed pressure

COCKPIT REALITY

On your first solo radio call your mouth goes dry and your mind blanks. The pilots who stay calm are the ones who *drilled the pattern* until it was automatic. The quizzes and revision cards exist for that — not merely to pass the exam, but so the right words come out when the frequency is busy and the pressure is real.

1.10 Exam-day strategy & common failure causes

The honest pass routine

Aim for **8/10 on every chapter's self-test before moving on**; re-drill the "Numbers to Memorise" tables until automatic; and **speak the phraseology out loud daily** — silent reading does not build the spoken fluency the practical demands.

On the day

Theory: read every question fully before answering; do the ones you know first; never leave objective questions blank; watch the units. Practical: **listen, then press, pause, speak, release**; control your pace; use full standard phraseology and read back correctly; if you fumble, say "**correction**" and continue calmly.

> **The five most common reasons candidates fail**
>
> 1. Neglecting the **practical/oral**.
> 2. Guessing **exact regulatory figures** (validity, pass mark, frequencies).
> 3. Confusing look-alike facts (ICAO vs IATA, QNH vs QFE, 121.5 vs 243.0).
> 4. Poor **readability** — too fast, mumbled, wrong pronunciation.
> 5. Freezing on **distress phraseology**.
>
> Every one is fixed by drilling, not cramming.

☆ Numbers to memorise

Essential Facts for Chapter 1

Fact	Value
Licence	RTR(A) — Radio Telephone Operator (Restricted), Aeronautical
Service authorised	Aeronautical Mobile Service — primarily AMS(R)
Aeronautical station	A land station in the AMS
Aircraft station	A mobile station in the AMS aboard an aircraft
Governing Act / Rules	Bharatiya Vayuyan Adhiniyam 2024 / RTR (Restricted) Rules 2025
Syllabus source	CAR Section 7, Series G, Part VI
International foundations	ICAO Annex 10 (telecom standards) · ITU Radio Regulations (spectrum)
Pass mark / validity / attempts	*Author to insert from current CAR*

Question bank

PART A — MCQS (CLICK AN OPTION TO CHECK)

1. "RTR(A)" stands for:

> Radio Transmission Rating (Advanced)
>
> Radio Telephone Operator (Restricted) — Aeronautical
>
> Registered Telecom Radio (Aircraft)
>
> Radar Tracking Receiver (Airborne)

```
Answer: Radio Telephone Operator (Restricted), Aeronautical — the operator's licence for the aircraft radio.
```

2. The "Restricted" in RTR(A) means the licence is limited to:

> Operation within the aeronautical mobile service
>
> Daylight operations only
>
> Ground operations only
>
> Any radio service worldwide

Answer: Operation within the aeronautical mobile service. It confines the privilege to the aeronautical mobile service — an aircraft radio operator, not a general operator.

3. The Aeronautical Mobile Service is a service between:

> Two land stations only
>
> Aircraft stations and aeronautical stations (and between aircraft stations)
>
> Ships and aircraft only
>
> Satellites and ground

Answer: Aircraft stations and aeronautical stations (and between aircraft stations). AMS is between aircraft stations and aeronautical (land) stations, or between aircraft stations; survival-craft/ELT may participate.

4. An "aeronautical station" is:

> A station aboard an aircraft
>
> A land station in the aeronautical mobile service
>
> A satellite
>
> An emergency beacon

Answer: A land station in the aeronautical mobile service. An aeronautical station is a land station; the airborne one is the aircraft station.

5. "Radiotelephony" is radiocommunication primarily for the exchange of:

> Data packets
>
> Morse code
>
> Information in the form of speech
>
> Radar returns

Answer: Information in the form of speech. Radiotelephony = communication primarily by speech.

6. AMS(R) is reserved for communications relating to:

> Company business
>
> Safety and regularity of flight along civil air routes
>
> Passenger entertainment
>
> Military operations only

Answer: Safety and regularity of flight along civil air routes. The "R" = Route; AMS(R) carries safety/regularity-of-flight traffic on civil routes.

7. The document that lists the examinable RTR(A) syllabus is:

> ICAO Annex 8
>
> The Aircraft Rules, 1937
>
> CAR Section 7, Series G, Part VI
>
> The ITU Radio Regulations

Answer: CAR Section 7, Series G, Part VI. The DGCA's CAR Section 7, Series G, Part VI carries the syllabus.

8. The correct top-to-bottom legal order is:

> CAR → Rules → Act
>
> Act → Rules → CAR
>
> Rules → CAR → Act
>
> Act → CAR → Rules

Answer: Act → Rules → CAR. "A Real Carrier" — Act (top statute) → Rules (made under it) → CAR (operationalises into a syllabus).

9. The principal civil aviation statute currently in force in India is the:

> Aircraft Act, 1934
>
> Bharatiya Vayuyan Adhiniyam, 2024
>
> Telegraph Act, 1885
>
> Carriage by Air Act

Answer: Bharatiya Vayuyan Adhiniyam, 2024. The 2024 Act replaced the older Aircraft Act framework.

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10. Which international instrument allocates the radio-frequency spectrum aviation may use?

> ICAO Annex 10
>
> The ITU Radio Regulations
>
> CAR Section 7
>
> The Chicago Convention

Answer: The ITU Radio Regulations. The ITU allocates spectrum; ICAO Annex 10 then standardises aviation's use of it.

11. The examination is passed only when the candidate clears:

> The theory paper alone
>
> The practical alone
>
> Both the theory and the practical/oral
>
> Either one

Answer: Both the theory and the practical/oral. Both halves must be passed (within the validity window) to be licensed.

12. Examiners most often fail candidates on:

> The application form
>
> The practical/oral phraseology test
>
> The medical
>
> The theory paper

Answer: The practical/oral phraseology test. Live phraseology under pressure trips up the under-prepared — drill it as hard as theory.

13. The RTR(A) authorises the holder to:

> Install and maintain aircraft transmitters
>
> Operate the aircraft radio and communicate with ATS
>
> Act as an air traffic controller
>
> Issue NOTAMs

Answer: Operate the aircraft radio and communicate with ATS. It is the operator's licence — communication, not engineering or control.

14. ICAO Annex 10 deals with:

> Rules of the Air
>
> Operation of Aircraft
>
> Aeronautical Telecommunications
>
> Aerodromes

Answer: Aeronautical Telecommunications. Annex 10 — Aeronautical Telecommunications — is the international standard behind the RTR syllabus.

15. The best preparation balance for the RTR(A) is:

> 100% theory, nothing spoken
>
> Theory and spoken phraseology drilled in parallel from the start
>
> Practical only in the final two days
>
> Memorise the question bank, ignore concepts

Answer: Theory and spoken phraseology drilled in parallel from the start. Run theory and live transmission practice together throughout; never defer the practical.

PART B — ORAL / VIVA (TAP TO REVEAL MODEL ANSWERS)

▶ **What does the RTR(A) licence permit, and what does "Restricted" mean?**

▶ **Define the Aeronautical Mobile Service.**

▶ **What is the difference between an aeronautical station and an aircraft station?**

▶ **Define radiotelephony.**

▶ **Name the Act, the Rules, and the syllabus document governing the RTR licence.**

▶ **What is the difference between AMS(R) and AMS(OR)?**

▶ **How is the RTR(A) examination structured?**

▶ **Which two international bodies underpin the system, and what does each do?**

60-SECOND REVISION CARD

- **RTR(A)** = operator's licence for the aircraft radio, restricted to the **Aeronautical Mobile Service** (mainly AMS(R)).
- **Aeronautical station** = land; **Aircraft station** = airborne; **Radiotelephony** = communication by speech.
- Exam = **theory** (regs / radio principles / RTF) + **practical & oral**; pass **both**. The oral fails most people.
- Law stack: **Act 2024** → **Rules 2025** → **CAR Sec 7**; foundations **ICAO Annex 10 + ITU Radio Regs**.
- Insert & memorise the exact **pass mark, validity, attempts, age** from the current CAR.

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CHAPTER 2 · SYLLABUS II

Who Governs the Airwaves

ITU, ICAO, DGCA and the Chicago Convention — the chain of organisations that turns one global agreement into the exact rules you obey on the radio.

Air Traffic Control operations coordinating flight signals across international aviation boundaries.

SYLLABUS MAP

`Part I (i) International Telecommunication Convention & Radio Regulations · International & national aviation bodies`

BY THE END OF THIS CHAPTER YOU WILL BE ABLE TO…

- Explain what the Chicago Convention is and what it created.
- Describe the roles of ICAO, the ITU and the DGCA — and how they differ.
- State what ICAO Annexes are and identify Annex 10 as the radiotelephony Annex.
- Distinguish a regulator (ICAO/DGCA) from a trade body (IATA).
- Trace the chain of authority from a global standard down to your microphone.

2.1 Why the world needed a shared rule-book

2. 2 The Chicago Convention (1944)

2.3 ICAO — the global standard-setter
2.4 ITU — the keeper of the spectrum
2.5 India's regulators — DGCA, AAI & WPC
2.6 ICAO vs IATA — regulator or club?
2.7 How it all fits together
2.8 Exam-day specifics: answering "Who publishes?"
2.9 Case study: a flight from Delhi to Dubai
2.10 Question bank

2.1 Why the world needed a shared rule-book

IN PLAIN TERMS

An aircraft that takes off in Delhi and lands in Dubai crosses the airspace of several sovereign nations in a single flight. If each country invented its own radio frequencies, its own phraseology, its own units and its own licences, international flying would be impossible — or lethal. So the world agreed, decades ago, to a shared rule-book. The organisations in this chapter are the bodies that write that rule-book and the chain that connects a global standard to the words you speak on the radio.

WHY IT WORKS THIS WAY

No single country can command another. So aviation is governed by agreement: nations sign a treaty, a UN body turns that treaty into technical standards, and each country then writes those standards into its own national law and enforces them. Understanding this three-step chain — treaty → international standards → national law — is the key to the whole chapter.

2.2 The Chicago Convention (1944)

IN PLAIN TERMS

The Chicago Convention is the founding treaty of international civil aviation — the document that made worldwide flying possible by agreeing common principles between nations. It also created ICAO to look after those principles.

DEFINITION — THE CHICAGO CONVENTION

The **Convention on International Civil Aviation**, signed at **Chicago on 7 December 1944** by 52 States. It established the principles and arrangements for safe and orderly international air transport — and created the **International Civil Aviation Organization (ICAO)**. India was an original signatory.

The 1944 Chicago Convention layout, standardizing global aviation rules and establishing ICAO.

EXAM TRAP

Memorise the date and the headline fact: **Chicago, 7 December 1944**, and **"it created ICAO."** Don't confuse the signing in 1944 with ICAO becoming operational later (1947, after the treaty entered into force).

▶ 🔍 Deep dive — the Freedoms of the Air

2.3 ICAO — the global standard-setter

IN PLAIN TERMS

ICAO is the United Nations agency for civil aviation. It writes the technical standards that let aircraft, crews and controllers from any country work together safely. Those standards live in documents called Annexes to the Chicago Convention.

Website
YouTube
Instagram
Telegram channel
Telegram community
Subject study group

International Civil Aviation Organization (ICAO) headquarters, setting international flight safety rules in Montreal.

DEFINITION — ICAO

The **International Civil Aviation Organization** — a specialised agency of the United Nations, headquartered in **Montreal, Canada**. It develops **Standards and Recommended Practices (SARPs)**, published as **19 Annexes** to the Convention.

DEFINITION — STANDARD VS RECOMMENDED PRACTICE

A **Standard** is a specification whose uniform application is recognised as *necessary* for safety; States must comply or file a **difference**. A **Recommended Practice** is *desirable* but not mandatory.

THE 19 ICAO ANNEXES — KNOW ANNEX 10

The most-tested fact in this chapter: **Annex 10 = Aeronautical Telecommunications**, the home of radiotelephony. Learn the full list, but lock Annex 10.

The 19 Annexes to the Chicago Convention

#	Annex — subject
1	Personnel Licensing
2	Rules of the Air
3	Meteorological Service for International Air Navigation
4	Aeronautical Charts
5	Units of Measurement
6	Operation of Aircraft
7	Aircraft Nationality and Registration Marks
8	Airworthiness of Aircraft
9	Facilitation
10 ☆	**Aeronautical Telecommunications — the radiotelephony Annex**
11	Air Traffic Services
12	Search and Rescue
13	Aircraft Accident and Incident Investigation
14	Aerodromes
15	Aeronautical Information Services
16	Environmental Protection
17	Security — Safeguarding against Acts of Unlawful Interference
18	The Safe Transport of Dangerous Goods by Air
19	Safety Management

MNEMONIC — ANNEX 10

"TEN = TELECOM." Annex **10** is Aeronautical **Tele**communications. The two RTR favourites: Annex **2** = Rules of the Air, Annex **10** = Telecom. (Remember the pair "2 & 10".)

ANNEX 10 HAS FIVE VOLUMES

Vol I — Radio Navigation Aids · Vol II — Communication Procedures · Vol III — Communication Systems · Vol IV — Surveillance Radar & ACAS · Vol V — Aeronautical Radio Frequency Spectrum Utilization. The RTR syllabus draws mainly on **Volumes I & II**.

2.4 ITU — the keeper of the spectrum

IN PLAIN TERMS

Radio waves are a shared natural resource. The ITU is the UN body that divides up the radio spectrum worldwide so that aviation, broadcasting, maritime, mobile phones and the military don't transmit on top of each other.

Global radio frequency allocation is partitioned by the ITU, visualized here as orbital spectrum coverage over a borderless globe.

DEFINITION — ITU

The **International Telecommunication Union** — the United Nations specialised agency for information and communication technologies, headquartered in **Geneva**. It allocates the global radio-frequency spectrum and satellite orbits and publishes the **Radio Regulations**. It is the oldest UN agency (origins in 1865).

WHY TWO BODIES?

Think of it as land and buildings. The **ITU owns the land** — it decides which frequency bands aviation may use at all. **ICAO designs the building on that land** — it decides exactly how aviation uses its allocated bands (which frequency is the emergency channel, how phraseology works, and so on). ITU allocates; ICAO applies.

EXAM TRAP

"Which body allocates radio frequency spectrum?" → **ITU**, not ICAO. "Which body publishes aviation communication procedures?" → **ICAO (Annex 10)**. Examiners deliberately swap these.

2.5 India's regulators — DGCA, AAI & WPC

IN PLAIN TERMS

ICAO's standards are not law in India until India adopts them. The bodies below are how India turns global standards into national rules, runs its airspace, and licenses radio operators.

AAI Airport operations in India translating global guidelines into domestic flight safety procedures.

THE THREE YOU MUST KNOW

DGCA (Directorate General of Civil Aviation) — India's civil aviation regulator under the Ministry of Civil Aviation; it adopts ICAO standards into national rules and publishes the **Civil Aviation Requirements (CARs)**, including the RTR syllabus.

AAI (Airports Authority of India) — manages airports and provides air navigation services (ATS and communications/navigation/surveillance) across Indian airspace.

WPC (Wireless Planning & Coordination Wing) — the national radio regulator under the Ministry of Communications; it licenses the use of radio spectrum and apparatus.

2.6 ICAO vs IATA — regulator or club?

> **IN PLAIN TERMS**
>
> These two are constantly confused because the acronyms look alike. ICAO is a **regulator** made of governments. IATA is a **trade association** made of airlines.

Comparison: ICAO vs IATA

Feature	ICAO	IATA
What it is	UN specialised agency (States)	Trade association (airlines)
Members	Sovereign States	Airlines
Role	Safety standards & SARPs (Annexes)	Commercial coordination, fares, industry practice
HQ	Montreal	Montreal / Geneva
Airport codes	4-letter location indicator (e.g. VIDP)	3-letter code (e.g. DEL)

> **MNEMONIC**
>
> **ICAO = countries, IATA = airlines.** "C" in ICAO → Countries. And for codes: ICAO = **4** letters (VIDP), IATA = **3** letters (DEL).

2.7 How it all fits together

> **SEE THE FIGURE**
>
> From a global resource down to your transmit button, authority flows in a clear chain — spectrum from the ITU, aviation standards from ICAO, national adoption and licensing by India's bodies, and finally the RTR licence that authorises you.

Figure 2.1 — The chain of authority — from global agreement to your microphone.

COCKPIT REALITY

When you say "QNH 1013, runway two-seven" to a controller in any country, it works because of this chain: ITU reserved the band, ICAO standardised the phraseology and the Q-code in Annex 10, and India's DGCA wrote it into the rules you were tested on. You are the last link — and the only one the passengers can hear.

2.8 Exam-day specifics: answering "Who publishes?"

IN PLAIN TERMS

Examiners love to ask "Who publishes X?" A candidate who confuses a national document with an international one loses the mark. Here is the definitive list to lock down.

Publications & their Authors

Document / Publication	Published By
Radio Regulations (RR)	ITU (International Telecommunication Union)
Annex 10 (Aeronautical Telecommunications)	ICAO (International Civil Aviation Organization)
AIP (Aeronautical Information Publication) India	AAI (Airports Authority of India)
Civil Aviation Requirements (CARs)	DGCA (Directorate General of Civil Aviation)
RTR(A) Syllabus	DGCA (via CAR Section 7)

EXAM TRAP

Do not say the DGCA publishes the AIP. **AAI publishes the AIP**. The DGCA regulates.

2.9 Case study: a flight from Delhi to Dubai

PUTTING IT ALL TOGETHER

Imagine you are departing Delhi for Dubai. How do the bodies in this chapter impact your flight?

1. **WPC** granted the licence for the VHF radio installed in your aircraft.
2. You hold an **RTR(A)** issued under rules made by the Indian Government, governed by a syllabus from the **DGCA**.
3. You request start-up from Delhi Delivery on 121.900 MHz — a frequency allocated globally to aviation by the **ITU**.
4. You are talking to an Air Traffic Controller employed by the **AAI**.
5. You cross into Pakistan airspace, then Iran, then UAE. The reason the controllers in all those countries speak English and use the same phraseology you do is because they all signed the **Chicago Convention** and adopted **ICAO Annex 10**.

☆ Numbers to memorise

Essential Facts for Chapter 2

Fact	Value
Chicago Convention signed	7 December 1944, Chicago — by 52 States; created ICAO
ICAO	UN agency · HQ Montreal · 19 Annexes (SARPs)
Annex 10	Aeronautical Telecommunications (5 volumes; RTR uses Vol I & II)
Annex 2 / Annex 11 / Annex 15	Rules of the Air / Air Traffic Services / Aeronautical Information Services
ITU	UN agency · HQ Geneva · allocates spectrum · Radio Regulations · oldest UN agency
DGCA	India's regulator · issues CARs (RTR syllabus = CAR Sec 7, Series G, Part VI)
ICAO vs IATA	ICAO = States/regulator (4-letter codes); IATA = airlines/trade (3-letter codes)

Question bank

PART A — MCQS (CLICK AN OPTION TO CHECK)

1. The Chicago Convention of 1944 created which organisation?

> A. ICAO
>
> B. ITU
>
> C. IATA
>
> D. DGCA

```
Answer: A. The Convention on International Civil Aviation (Chicago, 7 Dec 1944) established ICAO.
```

2. Which ICAO Annex covers Aeronautical Telecommunications?

> A. Annex 2
>
> B. Annex 6
>
> C. Annex 10
>
> D. Annex 11

```
Answer: C. Annex 10 — "TEN = TELECOM." Annex 2 is Rules of the Air, Annex 6 Operation of Aircraft, Annex 11 ATS.
```

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3. Which body allocates the global radio-frequency spectrum?

> A. ICAO
>
> B. ITU
>
> C. DGCA
>
> D. IATA

```
Answer: B. The ITU allocates spectrum and publishes the Radio Regulations; ICAO then decides aviation's use of it.
```

4. ICAO is headquartered in:

> A. Geneva
>
> B. Montreal
>
> C. Chicago
>
> D. London

```
Answer: B. ICAO — Montreal. (The ITU is in Geneva.)
```

5. Which statement about IATA is correct?

> A. It is a UN regulatory agency
>
> B. It is a trade association of airlines
>
> C. It allocates radio spectrum
>
> D. It issues the RTR licence

```
Answer: B. IATA is an airline trade body — commercial, not regulatory. ICAO is the regulator of States.
```

6. An ICAO "Standard" differs from a "Recommended Practice" in that a Standard is:

> A. Necessary for safety; States must comply or file a difference
>
> B. Only a suggestion with no obligation
>
> C. Applicable only to IATA airlines
>
> D. Issued by the ITU

```
Answer: A. A Standard is necessary and binding unless a State files a difference; a Recommended Practice is desirable but optional.
```

7. In India, the document carrying the RTR(A) syllabus is published by:

> A. ICAO
>
> B. ITU
>
> C. DGCA (as a CAR)
>
> D. IATA

`Answer: C. The DGCA issues Civil Aviation Requirements; the RTR syllabus is CAR Section 7, Series G, Part VI.`

8. Which body publishes the Aeronautical Information Publication (AIP) in India?

> A. AAI
>
> B. DGCA
>
> C. ICAO
>
> D. WPC

`Answer: A. The Airports Authority of India (AAI) publishes the AIP. The DGCA is the regulator.`

PART B — ORAL / VIVA (TAP TO REVEAL MODEL ANSWERS)

▶ **What is the Chicago Convention and what did it create?**

▶ **What is the difference between ICAO and the ITU?**

▶ **Which ICAO Annex deals with radiotelephony, and what are its volumes?**

▶ **What is the difference between ICAO and IATA?**

▶ **Name the main Indian bodies in civil aviation and their roles.**

▶ **Who publishes the Radio Regulations?**

30-SECOND REVISION CARD

- **Chicago Convention, 7 Dec 1944** → created **ICAO** (HQ Montreal, 19 Annexes).
- **Annex 10 = Telecom** (5 vols; RTR uses I & II). Also know Annex 2, 11, 15.
- **ITU** (Geneva) allocates **spectrum**; ICAO applies it to aviation.

- **DGCA** adopts standards as **CARs**; AAI runs ATS; WPC licenses radio.
- **ICAO** = States/regulator (4-letter); **IATA** = airlines/trade (3-letter).
- **AAI** publishes the AIP.

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Licensing, Radio Apparatus & Minimum Equipment

A legal radio call needs two licences working together — one for the operator, one for the box — and the aircraft must carry a defined minimum of communication, navigation and surveillance equipment set by ICAO Annex 6. This chapter unpacks the regulatory machinery behind a single "Delhi Tower, good morning".

SYLLABUS MAP

Part I (vi) Licensing of radio apparatus · (vii) Minimum radio equipment per Annex 6 (ICAO & DGCA)

LEARNING OBJECTIVES — BY THE END OF THIS CHAPTER YOU WILL BE ABLE TO…

- Distinguish the operator's licence (RTR) from the aircraft radio station licence, and name the authority that issues each.
- Describe how radio apparatus is licensed in India (WPC, NFAP, SACFA) and why type approval matters.
- State the minimum communication, navigation and surveillance equipment Annex 6 requires, and how it scales with the operation.
- Explain the independence (no single-point-failure) requirement and apply it.
- Describe ELT and transponder carriage and the role of the Minimum Equipment List (MEL).
- Explain how India's CARs give Annex 6 domestic effect.

3.1 Why two licences, not one

3.2 The operator's licence (RTR)

3.3 Licensing the apparatus — WPC, NFAP, SACFA

3.4 Annex 6 — the minimum-equipment standard

3.5 Communication equipment required

3.6 Navigation & surveillance required

3.7 The independence rule

3.8 ELT & emergency equipment

3.9 The MEL — flying with an item u/s

3.10 How India applies Annex 6

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3.1 Why two licences, not one

FIRST PRINCIPLES — POLICING THE SPECTRUM AT TWO LEVELS

The radio-frequency spectrum is policed at two independent levels because two different things can go wrong. The **person** can be untrained — so the operator is licensed (the RTR). The **equipment** can be the wrong type, off-frequency, or interfering — so the apparatus is licensed too (the aircraft radio station licence). Remove either control and the system fails: a trained pilot on an unapproved, interfering transmitter is as dangerous as an approved radio in untrained hands. To make a single legal transmission, **both** authorisations must exist at the same time.

The two authorisations

Operator's licence — RTR(A): authorises a *person* to operate aeronautical radio equipment and use the airwaves.
Aircraft Radio Station Licence / installation approval: authorises the *radio apparatus installed in a particular aircraft* to transmit on assigned frequencies. Issued by the national radio authority and carried/registered with the aircraft.

EXAM TRAP

"What licences are needed to operate an aircraft radio?" — the complete answer is **both**: the operator's RTR *and* the aircraft radio station licence. Naming only one is the classic half-mark answer.

3.2 The operator's licence (RTR)

IN PLAIN TERMS

The RTR(A) is the certificate covered in Chapter 1 — the licence of the *person* at the microphone. It certifies that the operator knows the regulations, the radio theory, and the disciplined phraseology needed to use a shared, safety-critical frequency.

Where the RTR sits in the carriage rules

The aircraft may carry a fully approved, licensed radio installation, but it may only be **operated** by a person holding a valid radio operator's licence appropriate to the equipment — for civil aircraft radiotelephony, the **RTR(A)**. Operation by an unlicensed person is an offence even if the equipment itself is fully licensed.

Author check before publishing

Confirm the exact wording linking operator licensing to operation of aircraft radio under the RTR (Restricted) Rules, 2025, and any categories of person permitted to operate under supervision/exemption.

3.3 Licensing the apparatus — WPC, NFAP, SACFA

FIRST PRINCIPLES — WHY A RADIO CANNOT JUST TRANSMIT

A transmitter radiates energy into shared spectrum. If it is the wrong type, drifts off frequency, or radiates spurious emissions, it interferes with everyone else on the band — potentially blocking a distress call. So the State requires that each radio station be **type-approved** and **licensed**, and that its frequencies be **assigned** from the national plan so two stations don't clash.

Aircraft antennas are part of the licensed radio station apparatus and are governed by WPC and DGCA fitment regulations.

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The Indian apparatus-licensing machinery

WPC (Wireless Planning & Coordination) Wing, Ministry of Communications — the national radio regulator: it licenses the use of radio spectrum and apparatus and assigns frequencies.

NFAP (National Frequency Allocation Plan) — India's table of which services may use which bands, derived from the ITU Radio Regulations.

SACFA (Standing Advisory Committee on Frequency Allocation) — clears siting of transmitters/masts.

The **DGCA** separately regulates the **airworthiness and operational fitment** of the equipment in the aircraft.

Mnemonic — who does what

WPC licenses the box; DGCA approves the fitment; you (RTR) operate it. Spectrum & apparatus = Communications ministry; airworthiness & operation = Civil Aviation.

Author check before publishing

Confirm the exact current name and validity period of the **aircraft radio station licence** in India and whether the 2025 framework changed the issuing route. Insert the validity figure if examinable.

3.4 Annex 6 — the minimum-equipment standard

IN PLAIN TERMS

An aircraft may not set off carrying whatever radios the owner happens to fancy. ICAO **Annex 6 (Operation of Aircraft)** prescribes the *minimum* communication, navigation and surveillance equipment that must be carried — and the requirement **scales with the kind of flying**: more is required for IFR and night than for simple day-VFR.

Definition — ICAO Annex 6

Annex 6 — Operation of Aircraft. Part I covers international commercial air transport (aeroplanes); **Part II** international general aviation (aeroplanes); **Part III** helicopters. It prescribes, among much else, the minimum communication, navigation and surveillance equipment to be carried.

Figure 3.1 — ICAO Annex 6 minimum equipment scaling from Day VFR to IFR complex.

3.5 Communication equipment required

The core radio requirement (Annex 6)

An aircraft shall carry radio communication equipment capable of:

• conducting **two-way communication** with the appropriate aeronautical stations on the frequencies required by the authority;

• receiving **meteorological information at any time** during flight;

• conducting two-way communication **at any time** with at least one aeronautical station and such other stations/frequencies as prescribed for the route.

A dual VHF COM setup ensuring two-way communication and met broadcast reception at any time during flight.

WHY THE WORDING IS "TWO-WAY" AND "AT ANY TIME"

Notice the standard is written around **capability and availability**, not mere possession of a box. A receive-only set, or one that cannot reach a met broadcast en route, would not satisfy it. This is precisely why airliners carry **multiple independent COM radios** and can tune VOLMET/ATIS — the rule demands the capability to talk both ways and to get weather throughout the flight.

3.6 Navigation & surveillance required

Navigation & surveillance (Annex 6)

The aircraft shall also carry **navigation equipment** sufficient to proceed in accordance with its **flight plan** and the requirements of air traffic services; and, where required for the airspace, a **transponder (SSR/Mode S)** and other surveillance equipment. For **IFR or night** operations the standard rises to redundant communication and navigation equipment appropriate to the route, and for precision approaches the relevant ILS/radio-altimeter fit.

Cockpit reality — "RNP/PBN approval"

"Sufficient to proceed per flight plan" is why an aircraft must hold the right **navigation approval** (e.g. RNP/PBN, Chapter 12) for the route it files. File an RNP route without the equipment and approval, and you are below the Annex 6 minimum for that flight.

3.7 The independence rule

No single point of failure

Communication and navigation equipment shall be installed so that the **failure of any single unit** required for either communication or navigation purposes **will not result in the failure of another** unit required for the other purpose. In plain words: **one failure must not take out both your talking and your navigating.**

Worked example — applying the rule

Two VHF radios and two navigation receivers share a **single power bus** and a **single antenna coupler**.
Does this meet the independence rule? **No** — the shared bus (or coupler) is a single point whose failure disables comms *and* nav together.
Compliant design: separate power sources/buses and separate antenna feeds, so any one failure leaves the other function working. This is why airliners split avionics across independent buses and generators.

3.8 ELT & emergency equipment

Emergency Locator Transmitter

Annex 6 requires the carriage of an **ELT** (the number/type depending on the operation). It activates automatically on impact (or manually) and transmits on the **406 MHz** satellite distress frequency (with a coded, registered identity) and **121.5 MHz** for homing, to summon search and rescue. (Full detail in Chapter 22.)

Author check before publishing

Insert the exact Indian CAR figures: number/type of **ELTs** required by operation, mandatory **transponder (Mode S / ADS-B)** carriage threshold, and the precise minimum-equipment wording from the current DGCA CAR mirroring Annex 6.

3.9 The MEL — flying with an item unserviceable

IN PLAIN TERMS

If a required item is unserviceable, the aircraft is not automatically grounded. The **Minimum Equipment List** tells the crew whether, and under what conditions, the flight may still be made.

MEL & MMEL

The **MMEL (Master MEL)** is produced by the manufacturer/State of design and lists items that may be inoperative for a type. The operator's **MEL** is at least as restrictive and approved for that operator. It specifies any limitations and "O" (operational) or "M" (maintenance) procedures required to dispatch with the item inoperative. The Annex 6 minimum-equipment principle is exactly what the MEL enforces: **never depart below the legal minimum for the intended flight.**

3.10 How India applies Annex 6

IN PLAIN TERMS

Annex 6 is the global standard; the DGCA writes the Indian version into its Civil Aviation Requirements, so the minimum-equipment rule you are tested on is the Indian CAR that mirrors Annex 6 — confirmed at every pre-flight by checking the required radios are fitted, serviceable, and (if not) permitted by the MEL.

▶ 🔍 Deep dive — type approval & spurious emissions

☆ Numbers to memorise

Essential Facts for Chapter 3

Fact	Value
Licences needed	Operator's RTR(A) + Aircraft Radio Station Licence (apparatus)
Apparatus regulator (India)	WPC Wing, Ministry of Communications (NFAP, SACFA)
Fitment/airworthiness regulator	DGCA
Minimum-equipment standard	ICAO Annex 6 — Operation of Aircraft (Parts I/II/III)
Core radio capability	Two-way comms on required frequencies + receive met at any time
Navigation requirement	Sufficient to proceed per flight plan + ATS requirements (+ transponder)
Independence rule	Single failure must not disable both comms and navigation
Emergency beacon	ELT — 406 MHz (satellite) + 121.5 MHz (homing)
Dispatch with item u/s	Governed by the MEL (derived from the MMEL)

Question bank

PART A — MCQS (CLICK AN OPTION TO CHECK)

1. To legally operate an aircraft radio you require:

> Only the RTR(A)
>
> Only the aircraft radio station licence
>
> Both the RTR(A) and the aircraft radio station licence
>
> Neither, below 5,700 kg

```
Answer: Both the RTR(A) and the aircraft radio station licence. Both — the operator (RTR) and the installed apparatus (station licence) must be licensed.
```

2. The minimum radio equipment to be carried on an aircraft is prescribed in:

> ICAO Annex 10
>
> ICAO Annex 6
>
> ICAO Annex 2
>
> ICAO Annex 11

```
Answer: ICAO Annex 6. Annex 6 — Operation of Aircraft. Annex 10 is the telecom standards themselves.
```

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3. In India, radio spectrum and apparatus are licensed by:

> DGCA
>
> AAI
>
> WPC (Ministry of Communications)
>
> ICAO

Answer: WPC (Ministry of Communications). The WPC Wing; DGCA handles airworthiness/operational fitment.

4. India's table of which services may use which bands is the:

> MEL
>
> NFAP (National Frequency Allocation Plan)
>
> CAR Section 7
>
> AIP

Answer: NFAP (National Frequency Allocation Plan). The NFAP, derived from the ITU Radio Regulations, allocates Indian spectrum.

5. SACFA is concerned primarily with:

> Pilot licensing
>
> Clearing the siting of transmitters/masts
>
> Weather forecasting
>
> Accident investigation

Answer: Clearing the siting of transmitters/masts. The Standing Advisory Committee on Frequency Allocation clears transmitter siting.

6. Annex 6 Part I applies to:

> Helicopters
>
> International general aviation aeroplanes
>
> International commercial air transport (aeroplanes)
>
> Gliders only

Answer: International commercial air transport (aeroplanes). Part I = commercial air transport aeroplanes; Part II = international GA; Part III = helicopters.

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7. The core Annex 6 radio capability includes the ability to:

> Transmit only

> Conduct two-way communication and receive meteorological information at any time

> Receive broadcast entertainment

> Operate without an assigned frequency

Answer: Conduct two-way communication and receive meteorological information at any time. Two-way comms on required frequencies plus the ability to receive met information at any time.

8. The "independence" requirement means:

> Each radio must be a different brand

> A single failure must not disable both communication and navigation

> The pilot must operate independently of ATC

> All equipment must be battery-powered

Answer: A single failure must not disable both communication and navigation. Installation must ensure one unit's failure won't take out another required for the other function.

9. Two COM and two NAV sets sharing a single power bus:

> Always meet the independence rule

> May fail the independence rule because the shared bus is a single point of failure

> Need no antenna

> Are exempt from Annex 6

Answer: May fail the independence rule because the shared bus is a single point of failure. A shared bus whose failure kills both comms and nav violates the no-single-point-of-failure principle.

10. Navigation equipment under Annex 6 must be sufficient to:

> Reach any airport in the world

> Proceed in accordance with the flight plan and ATS requirements

> Replace the transponder

> Operate without approval

Answer: Proceed in accordance with the flight plan and ATS requirements. It must let the aircraft fly its filed plan and meet ATS requirements (e.g. PBN where required).

11. An ELT transmits on:

> 118.0 MHz only
>
> 406 MHz (satellite) and 121.5 MHz (homing)
>
> 1030 MHz
>
> 75 MHz

Answer: 406 MHz (satellite) and 121.5 MHz (homing). 406 MHz for satellite detection with coded ID, plus 121.5 MHz for homing.

12. If a required item is unserviceable before departure, the document that governs whether the flight may proceed is the:

> AIP
>
> NOTAM
>
> Minimum Equipment List (MEL)
>
> NFAP

Answer: Minimum Equipment List (MEL). The MEL (from the MMEL) states whether and how the aircraft may dispatch with the item inoperative.

13. The MMEL is produced by:

> The operator
>
> The manufacturer / State of design
>
> The crew
>
> The ITU

Answer: The manufacturer / State of design. The Master MEL comes from the manufacturer/State of design; the operator's MEL is at least as restrictive.

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14. Type approval of a radio mainly ensures it:

> Is the cheapest available

> Transmits within its assigned channel with low spurious emissions

> Needs no licence

> Works only on HF

Answer: Transmits within its assigned channel with low spurious emissions. Type approval confirms controlled frequency/power and acceptably low spurious emissions to avoid interference.

15. Which body regulates the airworthiness and fitment of the radio in the aircraft?

> WPC

> DGCA

> ITU

> IATA

Answer: DGCA. DGCA regulates airworthiness/fitment; WPC licenses the spectrum/apparatus.

16. Compared with day-VFR, IFR/night operations under Annex 6 generally require:

> Less equipment

> More — including redundant comms/nav appropriate to the route

> No transponder

> No navigation equipment

Answer: More — including redundant comms/nav appropriate to the route. The minimum scales up with the operation; IFR/night needs redundancy.

17. The requirement to "receive meteorological information at any time" means a receive-only set that cannot reach a met broadcast en route:

> Is sufficient

> Does not satisfy Annex 6

> Is required by Annex 10

> Is exempt for IFR

Answer: Does not satisfy Annex 6. The standard is about capability/availability — it must be able to obtain weather throughout the flight.

18. The complete answer to "what is needed to make a legal aircraft radio transmission" is:

> A serviceable radio
>
> A licensed operator
>
> A licensed operator (RTR) AND a licensed/approved aircraft radio station
>
> ATC permission only

```
Answer: A licensed operator (RTR) AND a licensed/approved aircraft radio station. Both the person and the apparatus must be licensed simultaneously.
```

PART B — ORAL / VIVA (TAP TO REVEAL MODEL ANSWERS)

▶ **What licences are required to operate the radio on an aircraft, and who issues each?**

▶ **Where is the minimum radio equipment to be carried prescribed, and how is it structured?**

▶ **State the basic communication requirement under Annex 6.**

▶ **Explain the independence requirement and give an example.**

▶ **What is an ELT and on what frequencies does it transmit?**

▶ **What is the MEL and why does it exist?**

60-SECOND REVISION CARD

- Two licences: **RTR(A)** (operator) + **aircraft radio station licence** (apparatus, via WPC).
- India: **WPC** = spectrum/apparatus (NFAP, SACFA); **DGCA** = airworthiness/fitment.
- Minimum-equipment standard = **ICAO Annex 6**; scales up for IFR/night.
- Core radio: **two-way comms** on required frequencies + **receive met any time**; nav per flight plan + transponder.
- **Independence:** one failure must not kill both comms and nav. **ELT** 406 + 121.5; dispatch u/s governed by the **MEL**.

Q-Codes, Abbreviations & Definitions

Three-letter shorthand born in the Morse era still runs the cockpit today. The pressure and bearing Q-codes are among the most reliable marks on the paper and the viva — and a wrong altimeter setting is one of aviation's oldest killers. This chapter builds them from the physics up.

SYLLABUS MAP

Part I (ii) General & Aeronautical 'Q' Code signals and other abbreviations (ICAO Annex 10, Vol I & II)

LEARNING OBJECTIVES — BY THE END OF THIS CHAPTER YOU WILL BE ABLE TO…

- Explain what a Q-code is, where it came from, and why aviation keeps a few.
- Define and apply QNH, QFE, QNE and QFF, and explain the altimetry behind each.
- Explain transition altitude/level and when each pressure setting is used.
- Distinguish the bearing codes QDM, QDR, QTE, QUJ (and QGH) and solve bearing problems.
- Recall common operational Q-codes and standard abbreviations.
- State precisely the definitions of altitude, height, elevation and flight level.

4.1 What a Q-code is
4.2 The pressure-altimeter — how it works
4.3 QNH, QFE, QNE & QFF
4.4 Transition altitude & level
4.5 The bearing Q-codes
4.6 Working bearing problems
4.7 Other operational Q-codes
4.8 Standard abbreviations
4.9 Altitude · height · elevation · flight level

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4.1 What a Q-code is

FIRST PRINCIPLES — THREE LETTERS FOR A WHOLE SENTENCE

Q-codes were invented for **Morse telegraphy**, where every character costs time. A three-letter group beginning with **Q** was agreed to stand for a complete phrase, so a long sentence could be sent as three letters. Cleverly, the same code can be a **question or its answer** depending on context: "QFE?" asks the field pressure; "QFE 1004" answers it. Voice radio later made full sentences cheap, so most Q-codes faded — but a handful proved so **compact and unambiguous** (and impossible to mishear as ordinary words) that aviation kept them: above all the **pressure settings** and a few **bearings**.

Definition — Q-code

A standardised three-letter code, each group beginning with **Q**, representing a fixed phrase, listed in ICAO documentation (Annex 10 and related). It is international and language-independent, and may be used as a question or an answer.

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Navigation and pressure instruments rely heavily on correct Q-code settings like QNH and QDM for safe operation.

4.2 The pressure-altimeter — how it works

FIRST PRINCIPLES — PRESSURE AS A PROXY FOR HEIGHT

Atmospheric pressure **falls with height** — roughly **1 hPa per 27–30 ft** near sea level. A pressure altimeter is simply an aneroid barometer with its scale marked in feet: it measures the outside static pressure and displays the corresponding altitude. But "the corresponding altitude" depends entirely on **what pressure you call the datum**. The **subscale (Kollsman window)** lets the pilot set that datum. Setting a different datum slides the whole scale up or down — which is exactly what the pressure Q-codes select.

4.3 QNH, QFE, QNE & QFF

Pressure Q-Codes

Code	What you set on the subscale	Altimeter then reads
QNH	Pressure reduced to mean sea level for the area	**Altitude** above mean sea level — and the aerodrome **elevation** when on the ground
QFE	The actual pressure at the aerodrome / runway datum	**Height** above that datum — **zero** on the runway
QNE	The standard setting **1013.2 hPa** (29.92 inHg)	**Pressure altitude** / flight level
QFF	The *true* mean-sea-level pressure (meteorological)	Used by met services; not normally set for flying

The rule of thumb to carry

≈ 1 hPa = 27 ft near sea level (often rounded to 30 ft). So a **1 hPa** error in the subscale setting puts the altimeter out by ≈ 27 ft; a **10 hPa** error ≈ 270 ft. Always set the *correct* datum — this small number is why a wrong setting is so dangerous near the ground.

Worked example — a mis-set subscale

QNH is 1013 hPa but you mistakenly set **993 hPa** (20 hPa low).
Error ≈ 20 hPa × 27 ft = **540 ft**. With the lower pressure set, the altimeter **over-reads** — it shows you higher than you are, so you are actually ~540 ft **lower** than indicated. On an approach in cloud, that is the difference between clearing terrain and not.
Memory aid: "High to Low, look out below." Flying from high pressure to low (or setting too low a value) makes you lower than indicated.

Mnemonics

QFE = Field Elevation → reads **zero on the Field**. **QNH = Nautical Height** → reads height above the **sea** (altitude AMSL). **QNE = Normal/National standard** → the fixed **1013** setting for flight levels.

Exam trap — the word matters

"Altitude" vs "height" is the catch. **QNH** → **altitude** (above sea level). **QFE** → **height** (above the airfield). Naming the right code but the wrong word loses the mark. And QNE is a **fixed value** (1013.2), not a number passed by ATC.

▶ 🔍 Deep dive — QNH vs QFF

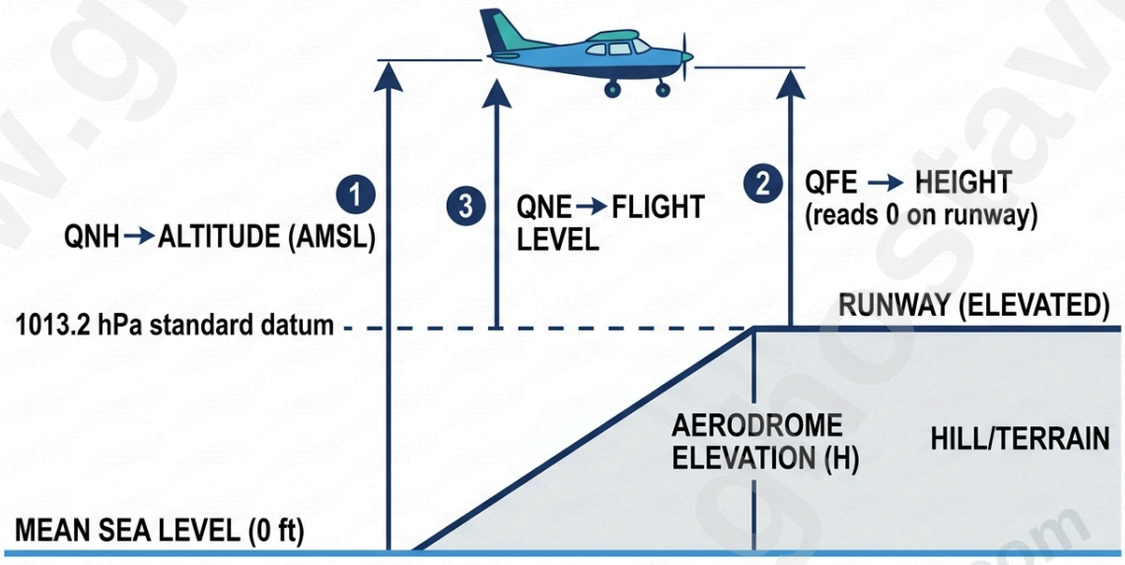

Figure 4.1 — QNH, QFE and QNE altimeter datums visualized relative to sea level, runway, and the standard pressure surface.

4.4 Transition altitude & level

IN PLAIN TERMS

Low down, near terrain, everyone wants altitude above the actual sea-level pressure (QNH). High up, for traffic separation, everyone wants a *common* datum so two aircraft's altimeters agree — so all switch to the standard 1013.2 (QNE) and call levels "flight levels". The switch happens at a defined boundary.

The three terms

Transition Altitude — the altitude at or below which vertical position is expressed as **altitude on QNH. Transition Level** — the lowest **flight level (on 1013.2)** available above the transition altitude. **Transition Layer** — the airspace between the two. **Climbing:** set 1013 (QNE) passing the transition **altitude**. **Descending:** set QNH passing the transition **level**.

Mnemonic

Climb → Standard at the Altitude; Descend → QNH at the Level. Going up you reach the altitude first; coming down you reach the level first.

4.5 The bearing Q-codes

IN PLAIN TERMS

These four answer "which way?" relative to a station. Two are **magnetic** (QDM, QDR), two are **true** (QUJ, QTE); two point **to** the station, two point **from** it.

Bearing Q-Codes

Code	Meaning	Reference
QDM	Magnetic **heading to steer** (zero wind) **to** reach the station	Magnetic · TO
QDR	Magnetic **bearing from** the station (the radial)	Magnetic · FROM
QUJ	**True** bearing **to** the station	True · TO
QTE	**True** bearing **from** the station	True · FROM

The reciprocal relationships

QDR = QDM ± 180° (the bearing from is the reciprocal of the heading to). Likewise **QTE = QUJ ± 180°**. And magnetic ↔ true differs by the local **variation**: True = Magnetic ± Variation (East least, West most → "Variation East, magnetic least").

Mnemonic

QDM = Direction to Mother — the magnetic heading home *to* the station. Its reciprocal **QDR** goes away. Letter cue: the **M**agnetic pair share "QD" (QD**M**/QDR); the **T**rue pair are QUJ/Q**T**E.

▶ 🔍 Deep dive — QGH, QGE & other navigation Q-codes

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4.6 Working bearing problems

Understanding QDM (TO) and QDR (FROM) is fundamental for VOR and NDB navigation.

> **Worked example 1 — heading to a station**
>
> An ADF gives QDM **090°** to an NDB.
>
> QDR (radial you are on) = 090 ± 180 = **270°**.
>
> So you are on the 270° radial (west of the station) and would steer 090° in nil wind to reach it.

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> **Worked example 2 — magnetic to true**
>
> QDM is **090°** and the local magnetic variation is **5° West**.
> True bearing to the station (QUJ): with West variation, True = Magnetic − Variation = 090 − 5 = **085°**.
> QTE (true from) = 085 ± 180 = **265°**. ("Variation West, magnetic best" — the magnetic figure is the larger.)

4.7 Other operational Q-codes

Other Operational Q-codes

Code	Meaning
QSY	Change frequency (to …)
QAM	The latest meteorological report (for …)
QFE / QNH / QNE	Pressure settings (see 4.3)
QDM / QDR / QUJ / QTE	Bearings (see 4.5)
QGH	Controlled descent through cloud (DF let-down)
QTF	Position (fix) from DF bearings

4.8 Standard abbreviations you must read on sight

Standard Aviation Abbreviations

Abbr.	Meaning	Abbr.	Meaning
UTC	Coordinated Universal Time ("Zulu")	**AMSL/AGL**	Above Mean Sea Level / Above Ground Level
ATIS	Automatic Terminal Information Service	**VOLMET**	En-route met broadcast
METAR/TAF	Weather report / aerodrome forecast	**RVR**	Runway Visual Range
CAVOK	Ceiling and visibility OK (defined)	**WILCO**	Will comply
ROGER	Received all your last transmission	**SELCAL**	Selective Calling (tone alert)
ETA/ETD	Estimated Time of Arrival / Departure	**QDM**	Magnetic heading to a station

4.9 Altitude · height · elevation · flight level

The four vertical definitions — recite precisely

Altitude — the vertical distance of a level, point or object measured **from mean sea level** (read on QNH).
Height — the vertical distance of a level, point or object measured **from a specified datum**, e.g. the aerodrome (read on QFE).
Elevation — the vertical distance of a *point on or affixed to the surface* above mean sea level.
Flight Level (FL) — a surface of constant atmospheric pressure related to the standard datum **1013.2 hPa** (read on QNE); e.g. FL100 = 10,000 ft pressure altitude.

Cockpit reality — three settings in one flight

On departure you fly on **QNH** reading altitude; passing the transition altitude you wind the subscale to **1013** (QNE) and read flight levels; arriving at a small field you might use **QFE** so the altimeter reads height above the runway. Three codes, three readings — set the wrong one and your "1,000 ft" may be 1,000 ft lower than you think (recall 4.2).

☆ Numbers to memorise

Essential Facts for Chapter 4

Fact	Value
Pressure lapse	≈ 1 hPa per 27 ft near sea level
QNH	Set → altimeter reads ALTITUDE (AMSL); reads elevation on the ground
QFE	Set → reads HEIGHT above aerodrome; zero on the runway
QNE	Standard 1013.2 hPa / 29.92 inHg → FLIGHT LEVEL
QFF	True MSL pressure — meteorological, not flown
Transition	Climb: set 1013 at transition ALTITUDE · Descend: set QNH at transition LEVEL
QDM / QDR	Magnetic heading TO / bearing FROM = QDM ± 180°
QUJ / QTE	True bearing TO / FROM the station
QSY / QGH	Change frequency / controlled descent (DF let-down)

Question bank

PART A — MCQS (CLICK AN OPTION TO CHECK)

1. With QNH set, the altimeter reads:

> Altitude above mean sea level
>
> Height above the aerodrome
>
> Flight level
>
> Zero on the runway

```
Answer: Altitude above mean sea level. QNH → altitude (AMSL); on the ground it reads the aerodrome elevation.
```

2. With QFE set, on the runway the altimeter reads:

> The aerodrome elevation
>
> Zero
>
> 1013 hPa
>
> The flight level

```
Answer: Zero. QFE is the aerodrome datum pressure, so it reads height above it — zero on the field.
```

3. The standard pressure setting QNE is:

> 1003.2 hPa
>
> 1013.2 hPa (29.92 inHg)
>
> 1023.2 hPa
>
> 760 hPa

```
Answer: 1013.2 hPa (29.92 inHg). QNE = 1013.2 hPa / 29.92 inHg — used for flight levels.
```

4. Approximately how much does 1 hPa correspond to in altitude near sea level?

> 2.7 ft
>
> 27 ft
>
> 270 ft
>
> 1 ft

```
Answer: 27 ft. ≈ 1 hPa per 27 ft near sea level — a 10 hPa error ≈ 270 ft.
```

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5. You set 993 hPa when the QNH is 1013 hPa. The altimeter will:

> Read correctly
>
> Under-read (show you lower than you are)
>
> Over-read (show you higher than you are — you are lower)
>
> Read zero

Answer: Over-read (show you higher than you are — you are lower). Setting too low a value over-reads: "High to Low, look out below" — you are about 540 ft lower than indicated.

6. QFF differs from QNH in that QFF is:

> The aerodrome pressure
>
> The true MSL pressure (meteorological), not normally flown
>
> The standard 1013 setting
>
> A bearing

Answer: The true MSL pressure (meteorological), not normally flown. QFF gives the true sea-level pressure for met charts; QNH (using ISA) makes the altimeter read field elevation.

7. When climbing, you set 1013 hPa (standard) on passing the:

> Transition level
>
> Transition altitude
>
> Transition layer top
>
> Aerodrome elevation

Answer: Transition altitude. Climb → standard at the transition ALTITUDE; descend → QNH at the transition LEVEL.

8. QDM is the:

> Magnetic heading to steer (zero wind) to the station
>
> Magnetic bearing from the station
>
> True bearing from the station
>
> Distance to the station

Answer: Magnetic heading to steer (zero wind) to the station. QDM — magnetic heading TO the station ("Direction to Mother"). QDR is its reciprocal, FROM.

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9. If QDM is 090°, the corresponding QDR is:

> 090°
>
> 180°
>
> 270°
>
> 000°

Answer: 270°. QDR = QDM ± 180°, so 090° → 270°.

10. QTE is a:

> Magnetic heading to the station
>
> True bearing from the station
>
> Pressure setting
>
> Request to change frequency

Answer: True bearing from the station. QTE — True bearing FROM the station. QUJ is True bearing TO.

11. QDM 100° with variation 6° East gives a true bearing to the station (QUJ) of:

> 094°
>
> 106°
>
> 100°
>
> 286°

Answer: 106°. Variation East → True = Magnetic + Variation = 100 + 6 = 106°. ("Variation East, magnetic least.")

12. QSY means:

> Say again
>
> Change frequency
>
> Standby
>
> Report position

Answer: Change frequency. QSY — change your frequency to the one specified.

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13. QGH describes a:

> Pressure setting
>
> Controlled descent through cloud using DF bearings
>
> Met report request
>
> True bearing

Answer: Controlled descent through cloud using DF bearings. QGH — a DF let-down / controlled descent through cloud on ground-station bearings.

14. "Altitude" is measured from:

> The aerodrome
>
> Mean sea level
>
> The 1013.2 datum
>
> The terrain below

Answer: Mean sea level. Altitude = vertical distance from mean sea level (QNH). Height is from a specified datum (QFE).

15. A flight level is referenced to:

> The local QNH
>
> The aerodrome QFE
>
> The standard pressure 1013.2 hPa
>
> True sea-level pressure (QFF)

Answer: The standard pressure 1013.2 hPa. Flight levels are surfaces of constant pressure referenced to 1013.2 hPa (QNE).

16. "Elevation" applies to:

> An aircraft in flight
>
> A point on or affixed to the surface, above MSL
>
> A pressure setting
>
> A radio bearing

Answer: A point on or affixed to the surface, above MSL. Elevation is the height of a surface point above MSL (e.g. aerodrome elevation).

17. A Q-code can function as:

> Only a question
>
> Only an answer
>
> Either a question or an answer, depending on context
>
> A frequency

```
Answer: Either a question or an answer, depending on context. The same three-letter code serves as the question and its reply.
```

18. Descending, you change from 1013 to QNH on passing the:

> Transition altitude
>
> Transition level
>
> Aerodrome elevation
>
> Middle marker

```
Answer: Transition level. Descend → QNH at the transition LEVEL; climb → standard at the transition ALTITUDE.
```

PART B — ORAL / VIVA (TAP TO REVEAL MODEL ANSWERS)

▶ **What is a Q-code and why does aviation still use a few?**

▶ **Explain QNH, QFE and QNE and what the altimeter reads on each.**

▶ **Why is the standard setting 1013.2 used above the transition altitude?**

▶ **What is the difference between QDM and QDR, and how are they related?**

▶ **Define altitude, height, elevation and flight level.**

▶ **If you set too low a pressure on the subscale, what happens, and what is the saying?**

▶ **What does QGH mean?**

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60-SECOND REVISION CARD

- **QNH** → altitude (AMSL, elevation on ground). **QFE** → height (0 on runway). **QNE** → flight level (1013.2 hPa). **QFF** → true MSL pressure (met).
- **≈ 1 hPa = 27 ft**; too-low setting over-reads — "High to Low, look out below."
- **Climb:** 1013 at transition **altitude**; **Descend:** QNH at transition **level**.
- **QDM** mag TO; **QDR** mag FROM = QDM ± 180°; **QUJ/QTE** true TO/FROM; **QGH** DF let-down; **QSY** change freq.
- Altitude (MSL) · Height (datum) · Elevation (surface point) · FL (1013.2).

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FIRs, Frequencies & Aeronautical Information

Every flight is built on information someone else collected, published and kept current — the elevation of a runway, the frequency of a beacon, the fact that a taxiway is closed tonight. This chapter covers how the airspace is divided, the frequencies you talk and navigate on, and the information system — AIP, AIC, NOTAM — that keeps it all current.

SYLLABUS MAP

Part I (viii) FIRs in India & principal comms/nav facilities · (x) Notices to airmen (NOTAM/AIS)

LEARNING OBJECTIVES — BY THE END OF THIS CHAPTER YOU WILL BE ABLE TO…

- Define an FIR and the related airspace structures (UIR, CTA, CTR, TMA), and the services provided within them.
- Name India's principal FIRs and decode an ICAO four-letter location indicator.
- Outline the ICAO airspace classes A–G and what each provides.
- State the principal communication, emergency and navigation frequencies used in India.
- Describe the AIS and the Integrated Aeronautical Information Package (AIP, AIC, NOTAM) and the NOTAM Q-line.
- Explain the AIRAC cycle and the AFTN with its message priorities.

5.1 How airspace is divided
5.2 Indian FIRs & the services
5.3 ICAO location indicators
5.4 Airspace classes A–G
5.5 Principal communication & emergency frequencies
5.6 Principal navigation frequencies
5.7 The AIS & the IAIP
5.8 NOTAM & the Q-line
5.9 AIRAC & AFTN

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Airspace is divided into blocks so ATC can manage traffic safely and efficiently.

5.1 How airspace is divided

FIRST PRINCIPLES — WHY AIRSPACE IS CARVED UP

Airspace is too vast and varied for one office, one service and one set of rules. So it is divided two ways: **horizontally and vertically** into named blocks, and **by class** according to how much control and service is provided. The largest block is the **Flight Information Region (FIR)**; within it sit smaller, more tightly controlled volumes around the busy parts.

The airspace volumes

FIR — Flight Information Region: a volume within which **Flight Information Service (FIS)** and **Alerting Service** are provided.

UIR — Upper Information Region: the FIR's upper-airspace counterpart.

CTA — Control Area: controlled airspace from a specified lower limit upwards (airways are CTAs).

TMA — Terminal Control Area: a CTA at the confluence of routes near major aerodromes.

CTR — Control Zone: controlled airspace from the **surface** upwards around an aerodrome.

ATZ — Aerodrome Traffic Zone.

Definition — Flight Information Region (FIR)

An airspace of defined dimensions within which **Flight Information Service** (useful information for the safe and efficient conduct of flight) and **Alerting Service** (notifying the appropriate organisations of aircraft needing search and rescue, and assisting) are provided. An FIR may extend over the high seas (an **Oceanic** FIR).

5.2 Indian FIRs & the services

India's principal FIRs

India's airspace is divided into the FIRs of **Delhi, Mumbai, Kolkata and Chennai**, with the **Chennai (Oceanic)** region extending far over the Bay of Bengal and Indian Ocean. Each is managed by the **Airports Authority of India (AAI)**, which provides ATS, communications, navigation and surveillance within them.

Author check before publishing

Confirm the **complete current list** of Indian FIRs and any sub-FIRs (e.g. Guwahati), their boundaries and exact FIR location indicators from the current AIP. Boundaries/names change — quote the live AIP.

Cockpit reality — crossing an FIR boundary

As you cross from one FIR to the next, control is handed from one area control centre to another and you change frequency on instruction. Over the ocean, where there is no radar or VHF, the Chennai Oceanic FIR uses **HF and position reports** (and increasingly CPDLC/ADS) to keep aircraft separated.

5.3 ICAO location indicators

FIRST PRINCIPLES — A UNIQUE ADDRESS FOR EVERY PLACE

Flight plans, NOTAMs and messages must name aerodromes and stations **unambiguously** across the world, so ICAO assigns each a unique **four-letter code**. The letters are structured by region and territory so the code itself tells you roughly where you are.

How the four letters work

1st letter = **region**; 2nd letter = **State/territory** within the region; 3rd–4th = the **specific location**. For India the first letter is **V**, and the second splits the country by zone (e.g. **VI**… north, **VA**… west, **VE**… east, **VO**… south). These are **ICAO** codes (4 letters), distinct from the **IATA** 3-letter codes used on tickets and baggage.

Indian 4-Letter Indicators

Airport	ICAO (4-ltr)	IATA (3-ltr)	Zone letter
Delhi	VIDP	DEL	VI — north
Mumbai	VABB	BOM	VA — west
Kolkata	VECC	CCU	VE — east
Chennai	VOMM	MAA	VO — south

> **Mnemonic**
>
> **India starts with V.** The four metros by zone letter: **V**IDP, **V**ABB, **V**ECC, **V**OMM — "**I-A-E-O**" for north, west, east, south.

The four primary metro airports defining India's FIR zones: VIDP, VABB, VOMM, and VECC.

5.4 Airspace classes A–G

> **IN PLAIN TERMS**
>
> ICAO grades airspace from **A** (most controlled) to **G** (uncontrolled). The class decides whether you need a clearance, whether you're separated from other traffic, and what service you receive. You need to recognise the spread, not memorise every cell.

Airspace Classes A–G

Class	Flights	Separation / service (summary)
A	IFR only	All separated; ATC clearance required
B	IFR & VFR	All separated from all
C	IFR & VFR	IFR separated from IFR & VFR; VFR get traffic info on other VFR
D	IFR & VFR	IFR separated from IFR; traffic info on VFR
E	IFR & VFR	IFR separated from IFR; VFR not separated; controlled for IFR
F	IFR & VFR	Advisory service to IFR (where provided)
G	IFR & VFR	Uncontrolled — flight information service only

Mnemonic

A is for "All controlled"; G is for "Go as you please" (uncontrolled).
Control and separation steadily decrease from A to G.

Author check before publishing

Confirm which airspace classes India actually uses and their VMC minima from the current AIP/CAR, and insert exact figures if examinable.

5.5 Principal communication & emergency frequencies

IN PLAIN TERMS

Within India you talk to ATC on the VHF aviation band; long-range/oceanic uses HF; and a small set of emergency frequencies is guarded worldwide. The full spectrum is in Chapter 8; these are the ones the examiner expects you to name.

Communication & Emergency Frequencies

Use	Frequency / band
VHF ATC voice	118.0 – 136.975 MHz (AM, line-of-sight)
VHF emergency (international guard)	121.5 MHz
UHF military emergency	243.0 MHz (= 2 × 121.5)
ELT — satellite / homing	406 MHz / 121.5 MHz
SAR scene-of-search (aux)	123.1 MHz
Maritime distress (HF/MF)	2182 kHz
HF long-range / oceanic voice	3 – 30 MHz (SSB)

> **Exam trap**
>
> **121.5 MHz** is the international VHF emergency (guard) frequency — memorise it exactly. Don't confuse it with the UHF military guard **243.0 MHz**, and note **123.1 MHz** is the SAR auxiliary scene-of-search frequency.

5.6 Principal navigation frequencies

Navigation Frequencies

Aid	Frequency / band
NDB / ADF	LF/MF ≈ 190 – 1750 kHz
VOR / ILS localizer	VHF 108.0 – 117.975 MHz
ILS glide path	UHF 329 – 335 MHz
Marker beacons	75 MHz
DME	UHF 962 – 1213 MHz
SSR transponder	1030 MHz interrogate / 1090 MHz reply

> **Cockpit reality**
>
> These frequencies and the facilities available at each aerodrome — runway, elevation, beacon idents, frequencies — are exactly what you look up in the **AIP** before flying there. Which brings us to the information system itself.

5.7 The AIS & the IAIP

FIRST PRINCIPLES — FLIGHT RUNS ON PUBLISHED INFORMATION

You cannot personally survey every runway or measure every beacon. Someone collects that data, publishes it, and — critically — **keeps it current**. That is the job of the **Aeronautical Information Service (AIS)**, and its output is bundled as the **Integrated Aeronautical Information Package (IAIP)**. Different kinds of information change at different speeds, so the package has different components for permanent, long-temporary, and short-notice data.

The Integrated Aeronautical Information Package

AIP — Aeronautical Information Publication: the permanent reference manual, in three parts — **GEN** (general), **ENR** (en-route) and **AD** (aerodromes).

AIP Amendment — permanent changes to the AIP.

AIP Supplement — temporary changes of **long** duration.

AIC — Aeronautical Information Circular: explanatory/advisory information not meeting NOTAM or AIP criteria.

NOTAM — short-notice, time-critical changes (see 5.8).

Plus pre-flight and post-flight information bulletins.

Mnemonic — by how fast it changes

AIP = permanent · Supplement = long temporary · AIC = advisory · NOTAM = urgent/now. The faster the information changes, the further "down" this list it lives.

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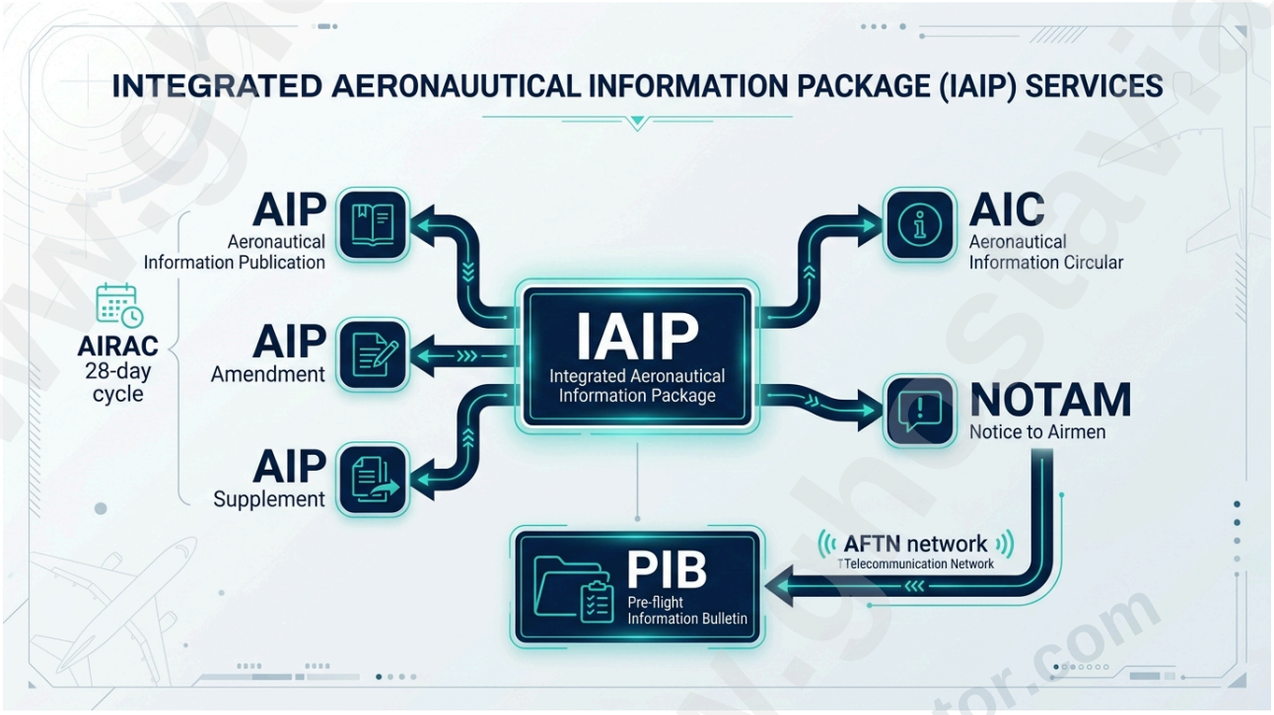

Figure 5.1 — The Integrated Aeronautical Information Package components branching down to time-critical NOTAMs.

5.8 NOTAM & the Q-line

Definition — NOTAM

A **Notice to Airmen** — a notice distributed by telecommunication containing information on the establishment, condition or change in any aeronautical facility, service, procedure or hazard, the timely knowledge of which is essential to personnel concerned with flight operations.

NOTAM types & relatives

NOTAMN — new · **NOTAMR** — replaces a previous NOTAM · **NOTAMC** — cancels one. Specialised: **SNOWTAM** (runway snow/ice/slush conditions, a special format) and **ASHTAM** (volcanic ash). The summarised set for a route is issued as a **PIB — Pre-flight Information Bulletin**.

The Q-line

Each NOTAM carries a coded **Q-line** that lets computers filter and route it: it encodes the FIR, a **Q-code subject & condition**, the traffic affected (I/V), the purpose, the scope, and the lower/upper limits and a radius. Below the Q-line, items A) to G) give the aerodrome/FIR, start and end times, schedule, and the plain-language text.

Worked example — reading a NOTAM

A NOTAM reads, in part: **"A) VIDP B) … C) … E) RWY 11/29 CLSD DUE WIP."**
A) = the aerodrome (Delhi, VIDP). B)/C) = valid-from / valid-to times (UTC).
E) = the plain text: **runway 11/29 closed due work in progress.**
So at Delhi, for the period in B)–C), you must plan around runway 11/29 being unavailable. This is exactly the kind of item the PIB surfaces before you fly.

Mnemonic

N-R-C: New, Replace, Cancel — a NOTAM is freshly issued, takes the place of an old one, or kills one.

5.9 AIRAC & AFTN

Definition — AIRAC

Aeronautical Information Regulation And Control — a system delivering operationally significant changes on common **effective dates** at fixed intervals of **28 days**, so that crews, databases and States all update **together** rather than at random.

AFTN — the network & its message priorities

NOTAMs, flight plans and service messages travel over the **AFTN — Aeronautical Fixed Telecommunication Network**, a worldwide ground network of fixed stations exchanging messages in a standard format with **priority indicators**. In descending priority: **SS** Distress, **DD** Urgency, **FF** Flight safety, **GG** Meteorological, **KK** Flight regularity (and lower categories for administrative/service messages).

Mnemonic — AFTN priority order

"Some Dogs Find Good Kennels" → **SS** Distress, **DD** Urgency, **FF** Flight-safety, **GG** Met, **KK** Regularity.

☆ Numbers to memorise

Essential Facts for Chapter 5

Fact	Value
FIR	Airspace with Flight Information Service + Alerting Service
Indian FIRs	Delhi, Mumbai, Kolkata, Chennai (+ Chennai Oceanic)
Location indicators	VIDP · VABB · VECC · VOMM (India = V, 4 letters)
Airspace classes	A (all controlled) → G (uncontrolled)
VHF ATC band	118.0 – 136.975 MHz
VHF / UHF emergency	121.5 MHz / 243.0 MHz · SAR aux 123.1
IAIP	AIP (GEN/ENR/AD) · Amendment · Supplement · AIC · NOTAM
NOTAM types	NOTAMN (new) · NOTAMR (replace) · NOTAMC (cancel)
AIRAC cycle	28 days
AFTN priorities	SS · DD · FF · GG · KK (distress → regularity)

Question bank

PART A — MCQS (CLICK AN OPTION TO CHECK)

1. Within an FIR, which services are provided?

> Air traffic control only
>
> Flight Information Service and Alerting Service
>
> Customs and immigration
>
> Refuelling and maintenance

```
Answer: Flight Information Service and Alerting Service. An FIR is the airspace within which FIS and Alerting Service are provided.
```

2. Controlled airspace extending from the surface upwards around an aerodrome is a:

> CTA
>
> TMA
>
> CTR (Control Zone)
>
> UIR

```
Answer: CTR (Control Zone). A CTR (control zone) is from the surface up; a CTA is from a specified level up.
```

3. The ICAO location indicator for Mumbai is:

> BOM
>
> VABB
>
> VIDP
>
> VOMM

Answer: VABB. VABB (ICAO, 4 letters). BOM is the IATA 3-letter code.

4. In an ICAO four-letter location indicator, the first letter denotes the:

> Specific airport
>
> Region
>
> Runway
>
> Airline

Answer: Region. 1st = region, 2nd = State/territory, 3rd–4th = location.

5. The most controlled ICAO airspace class is:

> Class A
>
> Class D
>
> Class F
>
> Class G

Answer: Class A. Class A is the most controlled (IFR only, all separated); G is uncontrolled.

6. Class G airspace is:

> IFR only, fully separated
>
> Uncontrolled, with flight information service only
>
> Reserved for the military
>
> Above FL600 only

Answer: Uncontrolled, with flight information service only. Class G is uncontrolled — no ATC separation, FIS only.

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7. The international VHF emergency frequency is:

> 243.0 MHz
>
> 118.0 MHz
>
> 121.5 MHz
>
> 406 MHz

Answer: 121.5 MHz. 121.5 MHz is the VHF guard; 243.0 MHz is the UHF military guard.

8. The SAR auxiliary scene-of-search frequency is:

> 121.5 MHz
>
> 123.1 MHz
>
> 243.0 MHz
>
> 2182 kHz

Answer: 123.1 MHz. 123.1 MHz is the auxiliary SAR frequency; 121.5 is the primary guard.

9. The VHF aeronautical communication band is:

> 108.0–117.975 MHz
>
> 118.0–136.975 MHz
>
> 329–335 MHz
>
> 3–30 MHz

Answer: 118.0–136.975 MHz. 118.0–136.975 MHz is the ATC voice band; 108–117.975 is VOR/ILS-LOC.

10. The permanent reference manual of a State's aeronautical data is the:

> NOTAM
>
> AIC
>
> AIP
>
> SNOWTAM

Answer: AIP. The AIP (parts GEN/ENR/AD) is the permanent publication.

11. The three parts of the AIP are:

> A, B, C
>
> GEN, ENR, AD
>
> N, R, C
>
> SS, DD, FF

Answer: GEN, ENR, AD. GEN (general), ENR (en-route), AD (aerodromes).

12. A temporary change of long duration to the AIP is published as a/an:

> NOTAM
>
> AIP Supplement
>
> AIC
>
> SNOWTAM

Answer: AIP Supplement. Long-duration temporary changes = AIP Supplement; short-notice = NOTAM.

13. A NOTAM that cancels a previous NOTAM is a:

> NOTAMN
>
> NOTAMR
>
> NOTAMC
>
> SNOWTAM

Answer: NOTAMC. N = new, R = replace, C = cancel.

14. Runway snow, ice and slush conditions are notified by a:

> ASHTAM
>
> SNOWTAM
>
> AIC
>
> PIB

Answer: SNOWTAM. SNOWTAM (special format) for runway contamination; ASHTAM for volcanic ash.

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15. The route NOTAM digest provided at briefing is the:

> TAF
>
> ATIS
>
> PIB (Pre-flight Information Bulletin)
>
> AIC

Answer: PIB (Pre-flight Information Bulletin). The PIB gathers the relevant NOTAMs for the route and aerodromes.

16. The AIRAC cycle interval is:

> 7 days
>
> 14 days
>
> 28 days
>
> 90 days

Answer: 28 days. AIRAC effective dates recur every 28 days worldwide.

17. NOTAMs and flight plans are carried over the:

> NFAP
>
> AFTN
>
> VOLMET
>
> SACFA

Answer: AFTN. The Aeronautical Fixed Telecommunication Network carries these messages with priority indicators.

18. The highest AFTN message priority is:

> FF (Flight safety)
>
> GG (Meteorological)
>
> SS (Distress)
>
> KK (Flight regularity)

Answer: SS (Distress). Descending priority: SS distress, DD urgency, FF flight-safety, GG met, KK regularity.

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PART B — ORAL / VIVA (TAP TO REVEAL MODEL ANSWERS)

▶ **What is an FIR and what services are provided in it?**

▶ **Distinguish a CTA, a CTR and a TMA.**

▶ **How is an ICAO four-letter location indicator structured?**

▶ **What is the difference between controlled and uncontrolled airspace classes?**

▶ **Name the components of the Integrated Aeronautical Information Package.**

▶ **What is a NOTAM, and what are its types?**

▶ **What is AIRAC?**

▶ **What is the AFTN and how are its messages prioritised?**

60-SECOND REVISION CARD

- **FIR** = airspace with FIS + Alerting. India: **Delhi, Mumbai, Kolkata, Chennai (+Oceanic)**. CTA/TMA/CTR nested inside.
- ICAO codes (4-ltr, India = V): **VIDP, VABB, VECC, VOMM**. Classes **A (all controlled) → G (uncontrolled)**.
- VHF ATC **118–136.975**; emergency **121.5/243.0**, SAR aux **123.1**.
- **IAIP** = AIP (GEN/ENR/AD) + Amendment + Supplement + AIC + NOTAM. NOTAM = **N/R/C**, SNOWTAM/ASHTAM.
- **AIRAC = 28 days; AFTN** carries messages, priorities **SS·DD·FF·GG·KK**.

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Meteorological Codes & the Pre-Flight Briefing Service

Weather is the single biggest variable in aviation, and the industry built a compact, coded language to describe it — terse strings that pack an entire weather picture into one line. This chapter teaches you to read those codes fluently and shows how the pre-flight briefing service puts that weather, and the NOTAMs, in your hands before you fly.

SYLLABUS MAP

Part I (ix) Meteorological codes, Pre-Flight briefing service and their usages

LEARNING OBJECTIVES — BY THE END OF THIS CHAPTER YOU WILL BE ABLE TO…

- Explain why aviation weather is coded and read a METAR/SPECI field by field.
- Interpret a TAF, its validity and change groups, and the TREND group.
- Identify SIGMET, AIRMET, PIREP and GAMET, and the ATIS/VOLMET broadcasts.
- State the CAVOK criteria and decode the cloud, weather and RVR groups.
- Describe the pre-flight briefing service and the pilot's information sequence to a go/no-go decision.

6.1 Why weather is coded
6.2 METAR & SPECI — field by field
6.3 Cloud, weather & RVR groups
6.4 TAF — the aerodrome forecast
6.5 TREND, SIGMET, AIRMET, PIREP
6.6 ATIS & VOLMET
6.7 CAVOK
6.8 The pre-flight briefing service

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Understanding coded weather accurately is critical for safe decision-making in the flight deck.

6.1 Why weather is coded

FIRST PRINCIPLES — LANGUAGE-PROOF, TRANSMISSION-PROOF WEATHER

Weather decisions are made by crews of every nationality, and the information often crosses a noisy radio or telex link. A free-text English forecast would be slow, ambiguous and easy to garble. So the world adopted a **fixed-order, internationally standardised code**: every report follows the same sequence of groups, so a pilot in Mumbai or Madrid reads it identically and a single transposed letter is obvious. Master the **order of the slots** and any report decodes itself.

The two families

Reports describe weather that *is* happening: **METAR** (routine, observed) and **SPECI** (special, off-schedule observation).

Forecasts describe weather that *will* happen: **TAF** (aerodrome forecast) and the appended **TREND** (a 2-hour landing forecast).

6.2 METAR & SPECI — field by field

> **IN PLAIN TERMS**
>
> A METAR is the routine observed weather at an aerodrome, issued at fixed intervals (commonly every 30 or 60 minutes). A SPECI uses the same format but is issued *off-schedule* when something changes significantly. Both follow the same fixed order.

Example METAR Breakdown

```
VABB 241230Z 27008KT 8000 -RA FEW020 SCT025 BKN090 32/26 Q1008 NOSIG
```

VABB	Mumbai — station (ICAO indicator)
241230Z	24th day of month, 1230 UTC (Zulu)
27008KT	wind 270° (TRUE) at 8 knots
8000	visibility 8000 metres
-RA	light rain (− light, + heavy)
FEW020 **SCT025** **BKN090**	few cloud at 2000 ft; scattered 2500 ft; broken 9000 ft
32/26	temperature 32 °C / dewpoint 26 °C
Q1008	QNH 1008 hPa (an "A2992" would be inches of mercury)
NOSIG	trend: no significant change expected

> **Mnemonic — read a METAR in order**
>
> **"Station, Time, Wind, Visibility, Weather, Cloud, Temp/Dewpoint, Pressure, Trend."** Always the same slots — once you know them, any METAR decodes itself.

> **Exam trap — wind reference**
>
> Wind in a METAR/TAF is in **degrees TRUE and knots**; the surface wind **passed by ATC on the radio** is in **degrees MAGNETIC**. Same wind, different reference — a favourite catch. A gust is shown as e.g. 27015G27KT; a variable direction as VRB.

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Worked example — extract the operational picture

From the METAR above: lowest **broken** (operational ceiling) is BKN090 = 9000 ft; visibility 8000 m; light rain; temp/dewpoint **32/26** — a 6 °C spread, so humid but not yet fog.
Why temp/dewpoint matters: when the dewpoint closes on the temperature, the air is near saturation and **fog/low cloud** becomes likely. A 1–2 °C spread on a cooling evening is a warning sign for the destination.

6.3 Cloud, weather & RVR groups

Cloud amount (oktas of sky covered)

FEW 1–2 oktas · **SCT** (scattered) 3–4 · **BKN** (broken) 5–7 · **OVC** (overcast) 8. Heights are in **hundreds of feet above aerodrome** (020 = 2000 ft). **NSC** = nil significant cloud; **SKC/NCD** = sky clear / no cloud detected.

Weather Groups

Group	Meaning	Group	Meaning
RA / SN	Rain / Snow	**FG / BR**	Fog / Mist
TS / SH	Thunderstorm / Shower	**DZ / GR**	Drizzle / Hail
CB / TCU	Cumulonimbus / Towering cumulus	**HZ / FU**	Haze / Smoke
+ / −	Heavy / Light prefix	**VC**	In the vicinity

RVR in a METAR

When visibility is low, an **RVR** group appears, e.g. **R27/0600** = runway 27, RVR 600 m. **Runway Visual Range** is the distance over which a pilot on the centreline can see the runway markings or lights — measured by transmissometers, and the governing visibility for low-visibility approaches (full RT detail in Chapter 21).

6.4 TAF — the aerodrome forecast

IN PLAIN TERMS

A TAF is the forecast for an aerodrome over a **validity period**, written in the same vocabulary as a METAR but with **change groups** that say "and then it will become…". A TAF begins with its issue time and the validity period (e.g. 2412/2512 = from the 24th 1200Z to the 25th 1200Z).

TAF Change Groups

Group	Meaning
FM (FROM)	A rapid, permanent change from the stated time onward
BECMG	A gradual (becoming) change over the stated period
TEMPO	Temporary fluctuations, each lasting under an hour, totalling less than half the period
PROB30 / PROB40	30% / 40% probability of the stated conditions

6.5 TREND, SIGMET, AIRMET & PIREP

Other Met Bulletins

Code	What it is
TREND	A 2-hour landing forecast appended to a METAR (NOSIG / BECMG / TEMPO)
SIGMET	Warning of weather **significant to all aircraft** — thunderstorms, severe turbulence/icing, volcanic ash, tropical cyclones, dust/sandstorms
AIRMET	Warning of weather significant to **low-level / lighter aircraft** operations
PIREP	A **pilot report** of conditions actually encountered in flight
GAMET	An area forecast for low-level flights

> **Mnemonic**
>
> **SIGMET = SIGnificant to all; AIRMET = lighter AIRcraft.** Both are warnings; PIREP is what a pilot actually *saw*.

6.6 ATIS & VOLMET

> **ATIS vs VOLMET**
>
> **ATIS** — Automatic Terminal Information Service: a continuous broadcast of **one aerodrome's** weather, runway-in-use and operational information, identified by a letter (Information "Alpha", "Bravo"…), so the frequency isn't tied up reading it to every aircraft.
> **VOLMET** — a scheduled broadcast of **en-route** meteorological information (METARs/TAFs/SIGMETs) for **several aerodromes**, for aircraft in flight, on HF and/or VHF.

Cockpit reality

You listen to ATIS before first contact and read back the letter ("…with Information Charlie") so the controller knows you have the current weather and runway. It saves a congested frequency from reading the same numbers a hundred times an hour — and confirms you and ATC share the same picture.

6.7 CAVOK

Definition — CAVOK ("Ceiling And Visibility OK")

Used when **all** of these hold simultaneously: visibility **10 km or more; no cloud below 5000 ft** (or below the highest minimum sector altitude, whichever is greater) and **no CB or TCU** at any level; and **no significant weather** at or near the aerodrome. When CAVOK applies, the visibility, cloud and weather groups are replaced by the single word.

Exam trap

CAVOK is not just "nice weather" — it has the three defined criteria above. A common wrong answer is "visibility 8 km"; the threshold is **10 km**. And a single CB, even with great visibility, **disqualifies** CAVOK.

6.8 The pre-flight briefing service

FIRST PRINCIPLES — FLYING THE PAPERWORK BEFORE THE AIRCRAFT

Before a flight a pilot must gather the **weather** and the **NOTAMs**, study the procedures, and **file a flight plan**. The service that makes this possible is the **pre-flight briefing service**, run by the Aeronautical Information Service at the aerodrome, increasingly through self-briefing terminals and online portals.

What the briefing service provides

At the AIS/briefing office (or via self-briefing terminals / the online flight-planning portal) a pilot obtains: the relevant **METARs and TAFs**; **SIGMET/AIRMET/GAMET** warnings; the route **NOTAMs** packaged as a **PIB (Pre-flight Information Bulletin)**; charts and AIP data; and the means to **file the ATS flight plan**.

A pilot's pre-flight information sequence

1. Obtain the **weather** — METAR/TAF for departure, destination and alternates; SIGMETs en route.
2. Obtain the **PIB** — NOTAMs affecting the route and aerodromes.
3. Study the **AIP/charts** for procedures, facilities and frequencies.
4. **File the flight plan** with ATS.
5. Make the **go / no-go** decision against the operating minima and fuel.

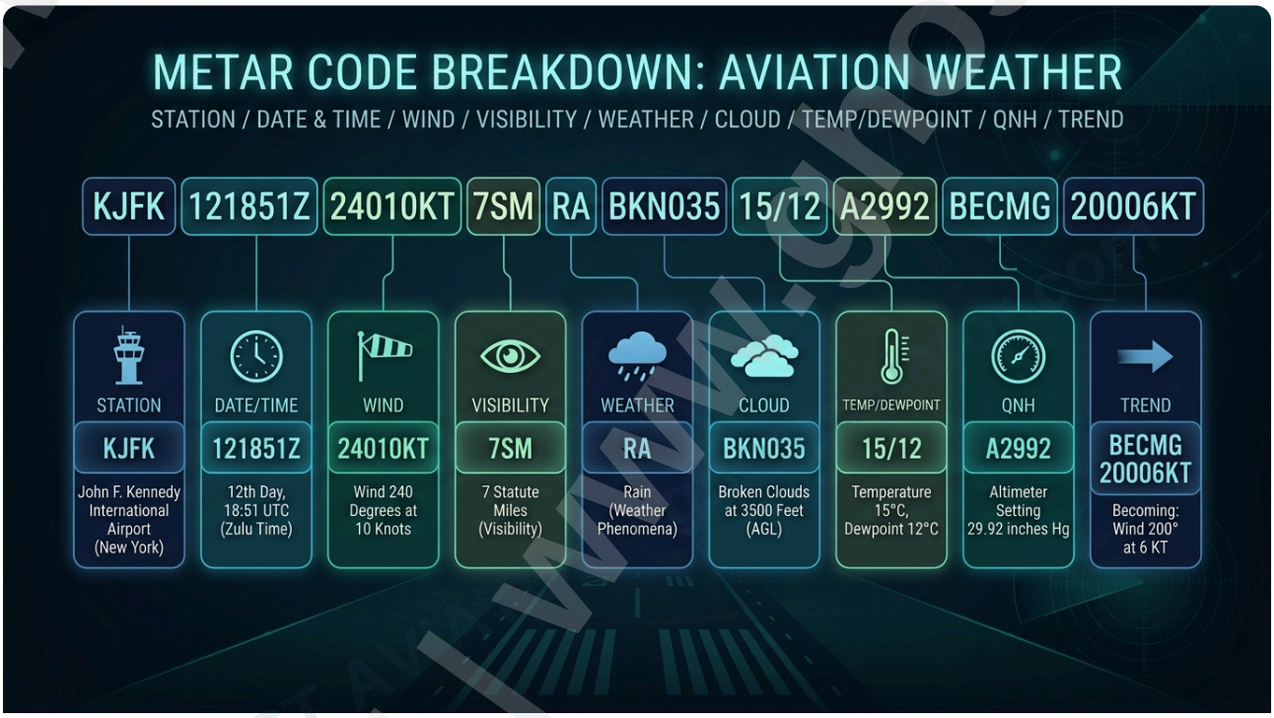

Figure 6.1 — Decoding a METAR — the slots in order.

☆ Numbers to memorise

Essential Facts for Chapter 6

Fact	Value
METAR / SPECI	Routine / special observed aerodrome report
METAR order	Station · Time(Z) · Wind · Vis · Weather · Cloud · Temp/Dew · QNH · Trend
Wind reference	METAR = degrees TRUE; ATC surface wind = degrees MAGNETIC
Cloud amounts	FEW 1–2 · SCT 3–4 · BKN 5–7 · OVC 8 oktas (hundreds of ft)
TAF change groups	FM · BECMG · TEMPO · PROB30/40
CAVOK	Vis ≥10 km · no cloud <5000 ft / no CB·TCU · no significant wx
SIGMET / AIRMET	Hazard to all aircraft / to low-level light aircraft
ATIS / VOLMET	One-aerodrome broadcast / en-route multi-aerodrome broadcast
PIB	Pre-flight Information Bulletin (route NOTAM digest)

Question bank

PART A — MCQS (CLICK AN OPTION TO CHECK)

1. A METAR is a:

> Routine observed aerodrome weather report
>
> Aerodrome forecast
>
> Pilot report
>
> NOTAM

```
Answer: Routine observed aerodrome weather report. METAR = routine observation; the forecast is the TAF; a special observation is a SPECI.
```

2. In a METAR, the wind is given in:

> Degrees magnetic, km/h
>
> Degrees true, knots
>
> Degrees magnetic, knots
>
> Degrees true, m/s

```
Answer: Degrees true, knots. METAR/TAF winds are degrees TRUE and knots; ATC passes surface wind in degrees magnetic.
```

3. "BKN" means cloud cover of:

> 1–2 oktas
>
> 3–4 oktas
>
> 5–7 oktas
>
> 8 oktas

Answer: 5–7 oktas. BKN (broken) = 5–7 oktas. OVC = 8, SCT = 3–4, FEW = 1–2.

4. "FEW020" indicates few cloud at:

> 20 ft
>
> 200 ft
>
> 2000 ft
>
> 20,000 ft

Answer: 2000 ft. Cloud heights are in hundreds of feet above aerodrome — 020 = 2000 ft.

5. "Q1008" in a METAR is the:

> Temperature
>
> QNH in hectopascals
>
> Visibility
>
> Cloud base

Answer: QNH in hectopascals. Q1008 = QNH 1008 hPa. An "A" prefix gives inches of mercury.

6. "R27/0600" in a report means:

> Runway 27 closed at 0600
>
> Runway 27 RVR 600 metres
>
> Wind 270° 6 kt
>
> QNH 600

Answer: Runway 27 RVR 600 metres. It is the Runway Visual Range group: runway 27, RVR 600 m.

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7. When temperature and dewpoint are close, you should expect an increased risk of:

> High pressure
>
> Fog or low cloud (air near saturation)
>
> Strong winds
>
> Clear skies

Answer: Fog or low cloud (air near saturation). A small temp/dewpoint spread means the air is near saturation, favouring fog/low cloud.

8. In a TAF, "TEMPO" indicates:

> A permanent change from a time
>
> Temporary fluctuations, each under an hour
>
> A 40% probability
>
> The validity period

Answer: Temporary fluctuations, each under an hour. TEMPO = temporary fluctuations each lasting under an hour; FM = permanent change; BECMG = gradual change.

9. A SIGMET warns of weather significant to:

> Only light aircraft
>
> All aircraft (e.g. thunderstorms, severe turbulence, volcanic ash)
>
> Ground vehicles
>
> Only IFR traffic at night

Answer: All aircraft (e.g. thunderstorms, severe turbulence, volcanic ash). SIGMET = significant to all aircraft; AIRMET is for low-level/lighter aircraft.

10. A report of conditions actually encountered in flight is a:

> TAF
>
> SIGMET
>
> PIREP
>
> ATIS

Answer: PIREP. A PIREP is a pilot report of actual conditions met in flight.

11. ATIS broadcasts information for:

> Several aerodromes en route
>
> One aerodrome (its weather and operational info)
>
> Only military fields
>
> The whole FIR

Answer: One aerodrome (its weather and operational info). ATIS = one aerodrome; VOLMET = several aerodromes for aircraft in flight.

12. CAVOK requires a visibility of at least:

> 5 km
>
> 8 km
>
> 10 km
>
> 1500 m

Answer: 10 km. CAVOK needs visibility ≥10 km, no cloud below 5000 ft / no CB·TCU, and no significant weather.

13. A single CB present with 15 km visibility:

> Still allows CAVOK
>
> Disqualifies CAVOK
>
> Requires a SPECI only
>
> Has no effect

Answer: Disqualifies CAVOK. Any CB or TCU disqualifies CAVOK, regardless of visibility.

14. The route NOTAM digest provided at briefing is the:

> TAF
>
> ATIS
>
> PIB (Pre-flight Information Bulletin)
>
> AIC

Answer: PIB (Pre-flight Information Bulletin). The PIB gathers the relevant NOTAMs for the route and aerodromes.

15. "-RA" in a METAR means:

> Recent rain
>
> Light rain
>
> Heavy rain
>
> Radar

Answer: Light rain. "–" = light, "+" = heavy; so –RA is light rain.

16. "NOSIG" appended to a METAR means:

> No signal
>
> No significant change expected (trend)
>
> Aerodrome closed
>
> Nil cloud

Answer: No significant change expected (trend). NOSIG is the trend forecast: no significant change in the next two hours.

17. The correct order of the first groups in a METAR is:

> Wind, station, time, visibility
>
> Station, time, wind, visibility
>
> Time, wind, station, cloud
>
> Pressure, temp, wind, station

Answer: Station, time, wind, visibility. Station → Time(Z) → Wind → Visibility → Weather → Cloud → Temp/Dewpoint → QNH → Trend.

18. The final step of the pre-flight information sequence is to:

> File the flight plan
>
> Read the ATIS
>
> Make the go/no-go decision against minima and fuel
>
> Request start-up

Answer: Make the go/no-go decision against minima and fuel. After weather, PIB, charts and filing the flight plan, the pilot makes the go/no-go decision.

PART B — ORAL / VIVA (TAP TO REVEAL MODEL ANSWERS)

▶ **What is the difference between a METAR, a SPECI and a TAF?**

▶ **Read out the order of the groups in a METAR.**

▶ **In which reference are METAR winds and ATC surface winds given?**

▶ **State the CAVOK criteria.**

▶ **What is the difference between SIGMET and AIRMET?**

▶ **What is the difference between ATIS and VOLMET?**

▶ **What does the pre-flight briefing service provide and in what sequence is it used?**

60-SECOND REVISION CARD

- **METAR** observed, **SPECI** special, **TAF** forecast (FM/BECMG/TEMPO/PROB). Order: Station·Time·Wind·Vis·Wx·Cloud·T/Dew·QNH·Trend.
- Wind: METAR **TRUE**/kt vs ATC **MAGNETIC**. Cloud **FEW 1–2 · SCT 3–4 · BKN 5–7 · OVC 8**.
- **CAVOK** = vis ≥10 km, no cloud <5000 ft / no CB·TCU, no significant wx (one CB disqualifies).
- **SIGMET** all aircraft · **AIRMET** light · **PIREP** pilot-seen · **ATIS** one field · **VOLMET** en-route.
- Briefing = weather + **PIB** (NOTAMs) + charts + **file flight plan** → **go/no-go**.

Electrical Fundamentals

A radio is, at heart, an electrical machine. Before you can understand how a transmitter pushes your voice 60 miles to a control tower, or why VHF behaves so differently from HF, you must be fluent in the quantities that describe all electricity — and the one equation that ties a circuit to a radio wavelength.

SYLLABUS MAP

Part II (i) Electrical units – volt, ampere, ohm, watt; wavelength, frequency & their relationship

LEARNING OBJECTIVES — BY THE END OF THIS CHAPTER YOU WILL BE ABLE TO…

- Define voltage, current, resistance and power with their units and symbols.
- Apply Ohm's Law and the power formulae, and solve numerical problems.
- Distinguish AC from DC, and explain RMS, peak, frequency and period.
- State the factors affecting resistance and combine resistors in series and parallel.
- Explain capacitance, inductance and reactance, and why they matter for tuning.
- Relate frequency, wavelength and the speed of radio waves and convert units.

7.1 The four quantities
7.2 Ohm's Law
7.3 Power
7.4 AC & DC
7.5 Resistance & materials
7.6 Series & parallel circuits
7.7 Capacitance, inductance & reactance
7.8 Frequency, period & wavelength

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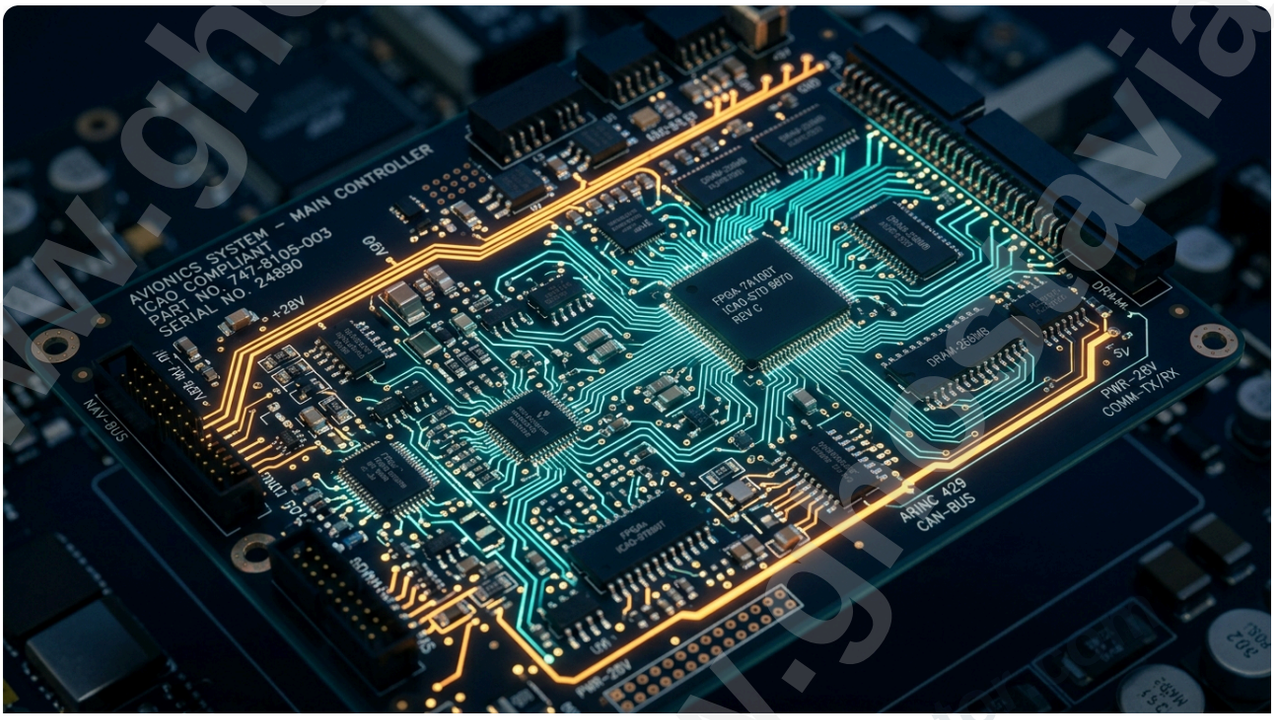
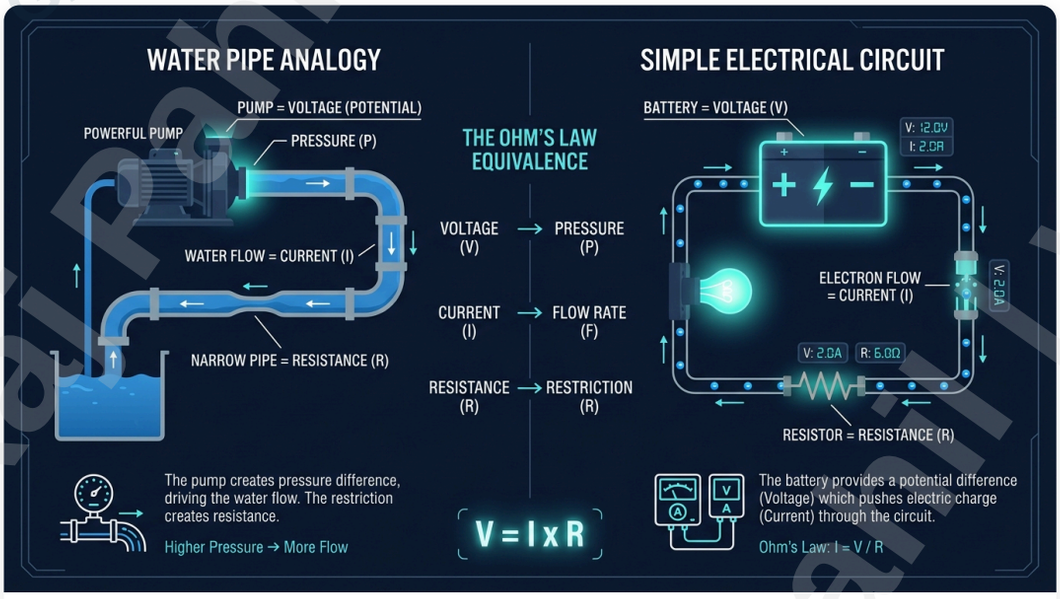


Understanding the core electrical principles is the first step to mastering radio transmission.

7.1 The four quantities

> **FIRST PRINCIPLES — THE WATER ANALOGY, MADE PRECISE**
>
> Picture electricity as water in a pipe. **Voltage** is the pressure pushing the water; **current** is how much water flows per second; **resistance** is how narrow the pipe is; **power** is the useful work the flow does. The analogy is exact enough to carry you through every formula in this chapter.



Voltage is the pump, Current is the flow, and Resistance is the pipe's narrow point.

Quantity	Symbol	Unit	Precise meaning
Voltage (EMF / p.d.)	V (or E)	Volt (V)	Energy given to / taken from each coulomb of charge — the "push"
Current	I	Ampere (A)	Rate of flow of charge: 1 A = 1 coulomb per second
Resistance	R	Ohm (Ω)	Opposition to current flow
Power	P	Watt (W)	Rate of doing work / using energy: 1 W = 1 joule per second

EMF vs potential difference

EMF (electromotive force) is the voltage a source *produces* (e.g. a battery's 12 V). **Potential difference (p.d.)** is the voltage *across a component* in the circuit. They differ by the voltage lost across the source's own **internal resistance** when current flows.

7.2 Ohm's Law

THE ONE EQUATION TO KNOW COLD

Push harder (more volts) and more current flows; narrow the pipe (more ohms) and less flows. The three lock together:

$$V = I \times R$$

Volts = Amps × Ohms · rearranged: I = V ÷ R and R = V ÷ I

Mnemonic — the VIR triangle

Draw a triangle: **V** on top, **I** and **R** below. Cover the one you want: cover V → I × R; cover I → V/R; cover R → V/I. "**V**ery **I**mportant **R**ule."

Worked example — Ohm's Law

A 28 V aircraft bus supplies a 14 Ω heater. Current = V/R = 28 ÷ 14 = **2 A**.

If the resistance halves to 7 Ω, current doubles to **4 A** (inverse relationship). If the bus sags to 24 V across the original 14 Ω, current = 24 ÷ 14 ≈ **1.71 A**.

7.3 Power

$$P = V \times I = I^2R = V^2 \div R$$

Watts = Volts × Amps · three forms, choose what you know

Worked example — power three ways

From 7.2: 28 V across 14 Ω drawing 2 A. **P = V × I** = 28 × 2 = **56 W**.

Check with **I²R** = 2² × 14 = 4 × 14 = **56 W**. Check with **V²/R** = 28² ÷ 14 = 784 ÷ 14 = **56 W**. All agree.

Units & efficiency

1 kW = 1000 W. **Efficiency** = useful output power ÷ input power (always < 100% because some energy becomes heat). A transmitter rated 50 W *output* draws more than 50 W from the bus because the power amplifier is not perfectly efficient.

7.4 AC & DC

Direct vs Alternating Current

DC (Direct Current) flows in one direction at a steady value — e.g. a battery.
AC (Alternating Current) reverses direction periodically, tracing a **sine wave**; the number of complete cycles per second is the **frequency** (hertz), and the time for one cycle is the **period** (T = 1/f). Every **radio-frequency signal** is alternating current.

Peak, peak-to-peak & RMS

An AC wave is described by its **peak** value, its **peak-to-peak** (twice peak), and its **RMS** (root-mean-square) — the equivalent DC value that would do the same heating work. For a sine wave: **RMS = 0.707 × peak** (and peak = 1.414 × RMS). Mains and aircraft AC voltages are quoted as RMS.

Cockpit reality — aircraft power

A typical aircraft has a **28 V DC** system (battery/bus feeding avionics) and a **115 V AC at 400 Hz** system (for many instruments and as the form every radio signal takes on the way to the antenna). 400 Hz is used rather than 50/60 Hz because transformers and motors can be smaller and lighter at the higher frequency — a real weight saving.

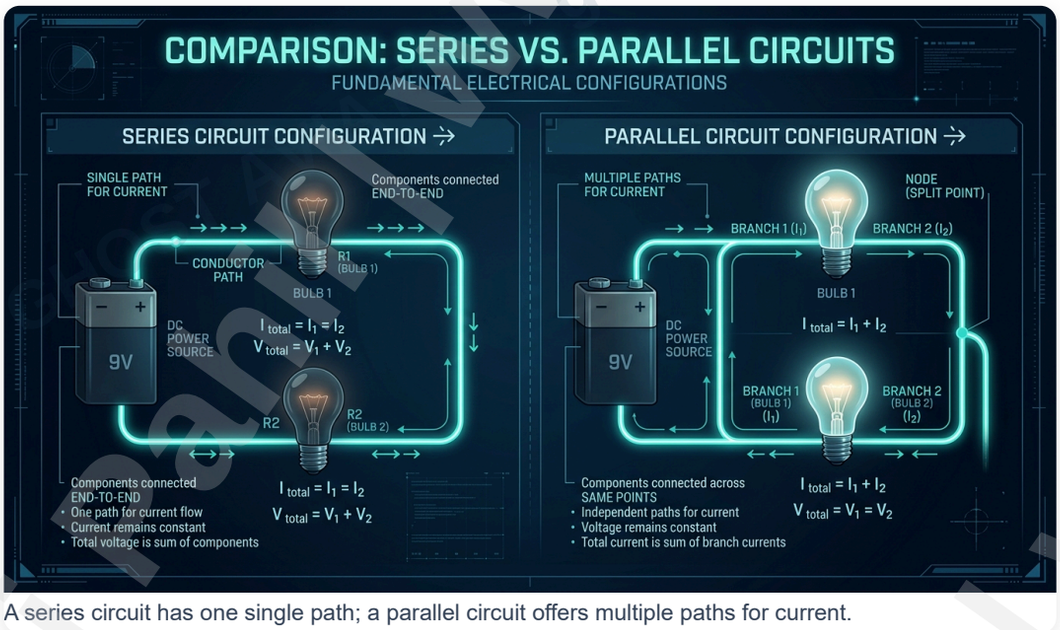
7.5 Resistance & materials

Material Types

Material type	Behaviour	Example
Conductor	Many free electrons — low resistance	Copper, aluminium, silver
Insulator	Almost no free electrons — very high resistance	PVC, rubber, glass, air
Semiconductor	In between; conductivity controllable — the basis of all radio electronics	Silicon, germanium (diodes, transistors)

> **What sets a conductor's resistance**
>
> Resistance increases with **length**, decreases with cross-sectional **area** (a thicker wire conducts better), depends on the **material** (its resistivity), and — for most metals — **increases with temperature**. Hence a long, thin, hot wire has the highest resistance.

7.6 Series & parallel circuits



A series circuit has one single path; a parallel circuit offers multiple paths for current.

Circuit Comparisons

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	Series	Parallel
Path	One single loop	Multiple branches
Current	Same through every component	Splits between branches
Voltage	Divides across components	Same across each branch
Total resistance	R = R+ + R&sub2; + ... (adds up)	1/R = 1/R+ + 1/R&sub2; + ... (less than the smallest)
If one opens	Whole circuit stops	Other branches keep working

Worked example — combining resistors

Series: 10 Ω + 20 Ω = **30 Ω**. Across 30 V the current = 30/30 = 1 A; the 10 Ω drops 10 V and the 20 Ω drops 20 V (voltage divides).
Parallel: two 10 Ω resistors → 1/R = 1/10 + 1/10 = 2/10, so R = **5 Ω** (half of one). For two equal resistors, parallel total = half the value; the current splits equally.

Mnemonic

Series = Same current; Parallel = same Pressure (voltage). Resistances *add* in series and *reduce* in parallel.

7.7 Capacitance, inductance & reactance

WHY A RADIO NEEDS MORE THAN RESISTORS

Resistors oppose current the same at any frequency. **Capacitors** and **inductors** oppose AC by an amount that *depends on frequency* — and that frequency dependence is exactly what lets a radio **tune** to one station and reject the rest.

The two reactive components

Capacitor — stores energy in an electric field; blocks DC, passes AC more easily as frequency rises (its reactance **falls** with frequency). Unit: **farad (F)**.
Inductor — stores energy in a magnetic field; passes DC, opposes AC more as frequency rises (its reactance **rises** with frequency). Unit: **henry (H)**.
Their frequency-dependent opposition is called **reactance (X)**, measured in ohms.

Resonance — the heart of tuning

Put an inductor and capacitor together and at one special **resonant frequency** their reactances cancel, giving a sharp peak in response. A **tuned LC circuit** selects that frequency and rejects others — this is how a receiver is "tuned" to a station and how a transmitter sets its operating frequency.

7.8 Frequency, period & wavelength

IN PLAIN TERMS

Every radio wave travels at the speed of light. Its **frequency** (cycles per second) and its **wavelength** (length of one cycle) are two views of the same wave — fix the speed and one decides the other.

$$c = f \times \lambda$$

speed = frequency × wavelength · c = 3 × 10&sup8; m/s · so λ = c/f and f = 1/T

Unit conversions

1 kHz = 1,000 Hz · 1 MHz = 1,000,000 Hz · 1 GHz = 1,000 MHz. Period and frequency are reciprocals: a 1 MHz signal has a period of 1 microsecond.

Worked example — wavelength of aviation VHF

A 120 MHz VHF signal: λ = c/f = (3 × 10&sup8;) ÷ (120 × 10&sup6;) = **2.5 m**.
A 300 MHz signal: λ = (3 × 10&sup8;) ÷ (300 × 10&sup6;) = **1 m**. A 3 MHz HF signal: λ = (3 × 10&sup8;) ÷ (3 × 10&sup6;) = **100 m**. Higher frequency → shorter wavelength.

Exam trap

Frequency and wavelength are **inversely proportional**: as frequency goes up, wavelength comes down. Mixing this round is the classic error, and it explains why high-frequency aerials are short and low-frequency aerials are long.

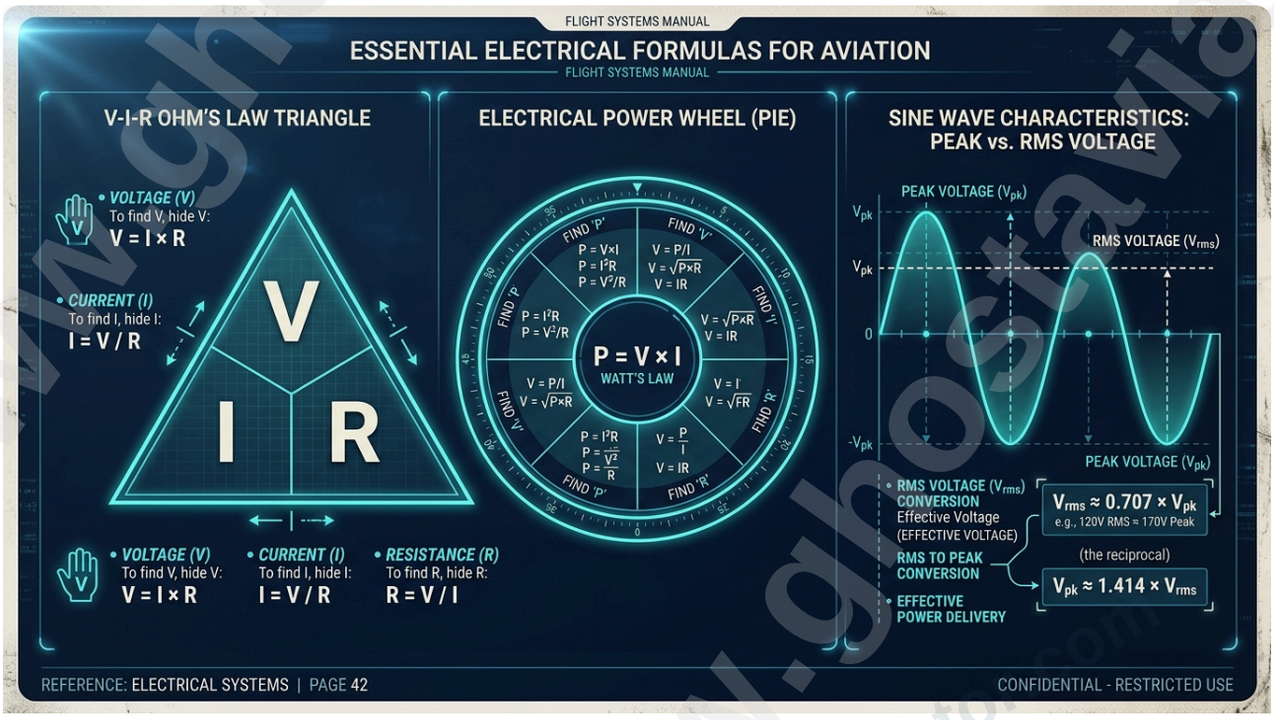

Figure 7.1 — Ohm's Law triangle · the power wheel · the sine wave.

☆ Numbers to memorise

Essential Facts for Chapter 7

Fact	Value
Units	Volt (V) · Ampere (A) · Ohm (Ω) · Watt (W) · Farad (F) · Henry (H)
Ohm's Law	V = I × R
Power	P = V × I = I²R = V²/R
RMS (sine)	RMS = 0.707 × peak · peak = 1.414 × RMS
Aircraft power	28 V DC · 115 V AC 400 Hz
Series / parallel	Series: R adds, same current · Parallel: R reduces, same voltage
Reactance	Capacitor falls with f · Inductor rises with f · LC resonance = tuning
Wave relationship	c = f × λ · c = 3 × 10&sup8; m/s · f = 1/T

Question bank

PART A — MCQS (CLICK AN OPTION TO CHECK)

1. The unit of electrical resistance is the:

> Volt
>
> Ampere
>
> Ohm
>
> Watt

Answer: Ohm. Resistance — ohm (Ω). Volt = voltage, ampere = current, watt = power.

2. One ampere is a flow of charge of:

> One volt per second
>
> One coulomb per second
>
> One ohm per second
>
> One joule per second

Answer: One coulomb per second. 1 A = 1 coulomb of charge per second.

3. Ohm's Law states:

> P = V × I
>
> V = I × R
>
> λ = f × c
>
> I = R × P

Answer: V = I × R. V = I × R. Power is a separate relationship.

4. A 24 V supply across a 6 Ω resistor draws:

> 2 A
>
> 4 A
>
> 6 A
>
> 144 A

Answer: 4 A. I = V/R = 24/6 = 4 A.

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5. The power dissipated by 4 A through 14 Ω is:

> 56 W
>
> 224 W
>
> 3.5 W
>
> 784 W

Answer: 224 W. $P = I^2R = 4^2 \times 14 = 16 \times 14 = 224$ W.

6. For a sine wave, the RMS value is:

> 1.414 × peak
>
> 0.707 × peak
>
> Equal to peak
>
> Twice peak

Answer: 0.707 × peak. RMS = 0.707 × peak (and peak = 1.414 × RMS).

7. A radio-frequency signal is a form of:

> Alternating current
>
> Direct current
>
> Static charge
>
> Resistance

Answer: Alternating current. Radio signals are AC — oscillating at the carrier frequency.

8. Typical aircraft AC power is:

> 230 V 50 Hz
>
> 115 V 400 Hz
>
> 12 V DC
>
> 28 V 60 Hz

Answer: 115 V 400 Hz. Aircraft use 115 V AC at 400 Hz (and 28 V DC); 400 Hz allows lighter components.

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9. Resistance of a wire increases when it is:

> Shorter and thicker
>
> Longer, thinner and hotter
>
> Made of silver
>
> Cooled

```
Answer: Longer, thinner and hotter. Resistance rises with length and temperature, falls with cross-sectional area.
```

10. Two 10 Ω resistors in series give a total of:

> 5 Ω
>
> 20 Ω
>
> 10 Ω
>
> 100 Ω

```
Answer: 20 Ω. Series resistances add: 10 + 10 = 20 Ω.
```

11. Two 10 Ω resistors in parallel give a total of:

> 20 Ω
>
> 5 Ω
>
> 10 Ω
>
> 0 Ω

```
Answer: 5 Ω. For two equal resistors, parallel total = half the value = 5 Ω.
```

12. In a series circuit, the current is:

> The same through every component
>
> Different in each component
>
> Zero
>
> Always alternating

```
Answer: The same through every component. Series = one path = same current; the voltage divides.
```

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13. The reactance of a capacitor as frequency increases:

> Decreases
>
> Increases
>
> Stays the same
>
> Becomes infinite

Answer: Decreases. Capacitive reactance falls with frequency (a capacitor passes high frequencies more easily).

14. A tuned LC circuit is used in a radio to:

> Generate DC
>
> Select one frequency and reject others (tuning)
>
> Increase resistance
>
> Store charge only

Answer: Select one frequency and reject others (tuning). At resonance an LC circuit responds sharply to one frequency — the basis of tuning.

15. The speed of radio waves in free space is approximately:

> 3 × 10&sup6; m/s
>
> 3 × 10&sup8; m/s
>
> 300 m/s
>
> 1013 m/s

Answer: 3 × 10&sup8; m/s. Radio waves travel at the speed of light ≈ 3 × 10&sup8; m/s.

16. As frequency increases, wavelength:

> Decreases
>
> Increases
>
> Stays the same
>
> Doubles

Answer: Decreases. Inversely proportional: c = f × λ with c fixed.

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17. The wavelength of a 150 MHz signal is:

> 0.5 m
>
> 2 m
>
> 20 m
>
> 200 m

Answer: 2 m. λ = c/f = 3×10&sup8; ÷ 150×10&sup6; = 2 m.

18. The period of a 1 MHz signal is:

> 1 second
>
> 1 millisecond
>
> 1 microsecond
>
> 1 nanosecond

Answer: 1 microsecond. T = 1/f = 1 ÷ 1,000,000 = 1 µs.

PART B — ORAL / VIVA (TAP TO REVEAL MODEL ANSWERS)

▶ **Name the four basic electrical quantities and their units.**

▶ **State Ohm's Law and the three forms of the power formula.**

▶ **What is the difference between AC and DC, and what is RMS?**

▶ **What factors affect the resistance of a conductor?**

▶ **How do resistances combine in series and in parallel?**

▶ **Why does a radio need capacitors and inductors, not just resistors?**

▶ **State the relationship between frequency and wavelength.**

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PART C — NUMERICAL PROBLEMS (TAP FOR WORKED SOLUTIONS)

▶ **P1. A 12 V battery with 0.5 Ω internal resistance drives a 5.5 Ω lamp. Find the current and the lamp voltage.**

▶ **P2. A transmitter draws 4 A from the 28 V bus. What input power, and if it is 60% efficient, what is the RF output?**

▶ **P3. Find the wavelength of the emergency frequency 121.5 MHz.**

▶ **P4. Three resistors 6 Ω, 3 Ω and 2 Ω in parallel — find the total.**

60-SECOND REVISION CARD

- Volt · Ampere · Ohm · Watt — pressure, flow, opposition, work. **V = I × R, P = VI = I²R = V²/R**.
- **AC** reverses (sine); RMS = 0.707 × peak. Aircraft: **28 V DC / 115 V 400 Hz AC**.
- Resistance: ↑ length/temperature, ↓ area. **Series** adds R (same current); **parallel** reduces R (same voltage).
- Capacitor reactance ↓ with f; inductor reactance ↑ with f; **LC resonance = tuning**.
- **c = f × λ**, c = 3 × 10&sup8; m/s; higher f → shorter λ; f = 1/T.

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Frequency Bands & Their Aviation Uses

Chapter 7 told you what a frequency is. This chapter tells you what each slice of the spectrum is used for — and this is where RTR papers harvest most of their marks. Examiners love precise numbers: which band carries ATC voice, which one the glide path, what the transponder replies on, where the emergency frequency sits.

SYLLABUS MAP

`Part II (ii) Radio frequency bands · principal aviation frequencies · polarization`

LEARNING OBJECTIVES — BY THE END OF THIS CHAPTER YOU WILL BE ABLE TO…

- Name the radio bands VLF to EHF with their frequency ranges, wavelengths and dominant propagation.
- Explain in depth why aviation uses VHF (AM) for short range and HF (SSB) for long range.
- Recall the principal aviation frequencies and what each does.
- Explain channel spacing (25 kHz / 8.33 kHz) and calculate channel capacity.
- Define polarization, state the polarization of aviation VHF, and relate antenna length to wavelength.

8.

8.1 The spectrum, band by band
8.2 VHF — the ATC workhorse
8.3 HF — reaching the ocean
8.4 The aviation frequency master table
8.5 Channel spacing & capacity
8.6 Polarization & antenna length

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The aircraft's antennas are the physical link between the electrical circuits inside and the invisible frequency bands outside.

8.1 The spectrum, band by band

> **FIRST PRINCIPLES — WHY THE SPECTRUM IS SPLIT INTO DECADES**
>
> The radio spectrum is divided into bands, each a factor of **ten** in frequency above the one below. This matters because the dominant way a wave travels — and therefore what it is useful for — changes with frequency. Low frequencies have long wavelengths and **bend around the Earth**; high frequencies have short wavelengths and travel in **straight lines**. Aviation lives mainly in the **HF, VHF and UHF** bands.

The Radio Spectrum Bands

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Band	Frequency	Wavelength	Dominant propagation	Typical aviation use
VLF	3–30 kHz	10–100 km	Ground/wave-guide	(Very long range; rare)
LF	30–300 kHz	1–10 km	Ground wave	NDB (low end)
MF	300 kHz–3 MHz	100 m–1 km	Ground wave	NDB; 2182 kHz maritime distress
HF	3–30 MHz	10–100 m	Sky wave	Long-range / oceanic voice (SSB)
VHF	30–300 MHz	1–10 m	Space (line-of-sight)	ATC voice, VOR, ILS LOC, markers
UHF	300 MHz–3 GHz	0.1–1 m	Line-of-sight	ILS glide path, DME, SSR, GPS, mil voice
SHF	3–30 GHz	1–10 cm	Line-of-sight	Radar, radio altimeter, satcom
EHF	30–300 GHz	1–10 mm	Line-of-sight	(Advanced / experimental)

Mnemonic — the band order

"Very Long Modems Help Various Useful Signals Excel" → VLF, LF, MF, HF, VHF, UHF, SHF, EHF. Each band is **×10** the previous.

8.2 VHF — the ATC workhorse

The VHF aviation communication band

118.0 – 136.975 MHz, using **Amplitude Modulation (AM)**, **vertically polarized**, propagating by **line-of-sight (space wave)**. Range is limited to roughly the radio horizon — higher you fly, farther you reach.

WHY AM AND WHY VHF

VHF gives clear, static-free, short-range communication ideal for an airport's busy frequencies. AM is used so that two overlapping transmissions produce a tell-tale **heterodyne squeal** — alerting everyone that two stations transmitted at once, rather than one silently capturing the channel as FM would (the "capture effect", Chapter 10). AM's "you'll hear the clash" behaviour is a **safety feature**.

8.3 HF — reaching the ocean

The HF band in aviation

3 – 30 MHz, using **Single Side Band (SSB)** for efficiency, propagating by **sky wave** (refracted off the ionosphere). This bounces signals thousands of miles — essential over oceans and polar routes where VHF line-of-sight fails. Crews change HF frequency between day and night (Chapter 9).

Exam trap

Tie the band to its job: **VHF = AM, short-range, line-of-sight; HF = SSB, long-range, sky wave.** Swapping AM/SSB or the propagation path is the classic trap.

8.4 The aviation frequency master table

Master Frequency Allocations

System / use	Frequency	Band
VHF ATC voice	118.0 – 136.975 MHz	VHF
VHF emergency (guard)	121.5 MHz	VHF
UHF military emergency	243.0 MHz	UHF
SAR scene-of-search (aux)	123.1 MHz	VHF
ELT (satellite / homing)	406 MHz / 121.5 MHz	UHF / VHF
VOR	108.0 – 117.975 MHz	VHF
ILS Localizer	108.10 – 111.95 MHz	VHF
ILS Glide Path	329 – 335 MHz	UHF
Marker beacons	75 MHz	VHF
DME	962 – 1213 MHz	UHF
SSR transponder	Interrogate 1030 / Reply 1090 MHz	UHF
NDB / ADF	≈ 190 – 1750 kHz	LF/MF
Radio altimeter	4200 – 4400 MHz	SHF
Maritime distress	2182 kHz	MF

8.5 Channel spacing & capacity

FIRST PRINCIPLES — FITTING MORE AIRCRAFT INTO ONE BAND

The VHF band is a fixed width, so the number of channels it holds depends on how closely they are spaced. Traditionally channels were **25 kHz** apart; to fit more in the crowded European band, **8.33 kHz** spacing was introduced — tripling the number of channels.

Worked example — channel capacity

The VHF band is 118.0 to 136.975 MHz, a width of **18.975 MHz = 18,975 kHz.**

At **25 kHz** spacing: 18,975 ÷ 25 ≈ **760 channels**. At **8.33 kHz** spacing: 18,975 ÷ 8.33 ≈ **2,280 channels** — roughly three times as many.

This is why 8.33-spaced frequencies read like **120.275** or **118.905** rather than ending in tidy tenths.

8.6 Polarization & antenna length

Definition & the aviation fact

Polarization is the plane of the wave's **electric field — vertical** or **horizontal**. The antenna must match it: a vertical wave needs a vertical antenna. **Aviation VHF communication is vertically polarized**, which is why the COM antenna (the blade) is mounted vertically.

Why higher-frequency aerials are short

An efficient antenna is sized to the wavelength — commonly a **half-wave** or **quarter-wave**. Since λ shrinks as frequency rises, VHF aerials are short blades while HF aerials are long wires or probes. From Chapter 7, a 120 MHz λ is 2.5 m, so a quarter-wave whip is about **0.6 m** — a practical blade. An HF λ of 100 m needs a long-wire or a tuned coupler.

Cockpit reality

The little vertical blade on the fuselage is your VHF COM antenna, matched to the vertical polarization of the signal. Mismatch the polarization and received signal strength drops sharply — one reason antennas are mounted exactly as they are.

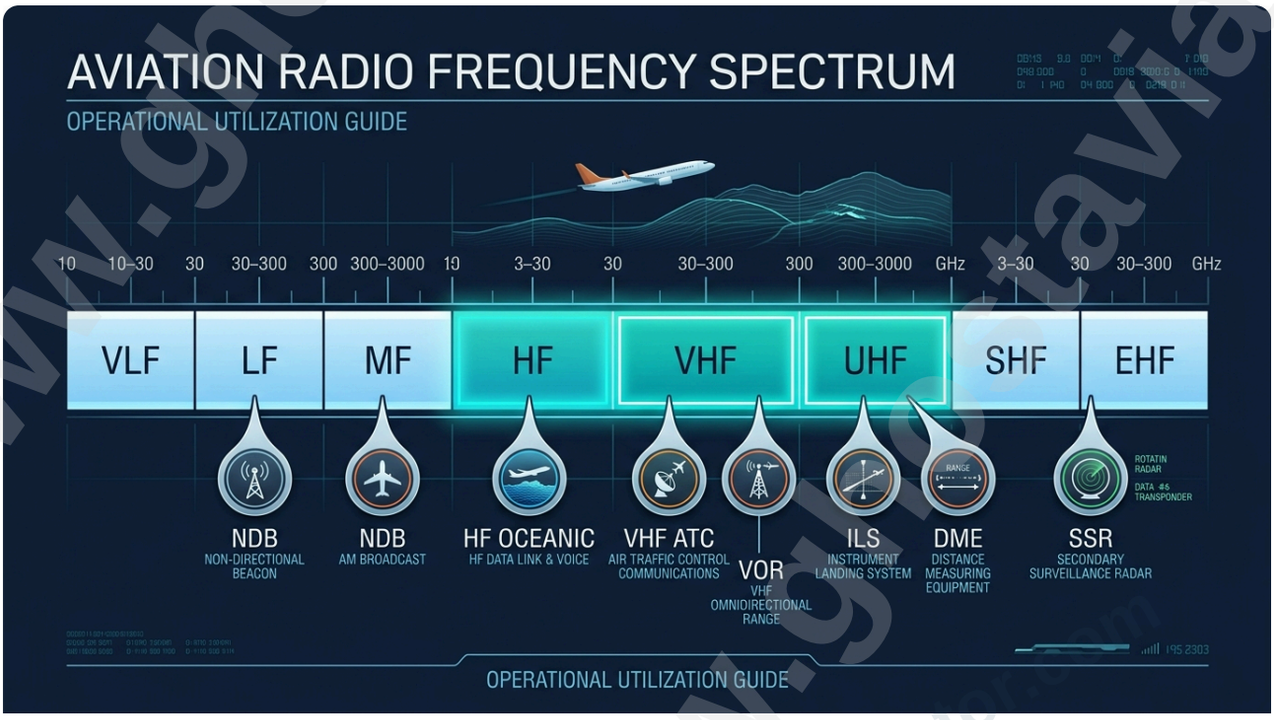

Figure 8.1 — The radio spectrum and where aviation lives.

☆ Numbers to memorise

Essential Facts for Chapter 8

Fact	Value
Band order (×10 each)	VLF · LF · MF · HF · VHF · UHF · SHF · EHF
VHF ATC voice	118.0–136.975 MHz · AM · vertical · line-of-sight
HF voice	3–30 MHz · SSB · sky wave · oceanic
Channel spacing	25 kHz (≈760 ch) and 8.33 kHz (≈2280 ch)
VOR / ILS LOC	108.0–117.975 / 108.10–111.95 MHz (VHF)
ILS GP / DME	329–335 MHz / 962–1213 MHz (UHF)
SSR transponder	1030 MHz interrogate · 1090 MHz reply
Emergency	121.5 (VHF) · 243.0 (UHF) · 406 (ELT) · 123.1 (SAR aux)
Aviation VHF polarization	Vertical

Question bank

PART A — MCQS (CLICK AN OPTION TO CHECK)

1. The VHF aeronautical communication band is:

> 108.0–117.975 MHz
>
> 118.0–136.975 MHz
>
> 3–30 MHz
>
> 329–335 MHz

Answer: 118.0–136.975 MHz. VHF ATC voice = 118.0–136.975 MHz. 108–117.975 is VOR/ILS-LOC.

2. Each radio band differs from the next by a factor of:

> 2
>
> 5
>
> 10
>
> 100

Answer: 10. Bands run in decades — each is ten times the previous in frequency.

3. Aviation VHF communication uses which modulation?

> AM
>
> FM
>
> SSB
>
> PCM

Answer: AM. VHF voice uses AM; HF long-range uses SSB.

4. Long-range oceanic voice communication uses:

> VHF AM
>
> HF SSB (sky wave)
>
> UHF
>
> VLF

Answer: HF SSB (sky wave). HF (3–30 MHz) with SSB propagates by sky wave for long range.

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5. The dominant propagation of NDB (LF/MF) is:

> Ground wave
>
> Sky wave
>
> Line-of-sight
>
> Satellite

Answer: Ground wave. LF/MF travels mainly by the ground (surface) wave.

6. An SSR transponder replies on:

> 1030 MHz
>
> 1090 MHz
>
> 121.5 MHz
>
> 75 MHz

Answer: 1090 MHz. Interrogation is 1030 MHz; the reply is 1090 MHz.

7. The ILS glide path operates in which band?

> VHF
>
> UHF (329–335 MHz)
>
> HF
>
> MF

Answer: UHF (329–335 MHz). The localizer is VHF; the glide path is UHF (329–335 MHz).

8. ILS marker beacons transmit on:

> 108 MHz
>
> 75 MHz
>
> 329 MHz
>
> 1090 MHz

Answer: 75 MHz. All markers share a 75 MHz carrier, differing in tone/code.

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9. Moving from 25 kHz to 8.33 kHz channel spacing roughly:

> Halves the number of channels
>
> Triples the number of channels
>
> Has no effect
>
> Doubles the band width

Answer: Triples the number of channels. 8.33 kHz is one-third of 25 kHz, so about three times as many channels fit.

10. Aviation VHF communication is:

> Horizontally polarized
>
> Vertically polarized
>
> Circularly polarized
>
> Unpolarized

Answer: Vertically polarized. Aviation VHF comms are vertically polarized — hence the vertical blade antenna.

11. Higher-frequency antennas are physically shorter because:

> They use less power
>
> Wavelength shrinks as frequency rises, and antennas are sized to the wavelength
>
> They are horizontally polarized
>
> They carry less data

Answer: Wavelength shrinks as frequency rises, and antennas are sized to the wavelength. Antennas are a fraction of a wavelength; shorter λ at high f means a shorter aerial.

12. The radio altimeter operates in the:

> VHF band
>
> HF band
>
> SHF band (≈4200–4400 MHz)
>
> LF band

Answer: SHF band (≈4200–4400 MHz). The radio altimeter works around 4200–4400 MHz in the SHF band.

13. The maritime distress frequency in the MF band is:

> 121.5 MHz
>
> 2182 kHz
>
> 406 MHz
>
> 243.0 MHz

Answer: 2182 kHz. 2182 kHz is the maritime distress frequency (MF).

14. The international VHF emergency frequency and the UHF military one are:

> 121.5 / 122.5 MHz
>
> 121.5 / 243.0 MHz
>
> 118.0 / 136.975 MHz
>
> 406 / 121.5 MHz

Answer: 121.5 / 243.0 MHz. 121.5 MHz (VHF guard) and 243.0 MHz (UHF military guard); 243 = 2 × 121.5.

15. VOR operates in the:

> VHF band (108.0–117.975 MHz)
>
> UHF band
>
> HF band
>
> MF band

Answer: VHF band (108.0–117.975 MHz). VOR is VHF, 108.0–117.975 MHz (sharing 108–111.95 with ILS-LOC).

16. DME operates in the:

> VHF band
>
> UHF band (962–1213 MHz)
>
> SHF band
>
> LF band

Answer: UHF band (962–1213 MHz). DME is UHF, 962–1213 MHz.

PART B — ORAL / VIVA (TAP TO REVEAL MODEL ANSWERS)

▶ **State the VHF communication band and its characteristics.**

▶ **Why is HF used for oceanic communication?**

▶ **Name the bands and uses for the ILS components.**

▶ **On what frequencies does an SSR transponder interrogate and reply?**

▶ **What is channel spacing, and why was 8.33 kHz introduced?**

▶ **What is the polarization of aviation VHF, and why does it matter?**

60-SECOND REVISION CARD

- Bands ×10: **VLF·LF·MF·HF·VHF·UHF·SHF·EHF**; aviation core = HF/VHF/UHF.
- **VHF 118–136.975 MHz, AM, vertical, line-of-sight; HF 3–30 MHz, SSB, sky wave, oceanic.**
- ILS: LOC **VHF** 108.10–111.95 · GP **UHF** 329–335 · markers **75 MHz**. DME **UHF** 962–1213. SSR **1030/1090**.
- Emergency **121.5 / 243.0 / 406**, SAR aux **123.1**. Channel spacing **25 / 8.33 kHz** (≈760 / 2280 ch).
- Aviation VHF **vertically polarized**; antenna length scales with wavelength.

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Radio Wave Propagation

A transmitter radiates energy; a receiver picks it up — but how the wave gets from one to the other decides everything about the radio. Will it hug the curve of the Earth, bounce off a layer 300 km up, or shoot in a straight line and stop dead at the horizon? The answer depends almost entirely on frequency.

SYLLABUS MAP

Part II (ii) Propagation · day/night frequencies · skip distance · fading · ground shadow

LEARNING OBJECTIVES — BY THE END OF THIS CHAPTER YOU WILL BE ABLE TO…

- Describe the three propagation paths — ground, sky and space wave — their mechanisms and bands.
- Explain the structure of the ionosphere and its daily, seasonal and solar variation.
- Define MUF, LUF and the optimum working frequency, and explain day/night frequency choice.
- Explain skip distance and dead space, with the geometry.
- Calculate line-of-sight range from the radio-horizon formula.
- Explain attenuation, fading and ground shadow.

9.1 The three paths
9.2 The ground wave
9.3 The ionosphere
9.4 The sky wave: MUF, LUF & OWF
9.5 Day & night frequencies
9.6 Skip distance & dead space
9.7 The space wave & radio horizon
9.8 Attenuation, fading & ground shadow

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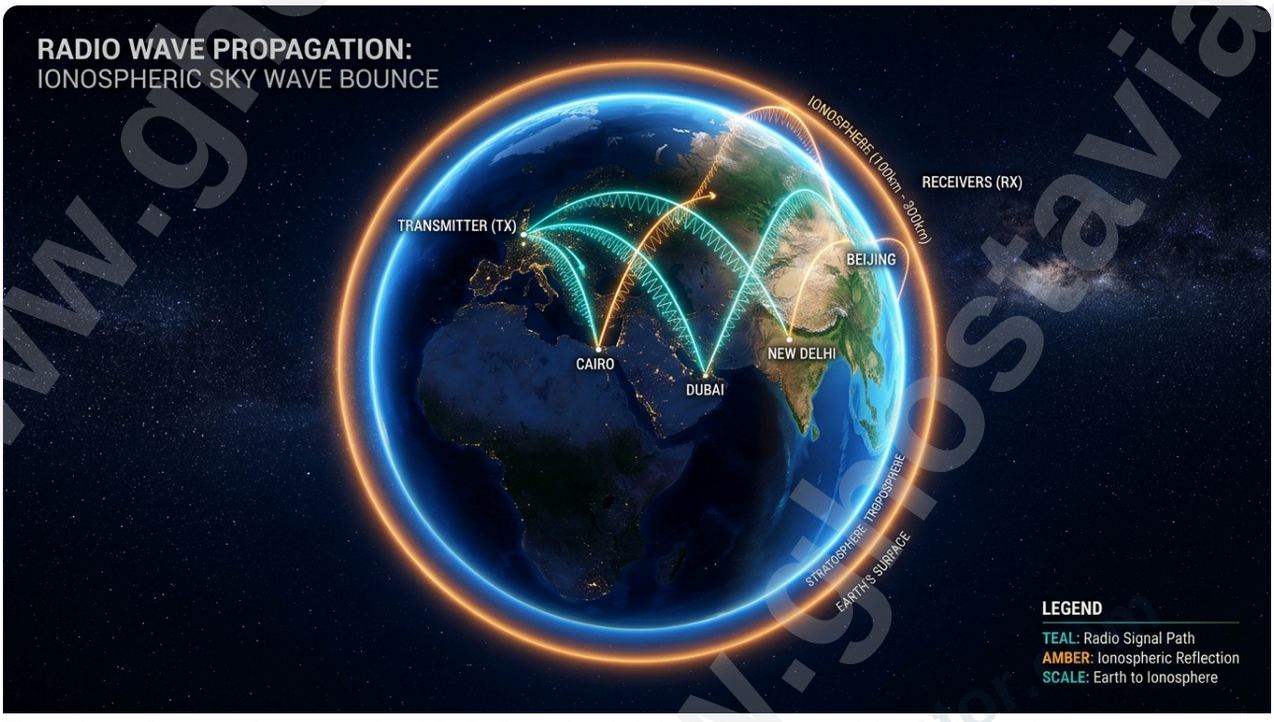


The path a radio wave takes across the globe is dictated entirely by its frequency and the physical environment.

9.1 The three paths a radio wave takes

Propagation Paths

Path	How it travels	Band	Used by
Ground (surface) wave	Follows the curve of the Earth, hugging the surface	LF / MF	NDB; maritime
Sky wave	Refracted (bent back) by the ionosphere	HF	Long-range / oceanic HF voice
Space wave	Straight line (line-of-sight) to the radio horizon	VHF & above	ATC VHF voice, VOR, ILS

> **Mnemonic**
>
> **Low hugs, High bounces, VHF shoots straight.** LF/MF follow the ground; HF bounces off the ionosphere; VHF and up go line-of-sight.

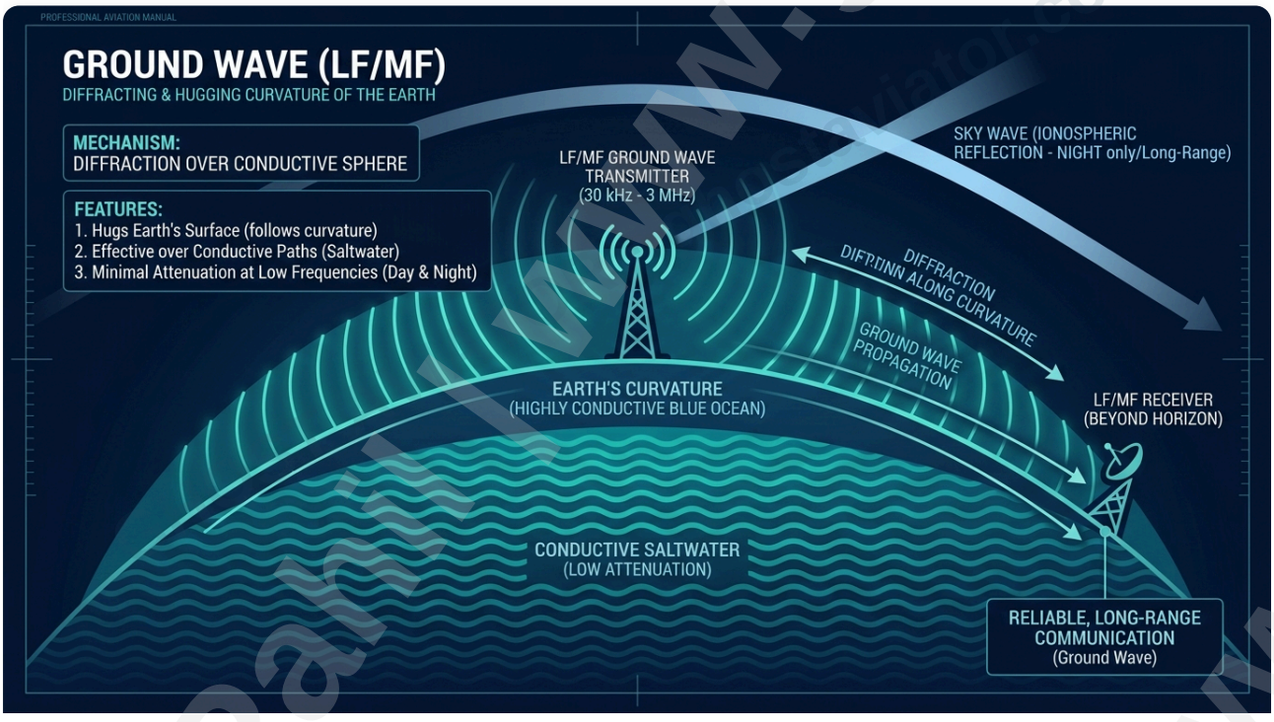
9.2 The ground wave

FIRST PRINCIPLES — WHY LOW FREQUENCIES FOLLOW THE EARTH

A vertically-polarized wave at low frequency induces currents in the conducting surface of the Earth; these slow the bottom of the wavefront, tilting it forward so it **diffracts around the curve** of the planet. The lower the frequency, the better it clings — and the farther it reaches.

What sets ground-wave range

Range grows with **transmitter power**, with **lower frequency**, and with the **conductivity of the surface** — so it is much greater over **sea water** (highly conductive) than over dry land or mountains. This is the path the NDB relies on, giving useful range beyond line-of-sight at low level.



The ground wave diffracts around the curvature of the Earth, achieving greatest range over highly conductive sea water.

9.3 The ionosphere

IN PLAIN TERMS

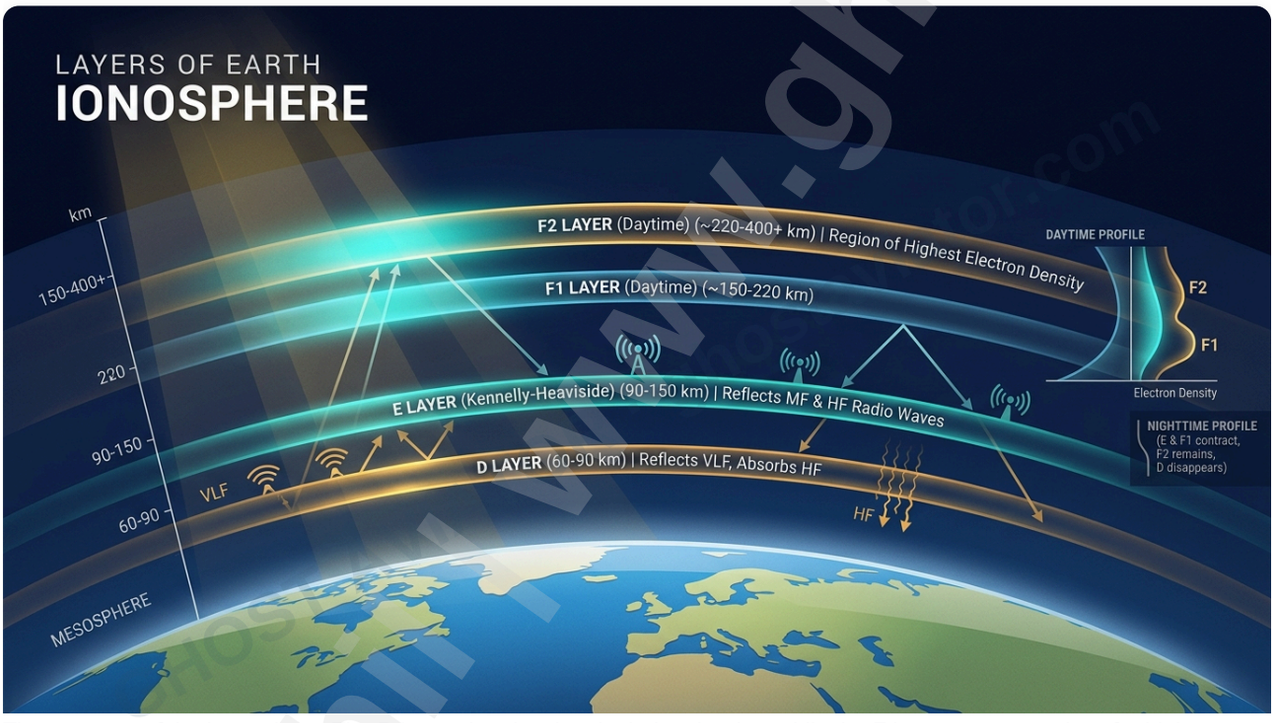
High above the Earth, solar radiation strips electrons from thin air, creating electrically charged **layers**. These act like a mirror for HF radio, bending the wave back to Earth and giving HF its enormous range.

The layers

D layer (lowest, ≈60–90 km) — forms only by **day** and **absorbs** lower HF frequencies. **E layer** (≈90–150 km). **F1 & F2 layers** (highest, ≈150–400 km) — the main HF-refracting layers; at night F1 and F2 merge into a single, weaker F layer.

Variation — why HF is never constant

Ionisation depends on the Sun, so it varies **diurnally** (strong by day, weak by night), **seasonally**, and with the **11-year solar cycle** (sunspots). The **critical frequency** is the highest frequency a layer will reflect at vertical incidence; above it the wave punches straight through.



The structure of the ionosphere: The D layer absorbs signals during the day, while the F layers act as a mirror for high-frequency (HF) sky waves.

9.4 The sky wave: MUF, LUF & OWF

Definitions

MUF — Maximum Usable Frequency: the highest frequency the ionosphere will still refract back to a given point. Above it, the wave escapes into space and is lost.
LUF — Lowest Usable Frequency: the lowest frequency that still arrives usable; below it, D-layer absorption kills it.
OWF — Optimum Working Frequency: typically about **85% of the MUF** — set safely below the MUF so normal ionospheric variation doesn't push the working frequency above it. The usable window lies **between LUF and MUF**.

Angle of incidence & hops

A wave striking the ionosphere at a **shallow (grazing) angle** is refracted back over a long distance; a steep angle may punch through. By bouncing between the ionosphere and the ground, a sky wave can make several **hops**, reaching right around the world.

9.5 Day & night frequencies

FIRST PRINCIPLES — WHY HF TRACKS THE SUN

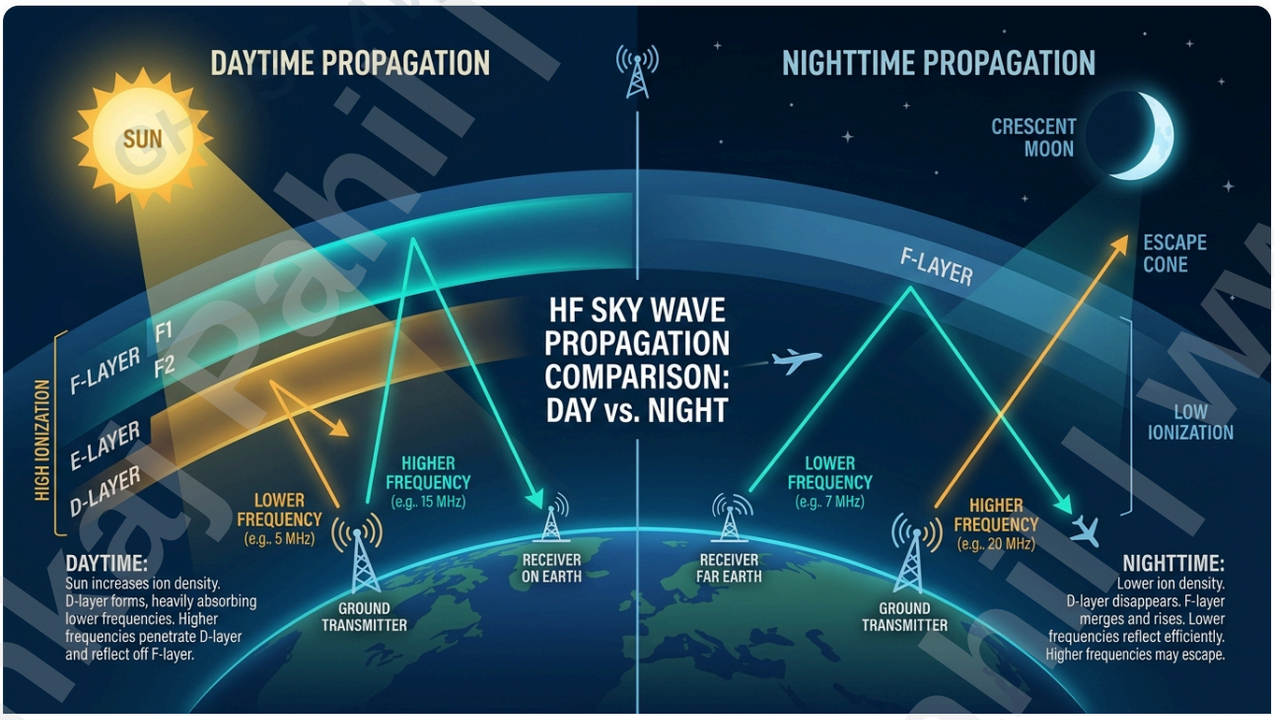
By **day** the D layer is present and absorbs lower frequencies, so you must use a **higher** HF frequency to get through. By **night** the D layer disappears and the ionosphere weakens, so a **lower** HF frequency works best.

The rule of thumb

Higher frequency by DAY, lower frequency by NIGHT for HF sky-wave communication. This is why long-haul crews change their HF frequency as they cross the day/night terminator.

Mnemonic

"Day = High, Night = Low." Sun up, frequency up.



By day, the absorbing D-layer forces crews to use higher frequencies. By night, the D-layer vanishes, allowing lower frequencies to reach the reflecting F-layer.

9.6 Skip distance & dead space

Definitions

Skip distance: the distance from the transmitter to the point where the **first sky wave** returns to Earth.
Dead space (skip zone): the silent ring *between* the end of ground-wave coverage and the point where the first sky wave lands — a region where neither wave reaches and reception fails.

Exam trap

Skip distance **increases** as you raise the frequency (toward the MUF) and as the ionosphere weakens at night — which is why dead space changes through the day. Above the MUF there is no sky-wave return at all.

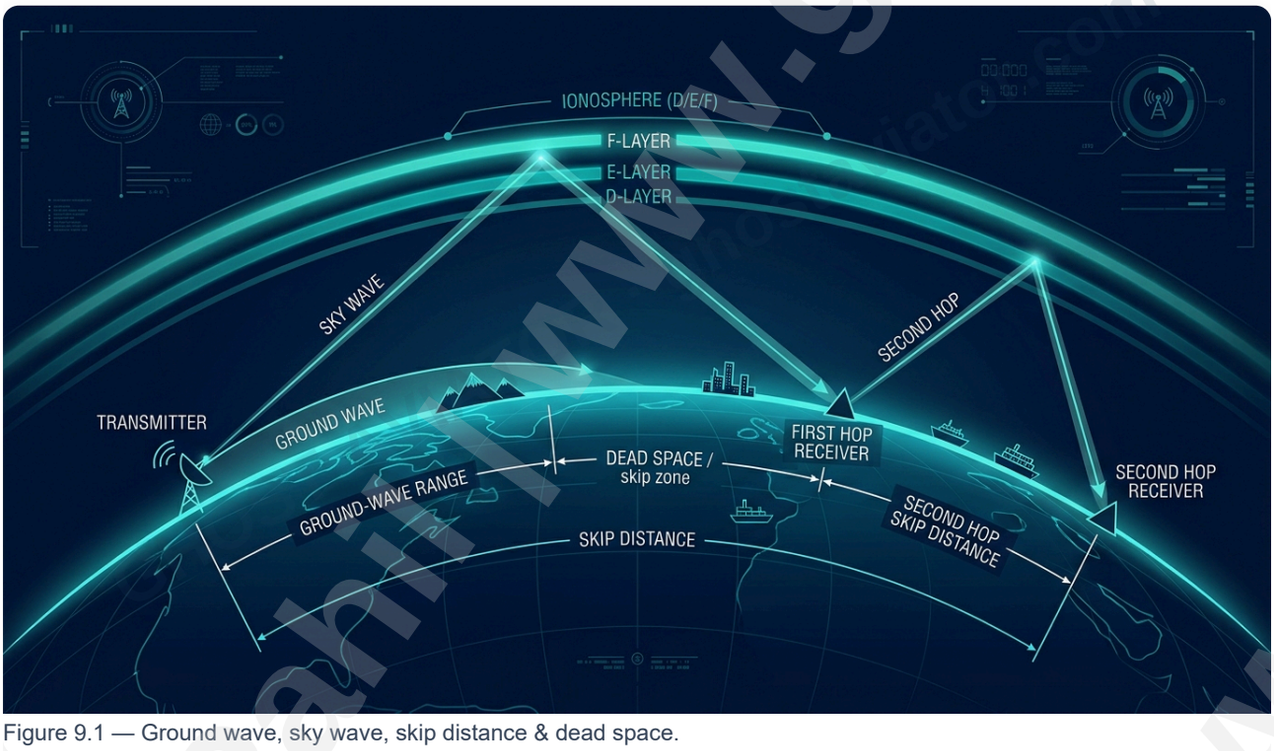


Figure 9.1 — Ground wave, sky wave, skip distance & dead space.

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9.7 The space wave & radio horizon

IN PLAIN TERMS

VHF and above travel in essentially straight lines (the **space wave**), so range is set by the **radio horizon** — and the higher you (or the station) are, the farther that horizon. This is why a VHF call that fails on the ground works once you climb.

$$d\ (\text{NM}) \approx 1.23 \times \sqrt{h}$$

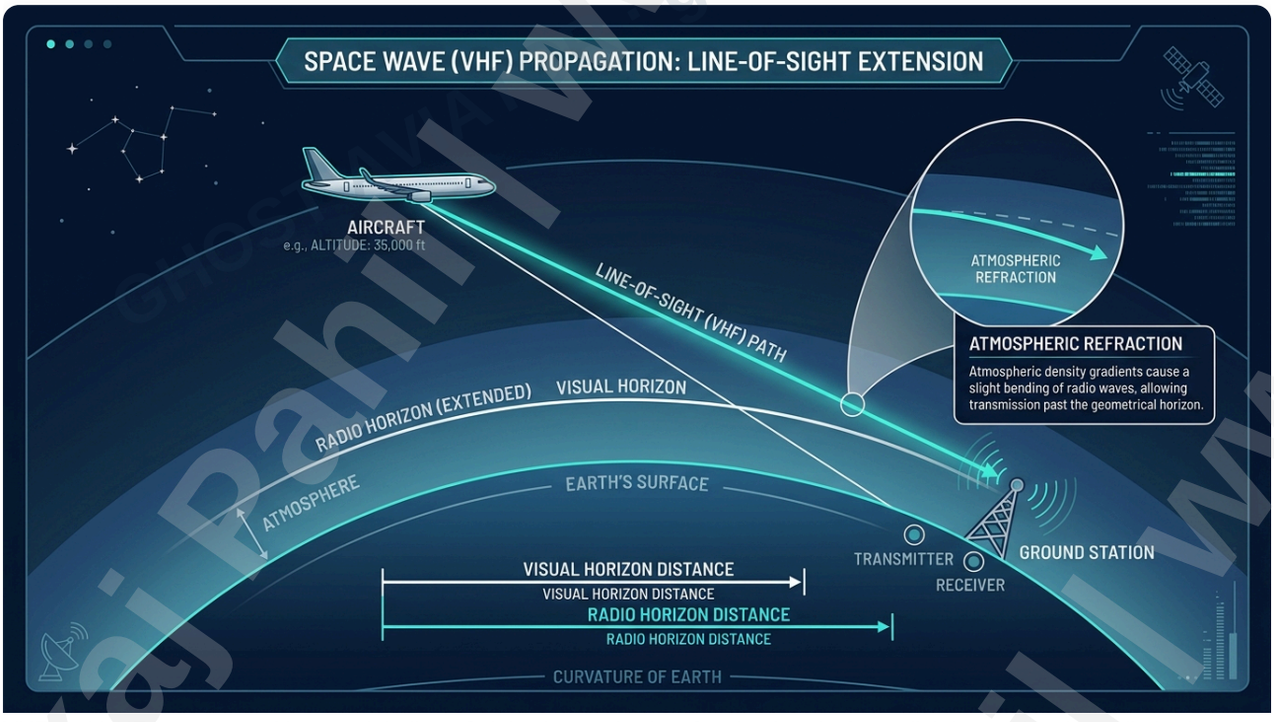
h = height in feet · the radio horizon is slightly beyond the visual horizon (atmospheric refraction)

Worked example — line-of-sight range

At **10,000 ft:** d ≈ 1.23 × √10,000 = 1.23 × 100 = **123 NM** to the radio horizon.

If the ground station antenna is at **400 ft:** its horizon ≈ 1.23 × √400 = 1.23 × 20 = ≈ **25 NM**. The **total** path is the sum: 123 + 25 ≈ **148 NM**.

Takeaway: **more height, more range** — the exact figure isn't usually required, but the principle and the formula are.



VHF and UHF waves travel in a straight line-of-sight. Atmospheric refraction bends them slightly, making the radio horizon ≈15% farther than the visual horizon.

▶ **Deep dive — ducting & super-refraction**

9.8 Attenuation, fading & ground shadow

Three signal enemies

Attenuation — the natural weakening of a signal with distance and absorption.
Fading — fluctuation in received strength, typically when two paths (ground + sky, or two sky paths) arrive together and interfere; common on HF at dusk.
Ground shadow — loss of a line-of-sight (space) wave behind a hill, building or terrain that blocks it.

Cockpit reality

Low on approach behind high ground, your VHF can drop out — that's ground shadow; climb a little and the line-of-sight clears. On HF over the ocean at night you'll hear the signal swell and fade — that's sky-wave fading, eased by selecting a better frequency.

☆ Numbers to memorise

Essential Facts for Chapter 9

Fact	Value
Ground wave	LF/MF, follows Earth's curve, better over sea (NDB)
Sky wave	HF, refracted by ionosphere (long range, multi-hop)
Space wave	VHF+, line-of-sight to radio horizon
Ionosphere layers	D, E, F1, F2 (D absorbs by day; F1/F2 merge at night)
HF day/night	Higher freq by day, lower by night
MUF / LUF / OWF	Max usable / Lowest usable / ≈85% of MUF
Dead space	Gap between ground-wave range and first sky-wave return
Radio horizon	d (NM) ≈ 1.23 × √h(ft)

Question bank

PART A — MCQS (CLICK AN OPTION TO CHECK)

1. An NDB relies mainly on which propagation path?

> Ground (surface) wave
>
> Sky wave
>
> Space wave
>
> Satellite

```
Answer: Ground (surface) wave. NDB (LF/MF) uses the ground wave that follows the Earth's curvature.
```

2. Ground-wave range is greatest over:

> Dry desert
>
> Sea water (high conductivity)
>
> Mountains
>
> Forest

```
Answer: Sea water (high conductivity). Higher surface conductivity (sea) gives much greater ground-wave range.
```

3. Long-range HF communication is achieved by:

> Ground wave
>
> Sky wave refracted by the ionosphere
>
> Line-of-sight space wave
>
> Direct wave only

```
Answer: Sky wave refracted by the ionosphere. HF sky waves refract off the ionosphere to travel thousands of miles.
```

4. Which ionospheric layer absorbs lower HF frequencies by day?

> D layer
>
> E layer
>
> F1 layer
>
> F2 layer

```
Answer: D layer. The D layer forms by day and absorbs lower frequencies; it disappears at night.
```

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5. For HF, the best night-time frequency compared with daytime is generally:

> Higher
>
> Lower
>
> The same
>
> Zero

Answer: Lower. The D layer disappears at night, so a lower frequency works best — "day high, night low."

6. A frequency above the MUF will:

> Be absorbed by the D layer
>
> Pass through the ionosphere into space
>
> Follow the ground
>
> Always reflect

Answer: Pass through the ionosphere into space. Above the Maximum Usable Frequency the wave is no longer refracted back — it escapes to space.

7. The optimum working frequency (OWF) is typically about:

> 50% of the MUF
>
> 85% of the MUF
>
> 110% of the MUF
>
> Equal to the LUF

Answer: 85% of the MUF. The OWF is set around 85% of the MUF so variation doesn't push it above the MUF.

8. The silent ring between ground-wave coverage and the first returning sky wave is the:

> Skip distance
>
> Dead space (skip zone)
>
> Radio horizon
>
> Fading zone

Answer: Dead space (skip zone). Dead space is the gap where neither ground nor sky wave is received.

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9. Skip distance generally increases when:

> Frequency is lowered
>
> Frequency is raised toward the MUF / at night
>
> Power is reduced
>
> The antenna is shortened

Answer: Frequency is raised toward the MUF / at night. Higher frequency and a weaker (night) ionosphere lengthen the skip distance.

10. The line-of-sight range from 10,000 ft is approximately:

> 12 NM
>
> 123 NM
>
> 1230 NM
>
> 40 NM

Answer: 123 NM. d ≈ 1.23 × √10,000 = 1.23 × 100 = 123 NM.

11. VHF reception is lost behind a hill on approach. This is:

> Sky-wave fading
>
> Skip distance
>
> Ground shadow
>
> Absorption by the D layer

Answer: Ground shadow. Terrain blocking the line-of-sight (space) wave creates a ground shadow.

12. Fading on an HF signal is usually caused by:

> High transmitter power
>
> Two paths interfering (e.g. ground + sky, or two sky paths)
>
> A short antenna
>
> Vertical polarization

Answer: Two paths interfering (e.g. ground + sky, or two sky paths). Fading is interference between signals arriving by different paths, common on HF.

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13. The ionosphere's ionisation varies with:

> Transmitter power only
>
> Time of day, season and the 11-year solar cycle
>
> Antenna length
>
> Aircraft altitude

Answer: Time of day, season and the 11-year solar cycle. It is driven by the Sun, so it varies diurnally, seasonally and with the sunspot cycle.

14. A sky wave can reach right around the world by:

> Following the ground
>
> Multiple hops between the ionosphere and the ground
>
> Travelling in a straight line
>
> Using a satellite

Answer: Multiple hops between the ionosphere and the ground. Repeated refraction and ground reflection make multiple hops possible.

15. The radio horizon is, compared with the visual horizon:

> Shorter
>
> Slightly farther (atmospheric refraction)
>
> Exactly the same
>
> Half as far

Answer: Slightly farther (atmospheric refraction). Atmospheric refraction bends the wave slightly, extending the radio horizon a little beyond the visual one.

16. Which path does VHF ATC voice use?

> Ground wave
>
> Sky wave
>
> Space wave (line-of-sight)
>
> All three equally

Answer: Space wave (line-of-sight). VHF uses the space wave — line-of-sight, range growing with altitude.

PART B — ORAL / VIVA (TAP TO REVEAL MODEL ANSWERS)

▶ **Name the three propagation paths and the bands that use them.**

▶ **Describe the ionosphere and how it varies.**

▶ **Define MUF, LUF and OWF.**

▶ **Why does the choice of HF frequency change between day and night?**

▶ **What is dead space, and how does it relate to skip distance?**

▶ **How is VHF line-of-sight range estimated, and what increases it?**

PART C — NUMERICAL PROBLEMS (TAP FOR WORKED SOLUTIONS)

▶ **P1. Find the radio horizon from 36,000 ft.**

▶ **P2. An aircraft at 25,000 ft talks to a station antenna at 100 ft. Estimate the maximum communication range.**

▶ **P3. If the MUF is 12 MHz, estimate the OWF.**

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- **Ground** (LF/MF, NDB, better over sea) · **Sky** (HF, ionosphere, multi-hop) · **Space** (VHF+, line-of-sight).
- Layers **D E F1 F2**; D absorbs by day. Ionisation varies day/season/solar cycle.
- HF: **day high, night low**. **MUF** above = lost · **LUF** below = absorbed · **OWF ≈ 85% MUF**.
- **Dead space** = gap before first sky wave; skip distance grows with frequency/at night.
- Radio horizon **d ≈ 1.23√h(ft)**; **ground shadow** blocks VHF behind terrain; **fading** = multi-path.

Modulation

Your voice is a slow, weak vibration in the air. A radio wave is a fast, powerful oscillation that can leap across a continent. Modulation is the marriage of the two — the trick that lets a feeble 1 kHz voice ride on a 120 MHz carrier and arrive intact at a control tower. Get it right and you understand why aviation insists on AM for VHF and SSB for HF.

SYLLABUS MAP

`Part II (iii) Systems for air-ground communication – carrier, AM/FM/SSB, the Tx/Rx chain`

LEARNING OBJECTIVES — BY THE END OF THIS CHAPTER YOU WILL BE ABLE TO…

- Explain what a carrier wave is and the two reasons modulation is needed.
- Describe AM in depth — sidebands, bandwidth, modulation depth and power distribution.
- Describe FM — deviation, bandwidth (Carson's rule), noise immunity and the capture effect.
- Compare DSB and SSB and calculate the power/bandwidth saving.
- Explain why aviation uses AM on VHF and SSB on HF, and read emission designators.
- Outline the transmitter chain and the superheterodyne receiver.

10.1 The carrier & why we modulate

10.2 AM — sidebands & depth

10.3 AM power distribution

10.4 FM

10.5 DSB & SSB

10.6 Why AM on VHF, SSB on HF

10.7 Emission designators

10.8 The Tx & Rx chains

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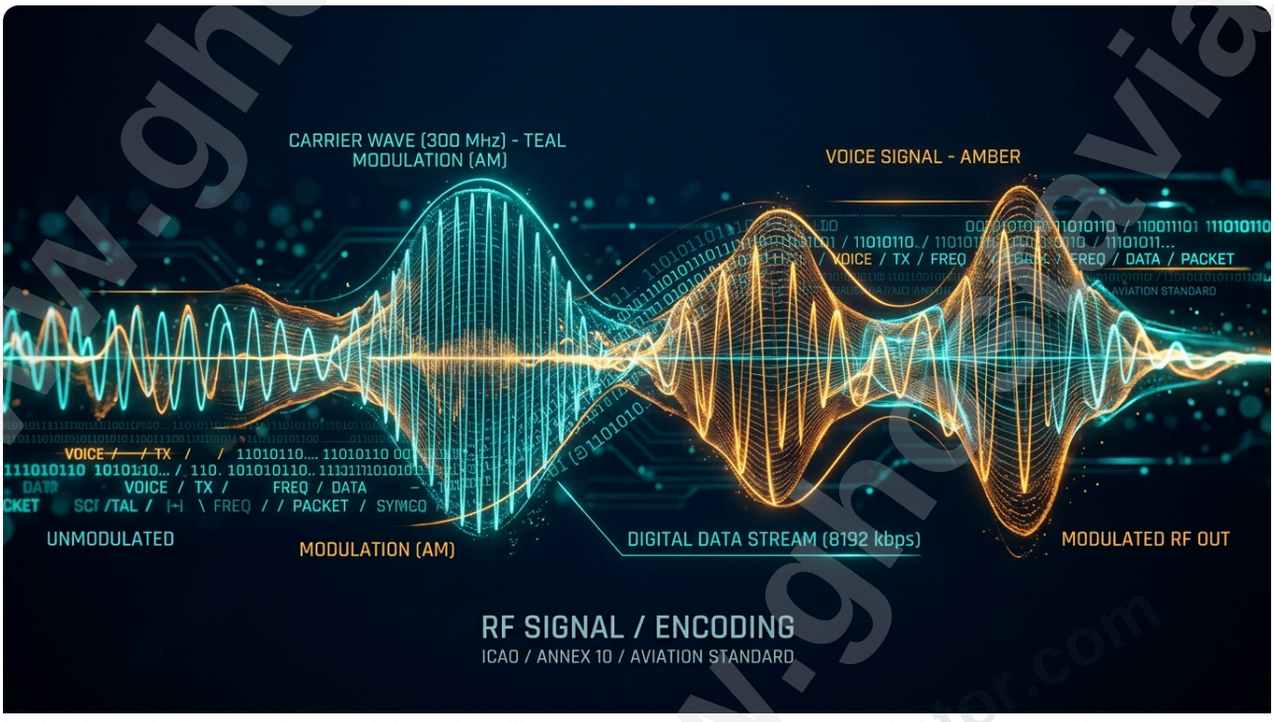


Modulation is the process of loading information—like your voice or digital data—onto a high-frequency carrier wave.

10.1 The carrier & why we modulate

FIRST PRINCIPLES — TWO REASONS WE CANNOT SEND THE VOICE DIRECTLY

You cannot simply connect a microphone to an antenna. **First**, an efficient antenna must be a fair fraction of a wavelength; a 1 kHz audio tone has a wavelength of 300 km, demanding an absurd aerial. **Second**, every voice would occupy the same low band, so no two stations could share the air. The solution is to put the voice onto a **high-frequency carrier**: the carrier needs only a short antenna, and each station can use a different carrier frequency. **Modulation** is the act of loading the voice onto the carrier.

Definition — carrier & modulation

The **carrier wave** is a high-frequency wave generated by the transmitter; on its own it carries **no information**. **Modulation** varies one property of the carrier — its **amplitude** or its **frequency** — in step with the low-frequency audio, so the information rides on the carrier and can be recovered at the receiver.

10.2 AM — sidebands & modulation depth

IN PLAIN TERMS

In Amplitude Modulation, the **height (amplitude)** of the carrier is varied in step with the voice, while its frequency stays fixed. The outline (envelope) of the wave traces your voice.

Sidebands & bandwidth

Mixing a carrier (fc) with an audio tone (fa) produces two new frequencies either side of the carrier: an **upper sideband** (fc + fa) and a **lower sideband** (fc − fa). The information lives in the sidebands; the carrier itself carries none.
The total AM bandwidth = 2 × highest audio frequency.

Modulation depth $m = V_m / V_c$

V_m = peak modulating (audio) voltage · V_c = carrier voltage · usually expressed as a %

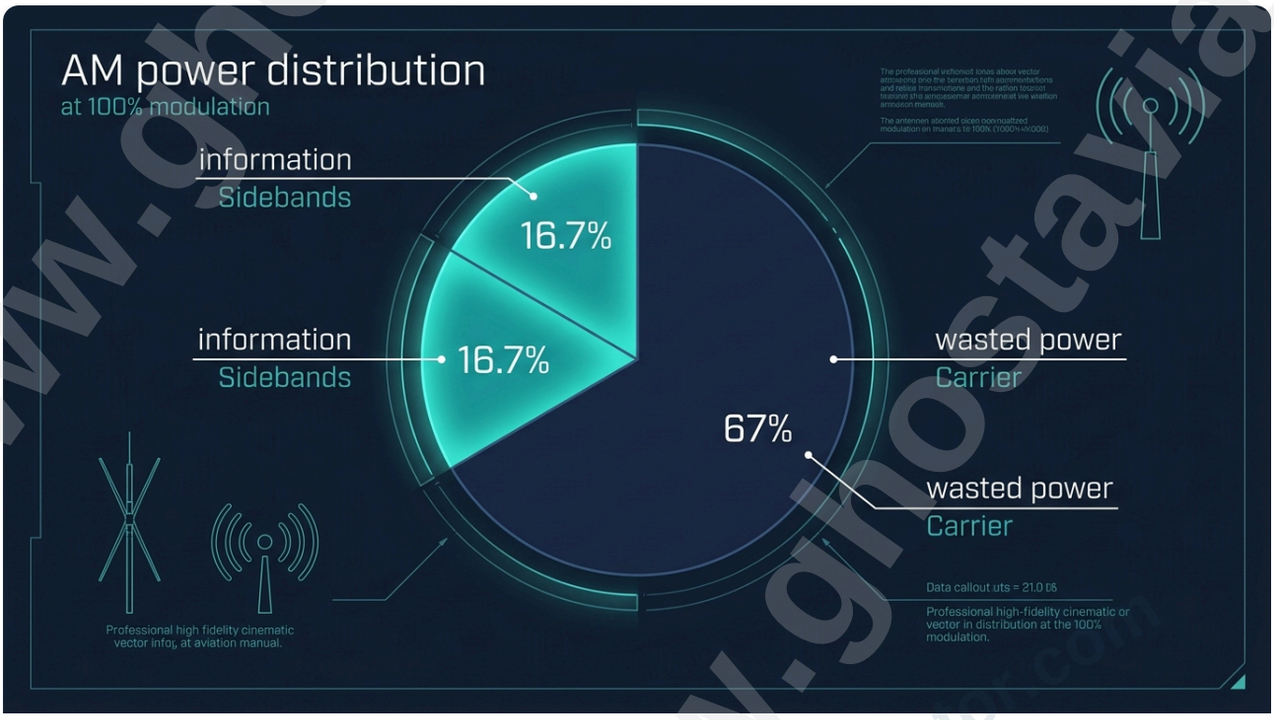
Modulation depth (index)

The **depth of modulation** is how strongly the carrier is varied. **Too little** (low %) wastes power and gives a weak, quiet signal; **over 100% (over-modulation)** distorts the audio and "splatters" energy into adjacent channels, causing interference.

10.3 AM power distribution

WHY AM IS POWER-INEFFICIENT

Even at full 100% modulation, **two-thirds of the transmitted power sits in the carrier**, which carries no information; only one-third is shared between the two sidebands. That waste is the motivation for SSB.



At 100% modulation depth, ≈67% of the transmitter's power is wasted in the carrier, leaving only ≈16.7% for each information-carrying sideband.

Worked example — where the power goes

At 100% modulation, of the total power: **~67%** is in the carrier, **~16.7%** in each sideband. The information-bearing power is only the two sidebands **≈ 33%**.

SSB throws away the carrier and one sideband, putting essentially **all** the transmitter's power into the single information-bearing sideband — a large efficiency gain, decisive over long HF distances.

10.4 FM — Frequency Modulation

IN PLAIN TERMS

In FM the carrier's **frequency** is varied in step with the voice while its **amplitude stays constant**. The amount of frequency swing is the **deviation**. Because most noise affects amplitude, FM is far more noise-resistant — which is why it is used for music broadcasting.

FM bandwidth — Carson's rule

FM bandwidth **≈ 2 × (deviation + highest audio frequency)**. FM generally needs **more bandwidth** than AM, which is part of why narrow aviation channels stayed with AM.

Exam trap — the capture effect

FM has a **capture effect**: the stronger of two signals captures the receiver and the weaker is silenced. In aviation that is dangerous — a second aircraft transmitting would be **silently lost. This is the key reason aviation does NOT use FM for VHF voice.**

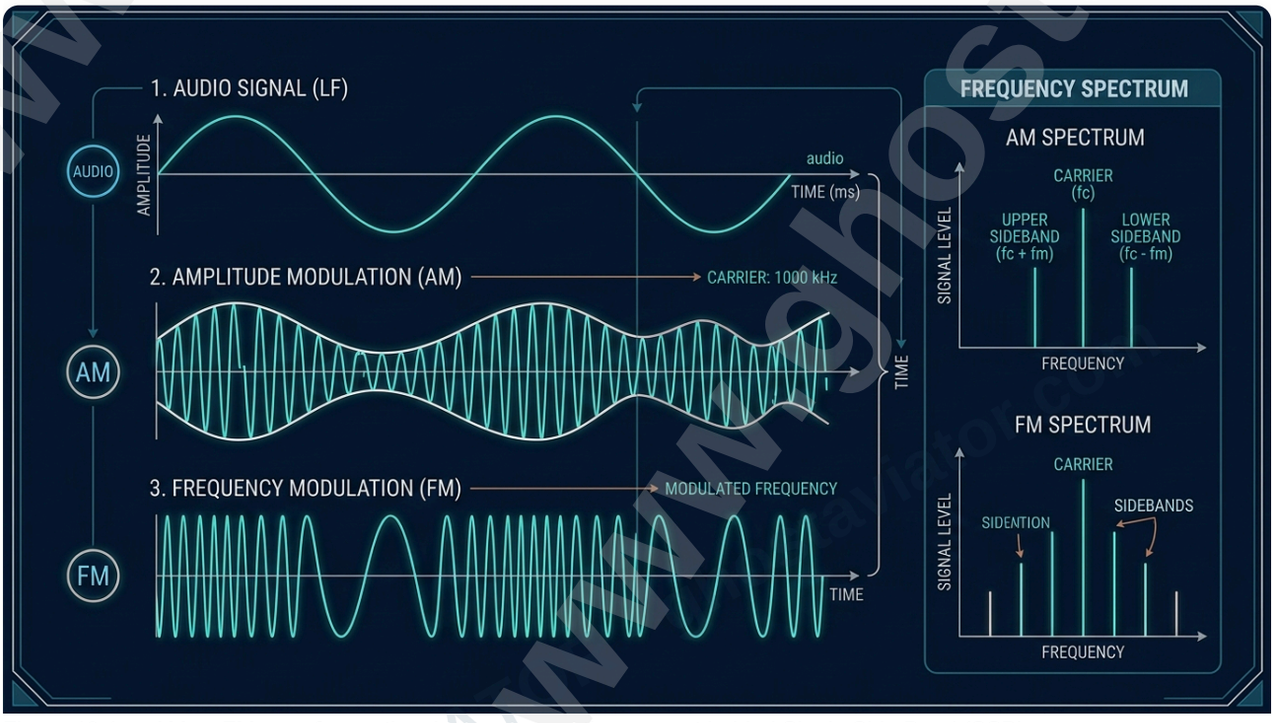


Figure 10.1 — AM vs FM waveforms, the sidebands, and the concept behind Single Side Band (SSB).

10.5 DSB & SSB

Modulation Modes Compared

Mode	What is transmitted	Trade-off
DSB (full AM)	Carrier + both sidebands	Simple receiver; but wasteful — most power in the carrier
DSB-SC	Both sidebands, carrier suppressed	Saves the carrier power; needs carrier reinsertion
SSB	**One sideband only; carrier & other sideband suppressed**	**Far more power- and bandwidth-efficient — ideal for HF long range**

10.6 Why AM on VHF, SSB on HF

THE TWO DESIGN DECISIONS, SIDE BY SIDE

VHF voice = AM: if two aircraft transmit together on the same VHF frequency, AM produces an audible **heterodyne squeal** — everyone hears that two stations "stepped on" each other and can re-transmit. With FM's capture effect, one would simply be lost. AM's "you'll hear the clash" behaviour is a **safety feature**.

HF voice = SSB: over oceanic distances, efficiency and bandwidth matter most, so HF uses the power- and bandwidth-saving SSB.

Why SSB for HF

SSB puts all the transmitter's power into the one sideband that carries information and uses **half the bandwidth** of full AM. Over thousands of miles of HF that efficiency is decisive — so **HF aviation voice uses SSB**. The receiver must reinsert a local carrier to demodulate it.

Cockpit reality

That brief squeal you sometimes hear after a call is two aircraft transmitting together. Because VHF is AM, you know it happened and can say "say again, two together." That awareness is exactly what AM buys aviation — and what FM would hide.

10.7 Emission designators

The ICAO/ITU three-symbol code

An emission is classified by three symbols: **type of modulation · nature of the signal · type of information**. The two you must recognise:
A3E = amplitude-modulated double-sideband telephony (aviation VHF voice)
J3E = single-sideband suppressed-carrier telephony (aviation HF voice).

10.8 The transmitter & receiver chains

Transmitter — voice to airwaves

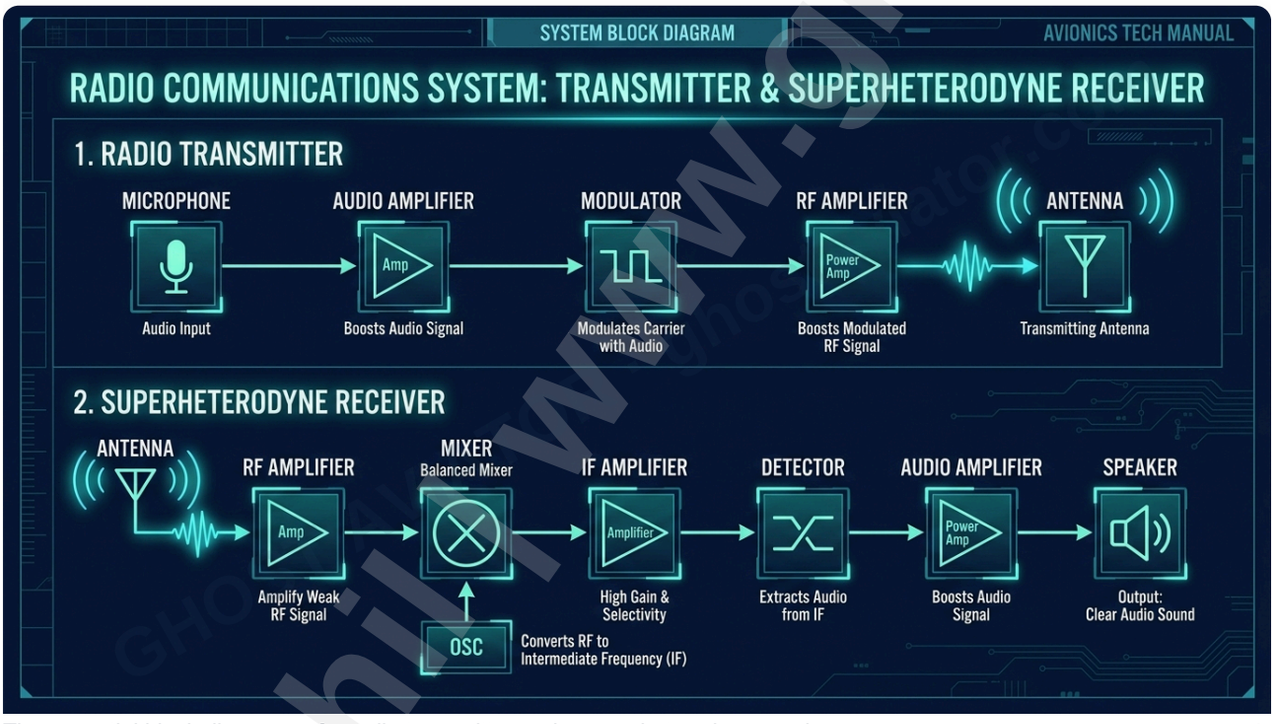
Microphone → audio amplifier → **oscillator** (generates the carrier) → **modulator** (mixes audio onto the carrier) → **RF power amplifier** → antenna.

Receiver — airwaves to voice (superheterodyne)

Antenna → RF amplifier → **mixer + local oscillator** (shifts the signal to a fixed **intermediate frequency**, IF) → IF amplifier (most gain/selectivity) → **detector / demodulator** (recovers the audio) → audio amplifier → speaker/headset. **AGC/AVC** holds the output level steady (Chapter 11).

Why "superheterodyne"

Converting every incoming signal to one fixed IF means the gain and selectivity stages are always tuned to the same frequency — far easier to make highly selective than tuning every stage to each station. It is the architecture of virtually every aviation receiver.



The essential block diagrams of a radio transmitter and a superheterodyne receiver.

☆ Numbers to memorise

Essential Facts for Chapter 10

Fact	Value
Carrier	High-frequency wave; carries no info until modulated
AM	Amplitude varied; bandwidth = 2 × highest audio; info in sidebands
Modulation depth	m = Vm/Vc; over 100% → distortion & splatter
AM power (100%)	≈ 67% carrier, ≈ 33% sidebands (the only useful part)
FM	Frequency varied; noise-resistant; capture effect; Carson BW = 2(dev + fmax)
SSB	One sideband, carrier suppressed; efficient; HF voice
Aviation choice	VHF voice = AM (A3E) · HF voice = SSB (J3E)
Receiver	Superheterodyne — converts to a fixed IF

Question bank

PART A — MCQS (CLICK AN OPTION TO CHECK)

1. In amplitude modulation, what is varied?

> The amplitude of the carrier
>
> The frequency of the carrier
>
> The speed of light
>
> The antenna length

```
Answer: The amplitude of the carrier. AM varies the carrier's amplitude in step with the audio; frequency stays fixed.
```

2. One reason a voice cannot be transmitted directly is that it would require:

> Too little power
>
> An impractically long antenna (its wavelength is enormous)
>
> A satellite
>
> Frequency modulation

```
Answer: An impractically long antenna (its wavelength is enormous). Audio wavelengths are hundreds of km; modulating a high-frequency carrier allows a short antenna and frequency sharing.
```

3. The information in an AM signal is carried in the:

> Carrier
>
> Sidebands
>
> Antenna
>
> Local oscillator

Answer: Sidebands. The carrier carries no information; the sidebands do.

4. If the highest audio frequency is 3 kHz, the AM bandwidth is:

> 3 kHz
>
> 6 kHz
>
> 1.5 kHz
>
> 9 kHz

Answer: 6 kHz. AM bandwidth = 2 × highest audio frequency = 2 × 3 = 6 kHz.

5. Modulating beyond 100% depth causes:

> A clearer signal
>
> More range
>
> Distortion and interference to adjacent channels
>
> Lower frequency

Answer: Distortion and interference to adjacent channels. Over-modulation distorts the audio and splatters into neighbouring channels.

6. At 100% AM modulation, the proportion of power in the carrier is about:

> 33%
>
> 67%
>
> 100%
>
> 0%

Answer: 67%. About two-thirds of the power is in the carrier, which carries no information.

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7. In FM, what is held constant?

> Frequency
>
> Amplitude
>
> Wavelength of the audio
>
> Bandwidth

Answer: Amplitude. FM varies frequency while amplitude stays constant — hence its noise immunity.

8. FM is generally NOT used for VHF aviation voice because of its:

> Low cost
>
> Capture effect (weaker signal silenced)
>
> Wide range
>
> Use of sidebands

Answer: Capture effect (weaker signal silenced). FM's capture effect would silently suppress a second transmitting aircraft — unsafe.

9. Aviation VHF voice uses AM mainly because:

> It is cheaper
>
> Two simultaneous transmissions produce an audible heterodyne, alerting users
>
> It has the longest range
>
> It needs no antenna

Answer: Two simultaneous transmissions produce an audible heterodyne, alerting users. AM reveals when two stations transmit together (a squeal); FM's capture effect would silently lose one.

10. SSB is preferred for HF because it is:

> Simpler to receive
>
> More power- and bandwidth-efficient
>
> Immune to all fading
>
> Higher frequency

Answer: More power- and bandwidth-efficient. SSB suppresses the carrier and one sideband, concentrating power and halving bandwidth.

11. SSB uses how much bandwidth compared with full AM?

> Twice
>
> Half
>
> The same
>
> Four times

Answer: Half. Transmitting one sideband instead of two halves the bandwidth.

12. The emission designator for aviation VHF voice (AM telephony) is:

> J3E
>
> A3E
>
> F3E
>
> A1A

Answer: A3E. A3E = AM double-sideband telephony (VHF voice); J3E = SSB telephony (HF voice).

13. The aviation receiver architecture that converts every signal to a fixed intermediate frequency is the:

> Direct-conversion receiver
>
> Superheterodyne receiver
>
> Crystal set
>
> Regenerative receiver

Answer: Superheterodyne receiver. The superheterodyne mixes the signal down to a fixed IF for easy selectivity and gain.

14. In the transmitter chain, the stage that generates the carrier is the:

> Modulator
>
> Oscillator
>
> Power amplifier
>
> Antenna

Answer: Oscillator. The oscillator generates the carrier; the modulator then mixes the audio onto it.

15. Modulation depth m is defined as:

> Carrier voltage ÷ audio voltage
>
> Audio (modulating) voltage ÷ carrier voltage
>
> Frequency ÷ wavelength
>
> Power ÷ bandwidth

`Answer: Audio (modulating) voltage ÷ carrier voltage. m = Vm/Vc, usually expressed as a percentage.`

16. Compared with AM, FM generally requires:

> Less bandwidth
>
> More bandwidth
>
> No bandwidth
>
> The same bandwidth

`Answer: More bandwidth. By Carson's rule, FM bandwidth ≈ 2(deviation + fmax), usually more than AM.`

PART B — ORAL / VIVA (TAP TO REVEAL MODEL ANSWERS)

▶ **What is modulation, and why is it needed?**

▶ **Describe AM, its sidebands and bandwidth.**

▶ **Why does aviation use AM for VHF voice?**

▶ **Why is SSB used for HF?**

▶ **What is over-modulation and its effect?**

▶ **Outline the superheterodyne receiver.**

PART C — NUMERICAL PROBLEMS (TAP FOR WORKED SOLUTIONS)

▶ **P1. A carrier of 120.000 MHz is amplitude-modulated by a 2.5 kHz tone. Give the sideband frequencies and bandwidth.**

▶ **P2. A carrier of 10 V is modulated by a 6 V audio signal. Find the modulation depth.**

▶ **P3. An FM signal has a deviation of 5 kHz and a top audio of 3 kHz. Estimate its bandwidth.**

60-SECOND REVISION CARD

- **Carrier** = blank vehicle; modulation loads the voice (needed for a short antenna + frequency sharing).
- **AM** varies amplitude; **BW = 2 × fmax**; info in sidebands; ≈67% power wasted in the carrier.
- **FM** varies frequency, noise-resistant, wider bandwidth, has the **capture effect**.
- **SSB** = one sideband, carrier suppressed, half the bandwidth → **HF (J3E)**; **AM = VHF voice (A3E)**.
- Over-modulation (>100%) → distortion & splatter. Receiver = **superheterodyne** (fixed IF).

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Radio Equipment & Operation

Everything so far has been theory; this chapter is your hands on the equipment. It covers the modes you communicate in, the discipline of the push-to-talk button, the controls on the panel, the antennas on the airframe, the emergency beacon that calls for help when you cannot, and the tell-tale signs that a radio has failed.

SYLLABUS MAP

Part II (iii)(iv) Communication systems · mic/squelch/AVC/tuning · simplex/duplex · advantages & limitations of RT

LEARNING OBJECTIVES — BY THE END OF THIS CHAPTER YOU WILL BE ABLE TO…

- Distinguish simplex, duplex, half-duplex and broadcast, and state which aviation VHF uses.
- Apply correct push-to-talk technique and explain the danger of a stuck microphone.
- Describe microphones, headsets, ANR, the audio panel and intercom.
- Explain squelch, AVC/AGC, volume, gain and frequency selection.
- Describe aircraft antennas, the need for matching, and the ELT and ACARS.
- Troubleshoot a radio failure and state the advantages and limitations of RT.

11.1 Simplex, duplex & half-duplex
11.2 Push-to-talk technique
11.3 Microphones, headsets & ANR
11.4 The audio panel & intercom
11.5 Squelch, AVC, volume & tuning
11.6 Antennas & matching
11.7 ELT & ACARS
11.8 Radio failure & troubleshooting
11.9 Advantages & limitations of RT

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Understanding the controls on the COM panel and the audio selector is essential for reliable communication.

11.1 Simplex, duplex & half-duplex

> **FIRST PRINCIPLES — ONE FREQUENCY OR TWO?**
>
> The number of frequencies a link uses, and whether both ends can talk at once, defines the mode. Aviation VHF deliberately uses the simplest and most disciplined arrangement so that everyone on a shared channel hears everyone else.

Communication Modes

Mode	Frequencies	How it works	Example
Simplex	**One**	**Transmit OR receive, one at a time (PTT); all stations share the channel**	**VHF ATC voice**
Duplex	Two	Transmit AND receive simultaneously (different frequencies)	Telephone; some satcom
Half-duplex	Two	Two frequencies but still one-at-a-time at the user end	Some HF / relay links
Broadcast (Simplex)	One	One station transmits, others only listen	ATIS, VOLMET

The aviation fact

VHF aeronautical voice is simplex — a single frequency shared by all, with only one station transmitting at a time. **ATIS** is simplex broadcast (the ground transmits, aircraft only listen). This shared discipline is why listening-out and brevity matter so much.

11.2 Push-to-talk technique

The correct sequence

1. **Listen out** first — make sure the frequency is clear.
2. **Press** the PTT and pause a beat so the first word isn't clipped.
3. **Speak** at a steady pace, normal volume, microphone close to the lips.
4. **Release** the PTT fully when finished so you can hear the reply.

Exam trap — the stuck microphone

On simplex, **while you transmit you cannot hear**. A PTT held down or stuck (a "stuck mic") therefore **blocks the whole frequency for everyone** and the offender doesn't even know — nobody else can get a word in. Never hold the button needlessly, release it fully, and if a frequency goes silent suspect a stuck mic.

Worked example — two aircraft transmit together

Two aircraft press PTT at the same instant on the same simplex VHF frequency.

Because the channel carries **AM** (Chapter 10), the two carriers beat together and the controller hears a **heterodyne squeal** rather than either message. The controller says "two together, say again", and each aircraft re-transmits.

With FM's capture effect one would have been silently lost — which is exactly why aviation uses AM.

11.3 Microphones, headsets & ANR

What each does

A **microphone** converts your voice to an electrical signal; **noise-cancelling** (differential) types reject the steady cockpit roar so only your speech is transmitted. A **boom or oxygen-mask microphone** keeps it positioned consistently. The **headset** reproduces received audio; **ANR (Active Noise Reduction)** headsets generate an anti-phase signal to cancel low-frequency cabin noise, reducing fatigue and improving intelligibility.

Cockpit reality

Hold the mic **close and steady** and speak across it, not into it, to avoid "popping". A consistent mic position is half of good readability — the other half is a steady, unhurried pace.

11.4 The audio panel & intercom

Selecting who hears whom

The **audio control panel** selects which transmitter the PTT keys (the **transmit selector**) and which receivers you listen to (the **receive selectors** — COM1, COM2, NAV idents, etc.). The **intercom (interphone)** lets the crew talk to each other — and to cabin/PA — **without transmitting** on the radio.

Selecting the wrong transmitter is a common cause of "transmitting on the wrong box".

Exam trap

An intercom conversation is **internal** — it does not go out on the radio. Conversely, keying the PTT transmits on whichever COM the **transmit selector** is set to, regardless of which receivers you are monitoring.

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11.5 Squelch, AVC, volume & tuning

The panel controls

Squelch — mutes the constant background hiss when no signal is present, opening the audio only when a real signal arrives. Set it just past the point where the hiss disappears.
AVC / AGC (Automatic Volume / Gain Control) — automatically holds the received audio at a steady level despite changing signal strength, so a distant station and a near one sound similar.
Volume — sets the listening level.
Tuning / frequency select — picks the operating channel, usually with a **standby/active** pair so the next frequency can be set ready and flip-flopped in.

Exam trap — squelch vs AVC

Set squelch **too tight** and weak (distant) signals are muted and missed. **AVC does the opposite job to squelch** — it keeps a wanted signal's *level* constant; it does not mute noise. Don't confuse the two.

11.6 Antennas & matching

Matching the antenna to the wave

The **VHF COM antenna** is the vertical blade — sized to a fraction of the wavelength and matched to the vertically-polarized VHF signal. **HF** uses a long wire or a probe (often in the fin) with an **antenna coupler/tuner**, because its long wavelength makes a full-size aerial impossible. A well-matched antenna transfers nearly all the transmitter power to the air; a mismatch reflects power back (high **VSWR**), wasting it and risking the transmitter.

Worked example — why the VHF aerial is a small blade

At 120 MHz the wavelength is 2.5 m (Chapter 7). A common efficient length is a **quarter-wave:** ¼ × 2.5 m = **0.625 m** — a practical blade.
An HF signal at 3 MHz has λ = 100 m; a quarter-wave would be 25 m — impossible on an airframe, so HF uses a long-wire/probe and a coupler to make a short aerial behave as if resonant.

Cockpit reality — antenna shadowing

In a steep turn the airframe can shadow an antenna mounted on the opposite side, briefly weakening the signal. It is one reason aircraft carry **two** COM antennas/radios and why a momentary drop in a hard turn is normal, not a fault.

The short vertical blades on the fuselage are the VHF COM antennas, sized to match a quarter-wavelength of the signal.

11.7 ELT & ACARS

> **ELT — Emergency Locator Transmitter**
>
> A beacon that activates **automatically on heavy impact** (g-switch) or manually, transmitting on **406 MHz** — detected by the Cospas-Sarsat satellite system, carrying a coded, **registered identity** so the aircraft is known — and on **121.5 MHz** for short-range homing by rescuers. It is carried under Annex 6 (Chapter 3). Functional testing is restricted to defined windows to avoid false alerts. (Full detail in Chapter 22.)

Website
YouTube
Instagram
Telegram channel
Telegram community
Subject study group

ACARS & datalink

ACARS (Aircraft Communications Addressing and Reporting System) is a digital datalink for short **text** messages between aircraft and ground — clearances, position reports, weather, maintenance and operational messages — reducing voice workload. **CPDLC** (Controller–Pilot Data Link Communications) carries ATC clearances as text, used especially in oceanic airspace (Chapter 18).

11.8 Radio failure & troubleshooting

Quick checks before declaring failure

Hearing nothing / getting no replies? Work through: correct **frequency** selected (active, not standby)? · **volume** up and **squelch** not too tight? · correct **transmit selector** and **receive selectors** on the audio panel? · **headset/PTT** plugged in and the PTT not stuck? · **circuit breaker / power** in? · try the **second COM**. If still nothing, follow the communications-failure procedure (Chapter 20) and **squawk 7600**.

Mnemonic

"Frequency, Volume, Squelch, Selector, Plug, Power" — the six things to check before believing the radio has truly failed. Then squawk **7600**.

11.9 Advantages & limitations of RT

Advantages and Limitations of VHF RT

Advantages	Limitations
Instant, direct two-way voice	VHF range limited to line-of-sight
Simple, universal; no fixed infrastructure airborne	One-at-a-time on simplex — congestion on busy frequencies
Standard phraseology crosses language barriers	Subject to interference, fading and ground shadow
Immediate read-back confirms understanding	Voice can be misheard — discipline essential
Datalink (ACARS/CPDLC) offloads routine traffic	A stuck mic can block a whole channel

☆ Numbers to memorise

Essential Facts for Chapter 11

Fact	Value
Aviation VHF voice	Simplex — one frequency, one transmitter at a time (PTT)
ATIS / VOLMET	Simplex broadcast (listen only)
Squelch	Mutes background hiss until a real signal arrives
AVC / AGC	Keeps received audio level steady (opposite job to squelch)
VHF antenna	Vertical blade, ~quarter-wave; HF uses long-wire + coupler
ELT	406 MHz (satellite, registered ID) + 121.5 MHz (homing)
Radio-failure squawk	7600

Question bank

PART A — MCQS (CLICK AN OPTION TO CHECK)

1. Aviation VHF voice communication operates in which mode?

> Simplex
>
> Duplex
>
> Full-duplex
>
> Broadcast only

```
Answer: Simplex. A single shared frequency, one transmitter at a time — simplex (PTT).
```

2. ATIS is an example of:

> Duplex
>
> Simplex broadcast (listen only)
>
> Half-duplex
>
> Two-way simplex

```
Answer: Simplex broadcast (listen only). ATIS is a one-way broadcast; aircraft only listen.
```

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3. Duplex operation uses:

> One frequency, one at a time
>
> Two frequencies, allowing simultaneous transmit and receive
>
> No frequency
>
> Only HF

Answer: Two frequencies, allowing simultaneous transmit and receive. Duplex uses two frequencies so both ends can talk at once.

4. A "stuck microphone" on a simplex frequency:

> Improves the signal
>
> Blocks the whole frequency for everyone
>
> Only affects that aircraft
>
> Changes the frequency

Answer: Blocks the whole frequency for everyone. A held/stuck PTT transmits continuously, blocking the shared channel; the offender can't hear it.

5. The correct PTT sequence is:

> Speak, press, release, listen
>
> Listen, press & pause, speak, release fully
>
> Press and hold throughout the exchange
>
> Press and speak instantly

Answer: Listen, press & pause, speak, release fully. Listen out, press and pause, speak clearly, then release fully to hear the reply.

6. A noise-cancelling microphone:

> Increases transmitter power
>
> Rejects steady cockpit noise so mainly speech is transmitted
>
> Tunes the radio
>
> Replaces the antenna

Answer: Rejects steady cockpit noise so mainly speech is transmitted. It cancels the steady ambient roar, transmitting clearer speech.

7. The intercom (interphone) is used to:

> Transmit to ATC
>
> Let the crew talk to each other without transmitting on the radio
>
> Broadcast ATIS
>
> Tune the navigation receiver

Answer: Let the crew talk to each other without transmitting on the radio. Intercom is internal; it does not go out on the radio.

8. The function of squelch is to:

> Increase transmitter power
>
> Mute background noise until a signal is received
>
> Change frequency automatically
>
> Keep audio level constant

Answer: Mute background noise until a signal is received. Squelch silences the hiss when no signal is present; keeping level constant is AVC's job.

9. AVC (automatic volume control) acts to:

> Mute weak signals
>
> Hold the received audio at a steady level despite signal variation
>
> Boost transmit power
>
> Select the frequency

Answer: Hold the received audio at a steady level despite signal variation. AVC/AGC keeps the wanted signal's level steady as strength changes.

10. Setting the squelch too tight will:

> Increase range
>
> Mute weak (distant) signals so they are missed
>
> Boost the audio
>
> Have no effect

Answer: Mute weak (distant) signals so they are missed. Too-tight squelch keeps the audio closed to weak but wanted signals.

11. The VHF communication antenna is mounted:

> Horizontally
>
> Vertically, to match the vertically-polarized VHF signal
>
> Inside the cabin
>
> At 45°

Answer: Vertically, to match the vertically-polarized VHF signal. Aviation VHF is vertically polarized, so the blade is vertical.

12. A poorly matched antenna causes:

> More radiated power
>
> Power reflected back (high VSWR), wasted and risking the transmitter
>
> A lower frequency
>
> Better range

Answer: Power reflected back (high VSWR), wasted and risking the transmitter. A mismatch reflects power (high VSWR) instead of radiating it.

13. HF aerials need a coupler/tuner because:

> HF is line-of-sight
>
> HF wavelengths are too long for a full-size aerial on an airframe
>
> HF uses AM
>
> HF needs no antenna

Answer: HF wavelengths are too long for a full-size aerial on an airframe. A 3 MHz quarter-wave is 25 m; a coupler makes a short aerial behave as resonant.

14. An ELT transmits on:

> 118.0 MHz only
>
> 406 MHz (satellite) and 121.5 MHz (homing)
>
> 1030 MHz
>
> 75 MHz

Answer: 406 MHz (satellite) and 121.5 MHz (homing). 406 MHz for satellite detection with coded ID, plus 121.5 MHz for homing.

15. ACARS is best described as:

> A navigation aid
>
> A digital datalink for short text messages between aircraft and ground
>
> An emergency beacon
>
> A type of antenna

Answer: A digital datalink for short text messages between aircraft and ground. ACARS carries short text messages, reducing voice workload.

16. On total radio failure, the transponder code to select is:

> 7500
>
> 7600
>
> 7700
>
> 2000

Answer: 7600. 7600 indicates radio/communications failure (7500 hijack, 7700 emergency).

17. Before declaring a radio failure you should first check:

> The weather
>
> Frequency, volume, squelch, audio-panel selectors, PTT/plug and power
>
> The transponder code
>
> The fuel

Answer: Frequency, volume, squelch, audio-panel selectors, PTT/plug and power. Work the quick checks; many "failures" are a wrong frequency, tight squelch or wrong selector.

18. A limitation of VHF radiotelephony is:

> Unlimited range
>
> Range restricted to line-of-sight
>
> It needs no frequency
>
> It cannot be misheard

Answer: Range restricted to line-of-sight. VHF is line-of-sight; range grows with altitude but is horizon-limited.

PART B — ORAL / VIVA (TAP TO REVEAL MODEL ANSWERS)

▶ **What is the difference between simplex and duplex, and which does aviation VHF use?**

▶ **Describe correct PTT technique and the hazard of a stuck microphone.**

▶ **What do squelch and AVC do, and how do they differ?**

▶ **Why is the VHF antenna a short vertical blade while HF needs a coupler?**

▶ **What is an ELT and on what frequencies does it transmit?**

▶ **State two advantages and two limitations of RT.**

60-SECOND REVISION CARD

- **Simplex** (one freq, PTT) = aviation VHF; duplex = two freqs simultaneous; ATIS = simplex broadcast.
- **PTT:** listen → press & pause → speak → release fully. A **stuck mic** blocks the channel.
- **Squelch** mutes hiss; **AVC** steadies a wanted signal (opposite jobs). Intercom is internal.
- VHF antenna = **vertical blade (~¼ wave)**; HF = long-wire + coupler; mismatch = high VSWR.
- **ELT** 406 + 121.5; radio failure → check freq/vol/squelch/selector/plug/power → squawk **7600**.

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Navigation Aids

The longest and most technical chapter in the book. Every system here is an application of the propagation and frequency principles of Chapters 7–10. We build each aid from first principles — how the signal is generated, how the receiver extracts navigation information, what its errors are, and how it is flown — then test it with worked examples and a full question bank.

SYLLABUS MAP

Part II (iv) Radio-navigation aids · Part I (viii) navigation facilities & principal frequencies

LEARNING OBJECTIVES — BY THE END OF THIS CHAPTER YOU WILL BE ABLE TO…

- Explain the four kinds of navigation information radio provides (bearing, distance, vertical guidance, position) and the physical principle behind each.
- Describe the operating principle, frequency band, presentation, range and full error set of NDB/ADF, VOR, ILS and DME — and work numerical bearing/range problems.
- Explain SSR Modes A/C/S, the interrogation/reply pulse scheme, garbling, fruit and side-lobe suppression, and the squawk codes.
- Contrast primary and secondary radar and state the basic pulse-radar relationships.
- Describe GNSS ranging, the need for four satellites, the error budget, DOP, and the augmentation systems (ABAS/SBAS/GBAS) including GAGAN and NavIC.
- Explain PBN (RNAV vs RNP), ACAS/TCAS logic, the radio altimeter, GPWS/EGPWS and INS/IRS.

12.0 How radio creates navigation information

1 2.1 NDB & ADF

12.2 VOR

12.3 ILS

12.4 DME

12.5 SSR, transponders & Mode S

12.6 Primary radar & radar principles

12.7 GNSS — GPS, GAGAN, NavIC

12.8 PBN — RNAV & RNP

12.9 ACAS / TCAS

12.10 Radio altimeter & GPWS/EGPWS

12.11 INS / IRS

Modern navigation displays integrate data from VORs, DME, GNSS, and Inertial systems into a seamless picture.

12.0 How radio creates navigation information

> **THE UNIFYING IDEA**
>
> Every navigation aid in this chapter answers one of four questions, and each does so by exploiting a single measurable property of a radio wave:

The Four Pillars of Navigation Aids

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Question	Physical principle	Aids that use it
Which direction? (bearing)	Compare the **phase** or **amplitude** of signals to extract an angle, or find the null of a directional antenna	ADF (null), VOR (phase), ILS (modulation balance)
How far? (distance)	Measure the **time** a pulse takes to travel out and back; distance = ½ · c · t	DME, primary radar, radio altimeter
Where exactly? (position)	**Range from several known points** (multilateration / satellite ranging)	GNSS, DME/DME, INS (dead reckoning)
Am I high/low, left/right of path?	Balance of two overlapping **modulated lobes**	ILS localizer & glide path, PAR

Two foundational numbers

Radio waves travel at c = 3 × 10&sup8; m/s. A pulse therefore takes about **12.36 µs to travel one nautical mile and back** (the "radar mile"). This single fact underlies DME, radar and the radio altimeter. And bearing systems all reduce to measuring an **angle** — by phase, by amplitude balance, or by an antenna null.

Classifying any aid in the viva

For any aid, be ready to state four things in order: **(1) band, (2) principle, (3) what it gives the pilot, (4) its main errors/limitations.** Examiners build almost every nav-aid question from this skeleton.

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12.1 NDB & ADF

The ADF dial in the cockpit points directly to the NDB ground station, providing a relative bearing.

12.1.1 THE GROUND STATION (NDB)

IN PLAIN TERMS

The Non-Directional Beacon is the simplest radio aid: a ground transmitter that radiates equally in **all** directions (hence "non-directional"). It does nothing clever itself — all the navigation intelligence is in the aircraft's receiver, which determines the *direction* from which the beacon's signal arrives.

NDB — the facts

Band: LF/MF, ICAO range **190–1750 kHz** (most are 200–415 kHz).
Propagation: primarily **ground (surface) wave**, which follows the Earth's curvature — giving useful range beyond line-of-sight at low level. **Emission:** a carrier keyed with a **two- or three-letter Morse ident** (NON A1A keyed carrier, or NON A2A where a 1020 Hz tone is added so the ident is heard on a standard AM receiver).

Classes & range

Range depends on transmitter power and surface conductivity. As a rule of thumb: **Locator (L)** beacons (low power, used on approaches) ≈ 10–25 NM; **en-route** NDBs ≈ 50–100+ NM. Range is greater over **sea** (high conductivity) than over dry land or mountains.

12.1.2 THE AIRBORNE RECEIVER (ADF) — HOW THE BEARING IS FOUND

FIRST PRINCIPLES — THE LOOP & SENSE ANTENNA

The Automatic Direction Finder uses a **loop antenna** whose reception pattern is a **figure-of-eight** (a "polar diagram" with two lobes and two sharp **nulls** at right angles to the lobes). A null is used rather than a peak because a null is far sharper and easier to detect precisely. But a single figure-of-eight has **two** nulls 180° apart — so the loop alone cannot tell whether the station is ahead or behind (180° ambiguity).

To resolve this, the receiver adds an omnidirectional **sense antenna** (a circular pattern). Combining the loop's figure-of-eight with the sense antenna's circle produces a **cardioid** (heart-shaped) pattern with a **single** null. The receiver electronically rotates this pattern (a goniometer/servo) until it finds that single null, and drives the **ADF needle** to point at the station.

What the needle shows — relative vs magnetic bearing

On a fixed-card indicator the needle gives the **Relative Bearing (RB)** — the angle to the station measured *from the aircraft's nose*. To get the **Magnetic Bearing TO the station (QDM)** you add the heading:

Magnetic Bearing TO (QDM) = Relative Bearing + Magnetic Heading
(QDR = QDM ± 180°)

On an **RMI** (Radio Magnetic Indicator) the compass card is slaved to the heading, so the needle reads the magnetic bearing (QDM) directly off the head of the needle.

Worked example — ADF bearing

Aircraft magnetic heading **040°**; the ADF relative bearing reads **030°** (i.e. 30° to the right of the nose).
QDM (magnetic bearing to the NDB) = RB + HDG = 030 + 040 = **070°**.
QDR (magnetic bearing from the NDB) = 070 − 180 = **250°**.
If the sum exceeds 360°, subtract 360. E.g. HDG 350°, RB 040° → QDM = 390 − 360 = **030°**.

12.1.3 ADF/NDB ERRORS — THE FULL SET

ADF Errors

Error	Cause	When worst
Night effect	After dusk the ionosphere returns a **sky wave** that contaminates the ground wave; the two interfere and the needle wanders	Dusk & dawn, at longer ranges
Coastal (shoreline) refraction	The ground wave changes speed crossing the coast and **bends towards the coast**	Signal crossing coast at a shallow angle; near the coastline
Quadrantal error	The aircraft's own structure **re-radiates** the signal, biasing the loop	When the station is off the four "quadrantal" headings; reduced by calibration
Thunderstorm (static) effect	A CB's electrical discharges radiate strongly in LF/MF; the needle **points at the storm**	Near active CBs
Mountain / terrain effect	Reflections from high ground	Mountainous areas
Station interference	At night, distant co-channel beacons are heard via sky wave	Night

Exam trap

NDB/ADF is the **least accurate** common aid (system accuracy roughly **±5°**). Always confirm the **Morse ident** before using a beacon, and treat a wandering needle near dusk, coast or a CB with suspicion. Unlike VOR, the NDB gives **no failure flag** — a needle that has quietly drifted can mislead you.

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12.2 VOR — VHF Omnidirectional Range

The Horizontal Situation Indicator (HSI) or traditional VOR dial allows the pilot to select and fly specific radials based on phase comparison.

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12.2.1 THE PHASE-COMPARISON PRINCIPLE

FIRST PRINCIPLES — TWO 30 HZ SIGNALS

A VOR transmits **two** 30 Hz signals and lets the receiver compare their **phase difference** to determine the radial. One is a **reference** phase that is the *same in every direction*; the other is a **variable** phase whose timing depends on the bearing of the receiver from the station.

In a **conventional VOR (CVOR)**: the reference 30 Hz is transmitted as **frequency modulation** on a 9960 Hz sub-carrier (the same everywhere), while the variable 30 Hz is produced as **amplitude modulation** by physically/electronically **rotating a directional pattern at 30 revolutions per second**. The two signals are arranged to be **exactly in phase when the aircraft is due magnetic NORTH** of the station. As you move around the station, the variable phase slips relative to the reference by exactly the bearing angle — so the measured phase difference **is** the magnetic radial.

VOR — the facts

Band: VHF **108.0–117.975 MHz** (108.00–111.95 shared with ILS — VOR uses the frequencies with an **odd** first decimal e.g. 108.10; ILS uses even, e.g. 108.10 vs 108.15… check the chart). **Propagation:** line-of-sight (space wave). **Gives:** magnetic **radial TO or FROM** — 360 radials like spokes. **Ident:** 3-letter Morse, sometimes with voice.

DVOR — why most modern stations are Doppler

The **Doppler VOR (DVOR)** *reverses* the roles (reference = AM, variable = FM produced by Doppler shift from a large ring of antennas electronically "rotated"). The wide antenna ring makes DVOR far **less sensitive to site/terrain reflections**, so it is more accurate and easier to site. The cockpit indication is identical — the pilot cannot tell CVOR from DVOR.

12.2.2 READING THE INSTRUMENT — OBS, CDI & THE TO/FROM FLAG

IN PLAIN TERMS

You select a course with the **OBS (Omni Bearing Selector)**. The **CDI** (Course Deviation Indicator) needle shows how far left/right of that selected course you are, and the **TO/FROM** flag shows whether flying the selected course would take you *toward* or *away from* the station. Each dot of CDI deflection ≈ **2°**; full-scale (5 dots) ≈ **10°**.

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Worked example — CDI interpretation

You select OBS **090°** and get a **TO** flag with the needle **2 dots left**.
TO flag → flying 090° would take you to the station, so the station is roughly **east** of you (you are west of it, near the 270° radial).
Needle 2 dots left → the selected course is to your **left**; fly left (turn toward the needle) to intercept. 2 dots ≈ **4°** displacement from the 090° course line.
Golden rule: with a FROM flag the CDI senses correctly when your heading roughly matches the OBS; with a TO flag and the OBS set to the reciprocal, beware **reverse sensing**.

12.2.3 VOR ERRORS & RANGE

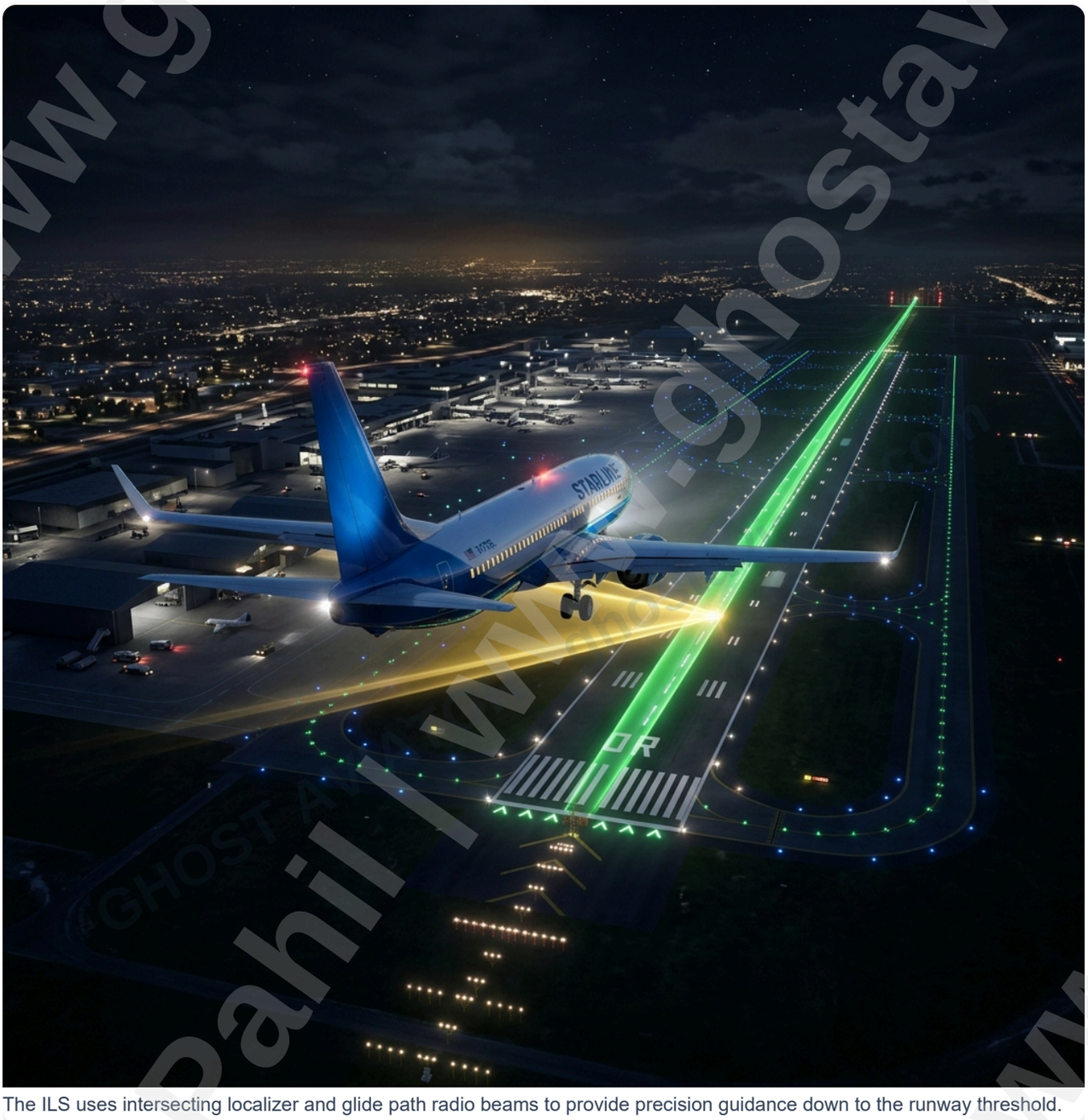
VOR Errors

Error / limitation	Detail
Site & terrain (scalloping)	Reflections cause the needle to oscillate ("scalloping"); much reduced on DVOR
Cone of confusion	Directly overhead the station the signal is ambiguous — the TO/FROM flag flickers and the needle is unreliable for a short time
Line-of-sight range	VHF — range grows with altitude; blocked by terrain
Aggregate accuracy	≈ **±5°** system (ground station ±1–2°), far better than NDB; a **failure flag** appears if the signal is unusable

VOR/DME, TACAN & VORTAC

A VOR is frequently co-located with a **DME** (VOR/DME) so a single station gives both bearing and distance — a complete fix. **TACAN** is the military equivalent (bearing + DME in UHF); a **VORTAC** co-locates a VOR with a TACAN so civil aircraft get VOR bearing + TACAN's DME.

12.3 ILS — Instrument Landing System



The ILS uses intersecting localizer and glide path radio beams to provide precision guidance down to the runway threshold.

> **IN PLAIN TERMS**
>
> The ILS is a **precision approach** aid giving the pilot both **azimuth** (left/right of the centreline) and **vertical** (above/below the glide path) guidance down to low height, plus distance checkpoints. It is built from three independent radio systems.

12.3.1 THE LOCALIZER — AZIMUTH GUIDANCE

FIRST PRINCIPLES — THE 90/150 HZ BALANCE

An antenna array at the **far end** of the runway radiates two overlapping lobes along the approach: the left side carries a tone modulated at **90 Hz**, the right side at **150 Hz**. On the exact centreline the two tones are received with **equal depth of modulation** — the **Difference in Depth of Modulation (DDM) is zero** — and the needle centres. Off to one side, one tone dominates and the needle deflects. The receiver literally measures the **balance** between two modulation tones.

Localizer — facts

Band: VHF **108.10–111.95 MHz** (odd-tenth + 0.05 frequencies). **Course sector:** adjusted so full-scale deflection corresponds to roughly **±2.5°** (course width ≈ runway-width dependent, typically 3–6° total). **Sensitivity:** about **4× more sensitive** than a VOR for the same needle movement — a small deviation moves the needle a lot.

Back course & reverse sensing

The localizer also radiates a **back course** off the approach end. Flying inbound on the back course (or a front course outbound) gives **reverse sensing** — the needle moves opposite to the correction needed — unless the system/HSI compensates. Know this exists.

12.3.2 THE GLIDE PATH — VERTICAL GUIDANCE

Glide path — facts

Band: UHF **329–335 MHz** (automatically paired with the localizer frequency — you tune one box). **Principle:** same 90/150 Hz balance but in the **vertical** plane — **90 Hz dominates above** the path, **150 Hz below**. **Nominal angle: 3°**; usable half-width ≈ ±0.7°.

False glide paths

The glide-path antenna pattern produces additional lobes, creating **false glide slopes** at higher angles (e.g. around 6°, 9°). They can be captured from above. Defence: intercept the glide path **from below** at the correct altitude after establishing on the localizer, and cross-check altitude/distance.

12.3.3 MARKER BEACONS & ILS CATEGORIES

Marker Beacons

Marker	Tone / code	Light	Typical position
Outer (OM)	400 Hz, low tone, dashes	Blue	≈ 3.9–5 NM (final approach fix)
Middle (MM)	1300 Hz, alternate dot-dash	Amber	≈ 0.5–0.6 NM (near CAT I DH)
Inner (IM)	3000 Hz, high tone, dots	White	Near threshold (CAT II/III)

All marker beacons share 75 MHz

Every marker — outer, middle, inner — transmits on a **75 MHz** carrier (VHF), radiating a narrow fan *upward*; the aircraft passes through it. Markers are increasingly replaced by **DME** for continuous distance. ICAO categories define the minima:

ILS Categories

Category	Decision Height	RVR (typical)
CAT I	≥ 200 ft	≥ 550 m
CAT II	100–200 ft	≥ 300 m
CAT IIIA	< 100 ft (or none)	≥ 200 m
CAT IIIB	< 50 ft (or none)	50–200 m
CAT IIIC	None	None (zero/zero)

Author check before publishing

Confirm exact CAT II/III DH/RVR values against the current DGCA CAR/ICAO Annex 10 & Annex 6, as national thresholds vary slightly. Also confirm protected-area (LSA/GSA critical & sensitive area) wording you wish to include.

12.4 DME — Distance Measuring Equipment

FIRST PRINCIPLES — TIMING A PULSE PAIR

The aircraft (the **interrogator**) transmits **pulse pairs** on a UHF frequency. The ground **transponder** receives them, waits a **fixed 50 µs system delay**, and replies with pulse pairs on a frequency offset by **63 MHz**. The airborne set measures the **total round-trip time**, subtracts the 50 µs, and converts the remaining time to distance using the radar mile (≈12.36 µs per NM round trip). Because the aircraft initiates and the reply is uniquely jittered, each aircraft recognises only its *own* replies.

DME — facts

Band: UHF **962–1213 MHz**. **Gives: slant range** (direct line-of-sight distance, not ground distance). **Capacity:** a ground transponder serves **≈ 100 aircraft**; beyond that it sheds the weakest (beacon saturation). Usually paired with a VOR or ILS so one selection drives both.

Worked example — slant vs ground range

You are at **6000 ft** (≈ 1 NM) overhead-ish a DME, reading **6 NM slant range**.
Ground distance = √(slant² − height²) = √(6² − 1²) = √35 **≈ 5.92 NM**.
Directly overhead (ground = 0) the DME reads your **height in NM**: 6000 ft ≈ 1 NM, so it reads ≈ 1 NM, not zero. **Slant-range error is greatest close-in and high**, and negligible far out at low level.

Cockpit reality — the DME arc

Holding a constant DME distance while turning lets you fly a **DME arc** around a station — a common way to join a final approach. The DME's continuous distance is also what replaces marker beacons on a modern ILS ("4 DME", "2 DME" checks against altitude).

12.5 SSR, transponders & Mode S

FIRST PRINCIPLES — INTERROGATION & REPLY

Unlike primary radar (which listens for a passive echo), Secondary Surveillance Radar is a **cooperative** system. The ground interrogator transmits on **1030 MHz**; the aircraft **transponder** detects the interrogation and actively **replies on 1090 MHz** with coded data. Because the reply is a strong, coded transmission rather than a faint echo, SSR gives reliable **identity and altitude** with far less power.

Modes & pulse scheme

Mode A (identity): interrogation pulses P1–P3 spaced **8 µs**; the reply is a 4096-code (octal) squawk framed between framing pulses **F1/F2 (20.3 µs apart)**. **Mode C** (altitude): P1–P3 spaced **21 µs**; reply encodes pressure altitude in 100 ft steps (Gillham code). **Mode S** (Select): each aircraft has a unique **24-bit address**, so the radar interrogates aircraft individually, carries a data-link, and eliminates the garbling problem. ADS-B (1090ES) is built on the Mode S reply.

SSR Problems and Solutions

Problem	What it is	Cure
Garbling	Two aircraft at similar range/bearing reply overlapping in time	Mode S selective addressing
FRUIT	Replies triggered by *other* interrogators received by this radar	Defruiters; Mode S
Side-lobe replies	Transponder replies to the radar's antenna side-lobes, giving false bearings	**Side-Lobe Suppression (SLS)** — a P2 control pulse

> **Squawk codes — memorise exactly**
>
> **7500** unlawful interference (hijack) · **7600** radio/communications failure · **7700** general emergency. Also: **7000** conspicuity (VFR, many regions), **2000** entering from a non-SSR area, and the **SPI / IDENT** pulse when ATC asks you to "squawk ident".

> **Mnemonic**
>
> **"75 taken alive · 76 nix the comms · 77 going to heaven."**

12.6 Primary radar & radar principles

Primary Surveillance Radar (PSR) transmits high-power pulses and listens for the faint, passive echo reflected off the aircraft skin.

IN PLAIN TERMS

Primary Surveillance Radar (PSR) is the original radar: transmit a pulse, listen for the **reflected echo** from the target, and time it. It needs **no equipment aboard** the aircraft, but gives only **range and bearing** — no identity, no altitude.

The pulse-radar relationships

Range = ½ · c · t (echo time there-and-back). **PRF** (pulse repetition frequency) sets the maximum unambiguous range — too high a PRF and a distant echo returns after the next pulse is sent (range ambiguity). **Pulse width** sets minimum range and range resolution; **beam width** sets bearing resolution. **MTI/Doppler** processing rejects stationary clutter to show moving targets.

Primary Radar Types

Radar	Role
ASR / TAR	Airport/terminal surveillance (approach control)
ARSR	En-route (area) surveillance — long range
SMR	Surface Movement Radar — ground traffic in poor visibility
PAR	Precision Approach Radar — controller-guided azimuth + elevation talkdown (Ch 17)
Weather radar	Airborne, ~9.3 GHz (X-band), detects precipitation/CB

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12.7 GNSS — GPS, GAGAN & NavIC

The GAGAN system uses geostationary satellites to broadcast corrections and integrity messages, augmenting the core GNSS signals over the Indian Flight Information Region.

12.7.1 HOW SATELLITE RANGING FIXES A POSITION

FIRST PRINCIPLES — PSEUDO-RANGING & THE FOURTH SATELLITE

Each satellite broadcasts its position and a precisely-timed code. The receiver measures the **time of flight** of the code → a **range** to that satellite, placing the aircraft on a sphere around it. Two satellites narrow it to a circle, three to two points — so in principle **3 satellites give a 3-D position**. But the receiver's own clock is cheap and slightly wrong, and a small clock error scales by c into a large range error. A **fourth satellite** is used to solve for that clock error as well — hence **four satellites are required for a 3-D fix.**

GPS — architecture

Space segment: ≥ 24 satellites in **6 orbital planes** at ≈ **20,200 km**, period ≈ **11 h 58 min**, inclination 55°. **Signals: L1 = 1575.42 MHz** (also L2 1227.6, L5 1176.45 MHz). **Control segment:** ground monitor/upload stations. **User segment:** the receiver. Other constellations: **GLONASS** (Russia), **Galileo** (EU), **BeiDou** (China), **NavIC/IRNSS** (India, regional — L5 & S-band, ~7 satellites).

12.7.2 ERROR BUDGET, DOP & INTEGRITY

GNSS Errors

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Error source	Note
Ionospheric / tropospheric delay	Largest natural error; partly modelled, dual-frequency cancels ionosphere
Satellite clock & ephemeris	Corrected by the navigation message
Multipath	Reflections off ground/structures near the antenna
Receiver noise	Internal
Geometry (GDOP/PDOP)	Satellites **spread wide** across the sky give a strong fix (low DOP); satellites bunched together give a weak fix (high DOP)

Augmentation — making GNSS good enough for approaches

ABAS (Aircraft-Based, e.g. **RAIM** — Receiver Autonomous Integrity Monitoring): the receiver cross-checks redundant satellites to detect a faulty one. **SBAS** (Satellite-Based): geostationary satellites broadcast wide-area corrections + integrity — **GAGAN** (India), WAAS (USA), EGNOS (Europe), enabling APV/LPV approaches. **GBAS** (Ground-Based, GLS): a local airport station for precision approaches.

GAGAN — the Indian SBAS

GAGAN (GPS Aided GEO Augmented Navigation) uses geostationary satellites to broadcast corrections and an **integrity** message over the Indian region, improving GPS accuracy to a few metres and enabling approach procedures with vertical guidance. **Integrity** — the system's ability to warn the pilot promptly when it should not be used — is what makes augmentation essential for IFR approaches.

12.8 PBN — RNAV & RNP

IN PLAIN TERMS

Performance-Based Navigation (PBN) frees aircraft from flying directly station-to-station: instead they fly defined paths to a stated **accuracy**. The key distinction the exam wants:

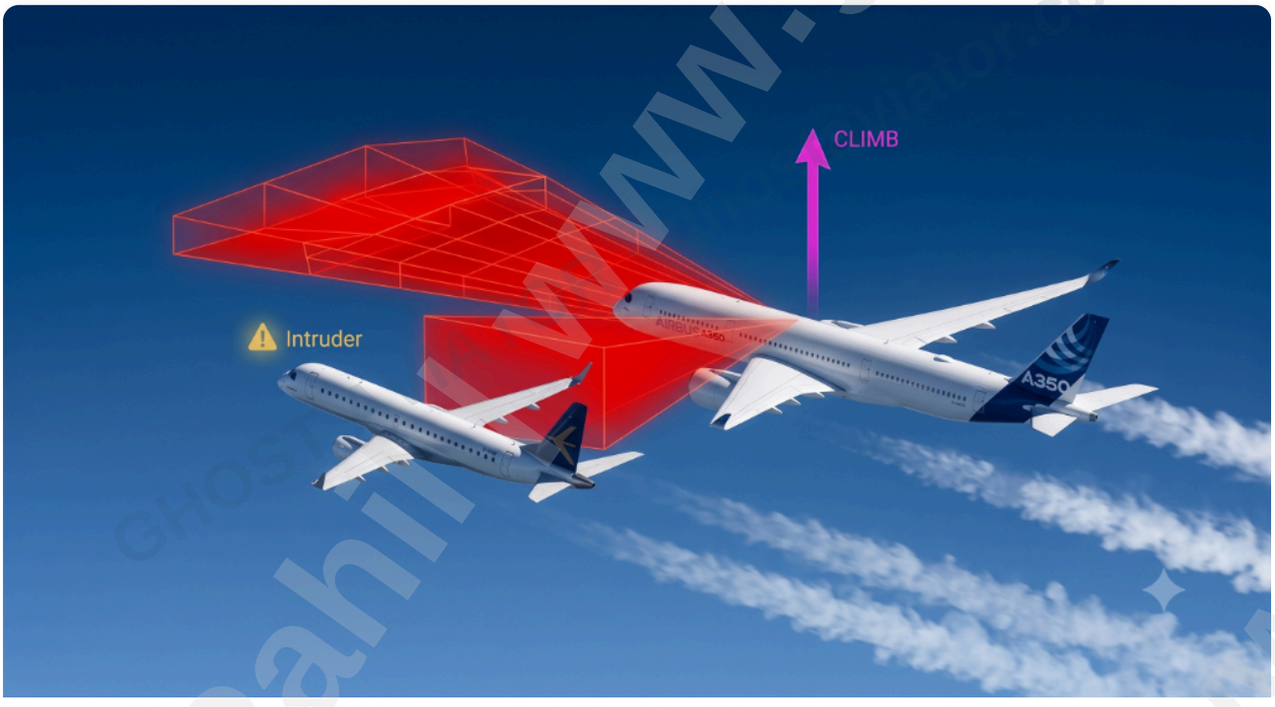
RNAV vs RNP — the one-line difference

RNAV = Area Navigation: fly any path within the coverage of the nav aids, to a stated lateral accuracy. **RNP** = RNAV **plus on-board performance monitoring and alerting** — the system continuously checks it is meeting the required accuracy and **warns the crew if it is not**. That self-monitoring is the whole difference.

PBN Specifications

Specification	Lateral accuracy (NM, 95%)	Phase
RNAV 10 / RNP 10	10	Oceanic / remote
RNAV 5	5	En-route continental
RNAV 1 / 2	1 / 2	Terminal
RNP 4	4	Oceanic (with monitoring)
RNP 1	1	Terminal (with monitoring)
RNP APCH / AR	0.3 → lower	Approach (AR = authorisation required, curved paths)

12.9 ACAS / TCAS



When a conflict is detected, TCAS II issues a coordinated Resolution Advisory (RA) commanding a vertical manoeuvre, such as "CLIMB", to ensure safe separation.

FIRST PRINCIPLES — USING TRANSPONDERS TO SEE TRAFFIC

ACAS (the ICAO term; TCAS II is the common implementation) **interrogates the transponders** of nearby aircraft — exactly like a tiny airborne SSR — and tracks their range, closure and altitude. It builds a picture independent of ground radar and warns of conflicts **in the vertical plane**.

TA then RA

Traffic Advisory (TA): "TRAFFIC, TRAFFIC" — **≈ 40 s** to closest approach; alerts the crew to look out, but commands nothing. **Resolution Advisory (RA): ≈ 25 s** to closest approach — a **vertical command** (climb/descend/maintain/level-off). When both aircraft have TCAS II, they **coordinate via Mode S** so one climbs and the other descends. **An RA is followed even if it conflicts with an ATC instruction**, and the crew advises ATC ("TCAS RA").

Limitations

TCAS only sees aircraft with an operating **transponder** (it cannot see non-transponding traffic). RAs are **vertical only** — never turn in response to an RA. It is a last-resort safety net, not a substitute for see-and-avoid or ATC separation.

12.10 Radio altimeter & GPWS/EGPWS

The radio altimeter measures true height above the surface, feeding the GPWS which issues immediate audio and visual warnings (e.g., "TERRAIN, PULL UP") when closure rates become dangerous.

Radio (radar) altimeter

Band: ≈ 4200–4400 MHz (SHF). **Technique: FM-CW** (frequency-modulated continuous wave) — the frequency difference between the transmitted and reflected signal is proportional to height. **Measures true height above the surface directly below (AGL)**, typically **0–2500 ft**, used on approach and for auto-land. It reads terrain height, not a barometric level.

GPWS modes (1–7) & EGPWS

The **Ground Proximity Warning System** uses radio-altimeter height and aircraft data to warn of dangerous situations: **1** excessive descent rate · **2** excessive terrain closure rate · **3** altitude loss after take-off · **4** unsafe terrain clearance (not in landing config) · **5** below glide slope · **6** altitude callouts / excessive bank · **7** wind shear. **EGPWS (Enhanced)** adds a worldwide **terrain database** and GPS position for **forward-looking** terrain alerting — warning of terrain ahead, not just below.

12.11 INS / IRS

FIRST PRINCIPLES — NAVIGATION WITH NO OUTSIDE SIGNAL

An Inertial Navigation System measures the aircraft's own **accelerations** (accelerometers) and **rotations** (gyroscopes), and mathematically integrates them from a **known start position** to compute present position continuously — needing **no external radio signal at all.** It is therefore completely self-contained and un-jammable.

Platform vs strapdown; drift; hybridisation

Older systems use a gimballed stable **platform**; modern **strapdown** IRS fix the sensors to the airframe and use **ring-laser or fibre-optic gyros**. Because errors accumulate through integration, the position **drifts** with time (typically **≈ 2 NM/hr**). **Schuler tuning** (an 84.4-minute period) prevents the system oscillating with vehicle motion. In practice the IRS is **blended with GNSS** (hybrid IRS/GPS) so GNSS bounds the drift while inertial provides smooth, high-rate, jam-resistant data.

☆ Numbers to memorise

Essential Facts for Chapter 12

Aid	Band / frequency	Principle	Gives / key figures
NDB / ADF	LF/MF 190–1750 kHz	Loop null + sense → cardioid	Relative bearing; ±5°; ground wave
VOR	VHF 108.0–117.975 MHz	30 Hz phase comparison (ref FM / var AM)	Magnetic radial; ±5°; in-phase at North
ILS LOC	VHF 108.10–111.95 MHz	90/150 Hz DDM balance (azimuth)	Centreline; ±2.5° full scale
ILS GP	UHF 329–335 MHz	90 above /150 below (vertical)	3° path; false slopes ~6°/9°
Markers	75 MHz (OM 400/MM 1300/IM 3000 Hz)	Vertical fan	Distance checkpoints
DME	UHF 962–1213 MHz	Pulse-pair timing; 50 µs delay; 63 MHz offset	Slant range; ~100 a/c; radar mile 12.36 µs/NM
SSR	1030 ↑ / 1090 ↓ MHz	Interrogate–reply; Mode A 8 µs / C 21 µs / S 24-bit	ID + altitude; 7500/7600/7700
GPS	L1 1575.42 MHz	Satellite pseudo-ranging	4 sats for 3-D; ~20,200 km; GAGAN augments
Radio altimeter	4200–4400 MHz	FM-CW	True AGL 0–2500 ft
INS/IRS	(no radio)	Accelerometers + gyros	Self-contained; drift ~2 NM/hr; Schuler 84.4 min

Question bank

PART A — MCQS (CLICK AN OPTION TO CHECK)

1. The ADF resolves the 180° ambiguity of the loop antenna by:

> Using a higher frequency
>
> Adding a sense antenna to form a cardioid with a single null
>
> Rotating the aircraft
>
> Comparing 90 and 150 Hz tones

```
Answer: Adding a sense antenna to form a cardioid with a single null. The omnidirectional sense antenna combines with the figure-of-eight loop to give a cardioid with one null.
```

2. Heading 350°(M), ADF relative bearing 040°. The QDM is:

> 310°
>
> 030°
>
> 050°
>
> 210°

Answer: 030°. QDM = RB + HDG = 040 + 350 = 390 − 360 = 030°.

3. A VOR's reference and variable 30 Hz signals are in phase when the aircraft is:

> Overhead the station
>
> Due magnetic north of the station
>
> On the 180° radial
>
> At the cone of confusion

Answer: Due magnetic north of the station. By design the phase difference is zero on the 360° (north) radial; it then equals the radial.

4. In a conventional VOR, the reference phase signal is carried as:

> 30 Hz AM
>
> 30 Hz FM on a 9960 Hz sub-carrier
>
> A 90 Hz tone
>
> A Morse code

Answer: 30 Hz FM on a 9960 Hz sub-carrier. CVOR: reference = FM on 9960 Hz sub-carrier (same in all directions); variable = AM from the rotating pattern.

5. The ILS localizer indicates centreline when:

> 90 Hz dominates
>
> 150 Hz dominates
>
> The depth of modulation of 90 and 150 Hz is equal (DDM = 0)
>
> The DME reads zero

Answer: The depth of modulation of 90 and 150 Hz is equal (DDM = 0). On the centreline the two tones are received equally — zero difference in depth of modulation.

6. ILS marker beacons all transmit on:

> 329–335 MHz
>
> 108.10–111.95 MHz
>
> 75 MHz
>
> 1030 MHz

Answer: 75 MHz. Outer/middle/inner markers share a 75 MHz carrier, differing in tone and code.

7. A false glide slope is typically encountered:

> Below the true 3° path
>
> Above the true path (e.g. ~6°), if intercepted from above
>
> On the localizer back course
>
> Only at night

Answer: Above the true path (e.g. ~6°), if intercepted from above. Antenna side-lobes create false slopes at higher angles; intercept the GP from below to avoid them.

8. DME measures distance by:

> Comparing two 30 Hz phases
>
> Timing the round trip of pulse pairs, less a 50 µs ground delay
>
> Measuring received signal strength
>
> Counting satellites

Answer: Timing the round trip of pulse pairs, less a 50 µs ground delay. The interrogator times the reply (minus the fixed 50 µs transponder delay) to derive slant range.

9. At 12,000 ft (2 NM) directly overhead a DME, the indicator reads approximately:

> 0 NM
>
> 2 NM
>
> 6 NM
>
> 12 NM

Answer: 2 NM. Overhead, ground range is zero so slant range equals height: 12,000 ft ≈ 2 NM.

10. An SSR transponder is interrogated on, and replies on, respectively:

> 1090 / 1030 MHz
>
> 1030 / 1090 MHz
>
> 329 / 335 MHz
>
> 75 / 75 MHz

Answer: 1030 / 1090 MHz. Ground interrogates on 1030 MHz; aircraft replies on 1090 MHz.

11. Side-Lobe Suppression in SSR prevents:

> Garbling
>
> FRUIT
>
> Replies to the antenna's side lobes giving false bearings
>
> Altitude errors

Answer: Replies to the antenna's side lobes giving false bearings. A P2 control pulse stops the transponder replying when interrogated via a side lobe.

12. The transponder code for radio communications failure is:

> 7500
>
> 7600
>
> 7700
>
> 7000

Answer: 7600. 7500 hijack, 7600 radio failure, 7700 emergency.

13. Which radar requires equipment carried aboard the aircraft to function?

> Primary radar
>
> Secondary radar
>
> Both
>
> Neither

Answer: Secondary radar. SSR needs a transponder; primary radar works on the passive echo from any target.

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14. A 3-D GNSS position fix requires a minimum of how many satellites?

> 2
>
> 3
>
> 4
>
> 6

Answer: 4. Three for position plus one to solve the receiver clock error = four.

15. The GPS L1 carrier frequency is:

> 1227.60 MHz
>
> 1575.42 MHz
>
> 1090 MHz
>
> 1176.45 MHz

Answer: 1575.42 MHz. L1 = 1575.42 MHz (L2 = 1227.6, L5 = 1176.45).

16. GAGAN is best described as:

> A ground-based primary radar
>
> A satellite-based augmentation system (SBAS) for the Indian region
>
> India's own satellite constellation for global coverage
>
> An inertial system

Answer: A satellite-based augmentation system (SBAS) for the Indian region. GAGAN broadcasts corrections + integrity over India via GEO satellites. (NavIC/IRNSS is the regional constellation.)

17. The essential difference between RNAV and RNP is that RNP adds:

> A higher cruising altitude
>
> On-board performance monitoring and alerting
>
> A second VOR
>
> Mode S

Answer: On-board performance monitoring and alerting. RNP = RNAV plus the requirement that the system monitors its own accuracy and alerts the crew.

18. A TCAS Resolution Advisory commands a manoeuvre in:

> The horizontal plane (turn)
>
> The vertical plane (climb/descend)
>
> Any plane
>
> Speed only

Answer: The vertical plane (climb/descend). RAs are vertical-only; never turn in response to an RA.

19. The radio altimeter operates around 4200–4400 MHz and indicates:

> Pressure altitude
>
> True height above the surface directly below (AGL)
>
> Altitude above mean sea level
>
> Slant range to a beacon

Answer: True height above the surface directly below (AGL). FM-CW radio altimeter gives AGL, typically 0–2500 ft.

20. A characteristic of INS used alone is that its position error:

> Is constant regardless of time
>
> Grows with time (drift), typically a few NM per hour
>
> Depends on satellite geometry
>
> Only occurs over the sea

Answer: Grows with time (drift), typically a few NM per hour. Integration accumulates error, so unaided INS drifts (~2 NM/hr); GNSS is used to bound it.

PART B — ORAL / VIVA (TAP TO REVEAL MODEL ANSWERS)

▶ **Explain how an NDB and ADF together give a bearing, and list the main errors.**

▶ **Describe the phase principle of a VOR.**

▶ **How does the ILS provide azimuth and vertical guidance?**

▶ **Why does DME read slant range, and when is the error significant?**

▶ **Explain SSR Modes A, C and S and the problems Mode S solves.**

▶ **Why are four satellites needed for a GNSS position, and what limits accuracy?**

▶ **What is the difference between RNAV and RNP?**

▶ **How does TCAS work and what are its limitations?**

▶ **Distinguish primary and secondary radar.**

▶ **What does the radio altimeter measure and how, and what uses it?**

PART C — NUMERICAL PROBLEMS (TAP FOR WORKED SOLUTIONS)

▶ **P1. Heading 120°(M), ADF relative bearing 250°. Find QDM and QDR.**

▶ **P2. At 9000 ft you read 3 NM on the DME. What is the approximate ground distance?**

▶ **P3. A radar echo returns 1235 µs after transmission. What is the target range?**

▶ **P4. You are on the 045° radial of a VOR, tracking inbound. What OBS and flag give correct sensing?**

60-SECOND REVISION CARD

- **ADF/NDB** LF/MF, loop-null+sense→cardioid, RB (QDM=RB+HDG), ±5°, many errors, no flag.
- **VOR** VHF, 30 Hz phase (ref FM/var AM), in-phase at North, radials, ±5°, cone of confusion, flag.
- **ILS** LOC VHF 90/150 DDM; GP UHF 329–335 (90 above/150 below, 3°, false slopes); markers 75 MHz; CAT I–III.
- **DME** UHF 962–1213, pulse timing, 50 µs delay, slant range, ~100 a/c.
- **SSR** 1030↑/1090↓, Mode A/C/S, SLS/garble/FRUIT, 7500/7600/7700.
- **GNSS** L1 1575.42, 4 sats for 3-D, DOP, augment ABAS/SBAS(GAGAN)/GBAS; NavIC=India constellation.
- **RNP**=RNAV+monitoring/alerting. **TCAS** vertical RA, Mode-S coordinated. **RadAlt** 4200–4400 FM-CW AGL. **INS** self-contained, drifts ~2 NM/hr.

The Language of the Air — Phonetics, Numbers & Time

From here the book turns from electronics to the spoken language of the air. A radio frequency is useless if "five" is heard as "nine" or "to" is mistaken for "two". So aviation built its own precise dialect: a spelling alphabet where every word sounds different, exact ways to speak numbers and time, and a small set of standard words that carry one meaning each.

SYLLABUS MAP

Part III (i) Transmitting technique · transmission of letters, numbers & time · standard words & phrases · test procedures

LEARNING OBJECTIVES — BY THE END OF THIS CHAPTER YOU WILL BE ABLE TO…

- Apply good transmitting technique (RSVP) and the listen-before-transmit discipline.
- Spell using the ICAO phonetic alphabet, with correct emphasis.
- Transmit numbers, frequencies, levels, headings, wind and pressure correctly, and apply the number rules.
- Transmit time in UTC.
- Use the standard words and phrases with their exact single meanings.
- Carry out a radio check and report readability on the 1–5 scale.

13.1 Transmitting technique (RSVP)

13.2 The phonetic alphabet

13.3 Transmission of numbers

13.4 The number rules

13.5 Levels, headings, wind & pressure

13.6 Time

13.7 Standard words & phrases

13.8 Radio check & readability

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Precision in radio telephony is critical. The phonetic alphabet ensures that call signs and critical data are never misunderstood.

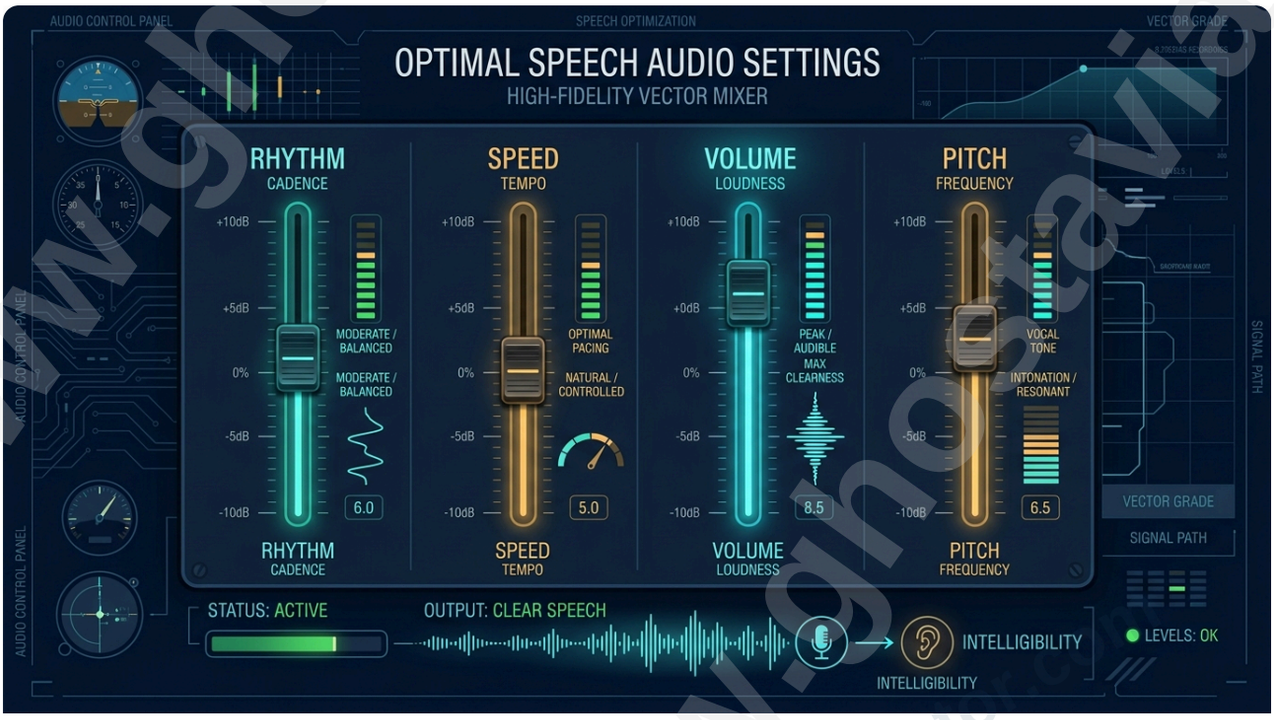
13.1 Transmitting technique (RSVP)

FIRST PRINCIPLES — CLARITY OVER SPEED

A transmission heard wrongly is worse than no transmission at all. The whole craft of radio is making yourself **understood the first time** on a noisy, shared channel. Before keying the microphone, **listen out** to be sure you won't block another station, then think what you will say.

RSVP — the four levers of good speech

- **R — Rhythm:** speak in a steady, even cadence; group the message into natural chunks.
- **S — Speed:** slightly slower than normal conversation, especially for numbers.
- **V — Volume:** constant, normal level; don't shout or trail off.
- **P — Pitch:** keep it level; a slightly higher, clear pitch carries better than a mumble.



Like an audio mixer, balancing Rhythm, Speed, Volume, and Pitch is the key to perfect RT clarity.

> **Exam trap — the practical**
>
> The examiner is judging exactly this. The most common practical fails are **too fast, swallowed numbers**, and forgetting to **release the PTT**. Slow down on figures and pause a beat after keying so the first word isn't clipped.

13.2 The phonetic alphabet

> **IN PLAIN TERMS**
>
> Each letter is spoken as a whole word chosen to sound unlike every other — so "B" and "D", or "M" and "N", can never be confused. The **bold syllable** is stressed.

The ICAO Phonetic Alphabet

Website
YouTube
Instagram
Telegram channel
Telegram community
Subject study group

Letter	Word	Letter	Word	Letter	Word
A	**AL**-fa	J	**JU**-li-**ETT**	S	Si-**ER**-ra
B	**BRA**-vo	K	**KI**-lo	T	**TANG**-go
C	**CHAR**-lie	L	**LI**-ma	U	**U**-ni-form
D	**DEL**-ta	M	**MIKE**	V	**VIC**-tor
E	**ECH**-o	N	No-**VEM**-ber	W	**WHIS**-key
F	**FOX**-trot	O	**OS**-car	X	**X**-ray
G	**GOLF**	P	Pa-**PA**	Y	**YANG**-key
H	Ho-**TEL**	Q	Ke-**BECK**	Z	**ZOO**-loo
I	**IN**-di-a	R	**RO**-me-o		

Exam trap — spelling & pronunciation

Watch the unusual spellings: **Alfa** (not Alpha), **Juliett** (double-t), **Whiskey**, **X-ray**. Pronounce the stressed syllable: **AL**-fah, **BRAH**-VOH, **CHAR**-lee, **KEY**-LOH, etc. Examiners often ask you to spell your own registration.

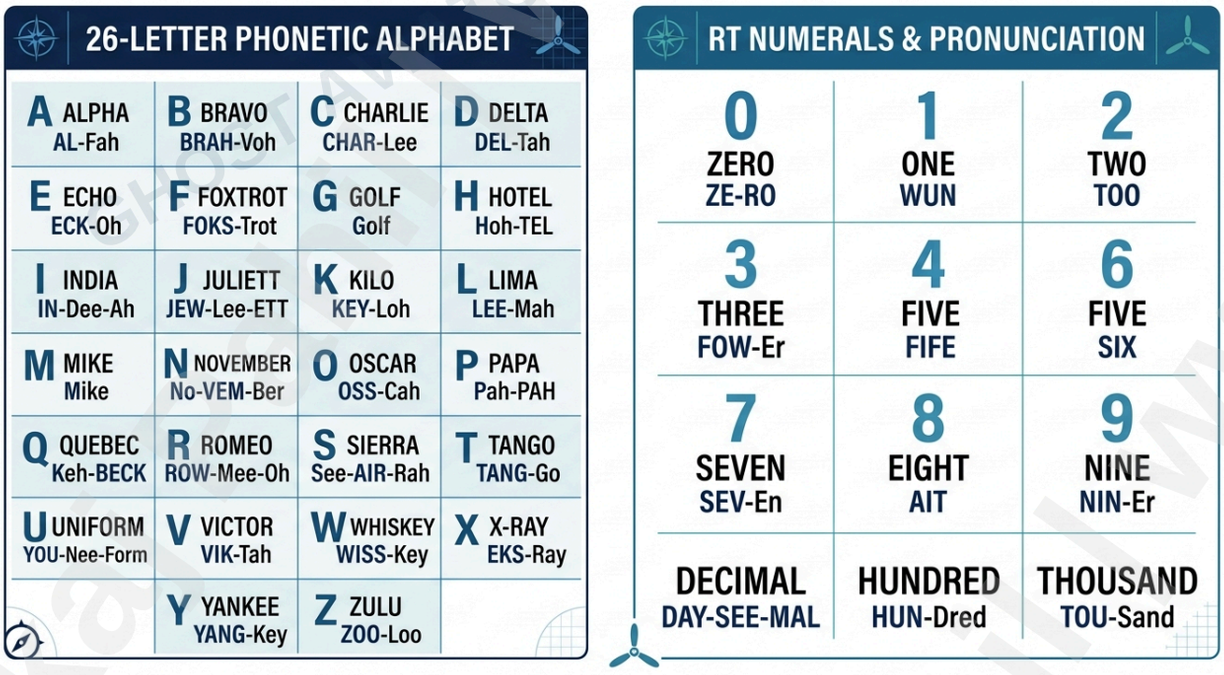


Figure 13.1 — The phonetic alphabet & number pronunciation reference card.

13.3 Transmission of numbers

Pronunciation of Numerals

Numeral	Pronounced	Numeral	Pronounced
0	ZE-RO	5	FIFE
1	WUN	6	SIX
2	TOO	7	SEV-en
3	TREE	8	AIT
4	FOW-er	9	NIN-er
Decimal	DAY-SEE-MAL	Hundred / Thousand	HUN-dred / TOU-SAND

Exam trap — the "safe" pronunciations

Three (**TREE**), five (**FIFE**), nine (**NIN-er**) and decimal (**DAY-SEE-MAL**) are deliberately altered so they cannot be confused with similar-sounding words or other languages. Use them exactly.

13.4 The number rules

The two rules

Rule 1 — digit by digit. Most numbers are transmitted by pronouncing **each digit separately**: aircraft call signs, headings, wind, pressure settings, transponder codes, frequencies, and flight levels.

Rule 2 — whole hundreds & thousands. Whole hundreds and whole thousands are transmitted by pronouncing each digit in the number of hundreds/thousands followed by **HUNDRED** or **THOUSAND** — used for altitudes, heights, cloud heights and visibility.

13.5 Levels, headings, wind & pressure

Applying the Number Rules

Item	How to say it	Example
Flight level	"Flight level" + each digit	FL280 → "flight level two eight zero"
Altitude / height	Whole thousands/hundreds	3,500 ft → "three thousand five hundred feet"
Heading	"Heading" + 3 digits	080 → "heading zero eight zero"
Wind	Direction (3 digits) + speed	"wind two seven zero degrees, one zero knots"
Pressure (QNH)	"QNH" + each digit (units optional)	1013 → "QNH one zero one three"
Frequency	Each digit incl. "decimal"	121.500 → "one two one decimal five"

Worked examples — numbers spoken correctly

Heading 270 → "heading two seven zero" (digit by digit). **Squawk 4321** → "squawk four three two one". **Altitude 2,500 ft** → "two thousand five hundred". **Flight level 100** → "flight level one zero zero" (digit by digit — note the difference).
Frequency 118.500 → "one one eight decimal five". **QNH 1009** → "QNH one zero zero nine".

Cockpit reality

A controller passing "descend flight level one zero zero, QNH one zero zero nine" expects an exact read-back. The discipline of digit-by-digit numbers is what prevents a "level 100" being flown as "level 1000".

13.6 Time

UTC — one clock for the world

Aviation uses **Coordinated Universal Time (UTC)**, spoken "Zulu". Time is normally given as the **minutes of the hour** only (e.g. "at three five"), unless there is any chance of confusion, when the **full hour and minutes** are given (e.g. "one three three five"). Each digit is pronounced separately.

Aviation relies on a single global time standard: Coordinated Universal Time (UTC), commonly referred to as "Zulu" time in RT.

> **Worked example**
>
> It is 1335 UTC. Normal: "time three five". If confusion is possible (e.g. near the hour): "time one three three five".

13.7 Standard words & phrases

Core Phraseology

Word / phrase	Single meaning
AFFIRM	Yes
NEGATIVE	No / not correct / not available
ROGER	I have received all of your last transmission (NOT "yes")
WILCO	I understand your message and will comply
STANDBY	Wait, I will call you
SAY AGAIN	Repeat all (or the specified part) of your last transmission
READ BACK	Repeat all of this message back to me exactly
CORRECTION	An error has been made; the correct version is …
DISREGARD	Ignore that transmission
CONFIRM	Have I correctly received …? / Confirm you have received …
MONITOR	Listen out on (frequency)
REQUEST	I should like to know … / I wish to obtain …
OUT	This exchange is ended; no reply expected

Exam trap — ROGER vs WILCO vs AFFIRM

A favourite: **ROGER** means "received", **not** "yes" and **not** "I will comply". If asked a yes/no question, answer **AFFIRM** or **NEGATIVE**; if instructed to do something, answer **WILCO**. "Over and out" is wrong — they are mutually exclusive; in modern RT you usually just end naturally.

13.8 Radio check & readability

The readability scale (1–5)

1 Unreadable · **2** Readable now and then · **3** Readable but with difficulty · **4** Readable · **5** Perfectly readable.

Transcript — a radio check

A/C Delhi Tower, VT-ABC, radio check on one one eight decimal one.

ATC VT-ABC, Delhi Tower, readability five.

A/C VT-ABC, readability five also.

☆ Numbers to memorise

Essential Facts for Chapter 13

Fact	Value
RSVP	Rhythm · Speed · Volume · Pitch
Altered numerals	3 = TREE · 5 = FIFE · 9 = NIN-er · decimal = DAY-SEE-MAL
Rule 1	Digit by digit: call signs, headings, wind, QNH, squawk, frequency, flight levels
Rule 2	Whole hundreds/thousands: altitude, height, cloud, visibility
Time	UTC (Zulu); minutes only, or full hours+minutes if confusion
ROGER / WILCO	Received / will comply (ROGER ≠ yes)
AFFIRM / NEGATIVE	Yes / No
Readability	1 unreadable → 5 perfectly readable

Question bank

PART A — MCQS (CLICK AN OPTION TO CHECK)

1. In the phonetic alphabet, the letter "J" is:

> Jupiter
>
> Juliet
>
> Juliett
>
> Jig

```
Answer: Juliett. "Juliett" — spelled with a double-t.
```

2. The numeral 9 is pronounced:

> Nine
>
> NIN-er
>
> Niner-zero
>
> Nin

```
Answer: NIN-er. 9 = "NIN-er", altered to avoid confusion with "no/nein".
```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3. The decimal point is spoken as:

> Point
>
> Dot
>
> DAY-SEE-MAL
>
> Comma

```
Answer: DAY-SEE-MAL. "Decimal" (DAY-SEE-MAL) — e.g. 118.1 = "one one eight decimal one".
```

4. Flight level 100 is transmitted as:

> "flight level one hundred"
>
> "flight level one zero zero"
>
> "flight level ten thousand"
>
> "level one hundred"

```
Answer: "flight level one zero zero". Flight levels are spoken digit by digit: "one zero zero".
```

5. An altitude of 3,500 ft is transmitted as:

> "three five zero zero"
>
> "three thousand five hundred"
>
> "flight level three five"
>
> "thirty-five hundred"

```
Answer: "three thousand five hundred". Altitudes use whole thousands/hundreds: "three thousand five hundred".
```

6. A heading of 080° is spoken as:

> "heading eighty"
>
> "heading zero eight zero"
>
> "heading eight hundred"
>
> "heading eight zero"

```
Answer: "heading zero eight zero". Headings are three digits, digit by digit: "zero eight zero".
```

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7. The frequency 121.500 is transmitted as:

> "one twenty-one point five"
>
> "one two one decimal five"
>
> "one hundred twenty-one five hundred"
>
> "one two one point five hundred"

Answer: "one two one decimal five". Digit by digit with "decimal": "one two one decimal five".

8. "ROGER" means:

> Yes
>
> I will comply
>
> I have received all your last transmission
>
> Standby

Answer: I have received all your last transmission. ROGER = received; it does NOT mean "yes" or "will comply".

9. The correct reply to an instruction you will carry out is:

> ROGER
>
> AFFIRM
>
> WILCO
>
> CONFIRM

Answer: WILCO. WILCO = "I understand and will comply".

10. The reply meaning "yes" is:

> ROGER
>
> AFFIRM
>
> WILCO
>
> CORRECT

Answer: AFFIRM. AFFIRM = yes; NEGATIVE = no.

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11. "STANDBY" means:

> Continue your approach
>
> Wait, I will call you
>
> Disregard
>
> Read back

```
Answer: Wait, I will call you. STANDBY = wait; the controller will call you.
```

12. Aviation time is given in:

> Local time
>
> UTC (Zulu)
>
> Indian Standard Time
>
> GMT+5:30

```
Answer: UTC (Zulu). Aviation uses UTC, spoken "Zulu".
```

13. Normally the time is transmitted as:

> The full hours and minutes always
>
> The minutes only, unless confusion is possible
>
> Local time with AM/PM
>
> Seconds only

```
Answer: The minutes only, unless confusion is possible. Minutes only is normal; the full hour and minutes are used if confusion could arise.
```

14. On the readability scale, "readable but with difficulty" is:

> 1
>
> 2
>
> 3
>
> 5

```
Answer: 3. 3 = readable but with difficulty; 5 = perfectly readable.
```

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15. A transponder code 7000 is read as:

> "seven thousand"
>
> "seven zero zero zero"
>
> "seventy hundred"
>
> "flight level seven zero"

`Answer: "seven zero zero zero". Squawk codes are digit by digit: "seven zero zero zero".`

16. The numeral 5 is pronounced:

> Five
>
> FIFE
>
> Fiver
>
> Fifty

`Answer: FIFE. 5 = "FIFE", altered for clarity.`

17. "SAY AGAIN" requests:

> A frequency change
>
> Repetition of all or part of the last transmission
>
> A read-back
>
> A radio check

`Answer: Repetition of all or part of the last transmission. SAY AGAIN asks the other station to repeat.`

18. Good transmitting technique is summarised by:

> Loud and fast
>
> RSVP — Rhythm, Speed, Volume, Pitch
>
> Brevity only
>
> High pitch only

`Answer: RSVP – Rhythm, Speed, Volume, Pitch. RSVP – steady rhythm, controlled (slightly slower) speed, even volume, level pitch.`

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PART B — ORAL / VIVA (TAP TO REVEAL MODEL ANSWERS)

▶ **Spell your registration VT-ABC using the phonetic alphabet.**

▶ **How are the numerals 3, 5 and 9 pronounced, and why are they altered?**

▶ **State the two number rules and give an example of each.**

▶ **What is the difference between ROGER, WILCO and AFFIRM?**

▶ **How is time transmitted in radiotelephony?**

▶ **What is the readability scale?**

60-SECOND REVISION CARD

- **RSVP:** Rhythm, Speed (slower for numbers), Volume, Pitch. Listen out, press & pause, release fully.
- Phonetics: Alfa…Zulu (watch Alfa, Juliett, Whiskey, X-ray). Numerals: **TREE, FIFE, NIN-er, DAY-SEE-MAL**.
- **Rule 1** digit-by-digit (call signs, headings, wind, QNH, squawk, freq, FL); **Rule 2** hundreds/thousands (altitude/height/cloud/vis).
- Time = **UTC (Zulu)**, minutes only normally. **ROGER**=received · **WILCO**=will comply · **AFFIRM/NEGATIVE**=yes/no.
- Readability **1–5**; radio check on initial contact.

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Call Signs & Establishing Communication

On a busy frequency a dozen aircraft may be listening at once, so every transmission must say who is calling and who is being called — unambiguously, every time. Call signs are that addressing system, and the rules around them are exact: which suffix names which controller, how an airline flight becomes a call sign, when you may shorten one, and what you must always read back.

SYLLABUS MAP

`Part III (i) Call signs · aeronautical & aircraft call signs · establishment, continuation & transfer · read-back · test procedures`

LEARNING OBJECTIVES — BY THE END OF THIS CHAPTER YOU WILL BE ABLE TO…

- Identify aeronautical (ground) station call signs by their suffix.
- Form the three types of aircraft call sign and apply the abbreviation rules correctly.
- Make a correct initial call and continue communication.
- Handle a transfer of communication.
- State the items that must always be read back.
- Carry out a radio-check (test) procedure.

14.1 Why call signs
14.2 Ground-station call signs
14.3 The three aircraft call-sign types
14.4 Abbreviation rules
14.5 Establishing communication
14.6 Transfer of communication
14.7 Mandatory read-back items
14.8 Test (radio-check) procedure

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The ATC tower is the hub of the airport's communication. Ground station suffixes like TOWER, GROUND, and DELIVERY explicitly define who you are talking to.

14.1 Why call signs

> **FIRST PRINCIPLES — ADDRESSING ON A PARTY LINE**
>
> VHF is a shared, simplex "party line": everyone hears everything. A call sign is the **address** that tells one specific station "this is for you" and tells everyone else "this is not for me". Every transmission therefore names **who is being called** and **who is calling**, so there can never be doubt about who must act.

14.2 Ground-station call signs — the suffix names the service

> **IN PLAIN TERMS**
>
> A ground station's call sign is the **place name + a suffix**, and the suffix tells you which service you have reached.

Ground Station Suffixes

Suffix	Service	Suffix	Service
DELIVERY	Clearance delivery	**CONTROL**	Area control centre
GROUND	Surface movement control	**RADAR**	Radar (general)
TOWER	Aerodrome control	**DIRECTOR**	Approach radar (final director)
APPROACH	Approach control	**PRECISION**	Precision approach radar (PAR)
DEPARTURE	Departure control	**INFORMATION**	Flight Information Service (FIS)
APRON	Apron management	**RADIO**	Aeronautical station (HF/communications)

> **Mnemonic — the ground flow**
>
> A departing flight typically walks through **Delivery → Ground → Tower → Departure/Radar → Control**; an arrival comes back **Control → Approach/Director → Tower → Ground**. Learn the suffixes in that running order.

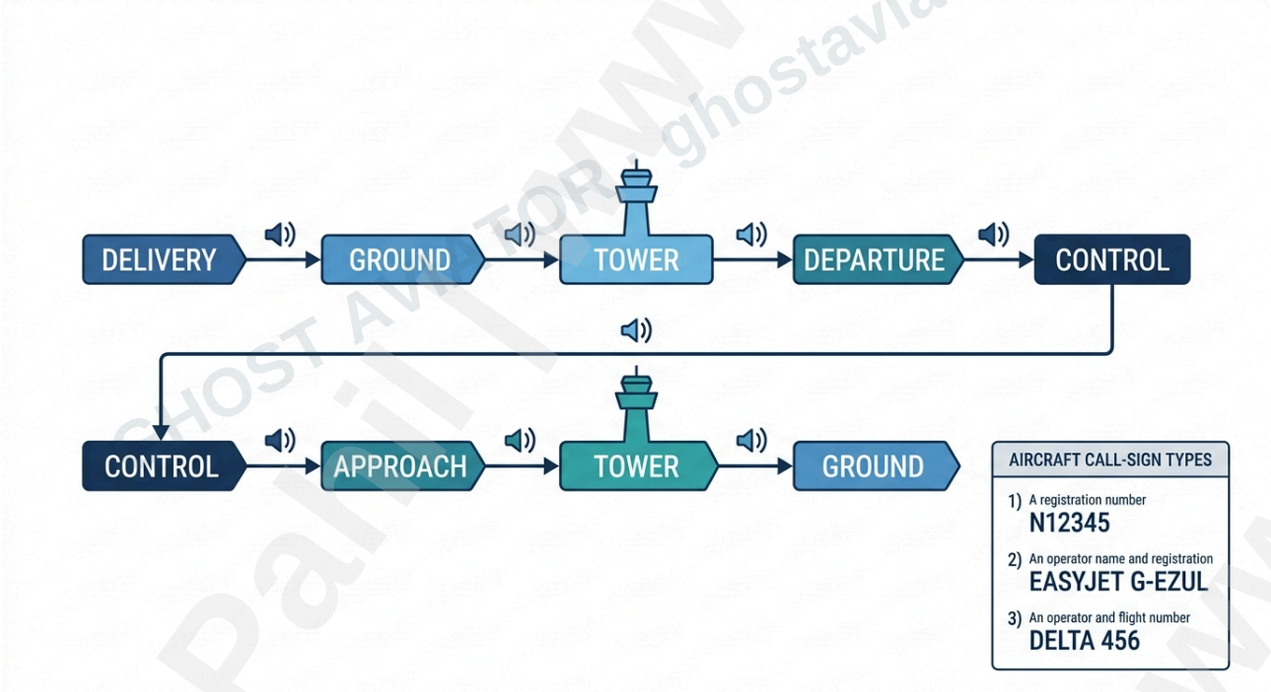

Figure 14.1 — Who you talk to — the ground-station flow.

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14.3 The three aircraft call-sign types

Type 1 — registration / type

The aircraft's **registration** spelled phonetically: **VT-ABC** → "Victor Tango Alfa Bravo Charlie". A general-aviation aircraft may prefix the manufacturer/type for the first call: "Cessna Victor Tango Alfa Bravo Charlie".

Type 2 — operator + last characters of registration

The telephony designator of the operator followed by the last characters of the registration: e.g. operator "GHOSTAIR" + registration ...ABC → "Ghostair Alfa Bravo Charlie".

Type 3 — operator + flight number

The operator's telephony designator followed by the **flight identification** (number/letters): "Air India one zero one" for AIC101. This is the airline call sign you hear most. The flight number is spoken **digit by digit**.

The three types of aircraft call signs ensure there is a clear, standard way to address both small general aviation aircraft and large commercial airliners.

Exam trap

The telephony designator is **not** the IATA two-letter code; "AI" is written but spoken as the telephony name "**Air India**". And the flight number is read digit by digit — "one zero one", not "one hundred and one".

14.4 Abbreviation rules

When and how a call sign may be shortened

1. An aircraft call sign may be **abbreviated only after the ground station has first abbreviated it** — never on the pilot's initiative, and not at all if there is any risk of confusion with another similar call sign on frequency.
2. **Type 1** (registration) abbreviates to the **first character and at least the last two**: "VT-ABC" → "Victor Tango … Bravo Charlie" or "V…BC".
3. **Type 2/3** (operator + flight number) is **not** abbreviated — the flight number is kept in full.

Transcript — full then abbreviated

A/C Delhi Ground, Victor Tango Alfa Bravo Charlie, request taxi.

ATC Victor Tango Alfa Bravo Charlie, taxi to holding point Alfa One. (ground initiates full)

A/C Taxi holding point Alfa One, Victor Tango Bravo Charlie. (now abbreviated)

14.5 Establishing communication

The initial call

An initial call names **(1) who you are calling**, then **(2) who you are**, then keeps the first message brief. On a first contact you may state a short request; for a longer message, wait to be invited ("pass your message").

Transcript — establishing communication

A/C Delhi Approach, Ghostair Alfa Bravo Charlie.

ATC Ghostair Alfa Bravo Charlie, Delhi Approach, pass your message.

A/C Ghostair Alfa Bravo Charlie, Cessna 172, 15 miles north, FL080, inbound for landing, information Charlie.

ATC Ghostair Alfa Bravo Charlie, roger, descend FL060, QNH one zero one three.

A/C Descend FL060, QNH one zero one three, Ghostair Alfa Bravo Charlie.

Cockpit reality — listen before you call

Before that first call, listen out for a few seconds: you may learn the runway in use, the QNH and how busy the controller is, and you avoid transmitting over someone mid-clearance.

14.6 Transfer of communication

Being handed on

As you pass between sectors, ATC **transfers** you to the next frequency: "contact [station] on [frequency]". You read back the frequency, change over, and make a fresh initial call on the new frequency. "**Monitor**" means tune and listen out without calling immediately.

Transcript — transfer

ATC Ghostair Alfa Bravo Charlie, contact Delhi Tower one one eight decimal one.

A/C Delhi Tower one one eight decimal one, Ghostair Alfa Bravo Charlie.

A/C Delhi Tower, Ghostair Alfa Bravo Charlie, FL060 inbound. (new call on new frequency)

14.7 Mandatory read-back items

WHY READ-BACK EXISTS

A read-back closes the loop: the controller hears you repeat the instruction and can catch a mishearing before it becomes an incident. Certain safety-critical items **must always** be read back in full.

Always read back

- **ATC route clearances**
- clearances to **enter, land on, take off from, backtrack, cross or hold short of** any runway
- **runway in use**
- **level instructions**
- **heading** and **speed** instructions
- **altimeter (QNH/QFE)** settings
- **SSR (squawk) codes**
- **frequency changes**
- VDF information and transition levels.

Mnemonic

"Runways, Levels, Headings, Speeds, QNH, Squawk, Frequencies" — if it's one of these, read it back in full. End the read-back with your call sign.

A pilot's mental checklist for read-backs ensures no safety-critical instruction is ever glossed over.

14.8 Test (radio-check) procedure

Transcript — radio check

A/C Delhi Tower, VT-ABC, radio check on one one eight decimal one.

ATC VT-ABC, Delhi Tower, readability five.

A/C VT-ABC, readability five also, thank you.

Reporting readability

State the readability on the **1–5 scale** (1 unreadable … 5 perfectly readable). If readability is poor, the receiving station may say "readability three, you are broken/distorted", prompting a frequency or equipment check.

☆ Numbers to memorise

Essential Facts for Chapter 14

Fact	Value
Ground suffixes	Delivery · Ground · Tower · Approach · Departure · Director · Radar · Control · Information
Call-sign types	1: registration · 2: operator + reg · 3: operator + flight number
Abbreviation	Only after ATC abbreviates; Type 1 → first + last two; never abbreviate the flight number
Initial call	Who you call → who you are → brief message / request
Transfer	"Contact [station] [freq]" → read back freq → new call
Always read back	Runways, levels, headings, speeds, QNH, squawk, frequency changes
Radio check	Readability 1–5

Question bank

PART A — MCQS (CLICK AN OPTION TO CHECK)

1. The ground-station suffix "GROUND" denotes:

> Area control
>
> Surface movement control
>
> Approach control
>
> Flight information

```
Answer: Surface movement control. GROUND = surface movement control (taxi); TOWER = aerodrome control.
```

2. "DIRECTOR" is associated with:

> Apron management
>
> Approach radar (final vectoring)
>
> Clearance delivery
>
> Area control

```
Answer: Approach radar (final vectoring). Director provides final approach radar vectoring.
```

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3. The suffix for Flight Information Service is:

> CONTROL
>
> RADAR
>
> INFORMATION
>
> TOWER

Answer: INFORMATION. INFORMATION denotes the Flight Information Service.

4. An airline flight AIC101 is spoken as:

> "A-I-C one hundred one"
>
> "Air India one zero one"
>
> "AI one zero one"
>
> "Air India one hundred and one"

Answer: "Air India one zero one". Telephony designator + flight number digit by digit: "Air India one zero one".

5. An aircraft call sign may be abbreviated:

> Whenever the pilot wishes
>
> Only after the ground station has first abbreviated it
>
> Never
>
> Only on initial contact

Answer: Only after the ground station has first abbreviated it. The pilot may abbreviate only once ATC has done so, and not if confusion is possible.

6. Type 1 call sign VT-ABC abbreviates to:

> "VT-A"
>
> First character and at least the last two — e.g. "Victor … Bravo Charlie"
>
> "ABC"
>
> It cannot be abbreviated

Answer: First character and at least the last two. Registration call signs shorten to the first plus at least the last two characters.

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7. An airline (operator + flight number) call sign is abbreviated by:

> Dropping the operator name
>
> Shortening the flight number
>
> It is not abbreviated — the flight number is kept in full
>
> Using the IATA code

Answer: It is not abbreviated. Operator + flight-number call signs are not abbreviated.

8. An initial call should give, in order:

> Your request, then your call sign
>
> Who you are calling, then who you are
>
> The frequency, then your message
>
> Your altitude, then your heading

Answer: Who you are calling, then who you are. Name the station being called first, then your own call sign.

9. "Pass your message" means:

> Change frequency
>
> Go ahead with your message / request
>
> Standby
>
> Read back

Answer: Go ahead with your message / request. It invites the aircraft to transmit its full message.

10. On a transfer of communication, ATC says:

> "Squawk ident"
>
> "Contact [station] on [frequency]"
>
> "Report ready"
>
> "Cleared to land"

Answer: "Contact [station] on [frequency]". A transfer instructs you to contact the next unit on a stated frequency.

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11. "Monitor [frequency]" means:

> Call immediately
>
> Tune and listen out, without calling immediately
>
> Squawk the code
>
> Change runway

```
Answer: Tune and listen out, without calling immediately. MONITOR = listen out on the frequency without an immediate call.
```

12. Which of the following must always be read back?

> Traffic information
>
> A runway clearance (enter/land/take off/cross/hold short)
>
> A weather report
>
> A position report

```
Answer: A runway clearance. Runway clearances are mandatory read-back items.
```

13. A squawk code is:

> Not read back
>
> A mandatory read-back item
>
> Only read back at night
>
> Read back as a whole number

```
Answer: A mandatory read-back item. SSR codes must be read back, digit by digit.
```

14. A read-back should normally end with:

> "Over and out"
>
> Your call sign
>
> "Roger"
>
> The frequency

```
Answer: Your call sign. Finish the read-back with your own call sign for positive identification.
```

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15. A radio check reply of "readability three" means the signal is:

> Perfectly readable
>
> Readable but with difficulty
>
> Unreadable
>
> Readable

```
Answer: Readable but with difficulty. 3 = readable but with difficulty (1 unreadable, 5 perfectly readable).
```

16. The telephony designator spoken for an airline is:

> The IATA two-letter code
>
> The assigned telephony name (e.g. "Air India")
>
> The aircraft registration
>
> The flight number only

```
Answer: The assigned telephony name. The spoken designator is the telephony name, not the written two-letter code.
```

PART B — ORAL / VIVA (TAP TO REVEAL MODEL ANSWERS)

▶ **What does the suffix of a ground station's call sign tell you? Give examples.**

▶ **Describe the three types of aircraft call sign.**

▶ **When may an aircraft abbreviate its call sign, and how?**

▶ **What information does an initial call contain?**

▶ **List the items that must always be read back.**

▶ **How do you carry out a radio check?**

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- **Ground suffix = service:** Delivery·Ground·Tower·Approach·Departure·Director·Control·Information.
- **Call signs: 1** registration · **2** operator+reg · **3** operator+flight number (digit by digit).

- **Abbreviate only after ATC does;** Type 1 → first + last two; never shorten the flight number.
- **Initial call: who you call → who you are → brief message**. Transfer: "contact [stn] [freq]".
- **Always read back: runways, levels, headings, speeds, QNH, squawk, frequencies**; end with your call sign. Radio check = readability **1–5**.

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Aerodrome Control: Aircraft

This is the chapter the practical examiner listens for. Standard phraseology is a script — the same words, in the same order, for every phase of flight — so that a controller in Delhi and a pilot from anywhere understand each other instantly. Here we walk a flight from clearance and start, through taxi, take-off, the circuit, final approach and landing, the go-around, and vacating.

SYLLABUS MAP

Part III (iii) Aerodrome control – aircraft: departure info, start, push-back, taxi, take-off, circuit, final & landing, go-around, after landing

LEARNING OBJECTIVES — BY THE END OF THIS CHAPTER YOU WILL BE ABLE TO…

- Describe the aerodrome controller's responsibilities and the manoeuvring area.
- Use the correct phraseology for clearance delivery, start-up and push-back.
- Use taxi phraseology, including holding points, "give way" and conditional clearances.
- Use line-up and take-off phraseology and the circuit reporting points.
- Use final-approach, landing and go-around phraseology.
- Use after-landing and vacating phraseology.

15.1 The aerodrome controller

15.2 Clearance & departure information

15.3 Start-up & push-back

15.4 Taxi

15.5 Line-up & take-off

15.6 The traffic circuit

15.7 Final approach & landing

15.8 Go-around

15.9 After landing & vacating

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From engine start to the final runway vacation, strict adherence to standard phraseology is the foundation of aerodrome safety.

15.1 The aerodrome controller

IN PLAIN TERMS

Aerodrome control (**Tower**) is responsible for traffic on the **runways and in the aerodrome traffic circuit**, and for vehicles and aircraft on the **manoeuvring area**.
The **manoeuvring area** is the runways and taxiways used for take-off, landing and taxiing — but **not** the aprons. The **movement area** is the manoeuvring area **plus** the aprons.

Two definitions to keep straight

Manoeuvring area = runways + taxiways (for take-off, landing, taxiing); excludes aprons. **Movement area** = manoeuvring area + aprons.
Ground/Tower control the manoeuvring area; the apron may be managed separately ("Apron").

15.2 Clearance & departure information

WHY A CLEARANCE IS READ BACK IN FULL

An IFR departure clearance is a contract: it states **where** you are cleared, by what **route**, to what **level**, and the **squawk**. Because an error here propagates into controlled airspace, the whole clearance is a mandatory read-back.

Transcript — IFR clearance (Delivery)

A/C Delhi Delivery, Ghostair Alfa Bravo Charlie, stand two one, request IFR clearance to Mumbai, information Charlie.

ATC Ghostair Alfa Bravo Charlie, cleared to Mumbai via PAPA two departure, climb flight level one zero zero, squawk four three two one.

A/C Cleared to Mumbai via PAPA two departure, climb flight level one zero zero, squawk four three two one, Ghostair Alfa Bravo Charlie.

ATC Ghostair Alfa Bravo Charlie, read-back correct.

Departure information

Before taxi the pilot needs the **runway in use, surface wind, QNH** and any essential information — usually obtained from **ATIS** (Chapter 6) and confirmed on first contact ("…information Charlie"). The controller passes anything ATIS does not cover.

15.3 Start-up & push-back

Transcript — start & push

A/C Delhi Ground, Ghostair Alfa Bravo Charlie, stand two one, request start-up.

ATC Ghostair Alfa Bravo Charlie, start-up approved.

A/C Request push-back.

ATC Ghostair Alfa Bravo Charlie, push-back approved, facing south.

A/C Push-back approved, facing south, Ghostair Alfa Bravo Charlie.

Exam trap

"Start-up approved" permits engine start, not movement. **Push-back is a separate clearance**, and the controller may give the direction to face. Never push or taxi on a start-up clearance alone.

15.4 Taxi

Transcript — taxi

A/C Delhi Ground, Ghostair Alfa Bravo Charlie, request taxi.

ATC Ghostair Alfa Bravo Charlie, taxi to holding point Alfa One runway two seven via taxiway Bravo, QNH one zero one three.

A/C Taxi to holding point Alfa One runway two seven via Bravo, QNH one zero one three, Ghostair Alfa Bravo Charlie.

Key taxi phrases

"Hold position" — stop where you are. **"Give way to [traffic]"** — let the named traffic pass. **"Follow [traffic]"**. **"Hold short of runway 27"** / **"holding point Alfa One"** — stop at the marked point and do not enter the runway. A clearance to taxi **does not** include clearance to enter or cross a runway.

Exam trap — runway entry

Taxiing to a holding point is **not** permission to enter the runway. Entering, crossing or backtracking a runway each needs its own explicit clearance — and each is a mandatory read-back.

15.5 Line-up & take-off

Transcript — line-up & departure

A/C Delhi Tower, Ghostair Alfa Bravo Charlie, holding point Alfa One, ready for departure.

ATC Ghostair Alfa Bravo Charlie, line up runway two seven and wait.

A/C Line up runway two seven and wait, Ghostair Alfa Bravo Charlie.

ATC Ghostair Alfa Bravo Charlie, wind two seven zero degrees eight knots, runway two seven, cleared for take-off.

A/C Runway two seven, cleared for take-off, Ghostair Alfa Bravo Charlie.

Take-off phraseology — exact words

The word **"take-off"** is used **only** when issuing or cancelling a take-off clearance — at all other times say "departure". **"Cleared for take-off"** / read back **"cleared for take-off"**. To cancel: **"hold position, cancel take-off, I say again cancel take-off"**. An **"immediate take-off"** may be issued if traffic requires.

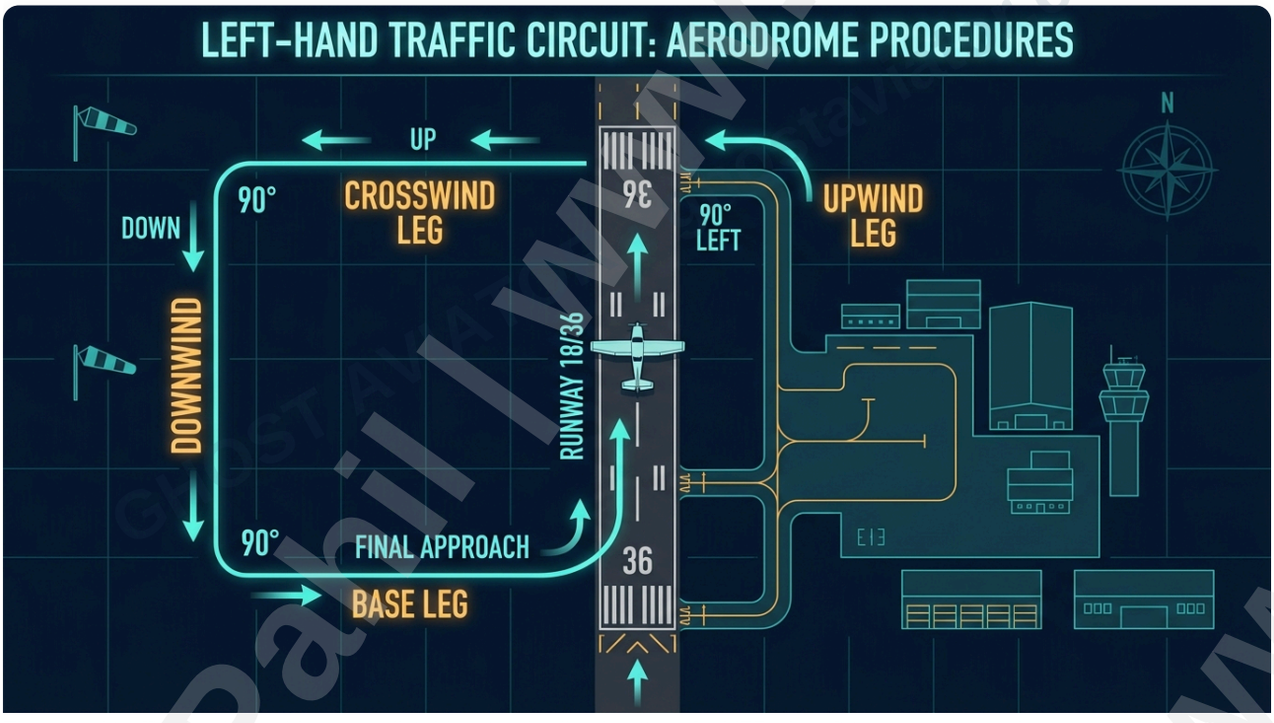
Worked example — conditional line-up

ATC: "Ghostair Alfa Bravo Charlie, **behind the landing Airbus**, line up runway two seven **behind**."
Correct read-back: "Behind the landing Airbus, line up runway two seven behind, Ghostair Alfa Bravo Charlie." The word "behind" is said **twice** — at the start and end — so the condition cannot be missed. Do not move until the Airbus has passed.

15.6 The aerodrome traffic circuit

IN PLAIN TERMS

The circuit is the standard rectangular pattern flown around a runway. Its legs — **upwind/crosswind, downwind, base, final** — are the reporting points the tower uses to sequence visual traffic.



The visual traffic circuit provides a predictable, standardized pattern for VFR aircraft to maneuver and sequence for landing.

Circuit Legs

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Leg	Where
Upwind / climb-out	Straight ahead after take-off
Crosswind	Turn (usually 90°) across the upwind end
Downwind	Parallel to the runway, opposite direction to landing
Base	Turn toward final, descending
Final	Lined up with the runway for landing

Transcript — joining & circuit reports

A/C Delhi Tower, VT-ABC, five miles north, for landing, information Charlie.

ATC VT-ABC, join left-hand downwind runway two seven, QNH one zero one three, report downwind.

A/C Join left-hand downwind runway two seven, QNH one zero one three, wilco, VT-ABC.

A/C VT-ABC, downwind.

ATC VT-ABC, number two, follow the Cessna on base, report final.

15.7 Final approach & landing

Transcript — final & landing

A/C VT-ABC, final.

ATC VT-ABC, wind two seven zero degrees one zero knots, runway two seven, cleared to land.

A/C Runway two seven, cleared to land, VT-ABC.

Landing variations

"Continue approach" — keep approaching; landing clearance not yet given.

"Land after the departing/landing [traffic]" — a conditional landing clearance behind preceding traffic. **"Cleared to land"** is the full clearance and a mandatory read-back.

15.8 Go-around

The go-around / missed approach

If the approach must be discontinued, the instruction is **"go around"** (said clearly, often twice). The pilot applies power, climbs, and flies the missed-approach procedure or the tower's instructions. A pilot may also initiate a go-around without being told — reporting **"going around"**.

Transcript — go-around

ATC VT-ABC, go around, I say again, go around, vehicle on the runway.

A/C Going around, VT-ABC.

ATC VT-ABC, climb straight ahead to two thousand feet, report downwind.

15.9 After landing & vacating

Transcript — vacating

ATC VT-ABC, vacate next left, contact Ground one two one decimal nine.

A/C Vacate next left, Ground one two one decimal nine, VT-ABC.

A/C Delhi Ground, VT-ABC, runway vacated, request taxi to apron.

ATC VT-ABC, taxi to stand two one via taxiway Bravo.

Exam trap — "runway vacated"

You are not clear of the runway until the whole aircraft is past the holding point. Report **"runway vacated"** only when fully clear — this releases the runway for the next aircraft, so accuracy matters.

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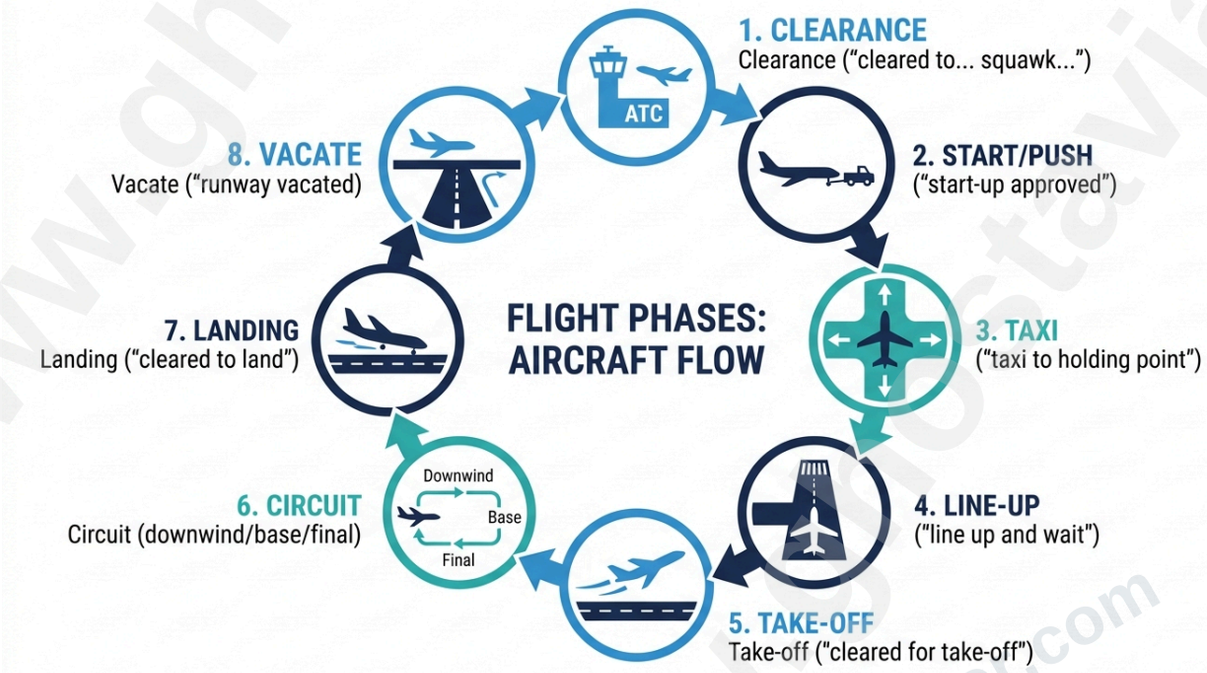

Figure 15.1 — A flight in phraseology: the critical communication nodes from clearance to vacating the runway.

☆ Numbers to memorise

Essential Facts for Chapter 15

Fact	Value
Manoeuvring vs movement area	Manoeuvring = runways+taxiways; Movement = + aprons
"Take-off"	Used only to issue/cancel a take-off clearance; else "departure"
Start vs push	"Start-up approved" ≠ movement; push-back is a separate clearance
Runway entry	Taxi to a holding point ≠ clearance to enter/cross/backtrack
Conditional clearance	Condition stated first & last ("behind … behind")
Circuit legs	Upwind · crosswind · downwind · base · final
Landing phrases	Continue approach / cleared to land / land after
Go-around	"Go around" (often twice); pilot may initiate "going around"

Question bank

PART A — MCQS (CLICK AN OPTION TO CHECK)

1. The manoeuvring area consists of:

> Aprons only
>
> Runways and taxiways (excluding aprons)
>
> Runways, taxiways and aprons
>
> The terminal building

Answer: Runways and taxiways (excluding aprons). Manoeuvring area = runways + taxiways; adding aprons makes it the movement area.

2. An IFR departure clearance must be:

> Acknowledged with "roger"
>
> Read back in full
>
> Read back only if requested
>
> Not read back

Answer: Read back in full. Route clearances are mandatory read-back items.

3. "Start-up approved" permits:

> Push-back and taxi
>
> Engine start only
>
> Take-off
>
> Crossing the runway

Answer: Engine start only. It allows engine start; push-back and taxi are separate clearances.

4. The word "take-off" is used:

> Throughout the taxi
>
> Only when issuing or cancelling a take-off clearance
>
> For any departure intention
>
> Only by the pilot

Answer: Only when issuing or cancelling a take-off clearance. Otherwise "departure" is used, to reserve "take-off" for the clearance itself.

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5. Taxiing to a holding point gives you:

> Clearance to line up
>
> Clearance to cross the runway
>
> No clearance to enter the runway — stop at the holding point
>
> Clearance for take-off

Answer: No clearance to enter the runway. A taxi clearance to a holding point does not authorise entering the runway.

6. "Give way to the Cessna" means:

> Follow the Cessna
>
> Let the Cessna pass before you continue
>
> Overtake the Cessna
>
> Stop permanently

Answer: Let the Cessna pass before you continue. Give way = yield to the named traffic, then continue.

7. "Line up runway 27 and wait" means:

> Take off immediately
>
> Enter the runway, line up, and hold for take-off clearance
>
> Hold short of the runway
>
> Vacate the runway

Answer: Enter the runway, line up, and hold for take-off clearance. You line up on the runway but do not take off until cleared.

8. A conditional clearance such as "behind the landing Airbus, line up behind" requires:

> Immediate line-up
>
> Waiting until the named Airbus has passed, reading the condition back
>
> A go-around
>
> Contacting Ground

Answer: Waiting until the named Airbus has passed, reading the condition back. You act only after the stated condition is satisfied, and read it back including "behind".

9. The downwind leg of the circuit is:

> Straight ahead after take-off
>
> Parallel to the runway, opposite to the landing direction
>
> Lined up for landing
>
> The turn onto final

Answer: Parallel to the runway, opposite to the landing direction. Downwind runs parallel to the runway in the opposite direction; base then turns toward final.

10. "Continue approach" means:

> You are cleared to land
>
> Keep approaching; landing clearance not yet given
>
> Go around
>
> Vacate the runway

Answer: Keep approaching; landing clearance not yet given. It is not a landing clearance; expect the landing clearance shortly.

11. "Cleared to land" is:

> Not read back
>
> A mandatory read-back item
>
> Read back only at night
>
> Optional

Answer: A mandatory read-back item. Landing clearances are mandatory read-back items.

12. The instruction to discontinue an approach is:

> "Orbit"
>
> "Go around"
>
> "Hold position"
>
> "Continue"

Answer: "Go around". "Go around" (often said twice) commands the missed approach.

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13. A pilot who initiates a missed approach reports:

> "Roger"
>
> "Going around"
>
> "Cancel landing"
>
> "Vacating"

Answer: "Going around". The pilot announces "going around" when initiating it.

14. You should report "runway vacated" when:

> The nose passes the holding point
>
> The whole aircraft is clear past the holding point
>
> You begin to slow
>
> You are on final

Answer: The whole aircraft is clear past the holding point. Only when fully clear of the runway, releasing it for the next aircraft.

15. To cancel a take-off clearance, ATC says:

> "Continue"
>
> "Hold position, cancel take-off, I say again cancel take-off"
>
> "Line up and wait"
>
> "Go around"

Answer: "Hold position, cancel take-off, I say again cancel take-off". The cancellation uses "cancel take-off", repeated for emphasis.

16. Surface wind passed with a take-off/landing clearance is in:

> Degrees true
>
> Degrees magnetic
>
> Knots only
>
> Metres per second

Answer: Degrees magnetic. ATC passes surface wind in degrees magnetic (the runway is also magnetic).

PART B — ORAL / VIVA (TAP TO REVEAL MODEL ANSWERS)

▶ **Define the manoeuvring area and the movement area.**

▶ **What does an IFR departure clearance contain, and how is it acknowledged?**

▶ **When is the word "take-off" used?**

▶ **What must you do on a conditional clearance, e.g. "behind the landing aircraft, line up behind"?**

▶ **Name the legs of the aerodrome traffic circuit.**

▶ **What is the difference between "continue approach" and "cleared to land"?**

60-SECOND REVISION CARD

- **Manoeuvring** = runways+taxiways; **movement** = + aprons. Clearance read back in full.
- **Start ≠ push ≠ taxi** (separate clearances). Taxi to holding point ≠ enter runway.
- **"Take-off"** only for the clearance; **line up and wait** → **cleared for take-off**. Conditional = "behind … behind".
- Circuit: upwind·crosswind·**downwind**·**base**·**final**. **Continue approach** / **cleared to land** / **land after**.
- **Go around** (twice); report **going around**; report **runway vacated** only when fully clear.

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Aerodrome Control: Vehicles

Aircraft are not the only traffic on the manoeuvring area. Fire trucks, "Follow-Me" cars, sweepers, tugs and inspection vehicles all move among the runways and taxiways — and a vehicle in the wrong place is one of the most dangerous causes of a runway incursion. This short but vital chapter covers the phraseology that keeps vehicles safely separated from aircraft.

SYLLABUS MAP

Part III (v) Aerodrome control – vehicles: movement instructions, crossing runways, vehicles towing aircraft

LEARNING OBJECTIVES — BY THE END OF THIS CHAPTER YOU WILL BE ABLE TO…

- Explain why vehicles on the manoeuvring area need ATC clearance and phraseology.
- Use vehicle call signs and basic movement instructions.
- Apply the strict phraseology for crossing and holding short of runways.
- Handle a vehicle towing an aircraft.
- Use "report vacated" and conditional instructions correctly.

16.1 Why vehicles talk to the tower

16.2 Vehicle call signs & movement

16.3 Crossing & holding short of runways

16.4 Vehicles towing aircraft

16.5 Conditional instructions & "report vacated"

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Vehicles on the manoeuvring area are subject to the same strict ATC control and read-back requirements as aircraft to prevent deadly incursions.

16.1 Why vehicles talk to the tower

FIRST PRINCIPLES — ONE CONTROLLER, ALL THE TRAFFIC

The manoeuvring area is a single shared space. Aircraft land and take off at high speed and cannot stop quickly; a vehicle that strays onto an active runway, or onto a taxiway in front of a taxiing aircraft, is a **runway/taxiway incursion** — a leading cause of serious accidents. So **every vehicle on the manoeuvring area is controlled by the aerodrome controller**, on the same frequency, using disciplined phraseology, exactly like an aircraft.

Who needs clearance

Any vehicle operating on the **manoeuvring area** (runways and taxiways) needs ATC clearance for its movements; vehicles confined to the **apron** are normally handled by apron management. The rule of thumb: if a vehicle could conflict with an aircraft taking off, landing or taxiing, the tower controls it.

16.2 Vehicle call signs & movement instructions

Call signs

Vehicles use a call sign that identifies their function and a number — e.g. **"Car 5", "Fire 1", "Sweeper 2", "Follow-Me"**. The controller addresses them just as an aircraft, and they read back instructions.

Transcript — vehicle movement

VEH Delhi Tower, Car 5, request to proceed along taxiway Bravo to the threshold of runway two seven.

ATC Car 5, proceed along taxiway Bravo, hold short of runway two seven.

VEH Proceed Bravo, hold short of runway two seven, Car 5.

Common vehicle phrases

"Proceed [route]" — move along the stated path. **"Hold position"** — stop.
"Hold short of [runway/point]" — stop at the marking, do not pass it.
"Give way to [aircraft/vehicle]". **"Vacate [runway/taxiway]"**. **"Report [position/vacated]"**.

16.3 Crossing & holding short of runways

THE CRITICAL CASE

A vehicle crossing a runway is the highest-risk vehicle movement, so the phraseology is strict and explicit. A vehicle **never** crosses or enters a runway without a specific clearance, and reports when clear so the controller knows the runway is released.

Transcript — crossing a runway

VEH Delhi Tower, Car 5, holding short of runway two seven, request to cross.

ATC Car 5, cross runway two seven, report vacated.

VEH Cross runway two seven, will report vacated, Car 5.

VEH Car 5, runway two seven vacated.

ATC Car 5, roger.

Exam trap

"Hold short of runway 27" is an instruction to **stop and wait** — it is **not** a clearance to cross. Crossing requires the explicit **"cross runway 27"**, read back in full, and the crossing is not complete until **"vacated"** is reported.

16.4 Vehicles towing aircraft

The tug & aircraft as one unit

When a vehicle (a **tug**) tows an aircraft, the combination is treated as a single moving unit on the manoeuvring area and is given **taxi/movement instructions** by the tower. The controller must know it is a tow (it may move slowly and turn within a large radius), and the tow is cleared along a route and to hold short or cross runways exactly like any other movement.

Transcript — towing an aircraft

VEH Delhi Ground, Tug 3, towing the Airbus from stand two one to the maintenance area, request tow.

ATC Tug 3, tow approved to the maintenance area via taxiway Bravo, hold short of runway two seven.

VEH Tow to maintenance via Bravo, hold short of runway two seven, Tug 3.

16.5 Conditional instructions & "report vacated"

Sequencing vehicles with aircraft

The controller often sequences a vehicle around aircraft movements: **"give way to the landing aircraft, then cross"**, or **"after the departing Airbus, cross runway 27"**. As with aircraft, a conditional instruction names the traffic, is read back including the condition, and is acted on only when that traffic has passed. Always **report when the runway/taxiway is vacated**.

Real-world relevance

Many of the world's worst runway accidents began with confusion over who was cleared onto the runway. Tight, explicit vehicle phraseology — hold short, cross, report vacated — exists precisely to make a vehicle on an active runway impossible to overlook.

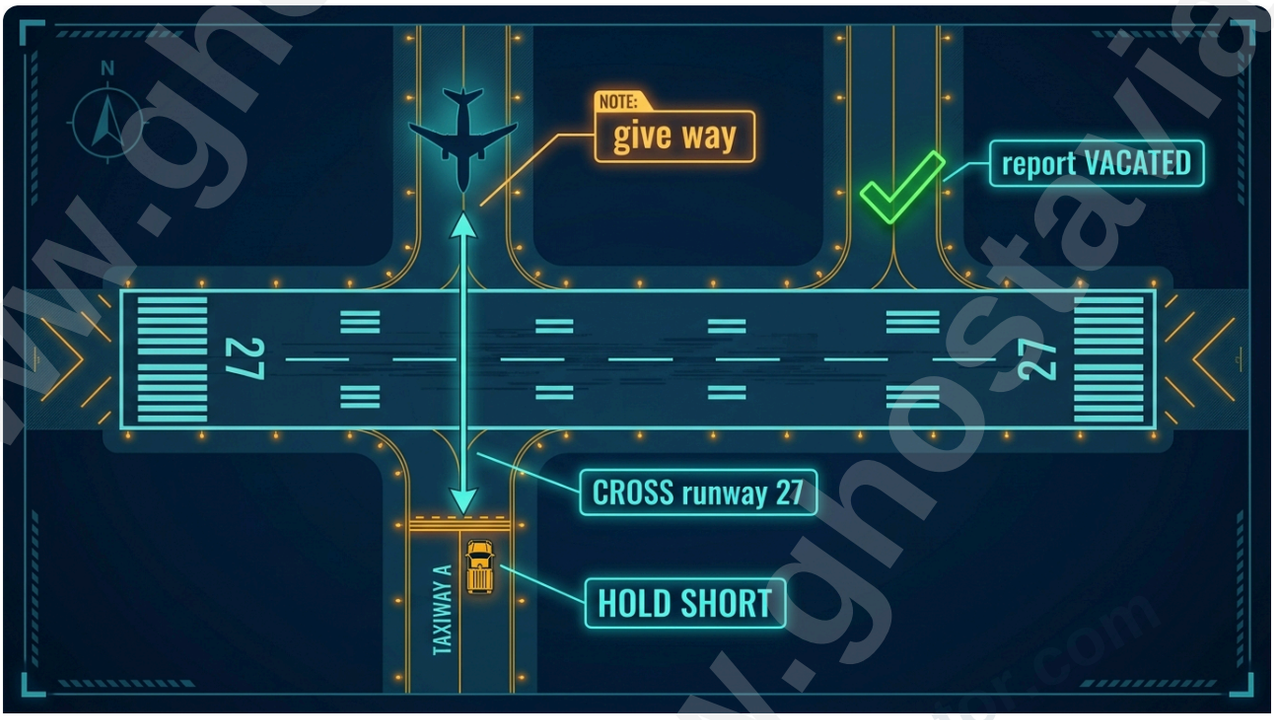

Figure 16.1 — A visual representation of the critical phraseology sequence required for a vehicle to safely cross an active runway.

☆ Numbers to memorise

Essential Facts for Chapter 16

Fact	Value
Who controls vehicles	Aerodrome controller (Tower/Ground) on the manoeuvring area
Vehicle call signs	Function + number (Car 5, Fire 1, Sweeper 2, Follow-Me, Tug 3)
Hold short	Stop at the marking — NOT a clearance to cross
Crossing a runway	Explicit "cross runway XX", read back, then "report vacated"
Towing aircraft	Tug + aircraft = one unit; given taxi/movement instructions
Conditional	"Give way to … then cross" / "after the … cross" — act only when condition met

Question bank

PART A — MCQS (CLICK AN OPTION TO CHECK)

1. Vehicles on the manoeuvring area are controlled by:

> No one
>
> The aerodrome controller (Tower/Ground)
>
> Area control
>
> The airline operations centre

Answer: The aerodrome controller (Tower/Ground). All vehicles on the manoeuvring area come under the aerodrome controller.

2. A vehicle in the wrong place on a runway is called a:

> Go-around
>
> Runway incursion
>
> Conditional clearance
>
> Transfer

Answer: Runway incursion. An unauthorised presence on the runway is a runway incursion — a major hazard.

3. A typical vehicle call sign is:

> "Ghostair 5"
>
> "Car 5" (function + number)
>
> "VT-CAR5"
>
> "Tower 5"

Answer: "Car 5" (function + number). Vehicles use a function name and number, e.g. Car 5, Fire 1.

4. "Hold short of runway 27" instructs the vehicle to:

> Cross the runway
>
> Stop at the marking and not enter the runway
>
> Proceed onto the runway and wait
>
> Vacate the taxiway

Answer: Stop at the marking and not enter the runway. Hold short = stop at the holding point; it is not a clearance to cross.

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5. To cross a runway a vehicle needs:

> No clearance if it is quick
>
> An explicit "cross runway XX" clearance, read back
>
> Only apron approval
>
> A go-around

Answer: An explicit "cross runway XX" clearance, read back. Crossing requires a specific clearance, read back in full.

6. After crossing, the vehicle should:

> Say nothing
>
> Report "runway vacated"
>
> Request take-off
>
> Change frequency

Answer: Report "runway vacated". Reporting "vacated" tells the controller the runway is clear again.

7. A vehicle towing an aircraft is:

> Not controlled by ATC
>
> Treated as a single moving unit and given taxi/movement instructions
>
> Always given priority over aircraft
>
> Handled only by the airline

Answer: Treated as a single moving unit and given taxi/movement instructions. The tug-and-aircraft combination is one unit under ATC movement instructions.

8. "Give way to the landing aircraft, then cross" means the vehicle should:

> Cross immediately
>
> Wait until the landing aircraft has passed, then cross
>
> Hold short permanently
>
> Vacate the area

Answer: Wait until the landing aircraft has passed, then cross. A conditional instruction is acted on only when the stated condition is satisfied.

9. Vehicles confined to the apron are normally handled by:

> Area control
>
> Apron management
>
> The Director
>
> No one

Answer: Apron management. Apron-only vehicles are usually managed by apron control, not the tower.

10. A "Follow-Me" vehicle is used to:

> Tow disabled aircraft only
>
> Guide an aircraft along a taxi route
>
> Inspect the runway surface
>
> Fight fires

Answer: Guide an aircraft along a taxi route. A Follow-Me car leads an aircraft along its taxi route to the stand or runway.

11. The phraseology for vehicles is:

> Casual and informal
>
> Standard and disciplined, like aircraft, with read-backs
>
> Spoken only by the controller
>
> Optional

Answer: Standard and disciplined, like aircraft, with read-backs. Vehicles use standard phraseology and read back instructions just like aircraft.

12. The most safety-critical vehicle movement is:

> Driving on the apron
>
> Crossing an active runway
>
> Parking at a stand
>
> Refuelling

Answer: Crossing an active runway. Crossing an active runway carries the highest incursion risk, hence the strict phraseology.

PART B — ORAL / VIVA (TAP TO REVEAL MODEL ANSWERS)

▶ **Why must vehicles on the manoeuvring area use ATC and standard phraseology?**

▶ **What is the difference between "hold short of runway 27" and "cross runway 27"?**

▶ **How is a vehicle towing an aircraft handled?**

▶ **How does a controller sequence a vehicle around aircraft?**

60-SECOND REVISION CARD

- All vehicles on the **manoeuvring area** are controlled by the **aerodrome controller**, with read-back phraseology.
- Call signs = function + number (**Car 5, Fire 1, Tug 3, Follow-Me**). Apron vehicles → apron management.
- **Hold short** = stop (not cross). Crossing needs explicit **"cross runway XX"**, read back, then **"report vacated"**.
- A **tug + aircraft** is one unit, given taxi/movement instructions.
- Conditional: **"give way … then cross"** — act only when the condition is met. Crossing an active runway is the highest-risk vehicle movement.

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Approach Control

Approach control owns the busy transition between the runway and the en-route world — sequencing arrivals down to final, launching departures up to the airways, and, where there is no ILS, talking an aircraft all the way to the runway by radar. This chapter covers IFR and VFR arrivals and departures, vectors to final, and the surveillance and precision radar approaches your original notes were missing.

SYLLABUS MAP

`Part III (vi) Approach control — IFR/VFR departures & arrivals, vectors to final, surveillance & precision radar approach`

LEARNING OBJECTIVES — BY THE END OF THIS CHAPTER YOU WILL BE ABLE TO…

- Describe approach control's role and where it sits between Tower and Area control.
- Use IFR and VFR departure and arrival phraseology.
- Be vectored to an ILS final and read back an approach clearance.
- Fly a Surveillance Radar Approach (SRA) talkdown.
- Fly a Precision Approach Radar (PAR) talkdown and apply its "do not acknowledge" rule.

17.1 What approach control does
17.2 IFR & VFR departures
17.3 IFR & VFR arrivals
17.4 Vectors to an ILS final
17.5 Surveillance Radar Approach (SRA)
17.6 Precision Approach Radar (PAR)
☆ Numbers to memorise
? Question bank

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Approach control uses surveillance radar to vector and sequence aircraft in the terminal area, bridging the gap between the aerodrome and the en-route airways.

17.1 What approach control does

IN PLAIN TERMS

Approach control ("Approach", "Radar", "Director") handles aircraft in the **terminal area** — climbing departures handed up from Tower until they reach the airways, and descending arrivals handed down from Area control until they are established on final and passed to Tower. It usually has radar (a surveillance display), which is why it can **vector** aircraft and provide radar approaches.

17.2 IFR & VFR departures

Transcript — IFR departure handed to approach

A/C Delhi Approach, Ghostair Alfa Bravo Charlie, passing two thousand, PAPA two departure.

ATC Ghostair Alfa Bravo Charlie, identified, climb flight level one zero zero, when ready direct PAPA.

A/C Climb flight level one zero zero, when ready direct PAPA, Ghostair Alfa Bravo Charlie.

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Transcript — VFR departure

A/C Delhi Approach, VT-ABC, leaving the zone to the north, VFR, two thousand feet.

ATC VT-ABC, roger, report leaving the zone, traffic is a helicopter two miles north.

17.3 IFR & VFR arrivals

Transcript — IFR arrival

A/C Delhi Approach, Ghostair Alfa Bravo Charlie, flight level one zero zero, inbound PAPA, information Delta.

ATC Ghostair Alfa Bravo Charlie, descend flight level six zero, expect ILS approach runway two seven.

A/C Descend flight level six zero, expect ILS runway two seven, Ghostair Alfa Bravo Charlie.

"Expect" vs a clearance

"Expect ILS approach runway two seven" is advance information so you can prepare — it is **not** the approach clearance. The clearance comes later: "cleared ILS approach runway two seven". VFR arrivals are given **join instructions** as in Chapter 15.

17.4 Vectors to an ILS final

WHY VECTORS

Radar lets the controller **vector** (assign headings to) arrivals so they intercept the ILS localizer at a sensible angle and spacing. The sequence is: turn onto a base heading, descend to the platform altitude, then a closing heading to intercept, then the approach clearance.

Transcript — vectors to ILS

ATC Ghostair Alfa Bravo Charlie, turn left heading two one zero, descend two thousand five hundred feet, QNH one zero one three.

A/C Left heading two one zero, descend two thousand five hundred feet, QNH one zero one three, Ghostair Alfa Bravo Charlie.

ATC Ghostair Alfa Bravo Charlie, closing the localizer from the left, report established. Turn left heading two five zero, cleared ILS approach runway two seven.

A/C Left heading two five zero, cleared ILS approach runway two seven, wilco, Ghostair Alfa Bravo Charlie.

A/C Ghostair Alfa Bravo Charlie, established ILS runway two seven.

ATC Ghostair Alfa Bravo Charlie, contact Tower one one eight decimal one.

17.5 Surveillance Radar Approach (SRA)

What an SRA is

A **Surveillance Radar Approach** is a non-precision approach in which the controller uses the surveillance radar to give the pilot **azimuth (heading) guidance** and a series of **ranges with advised heights**, but **no glidepath** — the pilot maintains the correct descent by comparing the radar range with the advised height. It usually **terminates at a fixed range** (commonly 2 NM or ½ NM) from touchdown, from where the pilot continues visually.

Transcript — SRA talkdown

ATC VT-ABC, this will be a surveillance radar approach runway two seven, terminating at two miles. Check your minima. Turn left heading two seven zero, closing from the right.

ATC VT-ABC, range six miles, height should be one thousand nine hundred feet.

ATC VT-ABC, range four miles, height should be one thousand three hundred feet. On track.

ATC VT-ABC, range two miles, height should be six hundred and fifty feet, approach completed, contact Tower.

Exam trap

On an SRA the pilot is **not** normally asked to acknowledge each range/height call — the controller transmits a steady stream. The pilot flies the heading and checks the advised heights against the ranges. There is **no electronic glidepath**; the height/range advisory *is* the descent guide.

17.6 Precision Approach Radar (PAR)

What a PAR is

A **Precision Approach Radar** gives the pilot **continuous azimuth AND elevation guidance** — the controller watches the aircraft against both the centreline and a precise glidepath, and "talks it down" with a constant patter of corrections almost to touchdown. It is a true precision approach (used mainly by military and some civil fields).

The "do not acknowledge" rule

Because the talkdown is continuous and time-critical, the controller instructs the pilot **"do not acknowledge further transmissions"**. The pilot then simply flies the corrections and does not reply, so the channel stays clear for the steady stream of guidance.

Transcript — PAR talkdown

ATC VT-ABC, this will be a precision approach runway two seven. Do not acknowledge further transmissions. Turn left heading two seven two.

ATC VT-ABC, approaching the glidepath … on the glidepath … on centreline.

ATC Slightly above the glidepath, coming down nicely … on the glidepath. On centreline.

ATC Going slightly left of centreline, turn right two degrees … on centreline. On the glidepath. Approaching decision height.

Mnemonic — SRA vs PAR

SRA = Steered + Range/height (no glidepath); **PAR = Precise (centreline + glidepath), continuous, don't acknowledge.** PAR gives *both* dimensions; SRA gives heading plus a height-check table.

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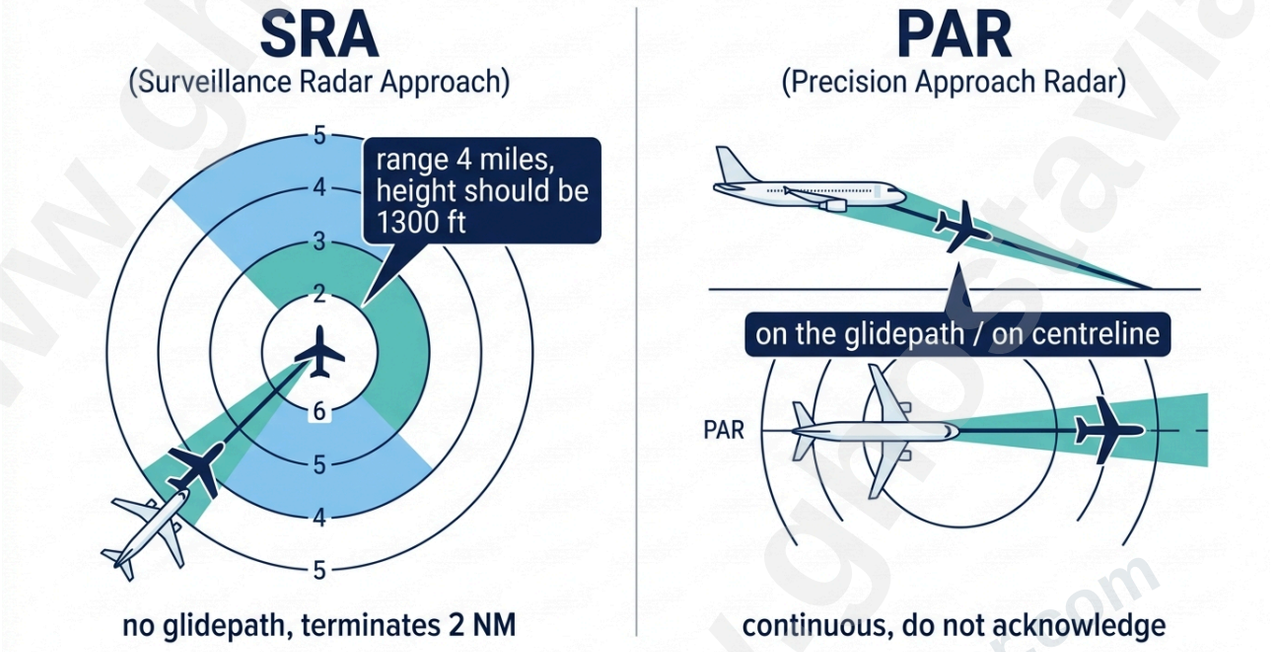

Figure 17.1 — SRA vs PAR: what the controller can see and say. SRA provides headings and step-down altitudes, while PAR provides a continuous glidepath talkdown.

☆ Numbers to memorise

Essential Facts for Chapter 17

Fact	Value
Approach control	Terminal area between Tower and Area control; usually radar-equipped
"Expect" approach	Advance info, NOT the approach clearance
Vectors to ILS	Base heading → descend platform → closing heading → "cleared ILS approach"
SRA	Heading + range/height advisories, NO glidepath; terminates ~2 NM (or ½ NM)
PAR	Continuous azimuth + elevation talkdown; "do not acknowledge further transmissions"

Question bank

PART A — MCQS (CLICK AN OPTION TO CHECK)

1. Approach control handles aircraft:

> Only on the runway
>
> In the terminal area between Tower and Area control
>
> Only over the ocean
>
> On the apron

```
Answer: In the terminal area between Tower and Area control. Approach sequences arrivals down to final and departures up to the airways.
```

2. "Expect ILS approach runway 27" is:

> The approach clearance
>
> Advance information, not a clearance
>
> A landing clearance
>
> A vector

```
Answer: Advance information, not a clearance. "Expect" prepares you; the clearance "cleared ILS approach" comes later.
```

3. To vector an aircraft, the controller assigns:

> A squawk
>
> Headings (and often levels)
>
> A frequency
>
> A call sign

```
Answer: Headings (and often levels). Vectoring is the assignment of headings using radar.
```

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4. A Surveillance Radar Approach provides:

> Azimuth and glidepath guidance

> Heading guidance plus range/height advisories, no glidepath

> No guidance at all

> Only a squawk

Answer: Heading guidance plus range/height advisories, no glidepath. SRA gives heading and a series of ranges with advised heights; there is no glidepath.

5. An SRA normally terminates at:

> The threshold

> A fixed range, commonly 2 NM (or ½ NM)

> 10 NM

> It never terminates

Answer: A fixed range, commonly 2 NM (or ½ NM). SRAs terminate at a set range, from where the pilot continues visually.

6. On an SRA, the pilot maintains the correct descent by:

> Following an electronic glidepath

> Comparing the radar range with the advised height

> Using GPS only

> Listening to the marker beacons

Answer: Comparing the radar range with the advised height. The advised height for each range is the descent guide; there is no glidepath signal.

7. A Precision Approach Radar provides:

> Heading only

> Continuous azimuth and elevation (centreline + glidepath) guidance

> Range only

> No talkdown

Answer: Continuous azimuth and elevation (centreline + glidepath) guidance. PAR gives both dimensions continuously — a true precision talkdown.

8. On a PAR, the controller instructs the pilot to:

> Acknowledge every transmission
>
> Not acknowledge further transmissions
>
> Squawk ident
>
> Change frequency each mile

Answer: Not acknowledge further transmissions. "Do not acknowledge further transmissions" keeps the channel clear for continuous guidance.

9. "Closing the localizer from the left, cleared ILS approach runway 27" tells the pilot to:

> Continue past the localizer
>
> Intercept the localizer and fly the ILS approach
>
> Go around
>
> Hold

Answer: Intercept the localizer and fly the ILS approach. It is the approach clearance once positioned to intercept the localizer.

10. A VFR arrival is given:

> A radar talkdown
>
> Join instructions (as for the circuit)
>
> An oceanic clearance
>
> A SELCAL check

Answer: Join instructions (as for the circuit). VFR arrivals receive join instructions, then are sequenced visually.

11. The platform/intercept altitude on vectors is given as a:

> Flight level only
>
> Descent to an altitude with QNH, read back
>
> Squawk
>
> Heading

Answer: Descent to an altitude with QNH, read back. Below the transition level the controller assigns an altitude on QNH, which is read back.

12. After establishing on the ILS, the aircraft is usually:

> Told to hold
>
> Transferred to Tower
>
> Told to squawk standby
>
> Given an oceanic clearance

Answer: Transferred to Tower. Approach hands the established aircraft to Tower for landing.

13. The key difference between SRA and PAR is that PAR also provides:

> A squawk
>
> Glidepath (elevation) guidance
>
> A frequency change
>
> Wind information

Answer: Glidepath (elevation) guidance. PAR adds continuous elevation/glidepath guidance that the SRA lacks.

14. On a PAR talkdown, "on the glidepath, on centreline" means the aircraft is:

> Too high and left
>
> Correctly positioned in both the vertical and the lateral plane
>
> Below the glidepath
>
> Off the radar

Answer: Correctly positioned in both the vertical and the lateral plane. It confirms correct position on both the glidepath and the runway centreline.

15. An IFR departure handed to Approach typically receives:

> A landing clearance
>
> A climb clearance and identification
>
> A SELCAL check
>
> Holding instructions only

Answer: A climb clearance and identification. Approach identifies the departure and issues climb instructions toward the airways.

16. The reason a controller can vector and provide radar approaches is:

> It has no radar

> Approach is usually radar-equipped (a surveillance display)

> It uses HF only

> It is on the apron

```
Answer: Approach is usually radar-equipped (a surveillance display). Radar at approach control enables vectoring and SRA/PAR.
```

PART B — ORAL / VIVA (TAP TO REVEAL MODEL ANSWERS)

▶ **What is approach control's role?**

▶ **What is the difference between "expect" an approach and being "cleared" for it?**

▶ **Describe a Surveillance Radar Approach.**

▶ **Describe a Precision Approach Radar and its key procedural rule.**

▶ **What is the difference between SRA and PAR?**

60-SECOND REVISION CARD

- **Approach** = terminal area, between Tower and Area; radar-equipped → vectoring & radar approaches.
- Vectors to ILS: base heading → descend to platform (QNH) → closing heading → **cleared ILS approach** → Tower.
- **"Expect"** ≠ clearance. VFR arrivals get join instructions.
- **SRA:** heading + range/height, **no glidepath**, terminates ~2 NM.
- **PAR:** continuous centreline + glidepath, **"do not acknowledge further transmissions"**.

Area Control & Oceanic

Once clear of the terminal area, an aircraft cruises through the airways under area control — reporting its position, holding when traffic demands, and joining, leaving or crossing routes on clearance. Out over the ocean, where there is no radar and no VHF, the same job is done by HF position reports, datalink and the procedural rules this chapter explains.

SYLLABUS MAP

Part III (vii) Area control – units, position/level info, joining/leaving/crossing airways, holding, ATS surveillance, ADS, oceanic control

LEARNING OBJECTIVES — BY THE END OF THIS CHAPTER YOU WILL BE ABLE TO…

- Describe area control units and the structure of a position report.
- Use level-information phraseology (maintain, climb/descend, cruise).
- Use joining, leaving and crossing-airway phraseology.
- Enter and fly an en-route hold, and understand the EAT/onward-clearance system.
- Explain ATS surveillance, ADS-C/ADS-B and CPDLC in area control.
- Describe oceanic control — the oceanic clearance, Mach-number technique and SELCAL.

18.1 Area control units
18.2 Position reporting
18.3 Level information
18.4 Joining, leaving & crossing airways
18.5 En-route holding
18.6 ATS surveillance & ADS
18.7 Oceanic control
☆ Numbers to memorise
? Question bank

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En-route oceanic control relies on procedural separation, position reporting, and datalink technologies since continuous VHF radar coverage is not available.

18.1 Area control units

IN PLAIN TERMS

The **Area Control Centre (ACC)**, with the call-sign suffix **"Control"**, provides air traffic control to aircraft **en route** in the airways and upper airspace of an FIR/UIR. A large FIR is split into **sectors**, and as you fly across it you are transferred from sector to sector.

18.2 Position reporting

The standard position report

Where there is no radar, the aircraft reports its progress at compulsory reporting points. The order is: **(1) position, (2) time, (3) level, (4) next point, (5) estimate for the next point** — and the name of the point after that. ("ABC, PAPA at four two, flight level three five zero, ROMEO five five, next SIERRA.")

Transcript — position report

A/C Chennai Control, Ghostair Alfa Bravo Charlie, PAPA at four two, flight level three five zero, estimating ROMEO five five, SIERRA next.

ATC Ghostair Alfa Bravo Charlie, Chennai Control, roger, report ROMEO.

Mnemonic — position report order

"Place, Time, Level, Next, Estimate (+ the one after)." Same order every time so the controller can write it straight onto the flight progress strip.

18.3 Level information

Phrase	Meaning
"Maintain FL350"	Stay at that level
"Climb / descend to FL…"	Change to the new level
"Climb to reach FL… by [point]"	Be at the level by the point
"Cruise climb"	Climb gradually as weight reduces (oceanic)
"Report leaving / reaching FL…"	Tell ATC at the stated event
"Expedite climb/descent"	Climb/descend at best rate

Exam trap

All level instructions are **mandatory read-backs**. "Climb FL350" and "maintain FL350" are different instructions — read back exactly what was said, and do not climb or descend on a "report" or "expect" alone.

18.4 Joining, leaving & crossing airways

Three airway interactions

Joining: an aircraft entering a controlled airway needs a clearance — "cleared to join airway [W] at [point], flight level …". **Leaving:** on leaving controlled airspace you are told "leaving controlled airspace, report …". **Crossing:** an aircraft crossing an airway is given a clearance to cross at a level/time keeping it separated from airway traffic.

Transcript — joining an airway

A/C Mumbai Control, VT-ABC, request join airway WHISKEY one zero at SIERRA, flight level one five zero.

ATC VT-ABC, cleared to join airway WHISKEY one zero at SIERRA, flight level one five zero, report SIERRA.

A/C Cleared to join WHISKEY one zero at SIERRA, flight level one five zero, will report SIERRA, VT-ABC.

18.5 En-route holding

WHY HOLD

When traffic ahead must be sequenced — for example a busy arrival airport — ATC may **hold** an aircraft in a racetrack pattern over a fix or beacon until it can be cleared onward.

Holding phraseology

"Hold at [fix], flight level …, expect onward clearance at [time]" or **"expect approach time [time] (EAT)"**. The clearance specifies the fix, the level, and the time you may expect to be released (the **Onward Clearance Time** or, for an arrival, the **Expected Approach Time, EAT**). The aircraft reads it back and flies the published hold.

Transcript — holding

ATC Ghostair Alfa Bravo Charlie, hold at PAPA, flight level one zero zero, expect approach time three five.

A/C Hold at PAPA, flight level one zero zero, expect approach time three five, Ghostair Alfa Bravo Charlie.

18.6 ATS surveillance & ADS

ATS surveillance — radar & ADS-B

In well-served continental airspace, area control uses **ATS surveillance** — primary/secondary radar and increasingly **ADS-B** (Chapter 12) — to see aircraft directly, reducing the need for position reports and allowing radar separation and vectoring.

ADS-C & CPDLC for remote/oceanic airspace

Where there is no radar, **ADS-C (Automatic Dependent Surveillance — Contract)** has the aircraft automatically send position reports over datalink at agreed intervals/events, and **CPDLC (Controller–Pilot Data Link Communications)** carries clearances and messages as text. Together they let an oceanic controller "see" and instruct aircraft without voice or radar.

18.7 Oceanic control

FIRST PRINCIPLES — CONTROL WITHOUT RADAR OR VHF

Over the ocean there is no VHF (line-of-sight) and traditionally no radar. So separation is **procedural**: aircraft are pre-cleared onto a track at a level and a **Mach number**, report their position by **HF voice or CPDLC**, and the controller keeps them apart by time and the Mach-number technique. This is why your original notes' "oceanic" was only a propagation footnote — the *control* procedure is here.

The oceanic clearance & Mach-number technique

Before entering oceanic airspace the aircraft obtains an **oceanic clearance**: the track, the flight level, and the **assigned Mach number**. Aircraft on the same track at the same level fly the same Mach number so the time-spacing between them stays constant — the **Mach-number technique**. **SELCAL** (Chapter 22) lets the crew stop monitoring HF static until called.

Transcript — oceanic clearance (often via HF/CPDLC)

A/C Oceanic Control, Ghostair Alfa Bravo Charlie, request oceanic clearance, estimating SIERRA at five five, requesting flight level three five zero, Mach decimal eight zero.

ATC Ghostair Alfa Bravo Charlie, cleared to destination via track SIERRA, flight level three five zero, Mach decimal eight zero, report SIERRA.

A/C Cleared via track SIERRA, flight level three five zero, Mach decimal eight zero, will report SIERRA, Ghostair Alfa Bravo Charlie.

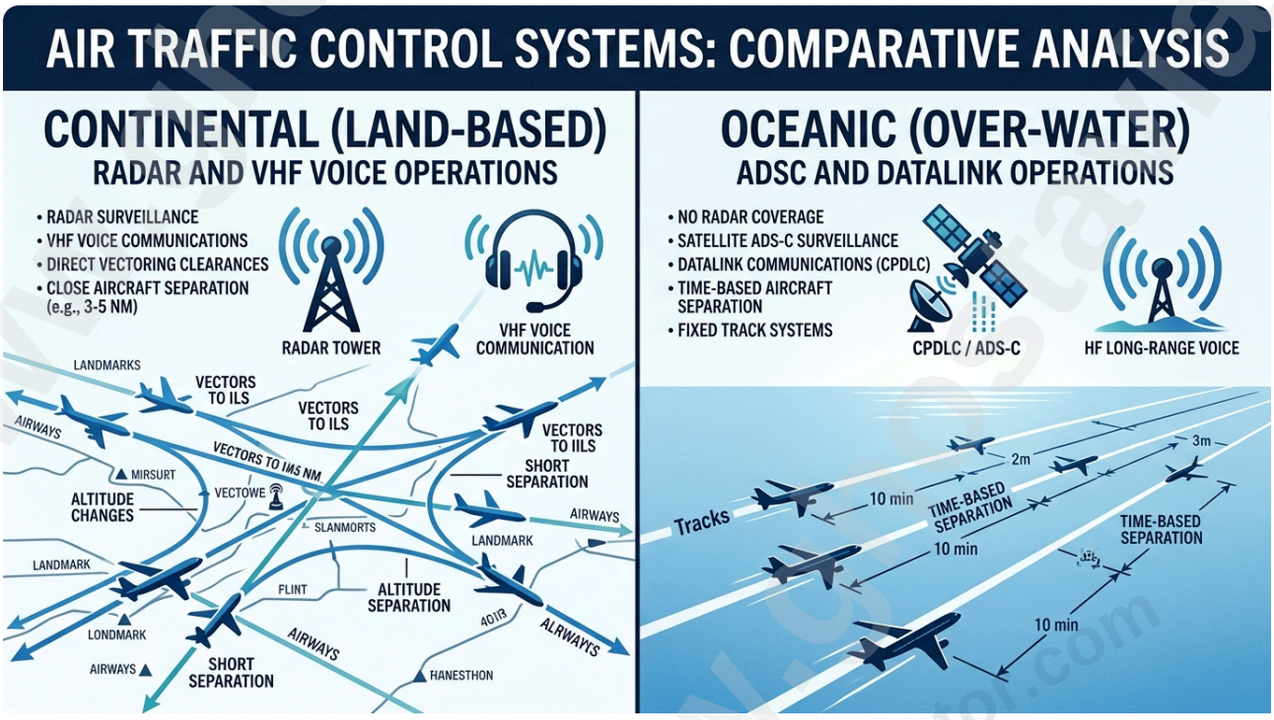

Figure 18.1 — Continental vs oceanic control: Continental airspace relies on VHF and radar vectoring. Oceanic relies on procedural tracks, CPDLC datalinks, and Mach-number spacing.

☆ Numbers to memorise

Essential Facts for Chapter 18

Fact	Value
Area Control Centre	Suffix "Control"; en-route control of the airways/upper airspace, by sector
Position report order	Position · time · level · next point · estimate (+ the one after)
Level instructions	Maintain / climb / descend — all mandatory read-backs
Airways	Join (clearance) · leave (report) · cross (level/time)
Holding	"Hold at [fix], FL…, expect approach time / onward clearance [time]"
ADS-C / CPDLC	Automatic datalink position reports / text clearances (oceanic-remote)
Oceanic	Procedural; oceanic clearance = track + level + Mach; HF/CPDLC; SELCAL

Question bank

PART A — MCQS (CLICK AN OPTION TO CHECK)

1. The call-sign suffix for an Area Control Centre is:

> Approach
>
> Tower
>
> Control
>
> Information

```
Answer: Control. "Control" denotes the Area Control Centre (en-route).
```

2. The correct order of a position report is:

> Level, position, time, next
>
> Position, time, level, next point, estimate
>
> Time, next, level, position
>
> Estimate, position, level

```
Answer: Position, time, level, next point, estimate. Position, time, level, next point and estimate (plus the point after).
```

3. "Maintain flight level 350" must be:

> Acknowledged "roger"
>
> Read back
>
> Not read back
>
> Read back only at night

```
Answer: Read back. Level instructions are mandatory read-back items.
```

4. To enter a controlled airway, an aircraft needs:

> No clearance
>
> A clearance to join
>
> Only a squawk
>
> A landing clearance

```
Answer: A clearance to join. Joining controlled airspace requires a clearance to join.
```

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5. An aircraft held over a fix is given:

> A take-off clearance
>
> A fix, level and an expected approach / onward clearance time
>
> A SELCAL code
>
> A runway

Answer: A fix, level and an expected approach / onward clearance time. Holding clearance gives the fix, level and EAT/onward-clearance time.

6. "Expect approach time three five" tells the pilot:

> To begin the approach now
>
> The time to expect release from the hold for approach
>
> To squawk 3500
>
> To climb to FL350

Answer: The time to expect release from the hold for approach. The EAT is the time the aircraft can expect to leave the hold for its approach.

7. ADS-C provides:

> Voice communication
>
> Automatic datalink position reports at agreed intervals/events
>
> A glidepath
>
> Engine data to ATC

Answer: Automatic datalink position reports at agreed intervals/events. ADS-C (Contract) sends automatic position reports over datalink.

8. CPDLC carries:

> Weather radar images
>
> Controller–pilot clearances and messages as text
>
> A glidepath signal
>
> The phonetic alphabet

Answer: Controller–pilot clearances and messages as text. CPDLC is a text datalink for ATC clearances and messages.

9. Over the ocean, separation is primarily:

> By primary radar
>
> Procedural (time/Mach), with HF/CPDLC reporting
>
> By VHF vectoring
>
> Not provided

Answer: Procedural (time/Mach), with HF/CPDLC reporting. No radar/VHF — separation is procedural, using position reports and the Mach-number technique.

10. An oceanic clearance specifies:

> Runway and squawk
>
> Track, flight level and assigned Mach number
>
> A heading only
>
> A holding fix

Answer: Track, flight level and assigned Mach number. The oceanic clearance gives the track, level and Mach number.

11. The Mach-number technique keeps aircraft separated by:

> Vectoring
>
> Maintaining a constant time spacing through assigned Mach numbers
>
> Squawk codes
>
> Radio checks

Answer: Maintaining a constant time spacing through assigned Mach numbers. Same Mach on the same track keeps the time interval between aircraft constant.

12. SELCAL allows an oceanic crew to:

> Vector themselves
>
> Stop continuously monitoring HF until selectively called
>
> Land without a clearance
>
> Change their Mach number

Answer: Stop continuously monitoring HF until selectively called. SELCAL alerts a specific aircraft, so the crew need not strain at HF static.

13. A large FIR is divided for control into:

> Runways
>
> Sectors
>
> Aprons
>
> Holding fixes only

Answer: Sectors. An ACC works the FIR in sectors, handing aircraft on between them.

14. "Report leaving flight level 350" requires the pilot to:

> Read back nothing
>
> Tell ATC when leaving FL350
>
> Maintain FL350
>
> Squawk ident

Answer: Tell ATC when leaving FL350. A "report leaving/reaching" instruction asks for a call at that event.

15. ATS surveillance in continental area control commonly uses:

> Only HF
>
> Radar and ADS-B
>
> SELCAL
>
> Light signals

Answer: Radar and ADS-B. Continental area control uses radar and increasingly ADS-B.

16. An aircraft crossing an airway is cleared to cross at:

> Any level it likes
>
> A level/time that keeps it separated from airway traffic
>
> Ground level
>
> The transition altitude only

Answer: A level/time that keeps it separated from airway traffic. A crossing clearance specifies a level/time providing separation.

PART B — ORAL / VIVA (TAP TO REVEAL MODEL ANSWERS)

▶ **What does an Area Control Centre do, and what is its suffix?**

▶ **State the order of a standard position report.**

▶ **How is an aircraft cleared to join, leave or cross an airway?**

▶ **What is contained in a holding clearance?**

▶ **How is air traffic controlled over the ocean without radar or VHF?**

▶ **What are ADS-C and CPDLC?**

60-SECOND REVISION CARD

- **Area Control** = "Control"; en-route, worked in sectors.
- Position report: **position · time · level · next · estimate**. Level instructions = mandatory read-back.
- Airways: **join** (clearance) · **leave** (report) · **cross** (level/time). Hold: fix + FL + **EAT/onward time**.
- Continental surveillance = **radar/ADS-B**; remote/oceanic = **ADS-C + CPDLC**.
- Oceanic = procedural: **oceanic clearance (track, level, Mach)**, HF/CPDLC, **Mach-number technique**, SELCAL.

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ATS Surveillance Phraseology

When a controller can see you on a surveillance display — radar or ADS-B — a whole vocabulary opens up: identifying you, turning you for spacing, warning you of conflicting traffic, ordering an avoiding turn, and managing your transponder. This chapter gathers the surveillance phraseology that appears across approach and area control.

SYLLABUS MAP

Part III (iv) General ATS surveillance — identification & vectoring · traffic info & avoiding action · SSR · radar assistance to comms-failure aircraft · alerting

LEARNING OBJECTIVES — BY THE END OF THIS CHAPTER YOU WILL BE ABLE TO…

- Use identification and vectoring phraseology and respond correctly.
- Receive traffic information and act on avoiding-action instructions.
- Use the full set of SSR (transponder) phraseology.
- Understand how radar assists an aircraft that has a communications failure.
- Recognise alerting and service-termination phraseology.

19.1 Identification
19.2 Vectoring
19.3 Traffic information & avoiding action
19.4 SSR / transponder phraseology
19.5 Radar assistance to comms failure
19.6 Alerting & termination of service
☆ Numbers to memorise
? Question bank

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When an aircraft is positively identified on a surveillance display, the controller can issue vectors, provide radar separation, and call out conflicting traffic using the clock-code system.

19.1 Identification

> **FIRST PRINCIPLES — THE CONTROLLER MUST BE SURE WHICH BLIP IS YOU**
>
> Before a controller can vector you or give you radar separation, they must **positively identify** your return on the display. They confirm it with "**identified**" (often with a position), or tell you "**not identified**". Common methods are a **squawk ident**, observing a turn, or correlating your reported position.

Phrase	Meaning
"Squawk [code]"	Set the transponder code
"Squawk ident"	Press IDENT to flash the return for identification
"Identified [position]"	You are identified on radar
"Not identified, resume own navigation"	You are not identified; navigate yourself
"For identification, turn left heading …"	Turn so the controller can identify you

19.2 Vectoring

Headings for a purpose

A **vector** is a heading the controller assigns — for sequencing, spacing, separation or to position for an approach. It usually states the **reason**: "turn left heading two seven zero **for spacing**". When the vectoring is finished, the controller says "**resume own navigation**" (with a position/route).

Transcript — identification & vector

ATC Ghostair Alfa Bravo Charlie, squawk four three two one and ident.

A/C Squawk four three two one, identing, Ghostair Alfa Bravo Charlie.

ATC Ghostair Alfa Bravo Charlie, identified five miles north of the field, turn right heading one two zero for sequencing.

A/C Right heading one two zero, Ghostair Alfa Bravo Charlie.

19.3 Traffic information & avoiding action

Traffic information

The controller describes conflicting traffic by its **clock-bearing, distance, direction of movement and (if known) level/type**: "traffic, eleven o'clock, five miles, opposite direction, indicating flight level three five zero". You reply "looking out" and, if you see it, "traffic in sight".

Avoiding action — act immediately

"**Avoiding action**" is an immediate instruction to turn (and/or climb/descend) to avoid a confliction — for example "**avoiding action, turn left immediately heading one eight zero**, traffic …". It is obeyed at once and read back; the reason follows the instruction so no time is lost.

Transcript — traffic & avoiding action

ATC VT-ABC, traffic, two o'clock, four miles, crossing left to right, type unknown.

A/C VT-ABC, looking out … traffic in sight.

ATC VT-ABC, avoiding action, turn left immediately heading three six zero, traffic now one o'clock, two miles.

A/C Avoiding action, left heading three six zero, VT-ABC.

Figure 19.1 — Reading traffic information: The clock code provides a relative bearing to the conflicting traffic based on the aircraft's nose being 12 o'clock.

19.4 SSR / transponder phraseology

Phrase	Meaning
"Squawk [code]"	Set the four-digit code
"Squawk ident"	Operate the IDENT feature
"Squawk Charlie" / "squawk Mode C"	Select altitude reporting (Mode C)
"Confirm squawk [code]"	Confirm the code set
"Reset squawk [code]"	Re-select the code (e.g. after a fault)
"Squawk standby"	Select standby (no replies)
"Stop squawk Charlie, wrong indication"	Switch off altitude reporting (faulty)
"Check altimeter setting and confirm level"	Mode C disagrees — verify QNH and level

> **Special codes (from Chapter 12)**
>
> **7500** hijack · **7600** radio failure · **7700** emergency · **7000** conspicuity (VFR) · **2000** entering from a non-SSR area. These are set without being told, in the relevant situation.

19.5 Radar assistance to a comms-failure aircraft

HELPING AN AIRCRAFT THAT CANNOT REPLY

If an aircraft has a **receiver-only** or partial communications failure but its transponder still works, the controller can still **see and assist** it. The controller transmits instructions **blind** and asks the pilot to acknowledge by transponder: "**reply not received, if you read [station], squawk ident**" or "**acknowledge by [a stated transponder action / a turn]**".

Cockpit reality

Even with no voice, a working transponder turns your aircraft from a guess into a tracked, instructable target — which is why you keep the transponder ON and squawk 7600 the moment a radio fails (full procedure in Chapter 20).

Transcript — radar assistance, receiver still working

ATC VT-ABC, if you read Delhi Approach, squawk ident.

ATC VT-ABC, ident observed, I will provide vectors. Turn left heading two seven zero, acknowledge with ident.

19.6 Alerting & termination of service

Position relative to known features

The controller may give your position relative to a feature for situational awareness or for an aircraft in difficulty: "position five miles east of [VOR]". When radar service ends — leaving cover, or at transfer — the controller says "**identification terminated [instructions]**" or "**radar service terminated, resume own navigation**".

Mnemonic — the surveillance flow

Identify → Vector → Inform/Avoid → (manage SSR) → Terminate. Every surveillance interaction runs through this loop.

☆ Numbers to memorise

Essential Facts for Chapter 19

Fact	Value
Identification	"Identified" / "not identified"; squawk ident, turn, or position correlation
Vector	Heading + reason ("for spacing/sequencing"); ends "resume own navigation"
Traffic info	Clock bearing · distance · direction · level/type
Avoiding action	Immediate turn/climb; instruction first, reason after; read back
SSR phrases	Squawk · ident · Charlie (Mode C) · confirm · reset · standby
Comms-failure assistance	Transmit blind; "if you read, squawk ident"; acknowledge by transponder
Termination	"Identification/radar service terminated, resume own navigation"

Question bank

PART A — MCQS (CLICK AN OPTION TO CHECK)

1. "Squawk ident" instructs the pilot to:

> Change frequency
>
> Operate the transponder IDENT feature to flash the return
>
> Set 7700
>
> Turn left

```
Answer: Operate the transponder IDENT feature to flash the return. IDENT momentarily highlights the return so the controller can identify it.
```

2. "Identified five miles north" means:

> You must descend
>
> The controller has positively identified your return
>
> You are not on radar
>
> You must squawk standby

```
Answer: The controller has positively identified your return. It confirms positive radar identification, often with a position.
```

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3. A vector is:

> A squawk code
>
> A heading assigned by the controller, usually with a reason
>
> A frequency
>
> A clearance to land

Answer: A heading assigned by the controller, usually with a reason. Vectoring is the assignment of headings for spacing, separation or positioning.

4. When vectoring ends, the controller says:

> "Squawk standby"
>
> "Resume own navigation"
>
> "Cleared to land"
>
> "Go around"

Answer: "Resume own navigation". "Resume own navigation" returns navigation responsibility to the pilot.

5. Traffic information is given as:

> A frequency and a level
>
> Clock bearing, distance, direction of movement, and level/type if known
>
> A squawk code
>
> A QNH

Answer: Clock bearing, distance, direction of movement, and level/type if known. e.g. "traffic, two o'clock, four miles, crossing left to right".

6. "Avoiding action, turn left immediately heading 180" must be:

> Considered and queried first
>
> Acted on at once and read back
>
> Ignored if VFR
>
> Acknowledged with "roger"

Answer: Acted on at once and read back. Avoiding action is an immediate manoeuvre, executed at once and read back.

7. "Squawk Charlie" instructs the pilot to:

> Set code 7700
>
> Select Mode C (altitude reporting)
>
> Squawk standby
>
> Ident

Answer: Select Mode C (altitude reporting). "Charlie"/Mode C turns on pressure-altitude reporting.

8. "Squawk standby" means:

> Reply normally
>
> Select standby — the transponder does not reply
>
> Set 2000
>
> Ident

Answer: Select standby — the transponder does not reply. Standby stops the transponder replying (e.g. on the ground).

9. If a controller suspects the Mode C readout is wrong, they may say:

> "Squawk ident"
>
> "Check altimeter setting and confirm level" / "stop squawk Charlie, wrong indication"
>
> "Resume own navigation"
>
> "Cleared to land"

Answer: "Check altimeter setting and confirm level" / "stop squawk Charlie, wrong indication". A Mode C disagreement prompts a level/altimeter check or stopping Mode C.

10. A controller can still assist an aircraft with a receiver-only failure if it has:

> No transponder
>
> A working transponder (so it can be tracked and instructed blind)
>
> A working ELT only
>
> Only HF

Answer: A working transponder (so it can be tracked and instructed blind). A working transponder lets the controller track it and pass blind instructions acknowledged by squawk.

11. To get a no-voice aircraft to confirm it can hear, the controller says:

> "Confirm visual"
>
> "If you read [station], squawk ident"
>
> "Squawk standby"
>
> "Resume own navigation"

Answer: "If you read [station], squawk ident". An ident in response confirms the pilot is receiving.

12. "Resume own navigation" returns to the pilot:

> The landing clearance
>
> Responsibility for navigating the aircraft
>
> The squawk code
>
> The frequency

Answer: Responsibility for navigating the aircraft. It ends ATC heading-vectoring and the pilot navigates again.

13. The transponder code for general emergency is:

> 7500
>
> 7600
>
> 7700
>
> 7000

Answer: 7700. 7700 emergency, 7600 radio failure, 7500 hijack.

14. "Traffic in sight" tells the controller:

> You will not avoid it
>
> You have visually acquired the reported traffic
>
> You have a radio failure
>
> You are identified

Answer: You have visually acquired the reported traffic. It confirms you can see the traffic the controller described.

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15. Radar service termination is phrased as:

> "Squawk ident"
>
> "Radar service terminated, resume own navigation"
>
> "Cleared to land"
>
> "Avoiding action"

```
Answer: "Radar service terminated, resume own navigation". When leaving cover or at transfer, the radar/identification service is terminated.
```

16. The general surveillance interaction sequence is:

> Land → taxi → park
>
> Identify → vector → inform/avoid → manage SSR → terminate
>
> Climb → cruise → descend
>
> Squawk → ident → land

```
Answer: Identify → vector → inform/avoid → manage SSR → terminate. Identify, vector, give traffic info/avoiding action, manage the transponder, then terminate.
```

PART B — ORAL / VIVA (TAP TO REVEAL MODEL ANSWERS)

▶ **How does a controller identify an aircraft on radar?**

▶ **What is a vector, and how is the service ended?**

▶ **How is traffic information given, and what is avoiding action?**

▶ **Give five SSR phraseology items and their meanings.**

▶ **How can a controller assist an aircraft with a communications failure?**

60-SECOND REVISION CARD

- **Identify** (squawk ident / turn / position) → "identified" or "not identified".
- **Vector** = heading + reason; ends "**resume own navigation**".
- Traffic = **clock · distance · direction · level/type**. **Avoiding action** = immediate, read back.
- SSR: squawk · ident · **Charlie (Mode C)** · confirm · reset · standby. Codes **7500/7600/7700/7000/2000**.

- Comms failure: "**if you read, squawk ident**", vectors acknowledged by transponder. Service ends "terminated, resume own navigation".

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Distress, Urgency & Communications Failure

This is the chapter that matters most on the worst day of a pilot's career. When something goes badly wrong, phraseology becomes a lifeline: the right word, spoken the right number of times, on the right frequency, summons the whole machinery of air traffic control and search-and-rescue to your aid. Here you learn the exact difference between MAYDAY and PAN, how to format a distress message, the silence procedures, the emergency descent, and what to do when the radio itself fails.

SYLLABUS MAP

Part III (viii) Distress & urgency · imposition/termination of silence · urgency · emergency descent · communications failure

LEARNING OBJECTIVES — BY THE END OF THIS CHAPTER YOU WILL BE ABLE TO…

- Distinguish distress (MAYDAY) from urgency (PAN-PAN) and their priorities.
- Compose and transmit a complete distress message.
- Apply the imposition and termination of silence (SEELONCE).
- Carry out and broadcast an emergency descent.
- Follow the communications-failure procedure, squawk 7600, and read light signals.

20.1 Distress vs urgency

20.2 The distress message

20.3 Acknowledgement & handling

20.4 Imposition & termination of silence

20.5 Urgency messages

20.6 Emergency descent

20.7 Communications failure

20.8 Light signals

☆ Numbers to memorise

? Question bank

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When an emergency occurs, precise standard phraseology cuts through the confusion and ensures ATC can immediately dispatch the necessary assistance.

20.1 Distress vs urgency

The two conditions

Distress — **MAYDAY** (spoken **three times**): a condition of being threatened by **grave and imminent danger** and requiring **immediate assistance**. It has **absolute priority over all other transmissions**.

Urgency — **PAN-PAN** (spoken **three times**): a condition concerning the **safety** of the aircraft or a person, but **not requiring immediate assistance**. It has priority over all communications except distress.

Mnemonic

MAYDAY = "M'aider" (help me) — **life in danger, help now**. **PAN = "Panne"** (breakdown) — **a problem, but not life-threatening yet**. Both are said three times. If in doubt, MAYDAY is never wrong.

Frequency & squawk

Make the call on the **frequency in use** (don't change away from the controller already helping you); if not in contact, use **121.5 MHz**. Squawk **7700** (Chapter 12) so you light up on every radar.

20.2 The distress message

The format — as much as time allows

1. **MAYDAY MAYDAY MAYDAY** · 2. name of the **station addressed** (if time/known) · 3. **aircraft call sign** · 4. **nature** of the emergency · 5. **intentions** of the pilot-in-command · 6. **position, level and heading** · 7. any other useful information (e.g. **persons on board**, fuel). Speak slowly and clearly; you may not get a second chance.

Transcript — a distress message

A/C MAYDAY MAYDAY MAYDAY, Delhi Approach, Ghostair Alfa Bravo Charlie, engine failure, attempting forced landing, eight miles north of the field, two thousand feet, heading two seven zero, three persons on board.

ATC Ghostair Alfa Bravo Charlie, Delhi Approach, MAYDAY received, wind two seven zero degrees eight knots, runway two seven, you are number one, all other traffic standby.

Mnemonic — what to pass

"Mayday × 3, who, who I am, what's wrong, what I'll do, where I am, anything else." Or remember it as **nature · intentions · position** as the three things ATC most needs.

20.3 Acknowledgement & handling

How the message is handled

The station addressed (or the first to hear it) **acknowledges** the distress, takes charge, gives the aircraft priority, and may impose silence on other traffic. Any station hearing a distress call that is not acknowledged should **relay** it and assist. The aircraft in distress (or the controlling station) is the **controlling station** for the distress traffic.

20.4 Imposition & termination of silence

Imposing silence

To keep the frequency clear for the distress, silence is imposed: the **controlling station** (or the aircraft in distress) may say **"STOP TRANSMITTING, MAYDAY"**; the standard signal is **SEELONCE MAYDAY** (silence, mayday) imposed by the station in control, or **SEELONCE DISTRESS** by another station. All other stations stop transmitting until released.

Terminating silence

When the distress is over, the controlling station announces the end and lifts the silence: "**DISTRESS TRAFFIC ENDED, SEELONCE FEENEE**" (silence finished). Normal working then resumes on the frequency.

Mnemonic — the French roots

SEELONCE = silence; **FEENEE** = fini (finished). "SEELONCE MAYDAY" imposes it; "SEELONCE FEENEE" ends it.

20.5 Urgency messages

PAN-PAN format

The format mirrors the distress message but begins **PAN-PAN PAN-PAN PAN-PAN**: station addressed, call sign, the nature of the urgency, intentions, position/level, and other information. A medical emergency, a minor technical problem, or a low-fuel situation that is concerning but not yet critical are typical urgency calls.

Transcript — an urgency message

A/C PAN-PAN PAN-PAN PAN-PAN, Delhi Approach, VT-ABC, a passenger seriously ill, request priority landing and medical assistance, six miles south, three thousand feet.

ATC VT-ABC, Delhi Approach, roger your PAN, cleared direct, descend two thousand feet, medical services alerted.

20.6 Emergency descent

FIRST PRINCIPLES — GET DOWN FAST, AND WARN EVERYONE BELOW

An **emergency descent** — typically after a **pressurisation failure** — requires the aircraft to descend rapidly to a safe (breathable) level. The danger is to **other traffic below**, so the procedure is as much about **warning them** as about the descent itself. This is the topic your original notes left out entirely.

The phraseology

The crew declares it (often as a MAYDAY): "**MAYDAY MAYDAY MAYDAY, Ghostair ABC, emergency descent**". ATC immediately **broadcasts to all other aircraft** in the area: "**ATTENTION ALL AIRCRAFT in the vicinity of [position], emergency descent in progress from flight level [X]**", and clears traffic out of the way. The descending aircraft turns off the airway if able and may continue to a lower level until cleared.

Transcript — emergency descent

A/C MAYDAY MAYDAY MAYDAY, Mumbai Control, Ghostair Alfa Bravo Charlie, pressurisation failure, emergency descent, leaving flight level three five zero.

ATC ATTENTION ALL AIRCRAFT in the vicinity of SIERRA, emergency descent in progress from flight level three five zero, all aircraft below flight level one zero zero leave the airway to the right.

ATC Ghostair Alfa Bravo Charlie, descend to flight level one zero zero, QNH one zero one three, report level.

20.7 Communications failure

The actions

1. **Squawk 7600** (radio failure).
2. **Try the basics** — frequency, volume, squelch, selector, PTT, alternate radio (Chapter 11).
3. **Listen out** — you may still be able to receive (a "transmitter failure"); follow instructions you hear and acknowledge by transponder/turns.
4. **Transmit blind** your intentions, prefixed "**transmitting blind**" (and "due to receiver failure" if you cannot hear).
5. **Continue in accordance with the last acknowledged clearance / the flight plan**, flying the cleared or expected levels and routing, and complete the flight, joining and landing using **light signals** if necessary.

Exam trap

Distinguish a **complete** failure (you can neither transmit nor receive) from a **transmitter-only** failure (you can still *hear*). If you can hear, comply with what you are told and acknowledge by transponder ident or turns — the controller can still help you (Chapter 19).

20.8 Light signals

WHEN THE RADIO IS GONE, THE TOWER USES A LAMP

If you arrive with no radio, the tower directs you with a **signal lamp**. Know these cold — they are the last line of communication.

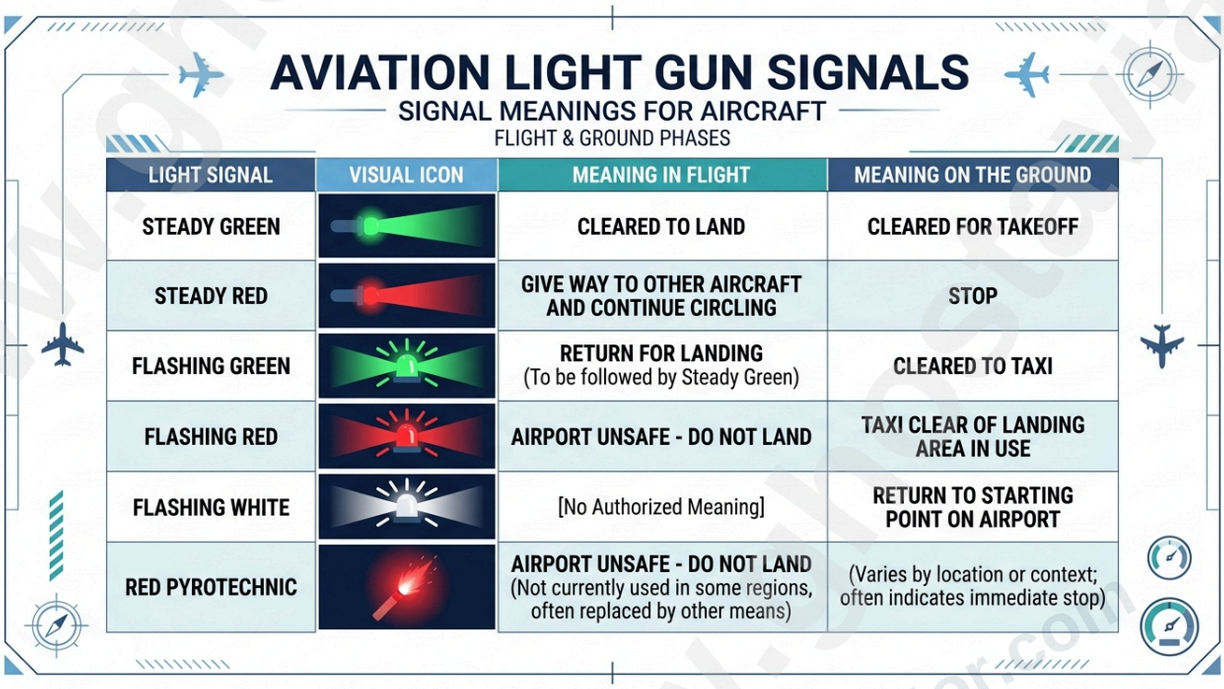

Figure 20.1 — Light Signals: When experiencing a total communications failure, the tower will use these visual signals to issue clearances.

Signal	Aircraft in flight	Aircraft on the ground
Steady green	Cleared to land	Cleared for take-off
Steady red	Give way to other aircraft and continue circling	Stop
Green flashes	Return for landing*	Cleared to taxi
Red flashes	Aerodrome unsafe — do not land	Taxi clear of the landing area in use
White flashes	Land at this aerodrome and proceed to apron*	Return to starting point on the aerodrome
Red pyrotechnic / flare	Notwithstanding any previous instructions, do not land for the time being	

Mnemonic

Green = go / good; Red = stop / danger. Steady = the clearance; flashing = an instruction to position. *Clearances to land/taxi will follow.

Author check before publishing

Confirm the exact light-signal wording and any nuances against the current ICAO Annex 2 / DGCA CAR, and confirm the precise comms-failure routing/level rules (IFR vs VFR) in Indian airspace.

☆ Numbers to memorise

Essential Facts for Chapter 20

Fact	Value
Distress	MAYDAY ×3 — grave & imminent danger, immediate assistance, absolute priority; squawk 7700
Urgency	PAN-PAN ×3 — safety concern, not immediate; priority over all except distress
Emergency frequency	121.5 MHz (if not already in contact)
Distress message	MAYDAY ×3 · station · call sign · nature · intentions · position/level/heading · POB/info
Silence	Impose: SEELONCE MAYDAY / "STOP TRANSMITTING MAYDAY"; end: DISTRESS TRAFFIC ENDED, SEELONCE FEENEE
Emergency descent	MAYDAY + "emergency descent"; ATC "ATTENTION ALL AIRCRAFT … emergency descent in progress"
Comms failure	Squawk 7600 · listen/transmit blind · continue per last clearance/flight plan · light signals
Light signals	Green = cleared (land/take-off); red = stop/give way; flashes = position

Question bank

PART A — MCQS (CLICK AN OPTION TO CHECK)

1. The distress signal is:

> PAN-PAN ×3
>
> MAYDAY ×3
>
> SECURITE ×3
>
> SOS ×3

`Answer: MAYDAY ×3. MAYDAY spoken three times signals distress.`

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2. Distress is a condition of:

> Minor inconvenience
>
> Grave and imminent danger requiring immediate assistance
>
> A passenger query
>
> A frequency change

Answer: Grave and imminent danger requiring immediate assistance. Distress = grave and imminent danger needing immediate help.

3. The urgency signal is:

> MAYDAY ×3
>
> PAN-PAN ×3
>
> SEELONCE
>
> 7700

Answer: PAN-PAN ×3. PAN-PAN three times signals urgency.

4. A distress call has priority:

> Only over VFR traffic
>
> Absolute, over all other transmissions
>
> Equal to urgency
>
> After area control

Answer: Absolute, over all other transmissions. Distress has absolute priority over everything; urgency over all except distress.

5. The transponder code for a general emergency is:

> 7500
>
> 7600
>
> 7700
>
> 7000

Answer: 7700. 7700 emergency; 7600 radio failure; 7500 hijack.

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6. If not already in contact, a distress call is made on:

> 123.45 MHz
>
> 121.5 MHz
>
> 243.0 MHz
>
> 2182 kHz

Answer: 121.5 MHz. 121.5 MHz is the international VHF emergency frequency.

7. The three things ATC most needs in a distress message are:

> Fuel, weather, route
>
> Nature of emergency, intentions, position
>
> Squawk, frequency, call sign
>
> Wind, QNH, runway

Answer: Nature of emergency, intentions, position. Nature, intentions and position — with call sign and any extra info as time allows.

8. To impose radio silence for a distress, the phrase is:

> "DISREGARD"
>
> "STOP TRANSMITTING, MAYDAY" / "SEELONCE MAYDAY"
>
> "STANDBY"
>
> "SEELONCE FEENEE"

Answer: "STOP TRANSMITTING, MAYDAY" / "SEELONCE MAYDAY". SEELONCE MAYDAY (or "stop transmitting, mayday") imposes silence.

9. Radio silence is lifted with:

> "SEELONCE MAYDAY"
>
> "DISTRESS TRAFFIC ENDED, SEELONCE FEENEE"
>
> "PAN-PAN"
>
> "ROGER"

Answer: "DISTRESS TRAFFIC ENDED, SEELONCE FEENEE". SEELONCE FEENEE (silence finished) ends the silence.

10. A passenger taken seriously ill, not yet life-threatening to the aircraft, is a:

> Distress (MAYDAY)
>
> Urgency (PAN-PAN)
>
> Routine call
>
> Radio failure

Answer: Urgency (PAN-PAN). A safety concern not requiring immediate assistance is an urgency (PAN-PAN); escalate to MAYDAY if it worsens.

11. An emergency descent is most often caused by:

> Engine fire
>
> Pressurisation failure
>
> A blocked pitot
>
> A radio failure

Answer: Pressurisation failure. Loss of pressurisation demands a rapid descent to a breathable level.

12. On an emergency descent, ATC broadcasts to other traffic:

> "SEELONCE FEENEE"
>
> "ATTENTION ALL AIRCRAFT … emergency descent in progress"
>
> "Squawk 7600"
>
> "Resume own navigation"

Answer: "ATTENTION ALL AIRCRAFT … emergency descent in progress". The all-stations broadcast warns and clears traffic below the descending aircraft.

13. On a complete communications failure, the pilot squawks:

> 7500
>
> 7600
>
> 7700
>
> 2000

Answer: 7600. 7600 indicates a radio/communications failure.

14. If you can still receive but not transmit, you should:

> Land immediately anywhere
>
> Comply with instructions heard and acknowledge by transponder/turns
>
> Switch off the transponder
>
> Ignore ATC

```
Answer: Comply with instructions heard and acknowledge by transponder/turns. A transmitter-only failure still lets you follow ATC and acknowledge by ident/turns.
```

15. A steady green light to an aircraft in flight means:

> Stop
>
> Cleared to land
>
> Do not land
>
> Return for landing

```
Answer: Cleared to land. Steady green in flight = cleared to land; on the ground = cleared for take-off.
```

16. Red flashes to an aircraft in flight mean:

> Cleared to land
>
> Aerodrome unsafe — do not land
>
> Cleared to taxi
>
> Give way

```
Answer: Aerodrome unsafe – do not land. Red flashes in flight warn that the aerodrome is unsafe – do not land.
```

17. Transmitting your intentions with no expectation of acknowledgement is prefixed:

> "PAN-PAN"
>
> "Transmitting blind"
>
> "SEELONCE"
>
> "Standby"

```
Answer: "Transmitting blind". "Transmitting blind" tells anyone listening you cannot receive a reply.
```

18. After a communications failure, an IFR aircraft should generally:

> Descend immediately to the ground
>
> Continue in accordance with the last clearance / flight plan and complete the flight
>
> Orbit until fuel runs low
>
> Squawk 7500

```
Answer: Continue in accordance with the last clearance / flight plan and complete the flight. Continue per the last acknowledged clearance and flight plan, using light signals to land.
```

PART B — ORAL / VIVA (TAP TO REVEAL MODEL ANSWERS)

▶ **What is the difference between distress and urgency?**

▶ **Give the format of a distress message.**

▶ **How is radio silence imposed and lifted?**

▶ **Describe the emergency-descent procedure on the radio.**

▶ **What are the steps on a communications failure?**

▶ **What does a steady red and a steady green light mean to an aircraft in flight?**

60-SECOND REVISION CARD

- **MAYDAY ×3** = distress (grave danger, immediate help, absolute priority, squawk **7700**); **PAN-PAN ×3** = urgency.
- Distress message: **nature · intentions · position** (+ call sign, POB). On **121.5** if not in contact.
- Silence: impose **SEELONCE MAYDAY**; end **DISTRESS TRAFFIC ENDED, SEELONCE FEENEE**.
- **Emergency descent**: MAYDAY + "emergency descent"; ATC "**ATTENTION ALL AIRCRAFT … emergency descent in progress**".
- Comms failure: **squawk 7600**, listen/transmit blind, continue per last clearance; **light signals** — green = cleared, red = stop.

Meteorological & Aerodrome Information on RT

Chapter 6 taught you to read coded weather; this chapter is about the weather and aerodrome information passed **by voice on the radio** at the critical moments of approach and landing — above all the Runway Visual Range and the state of the runway surface, the numbers that decide whether an approach may continue at all.

SYLLABUS MAP

`Part III (ix) Transmission of meteorological & other aerodrome information – RVR, runway surface conditions`

LEARNING OBJECTIVES — BY THE END OF THIS CHAPTER YOU WILL BE ABLE TO…

- Define RVR and distinguish it from meteorological visibility.
- Receive and read back RVR passed by ATC, including the three touchdown/mid/stop-end values.
- Receive runway surface-condition and braking-action information.
- Receive significant changes to wind, QNH and weather on the radio.

21.1 RVR — what and why

21.2 Transmission of RVR

21.3 RVR vs met visibility

21.4 Runway surface conditions

21.5 Passing changes in wind, QNH & weather

☆ Numbers to memorise

? Question bank

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When meteorological visibility drops, Runway Visual Range (RVR) becomes the governing factor for whether an aircraft is legally permitted to continue an approach.

21.1 RVR — what and why

Definition — Runway Visual Range

RVR is the distance over which the pilot of an aircraft on the centreline of a runway can see the **runway surface markings, or the lights delineating the runway or identifying its centreline**. It is measured close to the runway by **transmissometers** (or human observers) and is the **governing visibility** for low-visibility take-offs and approaches.

WHY RVR, NOT JUST "VISIBILITY"

General met visibility is measured away from the runway and may not reflect what a pilot actually sees down the runway in fog or at night. RVR measures the operationally relevant distance **at the runway itself**, including the effect of the runway lighting — which is why approach minima are expressed in RVR.

21.2 Transmission of RVR

Touchdown, midpoint & stop-end

For a long runway, RVR is measured and passed at up to **three points**: the **touchdown** zone, the **midpoint**, and the **stop-end**. When passed, the values are given in that order and identified, e.g. "**RVR runway two seven, touchdown six hundred metres, midpoint five fifty, stop-end five hundred**". RVR is given in **metres** and read back.

Transcript — RVR passed on approach

ATC Ghostair Alfa Bravo Charlie, RVR runway two seven, touchdown six hundred metres, midpoint five hundred and fifty metres, stop-end five hundred metres.

A/C RVR runway two seven, six hundred, five fifty, five hundred metres, Ghostair Alfa Bravo Charlie.

Exam trap

RVR is passed when the visibility or RVR is **at or below a stated value** (the reporting threshold). It is in **metres**, not the digit-group altitude rule — but still spoken digit by digit. The **touchdown** RVR is normally the controlling value for the approach.

21.3 RVR vs met visibility

	Meteorological visibility	RVR
Where measured	General aerodrome area / observer	Along the runway, near it
What it represents	Horizontal visibility in general	Distance a pilot sees runway markings/lights
Includes lighting?	No	Yes (lights help RVR be greater than met vis at night)
Used for	METAR, general picture	Approach/take-off minima

Cockpit reality

On a foggy approach the controller's RVR call is the number you compare against your minimum. If the touchdown RVR is below your minimum, you do not continue below the decision point unless the required visual references are in sight — the RVR transmission is therefore safety-critical, not just informational.

21.4 Runway surface conditions

What is passed

When the runway is contaminated, ATC passes the **surface condition** and, where assessed, the **braking action**: the type of contaminant (**water, slush, snow, ice**), its depth/coverage, and the braking action reported as **good · medium · poor** (or measured values). "**Runway two seven wet**" / "**standing water**" / "**braking action medium**".

Transcript — runway condition

ATC VT-ABC, runway two seven is wet, water patches, braking action reported medium to poor.

A/C Runway two seven wet, braking action medium to poor, VT-ABC.

Author check before publishing

Confirm the current runway-condition reporting format and braking-action terminology in use in Indian airspace (the global reporting format / RWYCC if adopted), and any standard depths/coverage wording.

21.5 Passing changes in wind, QNH & weather

Updates by voice

Where ATIS is not current or conditions change, ATC passes the **surface wind** (degrees magnetic, knots) with the landing/take-off clearance, an updated **QNH** (read back), and any **significant weather** — thunderstorms, wind shear, heavy precipitation. A wind-shear or significant-weather warning is prefixed "**caution**" (e.g. "caution, wind shear reported on final"). Significant changes are passed as they occur.

Transcript — updated information on approach

ATC VT-ABC, wind now two four zero degrees one eight knots, gusting two eight, QNH one zero zero nine, caution wind shear reported on short final.

A/C Wind two four zero one eight gusting two eight, QNH one zero zero nine, copied wind shear caution, VT-ABC.

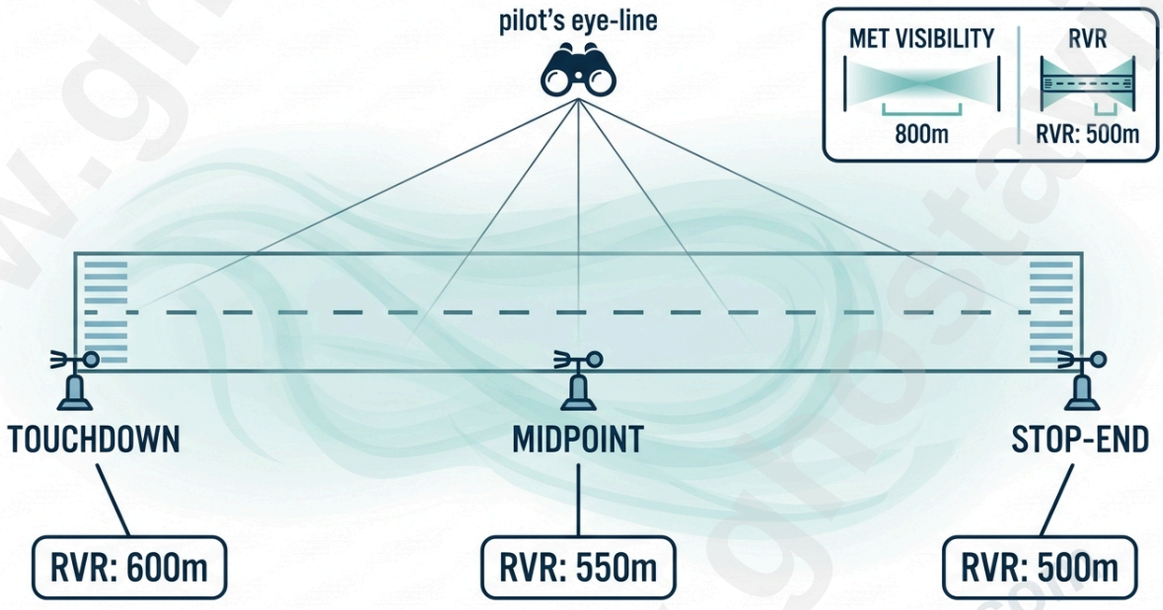

Figure 21.1 — RVR along the runway: Three transmissometers provide specific RVR values for the touchdown, midpoint, and stop-end segments of a long runway.

☆ Numbers to memorise

Essential Facts for Chapter 21

Fact	Value
RVR	Distance a pilot on the centreline sees runway markings/lights; in metres
Three RVR points	Touchdown · midpoint · stop-end (touchdown usually controlling)
RVR vs met vis	RVR at the runway, includes lighting; used for minima
Surface conditions	Wet / water / slush / snow / ice; braking action good/medium/poor
Updates	Wind (magnetic, knots), QNH (read back), significant weather "caution …"

Question bank

PART A — MCQS (CLICK AN OPTION TO CHECK)

1. RVR is the distance over which a pilot on the centreline can see:

> The control tower
>
> The runway markings, or the lights delineating/identifying the runway
>
> The horizon
>
> Other aircraft

```
Answer: The runway markings, or the lights delineating/identifying the runway. RVR is the visible distance of the runway markings or lights from the centreline.
```

2. RVR is measured by:

> The pilot's altimeter
>
> Transmissometers near the runway (or observers)
>
> Satellite
>
> The transponder

```
Answer: Transmissometers near the runway (or observers). Transmissometers (or human observers) measure RVR close to the runway.
```

3. RVR is given in:

> Nautical miles
>
> Metres
>
> Feet
>
> Kilometres only

```
Answer: Metres. RVR is reported in metres.
```

4. The three points at which RVR may be passed are:

> Start, middle, end of the taxiway
>
> Touchdown, midpoint and stop-end
>
> Apron, gate, runway
>
> Departure, cruise, arrival

```
Answer: Touchdown, midpoint and stop-end. Touchdown, midpoint and stop-end RVR for a long runway.
```

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5. The normally controlling RVR for an approach is the:

> Stop-end value
>
> Touchdown value
>
> Midpoint value
>
> Average of all three

Answer: Touchdown value. The touchdown RVR is normally the controlling value for the approach.

6. RVR differs from met visibility because RVR:

> Is measured far from the runway
>
> Is measured along the runway and includes the effect of runway lighting
>
> Ignores lighting
>
> Is only for take-off

Answer: Is measured along the runway and includes the effect of runway lighting. RVR is runway-specific and accounts for lighting; met visibility is general.

7. RVR is normally passed when the visibility/RVR is:

> Above 10 km
>
> At or below a stated reporting value
>
> Always, regardless of value
>
> Only at night

Answer: At or below a stated reporting value. RVR is reported when at or below the reporting threshold (low visibility).

8. "Braking action medium" describes the:

> Wind
>
> Runway surface friction condition
>
> RVR value
>
> QNH

Answer: Runway surface friction condition. Braking action (good/medium/poor) reports the runway surface condition.

9. A runway with standing water would be reported as:

> CAVOK
>
> A contaminated/wet surface condition
>
> A wind shear
>
> A NOTAM only

Answer: A contaminated/wet surface condition. Water/slush/snow/ice are surface conditions passed with braking action.

10. A wind-shear warning on final is prefixed:

> "MAYDAY"
>
> "Caution"
>
> "SEELONCE"
>
> "PAN-PAN"

Answer: "Caution". Hazard advisories such as wind shear are prefixed "caution".

11. An updated QNH passed on approach must be:

> Ignored if you have ATIS
>
> Read back
>
> Not read back
>
> Squawked

Answer: Read back. Altimeter settings are mandatory read-back items.

12. Surface wind passed with a landing clearance is in:

> Degrees true, km/h
>
> Degrees magnetic, knots
>
> Degrees true, knots
>
> Degrees magnetic, m/s

Answer: Degrees magnetic, knots. ATC passes surface wind in degrees magnetic and knots.

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13. "RVR runway 27 touchdown 600 metres" should be read back as:

> Not read back
>
> Read back including the runway and value
>
> Acknowledged with "roger" only
>
> Squawked

Answer: Read back including the runway and value. RVR passed to an aircraft is read back to confirm correct receipt.

14. At night, RVR is often:

> Lower than met visibility
>
> Greater than met visibility, because the runway lights are visible further
>
> The same as met visibility
>
> Not measured

Answer: Greater than met visibility, because the runway lights are visible further. Runway lighting lets a pilot see the runway further than general met visibility suggests.

PART B — ORAL / VIVA (TAP TO REVEAL MODEL ANSWERS)

▶ **What is RVR, and how is it measured?**

▶ **How is RVR transmitted for a long runway?**

▶ **How does RVR differ from meteorological visibility?**

▶ **What runway surface information is passed, and how is braking action reported?**

▶ **How are significant changes in wind, pressure or weather passed?**

60-SECOND REVISION CARD

- **RVR** = distance a pilot on the centreline sees runway markings/lights; in **metres**; the minima value.
- Three points: **touchdown · midpoint · stop-end** (touchdown controlling), read back.
- RVR is at the runway and **includes lighting** (often > met visibility at night).
- Surface: wet/water/slush/snow/ice + **braking action good/medium/poor**.

- Updates: wind (magnetic/kt), **QNH read back**, significant weather "**caution …**".

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Voice Weather Broadcasts

While data-link (D-ATIS/D-VOLMET) is now standard in modern cockpits, the exam focuses on the traditional voice broadcasts — VOLMET and ATIS — that continuously transmit met conditions on loop over VHF or HF. This chapter covers their structure, differences, and how they keep pilots updated without cluttering the control frequencies.

SYLLABUS MAP

`Part III (x) VOLMET & ATIS broadcasts`

LEARNING OBJECTIVES — BY THE END OF THIS CHAPTER YOU WILL BE ABLE TO…

- Define ATIS and VOLMET.
- Explain the sequence and contents of an ATIS broadcast.
- Explain the sequence and contents of a VOLMET broadcast.
- Understand how to use and acknowledge these broadcasts.

22.1 ATIS — the local picture

22.2 VOLMET — the regional picture

22.3 The broadcast sequence

22.4 Using the information

☆ Numbers to memorise

? Question bank

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Continuous voice broadcasts allow pilots to copy down weather and aerodrome information before contacting ATC, drastically reducing congestion on the main control frequencies.

22.1 ATIS — the local picture

Definition — Automatic Terminal Information Service

ATIS is the continuous broadcast of recorded non-control information in selected high-activity terminal areas. It contains **both meteorological and operational information** (runway in use, approach type, transition level) for a **single aerodrome**. It is updated when conditions change significantly, and each update is given an **alphabetical designator** (Alpha, Bravo, Charlie...).

Mnemonic — the purpose

The purpose of ATIS is **to relieve frequency congestion**. Because the pilot listens to the ATIS on a separate frequency before calling approach, the controller does not have to read the weather and runway details to every arriving aircraft.

22.2 VOLMET — the regional picture

Definition — VOLMET

VOLMET (meteorological information for aircraft in flight) is a continuous or scheduled voice broadcast of meteorological information for **several aerodromes in a region**. It contains **only met information** (METARs, TRENDs, and sometimes TAFs or SIGMETs) — no operational details like runway-in-use. It allows an aircraft in the cruise to check the weather at its destination and alternates well in advance.

Feature	ATIS	VOLMET
Scope	One specific aerodrome (terminal area)	Several aerodromes in a region
Contents	Weather **AND** operational info (runway, approaches, NOTAMs)	**ONLY** weather (METAR, TREND, TAF, SIGMET)
Updates	Designated by a letter (Information Alpha)	Usually updated every half hour (no letter)

22.3 The broadcast sequence

The standard order

Both ATIS and VOLMET follow the standard meteorological sequence (the same order as a METAR): **identification · time · surface wind · visibility/RVR · weather · cloud · temperature/dew point · QNH · trend**.

Transcript — typical ATIS

ATIS Delhi International Airport, information Charlie, zero eight three zero zulu. Expect ILS approach runway two seven, runway two seven and two eight in use. Wind two four zero degrees, one zero knots. Visibility three thousand metres. Haze. Cloud few three thousand feet. Temperature three two, dew point two zero. QNH one zero zero nine. Acknowledge receipt of information Charlie on initial contact.

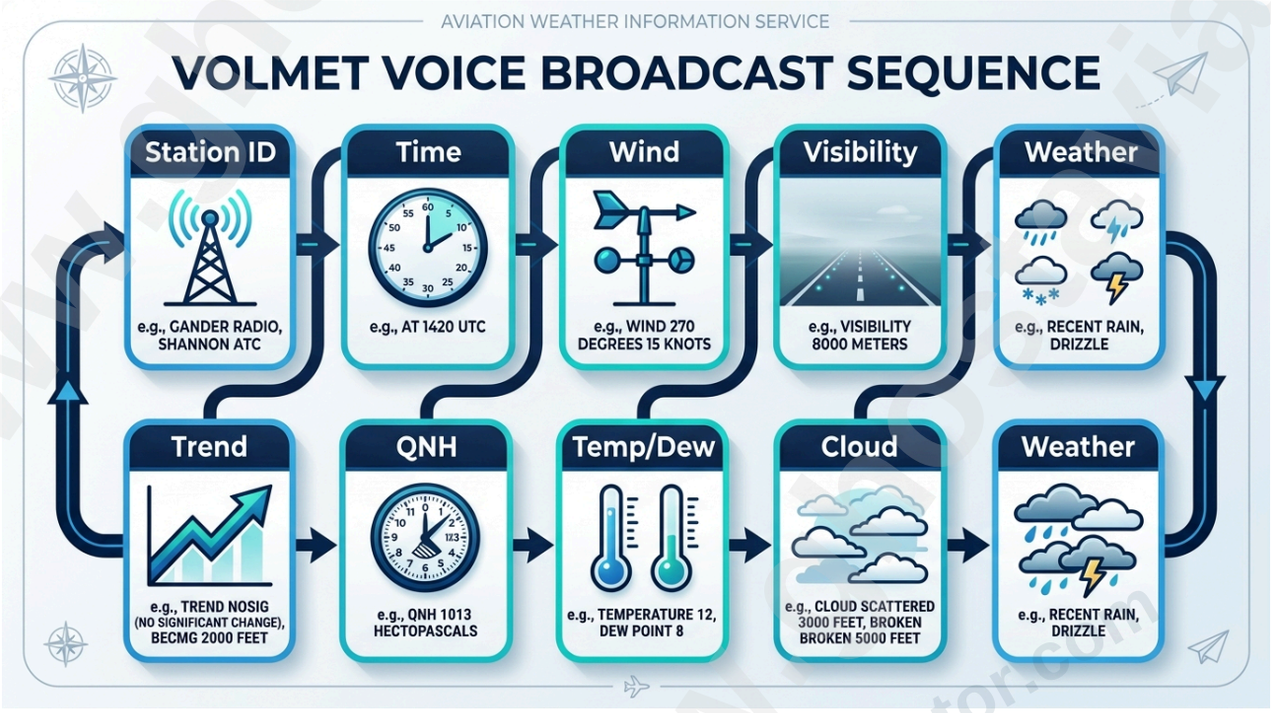

Figure 22.1 — The Sequence: The broadcast strictly follows the METAR order, allowing pilots to rapidly copy the data into standard forms on their kneeboards.

22.4 Using the information

Acknowledging ATIS

When making initial contact with Approach or Ground, the pilot states they have the ATIS by giving the **letter** and the **QNH** (to prove it is the correct current version). Example: "**Delhi Approach, Ghostair ABC, descending flight level one five zero, information Charlie, QNH one zero zero nine**." The controller then does not need to read the weather to them.

☆ Numbers to memorise

Essential Facts for Chapter 22

Fact	Value
ATIS	Automatic Terminal Information Service
ATIS covers	One aerodrome; met AND operational info (runways/approaches)
ATIS identifier	Alphabetical letter (Alpha, Bravo, etc.)
VOLMET covers	Several aerodromes; ONLY met info (no runways)
Sequence	ID · time · wind · vis · weather · cloud · temp/dew · QNH · trend
Pilot action (ATIS)	State information letter and QNH on initial contact with ATC

Question bank

PART A — MCQS (CLICK AN OPTION TO CHECK)

1. What does ATIS stand for?

> Aviation Terminal Information System
>
> Automatic Terminal Information Service
>
> Aerodrome Traffic Information Service
>
> Air Traffic Instruction Service

```
Answer: Automatic Terminal Information Service. ATIS is the Automatic Terminal Information Service.
```

2. An ATIS broadcast contains:

> Only meteorological information for several aerodromes
>
> Meteorological and operational information for a single aerodrome
>
> Control clearances
>
> Radar vectors

```
Answer: Meteorological and operational information for a single aerodrome. ATIS covers one aerodrome and includes operational data like runway-in-use.
```

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3. The primary purpose of ATIS is to:

> Provide radar separation

> Relieve frequency congestion on the ATC channels

> Clear aircraft to land

> Broadcast distress messages

Answer: Relieve frequency congestion on the ATC channels. By putting routine info on a separate broadcast, the main frequency is kept clear.

4. Successive ATIS broadcasts are identified by:

> A number

> An alphabetical letter (e.g. Information Alpha)

> The frequency

> A squawk code

Answer: An alphabetical letter (e.g. Information Alpha). Each new ATIS update gets the next letter in the alphabet.

5. VOLMET contains:

> Runway in use and approach types

> Only meteorological information for several aerodromes in a region

> Only information for a single aerodrome

> Control clearances

Answer: Only meteorological information for several aerodromes in a region. VOLMET provides only weather, for multiple stations.

6. On initial contact with ATC, a pilot proves receipt of the ATIS by stating:

> "Weather copied"

> The information letter and the QNH

> The surface wind

> "Roger ATIS"

Answer: The information letter and the QNH. "Information Charlie, QNH 1009" confirms you have the latest and correct version.

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7. The standard sequence of elements in an ATIS/VOLMET is:

> Randomised to keep pilots alert
>
> Wind, visibility, weather, cloud, temp/dew, QNH
>
> QNH, wind, cloud, visibility, temp
>
> Temp, wind, QNH, cloud, visibility

`Answer: Wind, visibility, weather, cloud, temp/dew, QNH. It follows the standard METAR sequence.`

8. If an aircraft hears the ATIS, does the controller need to read the weather to them?

> Yes, always
>
> No, not unless there is a significant change
>
> Only the wind
>
> Only the QNH

`Answer: No, not unless there is a significant change. That is the whole point of ATIS — to save the controller passing it.`

PART B — ORAL / VIVA (TAP TO REVEAL MODEL ANSWERS)

▶ **What is ATIS and what is its purpose?**

▶ **How does VOLMET differ from ATIS?**

▶ **What is the standard sequence of weather elements in a broadcast?**

▶ **How should a pilot acknowledge receipt of the ATIS?**

60-SECOND REVISION CARD

- **ATIS** = 1 aerodrome; met + operational info (runway); identified by a letter; relieves frequency congestion.
- **VOLMET** = regional (multiple aerodromes); ONLY met info.
- **Sequence**: wind · vis · weather · cloud · temp/dew · QNH.
- **Acknowledge ATIS** by stating the letter and QNH on initial contact.

Numbers to Memorise — Master Tables

Every figure, frequency, code and definition the examiner is most likely to demand, pulled together from all 22 chapters into one place. This is your final-week reference: read it, cover a column, and recite. If a number is here, it is worth knowing cold.

CONSOLIDATED FROM MODULES A, B & C

Pair this chapter with the Chapter 24 Question Bank for your final exam preparation.

HOW TO USE THIS CHAPTER

- Each table is a self-test: use the button below to blur the right-hand column, then tap/hover to reveal the value from the cue.
- The "Quick cards" at the end are the reference sheets to photograph and carry.
- Finish with the rapid-fire self-test, then move to the Chapter 24 question bank.

Enable Self-Test Mode (Blur Values)

A · Regulations, bodies & documents

B · Licensing, Annexes & equipment

C · Q-codes, pressure & definitions

D · FIRs, frequencies & AIS

E · Meteorology & RVR

F · Electrical & wave relationships

G · Frequency bands & propagation

H · Modulation & equipment

I · Navigation aids & squawks

J · Phraseology, numbers & procedures

☆ Quick cards

? Rapid-fire self-test

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When in doubt, consult the tables. These are the hard facts and figures you must know for both the written exam and the oral viva.

A Regulations, bodies & documents

Cue	Value
Chicago Convention	Signed 7 Dec 1944 (52 States); in force 4 Apr 1947; created ICAO; 96 Articles, 19 Annexes
Article 1	Complete & exclusive sovereignty over national airspace
Article 38	Notify ICAO of any differences from a Standard
ICAO	UN specialised agency · HQ Montreal · Assembly / Council(36) / ANC / Secretariat
ITU	UN agency · HQ Geneva · founded 1865 (oldest) · ITU-R/T/D · allocates spectrum (Radio Regs)
ITU Regions	Region 1 (Europe/Africa/ME), Region 2 (Americas), Region 3 (Asia-Pacific — India)
IATA	Airline trade association (not a regulator); 3-letter airport / 2-letter airline codes
Standard vs Recommended	Standard = necessary (comply or file difference); Recommended = desirable
India	DGCA (regulator/CARs) · AAI (ANS) · WPC (spectrum/NFAP) · BCAS (security) · AAIB (investigation)
Key ICAO docs	Annex 10 (telecom) · Doc 4444 PANS-ATM · Doc 8168 PANS-OPS · Doc 9432 RTF Manual

B Licensing, the 19 Annexes & equipment

Cue	Value
RTR(A)	Radio Telephone Operator (Restricted), Aeronautical — for the aeronautical mobile service
Two licences	Operator's RTR(A) + aircraft radio station licence (apparatus, via WPC)
Governing law	Bharatiya Vayuyan Adhiniyam 2024 · RTR (Restricted) Rules 2025 · CAR Sec 7, Series G, Part VI
Minimum equipment	ICAO Annex 6 (Operation of Aircraft); scales up for IFR/night
Independence rule	One failure must not disable both comms and navigation
Dispatch u/s	Governed by the MEL (from the MMEL)

#	Annex	#	Annex
1	Personnel Licensing	11	Air Traffic Services
2	Rules of the Air	12	Search and Rescue
3	Meteorological Service	13	Accident & Incident Investigation
4	Aeronautical Charts	14	Aerodromes
5	Units of Measurement	15	Aeronautical Information Services
6	Operation of Aircraft	16	Environmental Protection
7	Aircraft Nationality & Registration	17	Security
8	Airworthiness of Aircraft	18	Dangerous Goods
9	Facilitation	19	Safety Management
10 ☆	**Aeronautical Telecommunications** — 5 volumes (RTR uses Vol I & II)		

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C Q-codes, pressure & vertical definitions

Cue	Value
QNH	Set → altimeter reads ALTITUDE (AMSL); reads aerodrome elevation on the ground
QFE	Set → reads HEIGHT above aerodrome; zero on the runway
QNE	Standard 1013.2 hPa (29.92 inHg) → FLIGHT LEVEL
QFF	True MSL pressure (meteorological, not flown)
Pressure lapse	≈ 1 hPa per 27 ft near sea level
Transition	Climb: set 1013 at transition ALTITUDE · Descend: set QNH at transition LEVEL
QDM / QDR	Magnetic heading TO / bearing FROM station (QDR = QDM ± 180°)
QUJ / QTE	True bearing TO / FROM the station
QSY / QGH	Change frequency / controlled descent (DF let-down)
Altitude/Height/Elevation/FL	From MSL / from a datum / of a surface point / referenced to 1013.2 hPa

D FIRs, frequencies & AIS

Cue	Value
Indian FIRs	Delhi, Mumbai, Kolkata, Chennai (+ Chennai Oceanic)
ICAO location codes	VIDP Delhi · VABB Mumbai · VECC Kolkata · VOMM Chennai (India = V, 4 letters)
Airspace classes	A (all controlled) → G (uncontrolled)
VHF ATC band	118.0 – 136.975 MHz (AM, line-of-sight)
Emergency	121.5 MHz (VHF) · 243.0 MHz (UHF mil) · 406 MHz (ELT) · 123.1 (SAR aux)
Maritime distress	2182 kHz (MF)
IAIP	AIP (GEN/ENR/AD) · Amendment · Supplement · AIC · NOTAM
NOTAM types	NOTAMN (new) · NOTAMR (replace) · NOTAMC (cancel); SNOWTAM, ASHTAM
AIRAC	28-day cycle
AFTN priorities	SS Distress · DD Urgency · FF Flight-safety · GG Met · KK Regularity

E Meteorology & RVR

Cue	Value
METAR / SPECI / TAF	Observed / special observed / forecast
METAR order	Station · Time(Z) · Wind · Vis · Weather · Cloud · Temp/Dew · QNH · Trend
Cloud oktas	FEW 1–2 · SCT 3–4 · BKN 5–7 · OVC 8 (heights in hundreds of ft)
TAF change groups	FM · BECMG · TEMPO · PROB30/40
CAVOK	Vis ≥10 km · no cloud <5000 ft / no CB·TCU · no significant weather
SIGMET / AIRMET	Hazard to all aircraft / to low-level light aircraft
ATIS / VOLMET	One-aerodrome broadcast / en-route multi-aerodrome broadcast
RVR	Runway Visual Range (metres); points: touchdown · midpoint · stop-end
Wind reference	METAR = degrees TRUE; ATC surface wind = degrees MAGNETIC

F Electrical & wave relationships

Cue	Value
Units	Volt · Ampere · Ohm (Ω) · Watt · Farad · Henry
Ohm's Law	V = I × R
Power	P = V × I = I²R = V²/R
RMS (sine)	RMS = 0.707 × peak · peak = 1.414 × RMS
Aircraft power	28 V DC · 115 V AC 400 Hz
Series / parallel	Series: R adds, same current · Parallel: R reduces, same voltage
Wave relationship	c = f × λ · c = 3 × 10&sup8; m/s · f = 1/T (higher f → shorter λ)

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G Frequency bands & propagation

Band	Frequency	Propagation / use
LF/MF	30 kHz – 3 MHz	Ground wave — NDB
HF	3 – 30 MHz	Sky wave — oceanic voice (SSB)
VHF	30 – 300 MHz	Space wave (line-of-sight) — ATC voice, VOR, ILS LOC
UHF	300 MHz – 3 GHz	Line-of-sight — ILS GP, DME, SSR, GPS

Cue	Value
HF day/night	Higher frequency by DAY, lower by NIGHT (D layer absorbs by day)
MUF / LUF / OWF	Max usable / Lowest usable / Optimum ≈ 85% of MUF
Dead space	Gap between ground-wave range and first sky-wave return
Radio horizon	d (NM) ≈ 1.23 × √h(ft)
Channel spacing	25 kHz and 8.33 kHz
Aviation VHF polarization	Vertical

H Modulation & equipment

Cue	Value
AM	Amplitude varied; bandwidth = 2 × highest audio; info in sidebands; ~67% power in carrier
FM	Frequency varied; noise-resistant; capture effect (not used for VHF voice)
SSB	One sideband, carrier suppressed; efficient — HF voice
Aviation choice	VHF voice = AM (A3E) · HF voice = SSB (J3E)
Receiver	Superheterodyne (converts to a fixed IF)
Simplex	One frequency, one transmitter at a time (aviation VHF)
Squelch / AVC	Mutes background hiss / holds received audio level steady

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I Navigation aids & squawk codes

Aid	Band	Gives
NDB / ADF	LF/MF 190–1750 kHz	Relative bearing (±5°)
VOR	VHF 108.0–117.975 MHz	Magnetic radial TO/FROM
ILS LOC / GP	VHF 108.10–111.95 / UHF 329–335 MHz	Azimuth / glidepath
Markers	75 MHz	Distance to runway
DME	UHF 962–1213 MHz	Slant range
SSR	1030 ↑ / 1090 ↓ MHz	Identity (A) + altitude (C) + Mode S
GPS	L1 1575.42 MHz	4 satellites for a 3-D fix
Radio altimeter	4200–4400 MHz	True height (AGL)

Squawk	Meaning
7500	Unlawful interference (hijack)
7600	Radio / communications failure
7700	General emergency
7000 / 2000	Conspicuity (VFR) / entering from a non-SSR area

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J Phraseology, numbers & procedures

Cue	Value
Altered numerals	3 = TREE · 5 = FIFE · 9 = NIN-er · decimal = DAY-SEE-MAL
Number rules	Digit-by-digit (call signs, headings, FL, squawk, freq) · hundreds/thousands (altitude, height, cloud, vis)
Time	UTC (Zulu); minutes only, or full hours+minutes if confusion
ROGER / WILCO	Received / will comply (ROGER ≠ yes; AFFIRM/NEGATIVE = yes/no)
Readability	1 unreadable → 5 perfectly readable
Ground suffixes	Delivery · Ground · Tower · Approach · Departure · Director · Control · Information
Always read back	Runway/level/heading/speed clearances, QNH, squawk, frequency changes
Circuit legs	Upwind · crosswind · downwind · base · final
Distress / urgency	MAYDAY ×3 (grave danger) / PAN-PAN ×3 (safety, not immediate)
Silence	Impose: SEELONCE MAYDAY · End: DISTRESS TRAFFIC ENDED, SEELONCE FEENEE
Emergency descent	ATC "ATTENTION ALL AIRCRAFT … emergency descent in progress"
Comms failure	Squawk 7600 · continue per last clearance/flight plan · light signals
Wake categories	Super · Heavy (≥136t) · Medium (7–136t) · Light (≤7t)
VDF bearing classes	A ±2° · B ±5° · C ±10° · D worse
TCAS RA	Obey (vertical only), say "TCAS RA" then "clear of conflict"

☆ Quick cards

Phonetic alphabet

A Alfa	I India	Q Quebec
B Bravo	J Juliett	R Romeo
C Charlie	K Kilo	S Sierra
D Delta	L Lima	T Tango
E Echo	M Mike	U Uniform
F Foxtrot	N November	V Victor
G Golf	O Oscar	W Whiskey
H Hotel	P Papa	X X-ray

Y Yankee

Z Zulu

Numerals: 0 ZE-RO · 1 WUN · 2 TOO · 3 TREE · 4 FOW-er · 5 FIFE · 6 SIX · 7 SEV-en · 8 AIT · 9 NIN-er

Light signals (no radio)

Signal	In flight	On the ground
Steady green	Cleared to land	Cleared for take-off
Steady red	Give way, continue circling	Stop
Green flashes	Return for landing	Cleared to taxi
Red flashes	Aerodrome unsafe — do not land	Taxi clear of landing area
White flashes	Land & proceed to apron	Return to starting point
Red pyrotechnic	Do not land for the time being (notwithstanding previous instructions)	

> **Big mnemonics on one line**
>
> Bands: "**V**ery **L**ong **M**odems **H**elp **V**arious **U**seful **S**ignals Excel" · Annex 10 = **TEN=TELECOM** · Law = **Act→Rules→CAR** · Squawks **75 alive / 76 no comms / 77 heaven** · AFTN **S**ome **D**ogs **F**ind **G**ood **K**ennels · QDM = **Direction to Mother** · HF **Day high, Night low.**

Rapid-fire self-test

Click an option to check — one from each table

1. The Chicago Convention created ICAO and was signed in:

> 1944
>
> 1947
>
> 1919

Answer: 1944. Signed 7 Dec 1944; in force (ICAO created) 4 Apr 1947.

2. Annex 10 covers:

> Rules of the Air
>
> Aeronautical Telecommunications
>
> Aerodromes

Answer: Aeronautical Telecommunications. Annex 10 — Aeronautical Telecommunications (TEN = TELECOM).

3. QNE is the standard setting:

> 1003.2 hPa
>
> 1013.2 hPa
>
> 760 hPa

```
Answer: 1013.2 hPa. 1013.2 hPa (29.92 inHg) → flight levels.
```

4. The international VHF emergency frequency is:

> 243.0 MHz
>
> 121.5 MHz
>
> 406 MHz

```
Answer: 121.5 MHz. 121.5 MHz (VHF guard); 243.0 UHF military.
```

5. The AIRAC cycle is:

> 7 days
>
> 14 days
>
> 28 days

```
Answer: 28 days. Common effective dates every 28 days.
```

6. Broken cloud (BKN) is:

> 1–2 oktas
>
> 3–4 oktas
>
> 5–7 oktas

```
Answer: 5–7 oktas. FEW 1–2, SCT 3–4, BKN 5–7, OVC 8.
```

7. c = f × λ, where c is:

> 3 × 10♠ m/s (approx typo for 10^6)
>
> 3 × 10&sup8; m/s
>
> 300 m/s

```
Answer: 3 × 10&sup8; m/s. Speed of radio waves ≈ 3 × 10&sup8; m/s.
```

8. HF long-range voice uses:

> AM, ground wave
>
> SSB, sky wave
>
> FM, line-of-sight

```
Answer: SSB, sky wave. HF = SSB, sky wave; VHF = AM, line-of-sight.
```

9. An SSR transponder replies on:

> 1030 MHz
>
> 1090 MHz
>
> 121.5 MHz

```
Answer: 1090 MHz. Interrogate 1030, reply 1090 MHz.
```

10. The squawk for radio failure is:

> 7500
>
> 7600
>
> 7700

```
Answer: 7600. 7600 radio failure; 7500 hijack; 7700 emergency.
```

11. "ROGER" means:

> Yes
>
> I have received all your last transmission
>
> I will comply

```
Answer: I have received all your last transmission. ROGER = received; WILCO = will comply; AFFIRM = yes.
```

12. The distress signal, spoken three times, is:

> PAN-PAN
>
> MAYDAY
>
> SECURITE

```
Answer: MAYDAY. MAYDAY ×3 = distress; PAN-PAN ×3 = urgency.
```

FINAL-WEEK ROUTINE

- Recite each master table by covering the value column — aim for zero misses.
- Carry the phonetic, light-signal and squawk quick cards; glance daily.
- Then hammer the **Chapter 24 question bank** (297 real questions) until 80%+ everywhere.
- Say the phraseology and the oral answers out loud — the practical decides the result.

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Question Bank & Mock Exams

A complete practice set split the way the licence itself is examined: Part 1 — the written multiple-choice paper — and Part 2 — the oral/viva and practical R/T exam. Every question is compiled from real memory-based DGCA RTR(A) papers, cleaned, audited against current ICAO standards, and organised.

PART 1: Multiple-Choice Examination (418 Questions)

GENERAL

1. ICAO was created by which international convention?

> A. Warsaw Convention
>
> B. Geneva Convention
>
> C. Chicago Convention

```
Correct Answer: C
```

2. On which date was the Chicago Convention signed, leading to the creation of ICAO?

> A. 7th December 1944
>
> B. 4th April 1944
>
> C. 1st January 1945

```
Correct Answer: A
```

3. When did the Chicago Convention come into existence (become effective)?

> A. 7th December 1944
>
> B. 4th April 1947
>
> C. 30th June 1945

```
Correct Answer: B
```

4. Where is the headquarters of ICAO located?

> A. Geneva, Switzerland
>
> B. Montreal, Quebec, Canada
>
> C. New York, USA

```
Correct Answer: B
```

5. ICAO mainly develops which of the following to support global aviation safety?

> A. Local airfield rules
>
> B. Airport security codes
>
> C. Standards and Recommended Practices (SARPs)

```
Correct Answer: C
```

6. How many contracting states does ICAO currently have?

> A. 172
>
> B. 193
>
> C. 205

```
Correct Answer: B
```

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7. How many regional offices does ICAO maintain?

> A. 5
>
> B. 9
>
> C. 7

```
Correct Answer: C
```

8. How many geographic regions are maintained under ICAO?

> A. 6
>
> B. 9
>
> C. 12

```
Correct Answer: B
```

9. Which of the following is NOT one of ICAO's geographic regions?

> A. AFI – Africa-Indian Ocean
>
> B. MID – Middle East
>
> C. SEA – Southeast Asia

```
Correct Answer: C
```

10. ICAO office responsible for India falls under which regional office?

> A. AFI – Africa-Indian Ocean
>
> B. Bangkok (Asia and Pacific region)
>
> C. NAT – North Atlantic

```
Correct Answer: B
```

11. Which ICAO body is responsible for adopting Standards and Recommended Practices (SARPs)?

> A. ICAO Assembly only
>
> B. ICAO Secretariat
>
> C. ICAO Council

```
Correct Answer: C
```

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12. IATA is primarily known as what type of organization?

A. A government aviation regulator

B. A global trade organization for the air transport industry

C. A military aviation body

Correct Answer: B

13. What is the main objective of IATA?

A. Conduct aircraft manufacturing

B. Manage airport operations worldwide

C. Promote safe, reliable, secure, and economical air services

Correct Answer: C

14. What percentage of total global air traffic is approximately represented by IATA member airlines?

A. 70%

B. 85%

C. 83%

Correct Answer: C

15. How many airlines are members of IATA?

A. 220

B. Over 370

C. 150

Correct Answer: B

16. When was IATA founded?

A. 7th December 1944

B. 19th April 1945

C. 1st March 1946

Correct Answer: B

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17. Where was IATA founded?

> A. Montreal, Canada
>
> B. Geneva, Switzerland
>
> C. Havana, Cuba

```
Correct Answer: C
```

18. Where is the headquarters of IATA located?

> A. Montreal, Quebec, Canada
>
> B. New York, USA
>
> C. London, United Kingdom

```
Correct Answer: A
```

19. SARPs are issued by ICAO under which international agreement?

> A. Warsaw Convention
>
> B. Geneva Convention
>
> C. Chicago Convention

```
Correct Answer: C
```

20. Which ICAO body is responsible for adopting SARPs?

> A. ICAO Assembly
>
> B. ICAO Secretariat
>
> C. ICAO Council

```
Correct Answer: C
```

21. What does the acronym SARPs stand for?

> A. Safety and Regulation Procedures
>
> B. Standard Aviation Rule Policies
>
> C. Standards and Recommended Practices

```
Correct Answer: C
```

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22. How many annexes currently contain ICAO SARPs?

A. 12

B. 19

C. 24

Correct Answer: B

23. What is the primary purpose of SARPs?

A. To manage airline ticketing

B. To control airport finances

C. To ensure global uniformity in aviation standards

Correct Answer: C

24. What defines a Standard in ICAO SARPs?

A. The word MAY

B. The word SHOULD

C. The word SHALL

Correct Answer: C

25. Uniform application of SARPs is considered essential for what purpose?

A. Reducing aviation taxes

B. Ensuring safety and regularity of international air navigation

C. Promoting airline marketing practices

Correct Answer: B

26. If a Contracting State cannot comply with a Standard, what must it do according to the Chicago Convention?

A. Cancel all international flights

B. Notify the ICAO Council as per Article 38

C. Request permission from member airlines

Correct Answer: B

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27. Which Article of the Chicago Convention requires States to inform ICAO about differences from Standards?

A. Article 26

B. Article 12

C. Article 38

Correct Answer: C

28. Which word is used in ICAO documents to define a Recommended Practice?

A. SHALL

B. SHOULD

C. MAY

Correct Answer: B

29. Uniform application of Recommended Practices is considered what?

A. Legally mandatory

B. Not related to aviation safety

C. Desirable for the safety and regularity of international air navigation

Correct Answer: C

30. What is a Contracting State expected to do regarding Recommended Practices?

A. Endeavour to conform to them

B. Strictly enforce them by law

C. Ignore them unless convenient

Correct Answer: A

31. If a State does not comply with a Recommended Practice, what is required?

A. No compulsory notification to the ICAO Council

B. Mandatory report under Article 38

C. Cancellation of all operations

Correct Answer: A

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32. Recommended Practices are considered desirable for what aspect?

A. Airline pricing

B. Domestic tourism

C. Supporting the safety and regularity of international air navigation

Correct Answer: C

33. PANS-ATM (Doc 4444) is published by ICAO for what primary purpose?

A. To regulate aircraft manufacturing

B. To specify the operating procedures to be applied by air traffic services units

C. To train airport security staff

Correct Answer: B

34. Why is PANS-ATM required in addition to Annex 11?

A. Annex 11 is outdated

B. Annex 11 applies only to domestic flights

C. Annex 11 is not detailed enough for ATS operations

Correct Answer: C

35. PANS-ATM is primarily intended for which group?

A. Airline management teams

B. Aircraft dispatchers

C. Air Traffic Services (ATS) personnel

Correct Answer: C

36. Who else, besides ATS personnel, should be familiar with sections of PANS-ATM relating to Air Traffic Management?

A. Airport ground handling staff

B. Flight crews

C. Maintenance engineers

Correct Answer: B

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37. PANS-ATM provides more detailed guidance than which ICAO Annex?

A. Annex 14

B. Annex 2

C. Annex 11

Correct Answer: C

38. SUPPS are created when which of the following is necessary?

A. A State wants to publish its own national regulations

B. A specific region requires procedures different from worldwide procedures

C. Airlines request new route structures

Correct Answer: B

39. Who determines the need for regional procedures that differ from global ones?

A. Individual State airlines

B. ICAO Assembly

C. Regional Air Navigation Meetings

Correct Answer: C

40. SUPPS are approved for use in which areas?

A. Worldwide

B. Only within their respective geographic regions

C. Only within ICAO headquarters

Correct Answer: B

41. The status of SUPPS is similar to which other ICAO document type?

A. Annexes

B. PANS (Procedures for Air Navigation Services)

C. Airworthiness Manuals

Correct Answer: B

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42. DGCA is primarily responsible for which area in Indian civil aviation?

> A. Airport construction
>
> B. Airline marketing
>
> C. Aviation safety regulation

```
Correct Answer: C
```

43. DGCA regulates air transport services operating where?

> A. Only international flights
>
> B. Only domestic flights
>
> C. To/from/within/over India

```
Correct Answer: C
```

44. Where is the headquarters of DGCA located?

> A. Mumbai
>
> B. Bengaluru
>
> C. New Delhi

```
Correct Answer: C
```

45. Which of the following is a key function of DGCA?

> A. Manufacturing aircraft
>
> B. Registration of civil aircraft
>
> C. Setting fuel prices

```
Correct Answer: B
```

46. DGCA is responsible for the licensing of which personnel?

> A. Airport managers
>
> B. Pilots, aircraft maintenance engineers, and flight engineers
>
> C. Airline ground staff

```
Correct Answer: B
```

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47. Which aviation personnel licensing is also under DGCA's authority?

> A. Cabin crew
>
> B. Meteorologists
>
> C. Air Traffic Controllers

```
Correct Answer: C
```

48. DGCA certification of aerodromes includes which type of facilities?

> A. Cargo warehouses
>
> B. CNS/ATM facilities
>
> C. Duty-free shops

```
Correct Answer: B
```

49. DGCA regulates operations of flights belonging to which operators?

> A. Only Indian operators
>
> B. Only cargo operators
>
> C. Both Indian and foreign operators

```
Correct Answer: C
```

50. DGCA coordinates flexi-use of airspace between which groups?

> A. Airlines and passengers
>
> B. Civil and military air traffic agencies
>
> C. Travel agents and airlines

```
Correct Answer: B
```

51. Monitoring aircraft noise and emissions falls under which ICAO Annex?

> A. Annex 14
>
> B. Annex 11
>
> C. Annex 16

```
Correct Answer: C
```

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52. DGCA encourages indigenous design and manufacture of aircraft by acting as what?

A. A regulator

B. A catalytic agent

C. A sponsor

Correct Answer: B

53. DGCA coordinates with ICAO mainly for what purpose?

A. Hiring foreign pilots

B. Provision of more air routes through Indian airspace

C. Approving airport uniforms

Correct Answer: B

54. The International Telecommunication Union (ITU) operates under which global body?

A. European Union

B. Commonwealth Secretariat

C. United Nations Organization (UNO)

Correct Answer: C

55. Where is the headquarters of ITU located?

A. Montreal, Canada

B. Vienna, Austria

C. Geneva, Switzerland

Correct Answer: C

56. ITU was formed by combining which two earlier bodies?

A. ICAO + IATA

B. International Telegraph Union 1865 + International Radiotelegraph Convention 1906

C. WMO + IMO

Correct Answer: B

57. In which year was the ITU officially established in its modern form?

> A. 1954
>
> B. 1934
>
> C. 1984

```
Correct Answer: B
```

58. India signed the ITU convention during which 1982 conference?

> A. Geneva Conference 1971
>
> B. WARC Conference 1967
>
> C. 11th Nairobi Conference 1982

```
Correct Answer: C
```

59. When did the 1982 ITU convention become effective for India?

> A. 1st July 1982
>
> B. 1st April 1983
>
> C. 1st January 1984

```
Correct Answer: C
```

60. How many volumes are ITU regulations divided into?

> A. 2
>
> B. 3
>
> C. 4

```
Correct Answer: C
```

61. Which of the following is NOT one of the ITU regulation volumes?

> A. Articles
>
> B. Resolutions
>
> C. Manuals

```
Correct Answer: C
```

62. Which ITU regulation volume contains items like Q-codes and emission classes?

A. Appendices

B. Resolutions

C. Recommendations

```
Correct Answer: A
```

63. What is the primary purpose of ITU?

A. Regulating air traffic

B. Allocating radio spectrum and improving telecom efficiency

C. Granting aviation licenses

```
Correct Answer: B
```

64. WRC stands for what?

A. World Aeronautical Regulatory Committee

B. Wireless Aviation Radio Council

C. World Radiocommunication Conference

```
Correct Answer: C
```

65. What is the purpose of WARC?

A. To train pilots

B. To regulate airspace

C. To review global radio communication and ITU's working

```
Correct Answer: C
```

66. Which countries participate in WARC?

A. Only developed countries

B. Only ITU founders

C. All ITU member countries

```
Correct Answer: C
```

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67. Which ITU bureau handles radio communication matters?

A. ITU-T

B. ITU-D

C. ITU-R

Correct Answer: C

68. ITU-T deals with which area?

A. Radio navigation

B. Telecommunication standards

C. Space tracking

Correct Answer: B

69. ITU-D is responsible for what?

A. Dangerous goods regulations

B. Aviation safety development

C. Telecommunication development

Correct Answer: C

70. When was the Airports Authority of India (AAI) formed?

A. 1st January 1989

B. 1st April 1995

C. 1st March 2000

Correct Answer: B

71. AAI was formed by merging which two organizations?

A. DGCA + BCAS

B. Indian Airlines + Air India

C. International Airports Authority of India + National Airports Authority

Correct Answer: C

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72. Where is the headquarters of AAI located?

A. Mumbai

B. Kolkata

C. New Delhi

Correct Answer: C

73. Which of the following is a major objective of forming AAI?

A. Training pilots

B. Monitoring aircraft noise

C. Integrated development and modernization of airports

Correct Answer: C

74. AAI is responsible for the design and operation of which airports?

A. Only international airports

B. Only domestic airports

C. Both international and domestic airports

Correct Answer: C

75. AAI manages Indian airspace extending beyond territorial limits as accepted by whom?

A. Ministry of Defence

B. ICAO

C. ISRO

Correct Answer: B

76. Which of the following is included in AAI's responsibilities?

A. Certification of aircraft

B. Airline financial auditing

C. Construction and management of passenger terminals

Correct Answer: C

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77. AAI is responsible for developing which type of terminals?

A. Railway cargo terminals

B. Cargo terminals at international and domestic airports

C. Shipping terminals

Correct Answer: B

78. AAI expands and maintains which operational areas?

A. Cockpit instrumentation

B. Catering lounges

C. Runways, aprons, taxiways

Correct Answer: C

79. Which visual aids are provided by AAI?

A. Flight management computers

B. Airport lighting systems and markings

C. Aircraft weather radars

Correct Answer: B

80. Which area is NOT a responsibility of AAI?

A. Airport terminal development

B. Airspace management

C. Licensing of pilots

Correct Answer: C

81. CARs are issued by DGCA under which legal authority?

A. Aircraft Act 1950

B. Aircraft Rules 1937

C. ICAO Annex 6

Correct Answer: B

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82. The primary purpose of CARs issued by DGCA is to—?

A. Replace ICAO Standards completely

B. Provide information on AFTN operations

C. Meet ICAO requirements and standardize aviation procedures

Correct Answer: C

83. How many Sections are the CARs issued by DGCA divided into?

A. 5

B. 11

C. 21

Correct Answer: B

84. CAR Section 2 deals with—?

A. Licensing of engineers

B. Airworthiness requirements

C. Dangerous goods transport

Correct Answer: B

85. CAR Section 7 covers—?

A. Meteorology procedures

B. Flight crew standards, training, and licensing

C. Radio communication procedures

Correct Answer: B

86. CARs are primarily intended to—?

A. Provide military aviation regulations

B. Provide technical specifications for airport construction

C. Ensure standardized civil aviation procedures in compliance with ICAO

Correct Answer: C

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87. Which DGCA document governs pilot licensing standards?

A. CAR Section 3

B. CAR Section 7

C. CAR Section 11

Correct Answer: B

88. Which of the following is a responsibility covered under CAR Section 2?

A. Medical assessment of flight crew

B. Ensuring aircraft meet airworthiness and operational requirements

C. ATCO licensing

Correct Answer: B

89. CARs are primarily applicable to—?

A. ATC only

B. Meteorological departments only

C. All civil aviation stakeholders including operators, pilots, AMEs, and ATC

Correct Answer: C

90. Which ICAO Annex deals with the licensing of flight crew, ATCOs, and AMEs?

A. Annex 6

B. Annex 1

C. Annex 7

Correct Answer: B

91. The Rules of the Air are contained in—?

A. Annex 12

B. Annex 2

C. Annex 10

Correct Answer: B

92. Which Annex defines the Units of Measurement to be used in air and ground operations?

A. Annex 6

B. Annex 5

C. Annex 13

Correct Answer: B

93. Aircraft Nationality and Registration Marks fall under—?

A. Annex 8

B. Annex 9

C. Annex 7

Correct Answer: C

94. Which Annex defines the standards for Air Traffic Services (ATS)?

A. Annex 2

B. Annex 10

C. Annex 11

Correct Answer: C

95. Standards for Aerodrome design and operation come from—?

A. Annex 15

B. Annex 14

C. Annex 7

Correct Answer: B

96. Annex 10 mainly deals with which aspect of aviation?

A. Aerodrome design

B. Aircraft accident investigation

C. Aeronautical telecommunications

Correct Answer: C

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97. Which Annex 10 volume deals with actual airborne and ground communication systems (e.g., CPDLC, ACARS, SATCOM)?

> A. Volume I
>
> B. Volume III
>
> C. Volume V

Correct Answer: B

98. Which of the following systems is not covered under Annex 10 Volume I?

> A. VOR
>
> B. ILS
>
> C. TCAS

Correct Answer: C

99. CPDLC and ADS-C operational requirements appear under which Annex 10 volume?

> A. Volume III
>
> B. Volume II
>
> C. Volume IV

Correct Answer: A

100. Standard ELT battery endurance is

> A. 12 hours
>
> B. 24 hours
>
> C. 48 hours

Correct Answer: C

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101. Which situation typically triggers an INCERFA?

A. Aircraft declares PAN-PAN

B. Aircraft fails to land within 5 minutes

C. No RT contact and aircraft overdue by more than 30 minutes

Correct Answer: C

102. Which situation qualifies as DETRESFA?

A. Aircraft cannot establish initial contact

B. Fuel on board considered to be exhausted

C. Aircraft 10 minutes late in arrival

Correct Answer: B

103. The 4-letter ICAO location indicator for Indian stations always begins with which letter?

A. M

B. K

C. V

Correct Answer: C

104. What does the second letter of an ICAO location indicator represent?

A. AFSRA zone

B. Country / FIR

C. Station name

Correct Answer: B

105. The last two letters of a 4-letter ICAO location indicator denote

A. FIR

B. Actual aerodrome/station

C. Chart identification

Correct Answer: B

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106. Which organization publishes AERADIO containing Indian location indicators?

A. DGCA

B. AAI

C. IATA

Correct Answer: B

107. Location indicators are also published internationally in

A. ICAO Annex 10

B. ICAO Doc 7910

C. ICAO Doc 7030

Correct Answer: B

108. India has been divided into how many aeronautical regions for location indicator purposes?

A. 3

B. 6

C. 4 (Mumbai, Kolkata, Delhi, Chennai)

Correct Answer: C

109. Density altitude represents

A. The actual altitude of the aircraft

B. Altitude corrected for wind

C. The altitude in Standard Atmosphere corresponding to the observed density

Correct Answer: C

110. High density altitude means

A. High air density and better aircraft performance

B. Low air density and reduced aircraft performance

C. Low altitude and high pressure

Correct Answer: B

111. What is the maximum penalty for unauthorized use of radio communication equipment, as per the Indian Telegraph Act, 1885?

> A. Fine up to $1,000
>
> B. Imprisonment for up to 6 months or fine, or both.
>
> C. Imprisonment for up to 3 years or fine, or both.

```
Correct Answer: C
```

112. What is the ICAO standard pronunciation for the digit 9?

> A. NINE
>
> B. NINER
>
> C. NINAH

```
Correct Answer: B
```

113. What is the recommended range of speech rate (words per minute) for routine radio communication?

> A. 50 to 80 words per minute
>
> B. Not exceeding 100 words per minute
>
> C. 120 to 150 words per minute

```
Correct Answer: B
```

114. How are flight levels (FLs) above the Transition Altitude normally transmitted by radio?

> A. In feet above mean sea level (MSL).
>
> B. As the full number followed by 'THOUSAND FEET'.
>
> C. As the words FLIGHT LEVEL followed by the individual digits.

```
Correct Answer: C
```

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AIS, DOCUMENTS & MESSAGING

115. Why are standard words and phrases used in radiotelephony?

> A. To save time
>
> B. For formality
>
> C. To eliminate ambiguity and misunderstanding

```
Correct Answer: C
```

116. Regional Supplementary Procedures (SUPPS) are published in which ICAO document?

> A. Doc 4444
>
> B. Doc 7030/4
>
> C. Doc 8168

```
Correct Answer: B
```

117. What is the approval authority for Regional Supplementary Procedures (SUPPS)?

> A. National aviation authorities
>
> B. ICAO Secretariat
>
> C. ICAO Council

```
Correct Answer: C
```

118. SUPPS contain procedures that supplement which type of procedures?

> A. Military procedures
>
> B. Airport firefighting SOPs
>
> C. Worldwide air navigation procedures

```
Correct Answer: C
```

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119. Which of the following publications is issued by AAI under Aeronautical Information Services (AIS)?

A. ATPL question paper

B. Aviation fuel circular

C. AIP, NOTAMs, AIRACs, PIBs

Correct Answer: C

120. What type of information is published in the Aeronautical Information Publication (AIP)?

A. Temporary and short-term information

B. Long-lasting information essential to air navigation

C. Real-time weather updates

Correct Answer: B

121. Which document contains information that does NOT qualify for a NOTAM or inclusion in the AIP?

A. PIB

B. AIC

C. AIRAC

Correct Answer: B

122. Which of the following best describes a NOTAM?

A. Regular updates of ATC staffing

B. Telecommunicated notice of any change in facility, service, or hazard

C. Annual publication of navigation charts

Correct Answer: B

123. AIRAC is a system used for—?

A. Publishing daily weather forecasts

B. Emergency ATC coordination

C. Providing advance notice of significant changes on fixed common effective dates

Correct Answer: C

124. What is the primary purpose of an AIC (Aeronautical Information Circular)?

A. To replace NOTAMs

B. To provide non-urgent information relating to safety, administrative, or technical matters

C. To publish time-sensitive NOTAMs

Correct Answer: B

125. A NOTAM should be originated when the information—?

A. Has no operational significance

B. Can wait for the next AIC

C. Is critical and time-sensitive for flight operations

Correct Answer: C

126. Aeronautical Information Services (AIS) is included in—?

A. Annex 4

B. Annex 3

C. Annex 15

Correct Answer: C

127. What is the ICAO document number assigned to PANS-ATM?

A. Document 8168

B. Document 4444

C. Document 7300

Correct Answer: B

128. What is the total number of ITU member states?

A. 160

B. 193

C. 205

Correct Answer: B

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129. Aircraft operating within, to, or through an ADIZ must obtain which clearance prior to departure?

> A. ATC route clearance
>
> B. Flight Information Clearance
>
> C. Air Defence Clearance (ADC)

```
Correct Answer: C
```

130. Communication Procedures including radiotelephony phraseology are contained in which volume of Annex 10?

> A. Volume V
>
> B. Volume II
>
> C. Volume I

```
Correct Answer: B
```

131. Standard radiotelephony phraseology used for all RT communication is published in which volume?

> A. Volume II
>
> B. Volume I
>
> C. Volume V

```
Correct Answer: A
```

132. What is the full pronunciation for the number 5 when transmitting a series of digits in aeronautical mobile service communications?

> A. FIVE
>
> B. FIFE
>
> C. FIV

```
Correct Answer: B
```

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133. The time transmitted in the aeronautical mobile service should be given to the nearest

> A. Hour
>
> B. 10 minutes
>
> C. Minute

```
Correct Answer: C
```

134. The call sign of the ground station should normally appear at which point in a routine communication initiated by the aircraft?

> A. It is stated only at the end of the transmission.
>
> B. It is stated immediately following the aircraft's own call sign.
>
> C. It is stated first, before the aircraft's own call sign.

```
Correct Answer: C
```

135. Which of the following is the standard phraseology for saying 'Yes' or 'That is correct'?

> A. ROGER
>
> B. CONFIRM
>
> C. AFFIRM

```
Correct Answer: C
```

136. Which of the following ATC instructions are always mandatory readback items?

> A. Frequency change instructions.
>
> B. Taxi instructions.
>
> C. Altitude and flight level instructions are always mandatory readback items.

```
Correct Answer: C
```

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137. What is the correct way to express the time 0930 Coordinated Universal Time (UTC) to a ground station?

> A. ZERO NINER THREE ZERO ZULU
>
> B. ZERO NINER THREE ZERO HOURS
>
> C. TIME ZERO NINER THREE ZERO

Correct Answer: C

138. What is the correct phraseology for requesting the magnetic heading to be flown to reach the station?

> A. REQUEST QDR
>
> B. REQUEST QDM
>
> C. REQUEST QNH

Correct Answer: B

139. When an aircraft's assigned altitude or flight level is not to be read back, what phraseology should the controller use?

> A. ROGER
>
> B. READBACK NOT REQUIRED
>
> C. Altitude and flight level instructions are always mandatory readback items.

Correct Answer: C

140. What is the standard ICAO phonetic word for the letter 'X'?

> A. Xavier
>
> B. X-ray
>
> C. Xylophone

Correct Answer: B

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141. The procedural word (proword) used by an ATC unit to indicate a mistake has been made in the transmission and that the corrected version will follow immediately is

> A. NEGAT
>
> B. CORRECTION
>
> C. ERROR

```
Correct Answer: B
```

142. If an ATC unit transmits 'PASS YOUR MESSAGE', what does this request imply?

> A. The aircraft should now change frequency.
>
> B. The controller is ready to receive the full content of the previous message.
>
> C. The aircraft should stand by and wait for instructions.

```
Correct Answer: B
```

143. The safety signal SECURITÉ (spoken three times) has priority over which other class of message?

> A. Distress messages (MAYDAY)
>
> B. Urgency messages (PAN-PAN)
>
> C. All other communications except distress and urgency.

```
Correct Answer: C
```

144. What is the standard ICAO phonetic word for the number 3?

> A. THREE
>
> B. TREE
>
> C. TREE

```
Correct Answer: C
```

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145. The phonetic spelling for the letter 'V' in the ICAO phonetic alphabet is

> A. Vessel
>
> B. Volta
>
> C. Victor

```
Correct Answer: C
```

146. What is the proper phraseology for transmitting the number '10,000' feet in a standard radio communication?

> A. TEN THOUSAND
>
> B. ONE ZERO ZERO ZERO ZERO FEET
>
> C. ONE ZERO THOUSAND

```
Correct Answer: C
```

147. What is the minimum required signal strength (readability) for a successful two-way radiotelephone communication in the aeronautical mobile service?

> A. Readability 1 (Unreadable)
>
> B. Readability 2 (Readable now and then)
>
> C. Readability 3 (Readable, but with difficulty)

```
Correct Answer: C
```

148. In a situation involving a fire onboard the aircraft that is NOT immediately life-threatening but requires urgent action, which signal should the pilot transmit before the message?

> A. MAYDAY
>
> B. EMERGENCY
>
> C. PAN-PAN

```
Correct Answer: C
```

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149. If a ground station sends a message and receives no reply, what is the maximum number of times it may repeat the call sign of the station called, before giving the message 'TO ALL STATIONS'?

> A. Once
>
> B. Twice
>
> C. Three times

```
Correct Answer: C
```

150. When an Air Traffic Control (ATC) unit needs to cancel a previously issued clearance, which proword should they use to introduce the cancellation?

> A. DISREGARD
>
> B. CANCEL
>
> C. NEGATIVE

```
Correct Answer: B
```

151. What is the correct phraseology to indicate that the entire last message has been received and understood, and the instruction contained therein will be executed.

> A. ROGER
>
> B. AFFIRM
>
> C. WILCO

```
Correct Answer: C
```

152. According to the standard readability scale used in aeronautical radio communication, which rating signifies 'Readable but with difficulty'?

> A. Readability 1
>
> B. Readability 2
>
> C. Readability 3

```
Correct Answer: C
```

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153. What time standard is mandatory for use in all aeronautical mobile service communications?

> A. Local Time, specified by the controlling station.
>
> B. Coordinated Universal Time (UTC), often referred to as 'Zulu' time.
>
> C. Greenwich Mean Time (GMT).

```
Correct Answer: B
```

154. Which procedural word (proword) is used by a radiotelephone operator to indicate that a separation is made between portions of a message?

> A. OVER
>
> B. BREAK
>
> C. STANDBY

```
Correct Answer: B
```

155. What is the highest precedence category for a message transmitted in the aeronautical mobile service?

> A. Urgency messages, such as those concerning serious technical difficulties.
>
> B. Safety messages, such as those relating to important flight conditions.
>
> C. Distress messages, indicated by MAYDAY.

```
Correct Answer: C
```

156. What is the primary characteristic of the spoken word 'ALL STATIONS' when preceding a transmission from an Air Traffic Control (ATC) unit?

> A. The message is of a confidential nature and only authorized personnel should should listen.
>
> B. The message is directed to all aircraft stations within the receiving range of of the frequency or unit.
>
> C. It is a request for a radio check from every aircraft on the frequency.

```
Correct Answer: B
```

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157. Which information is mandatory to be given to ATC for pushback and engine start?

A. Route and cruising level

B. Fuel on board

C. Aircraft call sign, stand, request for pushback/engine start, ATIS

Correct Answer: C

158. What is the correct order of priority in radiotelephony?

A. Urgency → Distress → Flight safety

B. Regularity → Meteorological → Distress

C. Distress → Urgency → Flight safety → Meteorological → Regularity

Correct Answer: C

159. What is the correct order of a visual circuit pattern?

A. Downwind → Base → Crosswind → Final

B. Upwind → Crosswind → Downwind → Base → Final

C. Crosswind → Upwind → Final → Base

Correct Answer: B

160. Which of the following positions are reported in a visual circuit?

A. Taxi and holding point

B. Initial and overhead

C. Upwind, Crosswind, Downwind, Base, Final

Correct Answer: C

161. How is the call sign “AIC 200” read on RT?

A. Air India two zero zero

B. Air India Two Hundred

C. Air India two double zero

Correct Answer: A

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162. How much time before entering controlled airspace should clearance be requested?

> A. 5 minutes
>
> B. 10 minutes
>
> C. 15 minutes

```
Correct Answer: B
```

163. If transmitter power is increased four times, the range will

> A. Remain same
>
> B. Increase 4 times
>
> C. Double

```
Correct Answer: C
```

164. What is the correct term meaning “I have received your transmission”?

> A. Wilco
>
> B. Affirm
>
> C. Roger

```
Correct Answer: C
```

165. Which of the following is NOT an essential piece of information for a standard Position Report?

> A. Time over the reporting point.
>
> B. Position/Identification of the reporting point.
>
> C. Fuel on board in gallons.

```
Correct Answer: C
```

166. Define the term Readability Code 3.

> A. Unreadable
>
> B. Readable now and then
>
> C. Readable, but with difficulty

```
Correct Answer: C
```

167. How must the number '12,500' be spoken in radiotelephony?

A. ONE TWO FIVE ZERO ZERO

B. ONE TWO THOUSAND FIVE HUNDRED

C. TWELVE THOUSAND FIVE HUNDRED

Correct Answer: B

168. Which instruction from ATC must be read back by the pilot?

A. Weather conditions (METAR).

B. ATC route clearances.

C. Advisory information.

Correct Answer: B

169. What is the full phraseology for cancelling an urgent message if the situation has improved?

A. PAN-PAN TERMINATED

B. CANCELLING URGENCY

C. PAN-PAN OVER

Correct Answer: B

170. Under what condition is an aircraft permitted to use a shortened callsign?

A. The aircraft initiates the shortening after the first successful contact.

B. When the pilot deems the radio channel to be non-congested.

C. After the ground station (ATC) has initiated the use of the shortened callsign.

Correct Answer: C

171. If a message is classified as a Safety message, what does it primarily concern?

A. The safety of the destination runway.

B. The safety of navigation or the significant weather warnings.

C. The regular operational movements of the aircraft.

Correct Answer: B

172. What is the recommended maximum duration for a single transmission on the air?

> A. No more than 60 seconds.
>
> B. No more than 30 seconds.
>
> C. Only enough time to say the critical information.

Correct Answer: C

173. Security — safeguarding civil aviation against unlawful interference is covered under—?

> A. Annex 12
>
> B. Annex 17
>
> C. Annex 19

Correct Answer: B

174. Which ICAO Annex deals with the performance of distress monitoring, communication, coordination, and search and rescue?

> A. Annex 11
>
> B. Annex 12
>
> C. Annex 13

Correct Answer: B

175. The primary function described—distress monitoring, coordination, and medical evacuation— falls under

> A. ATS services
>
> B. AIS services
>
> C. Search and Rescue (SAR) services

Correct Answer: C

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176. In SAR, distress communication involves

A. Only VFR aircraft

B. Handling and relaying distress messages from any aircraft in need of assistance

C. Only communication between ATC units

Correct Answer: B

177. The term "distress monitoring" refers to

A. Monitoring runway lighting

B. Monitoring ATIS broadcasts

C. Monitoring and responding to distress alerts from aircraft

Correct Answer: C

178. Which of the following is NOT an ELT distress frequency?

A. 121.5 MHz

B. 118.1 MHz

C. 406 MHz

Correct Answer: B

179. Marine distress frequency associated with ELT-type beacons is

A. 3270 kHz

B. 2182 kHz

C. 1720 kHz

Correct Answer: B

180. The distress signal 'MAYDAY' must be preceded by which alerting call?

A. PAN-PAN (three times)

B. MAYDAY RELAY (three times)

C. MAYDAY (three times)

Correct Answer: C

181. What is the primary action required by a pilot who hears an urgency call (PAN-PAN) on the frequency?

> A. Immediately change frequency to clear the channel.
>
> B. Continue routine transmission, as it is only an urgency call.
>
> C. Cease any transmission likely to interfere with the urgency communication.

```
Correct Answer: C
```

182. The transmission of the distress message (MAYDAY message) should contain which three mandatory pieces of information, immediately after the distress signal?

> A. Aircraft color, number of people on board, and destination.
>
> B. Position, nature of distress, and pilot's name.
>
> C. Aircraft call sign, position, and nature of distress.

```
Correct Answer: C
```

183. The distress signal MAYDAY, when spoken three times, indicates what level of urgency for the aircraft and its occupants?

> A. An urgent situation concerning the safety of an aircraft or other vehicle, or of
>
> B. The aircraft is experiencing a minor technical fault that requires attention but is
>
> C. The aircraft is threatened by grave and imminent danger and requires

```
Correct Answer: C
```

184. What does the urgency signal “PAN PAN” indicate?

> A. Immediate danger requiring assistance
>
> B. Safety of aircraft/person on board, not requiring immediate assistance
>
> C. Radio failure

```
Correct Answer: B
```


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185. What is the international distress frequency?

A. 123.45 MHz

B. 118.00 MHz

C. 121.5 MHz

Correct Answer: C

186. What is the internationally recognized Distress signal?

A. PAN-PAN (spoken three times)

B. SECURITE (spoken three times)

C. MAYDAY (spoken three times)

Correct Answer: C

187. What is the internationally recognized Urgency signal?

A. PAN-PAN (spoken three times)

B. MAYDAY (spoken three times)

C. EMERGENCY (spoken three times)

Correct Answer: A

188. What action must an aircraft take immediately upon hearing the signal 'MAYDAY' on a frequency it is monitoring?

A. Reply to the distressed aircraft to offer assistance.

B. Cease all transmissions immediately and maintain a listening watch.

C. Transmit "SEELONCE FEENEE" to clear the frequency.

Correct Answer: B

189. Which are the two highest priority messages after Distress and Urgency calls?

A. Movement and Safety.

B. Meteorological and Regularity.

C. Safety and Regularity.

Correct Answer: C

190. What is the meaning of the phrase "SQUAWK 7700"?

A. Radio Communication Failure

B. Unlawful Interference

C. General Emergency

Correct Answer: C

METEOROLOGY, PRESSURE & RVR

191. What is the meaning of the phrase "SQUAWK 7600"?

A. General Emergency

B. Unlawful Interference

C. Radio Communication Failure

Correct Answer: C

192. ICAO is a specialized agency of which global organization?

A. International Court of Justice

B. World Trade Organization

C. United Nations

Correct Answer: C

193. CAR Section 2, Series ‘O’ specifically relates to—?

A. Operating minima for airports

B. Operational requirements for aircraft

C. Cabin safety regulations

Correct Answer: B

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194. Aeronautical Charts are published as per specifications in—?

A. Annex 4

B. Annex 11

C. Annex 15

Correct Answer: A

195. Operational temperature range of an ELT is

A. 0°C to 40°C

B. –20°C to +55°C

C. –40°C to +80°C

Correct Answer: B

196. In a METAR, the code VRB is used when

A. Wind speed is more than 15 kt

B. Wind direction cannot be determined

C. Winds are calm

Correct Answer: B

197. VRB appears in a METAR instead of wind direction when

A. Surface wind is above 20 kt

B. Wind is calm

C. Wind direction fluctuates so much that no definite direction exists

Correct Answer: C

198. A gust is defined as a sudden increase in wind speed lasting

A. More than 5 minutes

B. Less than one minute

C. Exactly one minute

Correct Answer: B

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199. For a gust to be included in a METAR/TAF, it must be at least

A. 5 kt above mean wind

B. 8 kt above mean wind

C. 10 kt above mean wind

Correct Answer: C

200. In METAR, how is a gust reported?

A. Appended after visibility

B. Shown as a separate line

C. Added after mean wind speed with letter G

Correct Answer: C

201. Example: METAR showing 12010G25KT indicates

A. Wind from 120° at 25 kt

B. Wind from 120° at 10 kt, gusting to 25 kt

C. Wind variable between 10–25 kt

Correct Answer: B

202. What is the standard phraseology for expressing visibility of 10 kilometers or more?

A. VISIBILITY ONE ZERO KILOMETRES

B. VISIBILITY ONE ZERO PLUS

C. VISIBILITY TEN KILOMETRES OR MORE or VISIBILITY MORE THAN NINE

Correct Answer: C

203. The Q-code QFE is used for setting the altimeter to read

A. Altitude above Mean Sea Level (MSL).

B. Pressure altitude (standard pressure 1013.2 hPa).

C. Height above the aerodrome elevation (Zero on the ground at the field

Correct Answer: C

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204. Which procedural word (proword) is used by a ground station to indicate that the aircraft is authorized to proceed under the conditions specified?

> A. STANDBY
>
> B. CLEARED
>
> C. ROGER

Correct Answer: B

205. What is the correct meaning of the Q-code QNH?

> A. The atmospheric pressure at aerodrome elevation.
>
> B. The atmospheric pressure at mean sea level (MSL) used for setting the
>
> C. The magnetic bearing from the station to the aircraft.

Correct Answer: B

206. What is the standard phraseology used by a mobile station to request the atmospheric pressure setting that will cause the altimeter to indicate elevation above the aerodrome reference point?

> A. REQUEST QNH
>
> B. REQUEST QFE
>
> C. REQUEST ELEVATION

Correct Answer: B

207. ATC gives an altimeter setting of 1007. How is it read?

> A. One zero zero seven
>
> B. QNH one zero zero seven
>
> C. One thousand seven

Correct Answer: B

208. The meaning of the abbreviation QNH is

A. Altimeter setting to read height above the aerodrome.

B. Altimeter setting to read height above Mean Sea Level (MSL).

C. The standard pressure of 1013.2 hPa.

Correct Answer: B

209. The meaning of the abbreviation QFE is

A. Altimeter setting to read height above MSL.

B. Standard atmospheric pressure setting.

C. Altimeter setting to read zero when on the ground (height above aerodrome elevation).

Correct Answer: C

210. What altimeter setting must be used when operating at or above the Transition Level (TL)?

A. QFE

B. QNH of the destination aerodrome

C. The standard pressure setting (1013.2 hPa)

Correct Answer: C

211. What is the full form of the abbreviation METAR and what information does it contain?

A. Medium Terminal Advisory Report; forecast weather.

B. Aviation Routine Weather Report; actual weather observation.

C. Meteorological Terminal Area Report; general climate info.

Correct Answer: B

212. Who grants Air Operator's Certificates (AOC) to Indian carriers?

A. BCAS

B. AAI

C. DGCA

Correct Answer: C

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213. Why is a single HF frequency NOT suitable for 24-hour aircraft communication?

A. HF radios cannot operate continuously

B. HF propagation varies due to day/night effects

C. VHF interference blocks HF signals

Correct Answer: B

214. Due to propagation problems on HF, aircraft are allotted—?

A. Only one HF channel

B. Two frequencies: Primary and Secondary

C. Three frequencies for every phase of flight

Correct Answer: B

215. The selection of primary and secondary HF frequencies depends mainly on—?

A. Pilot's preference

B. Time of departure, nature of flight (domestic/international), and route/area of flight

C. Colour of aircraft

Correct Answer: B

216. The system that allows ground stations to alert crew without constant listening on HF is—?

A. ADS-B

B. SELCAL

C. ELT

Correct Answer: B

217. HF propagation issues requiring two frequencies are mainly caused by—?

A. Airport congestion

B. Rain and snow only

C. Day and night variation in the ionosphere

Correct Answer: C

218. HF is mainly used along with SELCAL because—?

A. HF audio is extremely clear

B. Continuous listening on HF is difficult due to static and noise

C. HF requires no tuning

Correct Answer: B

219. In long-range communication, the secondary HF frequency is used when—?

A. The primary frequency is too busy

B. Pilot requests music

C. Propagation on the primary frequency is poor

Correct Answer: C

220. The primary purpose of SELCAL in aircraft communication is to—?

A. Improve HF signal strength

B. Alert pilots without continuous radio monitoring

C. Replace HF radios

Correct Answer: B

221. SELCAL is mainly used to reduce pilot workload because—?

A. HF is never used anymore

B. Continuous monitoring of HF causes static noise and fatigue

C. VHF is always unavailable

Correct Answer: B

222. SELCAL is a system used for communication—?

A. Aircraft-to-aircraft

B. Ground-to-air only

C. Satellite-to-air

Correct Answer: B

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223. In India, SELCAL signals are transmitted over—?

A. VHF only

B. HF and VHF

C. HF only

Correct Answer: C

224. A SELCAL code consists of—?

A. Four digits

B. Four letters

C. Two letters and two numbers

Correct Answer: B

225. SELCAL letters may include—?

A. A to Z

B. Only vowels

C. A to S, excluding I, O, N

Correct Answer: C

226. A valid SELCAL code must contain—?

A. Repeated letters

B. One pair of tones

C. Two pairs of letters in ascending order

Correct Answer: C

227. Which of the following SELCAL codes is correctly formed?

A. BP CO

B. AB CD

C. GG HI

Correct Answer: B

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228. How long does a complete SELCAL transmission take?

A. 5 seconds

B. Approximately 2.2 seconds

C. 10 seconds

Correct Answer: B

229. The pause between the two SELCAL tone pairs is—?

A. 1 second

B. 0.2 seconds

C. 2 seconds

Correct Answer: B

230. SELCAL cannot include repeated letters because—?

A. HF cannot transmit repeating tones

B. Each tone must be unique for identification

C. ICAO does not allow consonants

Correct Answer: B

231. A SELCAL code like “AD BC” is—?

A. Incorrect because D is not allowed

B. Incorrect because pairs must be alphabetical

C. Correct, because both pairs are in ascending order

Correct Answer: C

232. What happens in the cockpit when a valid SELCAL tone is received?

A. HF becomes unreadable

B. A light flashes or chime sounds to alert the crew

C. VHF disconnects

Correct Answer: B

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233. SELCAL-32 was introduced mainly to—?

A. Increase HF transmission power

B. Reduce cockpit noise

C. Increase the total number of available SELCAL codes

Correct Answer: C

234. SELCAL-32 allows the use of which additional characters?

A. W, X, Y only

B. T to Z, and digits 1 to 9

C. All letters A–Z including I and O

Correct Answer: B

235. Traditional SELCAL used only letters from—?

A. A to Z

B. A to F

C. A to S (excluding I, O, N)

Correct Answer: C

236. SELCAL-32 increases the number of possible SELCAL combinations to—?

A. 10,000

B. 1 million

C. More than 2 lakh unique codes

Correct Answer: C

237. What appears in the cockpit when a SELCAL alert is received?

A. Radio volume automatically increases

B. A chime sounds and a SELCAL light illuminates

C. The HF radio switches off

Correct Answer: B

238. Where is the SELCAL controller located?

A. In the avionics bay

B. On the flight deck

C. In the cabin

Correct Answer: B

239. After the crew responds to a SELCAL alert, they must—?

A. Change HF frequency

B. Press the reset button to arm the system for the next call

C. Switch to VHF communication

Correct Answer: B

240. Why were digits 1–9 added in SELCAL-32?

A. To simplify tone generation

B. To greatly expand the number of assignable SELCAL codes

C. To avoid confusion with Morse code

Correct Answer: B

241. In SELCAL-32, which of the following is a valid component?

A. Letter “O”

B. Number “7”

C. Letter “I”

Correct Answer: B

242. What is the operational advantage of SELCAL for the flight crew?

A. It eliminates the need for HF radios

B. Crew do not need to continuously monitor HF manually

C. It replaces CPDLC on oceanic routes

Correct Answer: B

243. Which organization handles the worldwide registration of SELCAL codes?

A. ICAO

B. Aviation Spectrum Resources

C. AAI

Correct Answer: B

244. Registration of SELCAL codes with Aviation Spectrum Resources is—?

A. Free of cost

B. Included with aircraft registration

C. A paid service

Correct Answer: C

245. SELCAL codes are assigned to—?

A. Aircraft operators

B. Individual aircraft

C. Flight crew members

Correct Answer: B

246. In a flight plan, where is the SELCAL code entered?

A. Item 15

B. Item 10

C. Item 18 using the “SEL/” indicator

Correct Answer: C

247. The purpose of Aviation Spectrum Resources maintaining SELCAL registration is to—?

A. Allocate HF frequencies

B. Monitor ATC radio calls

C. Ensure global uniqueness and avoid duplicate SELCAL codes

Correct Answer: C

248. When is a SELCAL code typically registered for an operator?

A. After every flight

B. At the time of acquiring or adding aircraft to their fleet

C. Before each departure

Correct Answer: B

249. Why is SELCAL registration important?

A. To increase aircraft MTOW

B. To prevent two aircraft from having the same SELCAL tones

C. To allow CPDLC operations

Correct Answer: B

250. Utilization and protection of the aeronautical radio frequency spectrum is covered in which volume of Annex 10?

A. Volume I

B. Volume V

C. Volume IV

Correct Answer: B

251. Frequency allocation rules ensuring no harmful interference in aeronautical bands are described in which Annex 10 volume?

A. Volume V

B. Volume IV

C. Volume I

Correct Answer: A

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252. Which frequency band is primarily used for long-distance, high-frequency (HF) aeronautical communications?

> A. VHF (Very High Frequency)
>
> B. UHF (Ultra High Frequency)
>
> C. HF (High Frequency)

```
Correct Answer: C
```

253. What is the maximum effective communication range of standard VHF (Very High Frequency) air- ground communication, which is essentially line-of-sight?

> A. 50 nautical miles (NM)
>
> B. Approximately 200 nautical miles (NM) at high altitudes.
>
> C. 2,000 nautical miles (NM)

```
Correct Answer: B
```

254. What is the correct phraseology ATC should use to instruct a pilot to change frequency from the current controlling agency to another?

> A. CHANGE TO NEW FREQUENCY
>
> B. REQUEST CONTACT
>
> C. CONTACT [CONTROL UNIT] [FREQUENCY]

```
Correct Answer: C
```

255. If impedance is reduced, what happens to current?

> A. Decreases
>
> B. Becomes zero
>
> C. Increases

```
Correct Answer: C
```

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256. What is wavelength?

A. Speed of wave

B. Distance between two successive crests or troughs

C. Number of cycles per second

Correct Answer: B

257. What is frequency?

A. Distance between waves

B. Wave speed

C. Number of cycles per second

Correct Answer: C

258. What is the wavelength of a 300 MHz frequency?

A. 3 metres

B. 30 cm

C. 1 metre

Correct Answer: C

259. What is the inter-pilot air-to-air communication frequency?

A. 121.5 MHz

B. 122.8 MHz

C. 123.45 MHz

Correct Answer: C

260. What is the standard VHF Emergency Frequency?

A. 121.5 MHz

B. 123.45 MHz

C. 118.0 MHz

Correct Answer: A

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261. What is the correct phraseology for requesting permission to change frequency?

A. REQUEST CHANGE TO [frequency/station].

B. CLEARED TO CHANGE FREQUENCY.

C. ROGER CHANGE FREQUENCY.

Correct Answer: A

262. What is the primary purpose of Selective Calling (SELCAL)?

A. To automatically select the best radio channel.

B. To provide an alternate communication means during RCF.

C. To alert a specific aircraft crew to an incoming radio call.

Correct Answer: C

263. Which radio wave property allows HF to travel long distances?

A. Line-of-sight propagation.

B. Skywave reflection off the Ionosphere.

C. Ground wave following the Earth's curvature.

Correct Answer: B

264. Why is VHF communication primarily line-of-sight?

A. The transmitter power is very low.

B. VHF waves do not bend significantly around the Earth's curvature or reflect off the ionosphere.

C. The frequency is too low.

Correct Answer: B

265. What does the term 'Simplex' operation mean in radiotelephony?

A. Two frequencies used simultaneously for both transmission and reception.

B. A single frequency used for both transmission and reception, but only one station transmits at a time.

C. The ability to transmit and receive simultaneously on a single frequency.

Correct Answer: B

266. What type of modulation is used for standard VHF voice communication in aviation?

> A. Amplitude Modulation (AM).
>
> B. Frequency Modulation (FM).
>
> C. Single Sideband (SSB).

```
Correct Answer: A
```

267. What is the function of the Squelch circuit in a radio receiver?

> A. To increase the volume of weak signals.
>
> B. To tune the receiver to the exact frequency.
>
> C. To suppress background receiver noise when no signal is present.

```
Correct Answer: C
```

268. The relationship between Wavelength (\lambda) and Frequency (f) is defined by

> A. \lambda = f / c
>
> B. \lambda = f \times c
>
> C. \lambda = c / f (where c is the speed of light)

```
Correct Answer: C
```

269. What is the primary advantage of using the Single Sideband (SSB) technique in HF communication?

> A. Superior clarity and audio fidelity.
>
> B. Elimination of all static and noise.
>
> C. Increased range and efficiency by reducing bandwidth and wasted power.

```
Correct Answer: C
```

270. The process of separating the voice signal from the carrier wave in the receiver is called

> A. Modulation
>
> B. Demodulation
>
> C. Rectification

```
Correct Answer: B
```

271. The most common radio wave propagation mode for VHF is

A. Skywave.

B. Surface wave.

C. Direct/Line-of-sight.

Correct Answer: C

272. The term 'Dead Zone' is a problem primarily associated with which frequency band?

A. VHF

B. UHF

C. HF

Correct Answer: C

273. The speed of a radio wave in a vacuum is approximately

A. 300 meters per second

B. 300 kilometers per second

C. 300,000 kilometers per second

Correct Answer: C

274. What is the primary limitation of VHF (Very High Frequency) radio for long-distance communication?

A. Susceptibility to sunspot activity.

B. Limited frequency bandwidth.

C. Line-of-sight propagation.

Correct Answer: C

275. The process of modifying a high-frequency carrier wave with a low-frequency signal (voice) is called

A. Demodulation

B. Coupling

C. Modulation

Correct Answer: C

276. Which band of frequencies is most affected by the changes in the Ionosphere between day and night?

A. VHF

B. UHF

C. HF

Correct Answer: C

277. What is a Transceiver?

A. A device that only transmits radio signals.

B. A device that only receives radio signals.

C. A unit combining a transmitter and a receiver.

Correct Answer: C

278. In the formula for wavelength (\lambda) and frequency (f), \lambda = c/f, what does 'c' represent?

A. Capacitance

B. Speed of the aircraft

C. Speed of light (radio wave)

Correct Answer: C

279. The term 'Dead Zone' or 'Skip Zone' is associated with which type of radio wave propagation?

A. Ground wave

B. Line-of-sight

C. Skywave (HF)

Correct Answer: C

280. Which type of antenna is commonly used for VHF communication on aircraft?

A. Long wire antenna

B. Blade or whip antenna

C. Loop antenna

Correct Answer: B

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281. What is the standard audio frequency range for human speech transmitted in radiotelephony?

A. 100 kHz to 300 kHz

B. 300 Hz to 3400 Hz

C. 10 kHz to 20 kHz

Correct Answer: B

282. What does AM stand for in the context of radio modulation?

A. Antenna Modulation

B. Amplitude Modulation

C. Analog Modulation

Correct Answer: B

283. The main advantage of using HF SSB (Single Sideband) for long-range communication over traditional AM is

A. Lower bandwidth and higher range for the same power.

B. Higher fidelity and clearer audio.

C. It is immune to lightning and atmospheric static.

Correct Answer: A

284. In an aircraft's VHF radio system, the inter-communication system used for communication between the cockpit and the cabin is called the

A. ATC Radio

B. Interphone or Cabin Announcing System

C. HF Radio

Correct Answer: B

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285. DGCA must align its rules and requirements with amendments made to what?

A. Airport Authority manuals

B. Airline SOPs

C. ICAO Annexes

Correct Answer: C

286. AAI provides which communication and navigation aids?

A. SONAR, GPS, HF-SDR

B. ILS, DVOR, DME, Radar

C. GPWS, TCAS, ELT

Correct Answer: B

287. Radio Navigational Aids such as VOR, DME, and ILS fall under which volume of Annex 10?

A. Volume I

B. Volume IV

C. Volume V

Correct Answer: A

288. Surveillance Radar and Collision Avoidance Systems (SSR, Mode S, TCAS) are included in which volume?

A. Volume II

B. Volume V

C. Volume IV

Correct Answer: C

289. Accuracy of modern 406 MHz ELT with GPS (NG ELT) is

A. 20 km

B. 5 km

C. 500 meters

Correct Answer: C

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290. In the event of a total communication failure (radio failure) on an IFR flight, the pilot should first set the transponder to which emergency code?

> A. 7500
>
> B. 7600
>
> C. 7700

```
Correct Answer: B
```

291. When a radiotelephone operator is told to 'STANDBY', what is the expected action?

> A. Immediately change frequency as instructed.
>
> B. End the transmission with 'OUT' and await further instructions.
>
> C. Wait for a short but unspecified period for a reply.

```
Correct Answer: C
```

292. DME operates in which frequency band?

> A. VHF
>
> B. HF
>
> C. UHF

```
Correct Answer: C
```

293. If experiencing RCF in IMC, the pilot should immediately set the transponder to which code?

> A. 7700
>
> B. 7600
>
> C. 7500

```
Correct Answer: B
```

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294. What is the function of the Transponder in air traffic control?

A. To transmit radio messages to ATC.

B. To reply to ground radar interrogation with a code and altitude information.

C. To receive all ATC instructions and display them digitally.

Correct Answer: B

295. Who is responsible for issuing Certificates of Airworthiness in India?

A. AAI

B. MoCA

C. DGCA

Correct Answer: C

296. Which authority in India investigates civil aviation accidents and incidents?

A. NTSB

B. AAI

C. Aircraft Accident Investigation Bureau (AAIB)

Correct Answer: C

297. Who approves operator training programmes for carriage of dangerous goods?

A. BCAS

B. ICAO

C. DGCA

Correct Answer: C

298. Provision of passenger facilities and information systems at airports falls under which organization?

A. DGCA

B. BCAS

C. AAI

Correct Answer: C

299. Provision of Air Traffic Services (ATS) at airports under its jurisdiction is done by whom?

A. DGCA

B. Indian Air Force

C. AAI

Correct Answer: C

300. What is the primary purpose of an Air Defence Identification Zone (ADIZ)?

A. To provide meteorological updates to aircraft

B. To define airspace where special communication and reporting procedures apply

C. To regulate airport parking stands

Correct Answer: B

301. How many ADIZ regions are designated in India?

A. 3

B. 4

C. 5

Correct Answer: C

302. The Western ADIZ of India lies along which boundary?

A. International border with Nepal

B. International border with Pakistan

C. LOC of China

Correct Answer: B

303. The South ADIZ covers which airspace?

A. Airspace over the Bay of Bengal

B. Airspace over territorial waters of Lakshadweep Islands

C. Airspace over the Rann of Kutch

Correct Answer: B

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304. Before entering an ADIZ from outside Indian FIR, aircraft must obtain ADC at least—?

A. 30 minutes prior to entry

B. 10 minutes prior to entering Indian airspace

C. Only after entering the FIR

Correct Answer: B

305. Which ADIZ lies along the Line of Control (LOC) with Pakistan and China?

A. East ADIZ

B. North ADIZ

C. West ADIZ

Correct Answer: B

306. Which ADIZ is a subordinate extension to the South ADIZ?

A. East ADIZ

B. Central ADIZ

C. South-East ADIZ

Correct Answer: C

307. ADC issued to an aircraft remains valid for—?

A. Only the first FIR

B. Only for the climb phase

C. The entire route

Correct Answer: C

308. The Central ADIZ corresponds to which international boundaries?

A. Bangladesh border only

B. Sri Lanka FIR boundary

C. International border with Nepal and LOC of China

Correct Answer: C

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309. The Pre-Flight Information Bulletin (PIB) contains—?

A. Radar service availability

B. HF/MF communication schedules

C. Current NOTAMs of operational significance

Correct Answer: C

310. Which publication is issued by or with the authority of the State?

A. AIP

B. NOTAM

C. AIRAC

Correct Answer: A

311. If a change in airspace procedures is to be implemented, which system ensures pilots get advance notice?

A. AIC

B. PIB

C. AIRAC

Correct Answer: C

312. PIB is prepared—?

A. After aircraft landing

B. Before the flight

C. Once every AIRAC cycle

Correct Answer: B

313. An aerodrome is defined as—?

A. Only a paved runway for aircraft operations

B. A terminal building for passenger movement

C. A defined area of land/water intended for arrival, departure, and surface movement of aircraft

Correct Answer: C

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314. Which statement correctly describes an aerodrome?

A. It includes land only, not water

B. It may include buildings, installations, and equipment

C. It must have customs and immigration facilities

Correct Answer: B

315. An airport is best described as—?

A. Any open field used for emergency landings

B. A landing/take-off area with runway and passenger facilities such as customs & immigration

C. A location used only for helicopter operations

Correct Answer: B

316. Which of the following is a key difference between an aerodrome and an airport?

A. Airports allow military aircraft only

B. Airports have additional facilities such as maintenance, terminal buildings, customs, and immigration

C. Aerodromes must have more than one runway

Correct Answer: B

317. Which of the following is true?

A. All aerodromes are airports

B. All airports are aerodromes, but not all aerodromes qualify as airports

C. Aerodromes require mandatory customs facilities

Correct Answer: B

318. Which area does the definition of aerodrome include?

A. Only concrete or asphalt surfaces

B. Land or water used for aircraft movements

C. Only areas monitored by ATC

Correct Answer: B

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319. An airport must include—?

A. Cargo handling only

B. Runway + aerodrome facilities + passenger facilities (customs, immigration, terminal)

C. ATC tower only

Correct Answer: B

320. Meteorological Services for International Air Navigation is covered under—?

A. Annex 4

B. Annex 9

C. Annex 3

Correct Answer: C

321. Operation of Aircraft is covered under—?

A. Annex 11

B. Annex 6

C. Annex 1

Correct Answer: B

322. Airworthiness requirements for aircraft are contained in—?

A. Annex 8

B. Annex 6

C. Annex 16

Correct Answer: A

323. Facilitation (customs, immigration, passenger processing) is included in—?

A. Annex 15

B. Annex 9

C. Annex 12

Correct Answer: B

324. Aeronautical Telecommunications is covered under—?

A. Annex 11

B. Annex 10

C. Annex 18

Correct Answer: B

325. Search and Rescue (SAR) procedures are covered under—?

A. Annex 12

B. Annex 17

C. Annex 18

Correct Answer: A

326. Aircraft Accident & Incident Investigation is addressed in—?

A. Annex 16

B. Annex 13

C. Annex 14

Correct Answer: B

327. Environmental protection requirements are detailed in—?

A. Annex 8

B. Annex 16

C. Annex 6

Correct Answer: B

328. The safe transport of dangerous goods by air is covered in—?

A. Annex 10

B. Annex 18

C. Annex 3

Correct Answer: B

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329. Safety Management System (SMS) is part of—?

A. Annex 2

B. Annex 11

C. Annex 19

Correct Answer: C

330. In India, IFR and VFR flights are permitted in which airspace classes?

A. Only Class D and E

B. Only Class C

C. All airspace classes

Correct Answer: C

331. Class C airspace in India provides which type of separation?

A. IFR–IFR only

B. IFR–IFR & IFR–VFR separation

C. IFR receives only traffic information

Correct Answer: B

332. Class D airspace in India is typically associated with which areas?

A. Only ATS routes

B. CTAs, TMAs, CTZs, and ATZs

C. FIR only

Correct Answer: B

333. What services are provided in Class D airspace?

A. IFR receives no separation

B. IFR–IFR separation; IFR–VFR traffic information

C. Only flight information service

Correct Answer: B

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334. Class E airspace in India generally applies to

> A. Inside CTAs and TMAs
>
> B. ATS routes outside CTAs and TMAs
>
> C. Military restricted airspace

```
Correct Answer: B
```

335. What ATC service does Class E airspace provide?

> A. No service to any flight
>
> B. Service only to VFR
>
> C. ATC service to IFR; traffic information to all flights

```
Correct Answer: C
```

336. Class F airspace in India provides which service to IFR flights?

> A. Full ATC separation
>
> B. Advisory service
>
> C. Only alerting service

```
Correct Answer: B
```

337. In Class F airspace, what service is provided to all aircraft upon request?

> A. Radar vectors
>
> B. FIS (Flight Information Service)
>
> C. Procedural separation

```
Correct Answer: B
```

338. Class G airspace in India is defined as

> A. Controlled airspace with VFR restrictions
>
> B. Standby-only airspace
>
> C. Uncontrolled airspace (other than D, E, and F)

```
Correct Answer: C
```

339. In Class G airspace, what service is provided?

A. Full ATC service

B. FIS to all aircraft

C. No services at all

Correct Answer: B

340. Which type of airspace in India does NOT permit flight under any circumstance?

A. Restricted Area

B. Danger Area

C. Prohibited Area

Correct Answer: C

341. A Prohibited Area in India is identified using which format?

A. FIR + (R) + Number

B. FIR + (D) + Number

C. FIR + (P) + Number

Correct Answer: C

342. VA(P)-2 refers to which of the following?

A. Danger Area No. 2 in Delhi FIR

B. Prohibited Area No. 2 in Mumbai FIR

C. Restricted Area No. 2 in Mumbai FIR

Correct Answer: B

343. Which statement correctly describes a Restricted Area?

A. Flights are always allowed without clearance.

B. Flights are allowed but with restrictions as notified.

C. Flights are never allowed.

Correct Answer: B

344. Restricted Areas in India are marked with which letter?

A. D

B. R

C. P

Correct Answer: B

345. VO(R)-81 indicates which of the following?

A. Danger Area No. 81 in Chennai FIR

B. Restricted Area No. 81 in Chennai FIR

C. Prohibited Area No. 81 in Chennai FIR

Correct Answer: B

346. A Danger Area contains what type of activity?

A. ATC training

B. Activities dangerous to aircraft operations

C. VIP movement only

Correct Answer: B

347. When must aircraft avoid a Danger Area?

A. At all times

B. Never

C. When the area is active, as notified by NOTAM

Correct Answer: C

348. VE(D)-72 refers to what?

A. Restricted Area in Kolkata FIR

B. Prohibited Area in Kolkata FIR

C. Danger Area No. 72 in Kolkata FIR

Correct Answer: C

349. What is the purpose of designating a Danger Area?

A. To legally ban all flights permanently

B. To allow only military aircraft

C. To caution pilots to avoid the area during active dangerous activities

Correct Answer: C

350. SAR operations may use which types of resources?

A. Only government aircraft

B. Only military ships

C. Public and private resources including aircraft, vessels, and installations

Correct Answer: C

351. Initial medical assistance during SAR operations is provided

A. Only after aircraft reaches the nearest airport

B. As part of SAR functions during rescue operations

C. Only by licensed doctors onboard aircraft

Correct Answer: B

352. Medical evacuation under SAR refers to

A. Transporting cargo from accident sites

B. Transporting airline crew only

C. Transporting injured persons to appropriate medical facilities

Correct Answer: C

353. SAR services may cooperate with which of the following?

A. Only local police

B. Only airport operator

C. Any cooperating aircraft, vessels, or installations

Correct Answer: C

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354. Coordination in SAR operations ensures

A. Only weather briefings are sent

B. Efficient use of resources to assist aircraft in distress

C. Only NOTAM issuance

Correct Answer: B

355. Search and rescue services are established primarily to ensure

A. Cargo movement efficiency

B. Safety of aircraft and people in distress situations

C. Compliance with customs procedures

Correct Answer: B

356. What is the primary purpose of an Emergency Locator Transmitter (ELT)?

A. To record flight data

B. To broadcast distinctive distress signals on designated frequencies

C. To measure altitude

Correct Answer: B

357. ELT can be activated

A. Only manually

B. Only automatically

C. Automatically by impact or manually by crew

Correct Answer: C

358. The ELT is generally located in which part of the aircraft?

A. Cockpit

B. Fuel tank area

C. Tail section

Correct Answer: C

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359. The ELT cockpit control panel has which 3 positions?

A. ON, STBY, TEST

B. AUTO, TEST, FAIL

C. ON, ARM, OFF

Correct Answer: C

360. At what impact force does the ELT automatically activate?

A. 1–2 g

B. 5–7 g

C. 10–12 g

Correct Answer: B

361. ELT is required to undergo maintenance at what interval?

A. Every 3 months

B. Every 6 months

C. Once every 12 months

Correct Answer: C

362. Accuracy of ELT on 121.5 / 243 MHz frequencies is

A. 1 km

B. 20 km

C. 100 m

Correct Answer: B

363. Accuracy of a 406 MHz ELT is approximately

A. 50 km

B. 5 km

C. 1 km

Correct Answer: B

364. Indian-registered ELTs begin with which identification code?

A. 406

B. 315

C. 419

Correct Answer: C

365. What is the main function of the Cockpit Voice Recorder (CVR)?

A. Record flight data

B. Record radar information

C. Record audio to and from flight crew

Correct Answer: C

366. How many channels of audio does a CVR record?

A. 2

B. 3

C. 4

Correct Answer: C

367. Which of the following is not a CVR audio channel?

A. Pilot

B. Co-pilot

C. Cabin crew

Correct Answer: C

368. CVR recording duration is

A. 10 or 25 hours

B. 2 hours or 25 hours

C. 5 hours

Correct Answer: B

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369. What is the colour of the CVR?

A. Black

B. Blue

C. International orange or yellow

Correct Answer: C

370. Why is the CVR mounted in the tail section of the aircraft?

A. Convenience

B. To reduce wiring

C. To minimise damage in case of crash

Correct Answer: C

371. What is the main function of the Flight Data Recorder (FDR)?

A. Record cockpit conversations

B. Record aircraft parameters during entire flight

C. Display weather data

Correct Answer: B

372. FDR is used for

A. Entertainment

B. Accident investigation & trend analysis

C. Cabin announcements

Correct Answer: B

373. The FDR records

A. Last 30 minutes

B. Last 2 hours

C. Last 10 or 25 hours

Correct Answer: C

374. FDR stores data on

A. Magnetic tape

B. Digital storage device

C. Paper strip

Correct Answer: B

375. What is the colour of the FDR?

A. Black

B. International orange or yellow (with reflective tape)

C. Grey

Correct Answer: B

376. Where is the FDR located in most aircraft?

A. Nose section

B. Wing

C. Rear of aircraft, usually under the fin

Correct Answer: C

377. What is attached to the front of the FDR unit?

A. ELT

B. Anti-collision light

C. Underwater Locating Device (ULD)

Correct Answer: C

378. What is the purpose of the ULD?

A. Detect weather

B. Emit underwater pings to help locate CVR/FDR

C. Display flight speed

Correct Answer: B

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379. Small aircraft may have CVR and FDR

A. Removed completely

B. Combined into one unit

C. Installed on wings

Correct Answer: B

380. What does the term INCERFA refer to?

A. A situation requiring immediate assistance

B. Uncertainty exists regarding the safety of an aircraft

C. Aircraft under unlawful interference

Correct Answer: B

381. What is ALERFA?

A. A situation where apprehension exists about the safety of an aircraft

B. Aircraft known to be safe

C. Aircraft has crashed

Correct Answer: A

382. Which example corresponds to ALERFA?

A. No communication for 30 minutes

B. Aircraft in circuit fails to land within 5 minutes of expected landing time

C. Aircraft is confirmed missing

Correct Answer: B

383. Which condition also leads to ALERFA?

A. Severe turbulence

B. Engine failure

C. Aircraft is known or believed to be under unlawful interference

Correct Answer: C

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384. What is DETRESFA?

A. A minor technical problem

B. Reasonable certainty that aircraft and occupants face grave and imminent danger

C. Loss of radar contact for 5 minutes

Correct Answer: B

385. Which of the following is a DETRESFA indication?

A. Aircraft known to be under hijack

B. Failure to make communication and aircraft efficiency is impaired

C. RT failure but aircraft is maintaining normal flight

Correct Answer: B

386. Which phase is most serious?

A. INCERFA

B. ALERFA

C. DETRESFA

Correct Answer: C

387. Arrange the phases in increasing order of seriousness

A. DETRESFA → INCERFA → ALERFA

B. INCERFA → ALERFA → DETRESFA

C. ALERFA → INCERFA → DETRESFA

Correct Answer: B

388. In “VOHS”, what does the “O” represent?

A. Operational status

B. Chennai FIR

C. Hyderabad station

Correct Answer: B

389. India lies in which Aeronautical Fixed Service Routing Area (AFSRA) zone?

A. C

B. V

C. Z

Correct Answer: B

390. Which of the following letters are not used as the first letter of AFSRA zones?

A. C, M, P

B. I, J, Q, X

C. A, B, D

Correct Answer: B

391. How many AFSRA zones exist worldwide?

A. 10

B. 15

C. 22

Correct Answer: C

392. The atmosphere is defined as

A. A layer of gases only above 50,000 ft

B. A spheroidal gaseous envelope surrounding a heavenly body

C. A thick layer of clouds around Earth

Correct Answer: B

393. Atmospheric pressure is

A. Mass per unit volume

B. Weight of air per cubic meter

C. Force per unit area exerted by the atmosphere

Correct Answer: C

394. Atmospheric pressure decreases with height because

A. Temperature increases with height

B. There is less air above as altitude increases

C. Humidity increases with height

Correct Answer: B

395. Density is defined as

A. Force per unit area

B. Mass per unit volume

C. Temperature per unit height

Correct Answer: B

396. Air density decreases when

A. Temperature decreases

B. Humidity decreases

C. Temperature increases

Correct Answer: C

397. Humidity refers to

A. Pressure in unit volume

B. Temperature in unit mass

C. Weight of water vapour in a unit volume of air

Correct Answer: C

398. Air with high humidity has

A. Higher density

B. Lower density

C. No effect on density

Correct Answer: B

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399. Which of the following affects air density?

A. Pressure

B. Temperature

C. All of the above

Correct Answer: C

400. Gusting is typically a result of

A. Temperature difference only

B. Sudden local turbulence or surface friction

C. High humidity only

Correct Answer: B

401. Which Q-code is used by the aircraft to ask the ground station: "What is my true bearing from you?"?

A. QDM

B. QTE

C. QDR

Correct Answer: B

402. Which procedural word is used to request permission for the intended action?

A. REQUEST

B. CLEARED

C. PERMIT

Correct Answer: A

403. The proword 'I SAY AGAIN' should be used when

A. The message being sent is extremely important.

B. The sender is repeating a specific part of the message for clarity.

C. The receiving station requested a repeat using 'SAY AGAIN'.

Correct Answer: B

404. Which of the following is the correct radiotelephony procedure for establishing initial contact with an ATC unit?

> A. Aircraft call sign, followed by the ground station call sign and current altitude.
>
> B. Ground station call sign, followed by the aircraft call sign, then position,
>
> C. Aircraft call sign only, followed by 'CHECKING IN'.

```
Correct Answer: B
```

405. If a pilot has been instructed to 'REPORT PASSING [FIX]', what is the primary purpose of this instruction?

> A. To notify the controller that the aircraft is performing a holding procedure.
>
> B. To request permission to proceed to the next waypoint.
>
> C. To allow the controller to confirm the aircraft's position and verify the

```
Correct Answer: C
```

406. What Q-code is used to request the magnetic bearing from the station to the aircraft?

> A. QDM
>
> B. QDR
>
> C. QTE

```
Correct Answer: B
```

407. If an ATC unit transmits 'SAY AGAIN', what is the pilot being requested to do?

> A. Change the transmitting frequency.
>
> B. Repeat all or part of the message.
>
> C. Speak slower and more clearly.

```
Correct Answer: B
```

408. What is the SI unit of mass?

> A. Gram
>
> B. Pound
>
> C. Kilogram

```
Correct Answer: C
```

409. Traffic information reported by aircraft is acknowledged when

A. Always

B. Requested by ATC

C. Collision risk exists

Correct Answer: C

410. What is the correct response to “Report FL 60”?

A. FL 60

B. Wilco

C. Roger, reporting flight level six zero

Correct Answer: B

411. What is the aerodrome control service called?

A. Mumbai Ground

B. Mumbai Dispatch

C. Mumbai Tower

Correct Answer: C

412. How is 108.1 read in radiotelephony?

A. One zero eight point one

B. One hundred eight decimal one

C. One zero eight decimal one

Correct Answer: C

413. What does FM stand for?

A. Field Modulation

B. Fast Modulation

C. Frequency Modulation

Correct Answer: C

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414. What must an aircraft do if it experiences a complete Radio Communication Failure (RCF) under IFR in VMC?

A. Climb to the highest minimum safe altitude.

B. Continue the flight in VMC, land at the nearest suitable aerodrome, and report by phone.

C. Proceed according to the last received clearance under IFR rules.

Correct Answer: B

415. When setting up a radio, what step must be taken before speaking into the microphone?

A. Depress the transmit switch fully and pause briefly before speaking

B. Wait 5 seconds after keying to allow the transmitter to stabilize.

C. Check the volume level.

Correct Answer: A

416. When two stations use two different frequencies for simultaneous two-way communication, this is known as

A. Simplex

B. Half Duplex

C. Full Duplex

Correct Answer: C

417. The electrical unit for measuring the rate of flow of current is the

A. Volt

B. Ohm

C. Ampere

Correct Answer: C

418. What is the main purpose of an Automatic Voice Announcing System (AVAS) at an aerodrome?

A. To automatically provide ATC clearances to pilots.

B. To broadcast the ATIS (Automatic Terminal Information Service) message.

C. To automatically acknowledge all pilot transmissions.

Correct Answer: B

PART 2: Viva / Oral & Practical Examination

PART 2 OF 2

Technical oral Q&A · rapid-fire orals · R/T phraseology drills · full practical flight papers

How the technical viva runs

The examiner fires short conceptual questions on radio principles, digital communication and procedures. Answer in one or two crisp sentences — say the definition, then one distinguishing fact. The model answers below are written in exactly that style.

💡

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1. What are Linear Block Codes?

Model answer: Forward error-correcting codes in which a block of k data bits is mapped to a longer codeword of n bits (n > k) by adding parity bits formed from linear (XOR) combinations of the data. The modulo-2 sum of any two valid codewords is itself a valid codeword. They let the receiver detect and correct a limited number of bit errors (e.g. Hamming codes).

2. What is Shannon Capacity?

Model answer: The theoretical maximum error-free data rate of a communication channel: $C = B \cdot \log_2(1 + S/N)$ bits per second, where B is the channel bandwidth and S/N the signal-to-noise power ratio. No coding scheme can carry information reliably above this limit.

3. What is a Matched Filter?

Model answer: A linear filter whose impulse response is the time-reversed copy of the known transmitted pulse. It maximises the signal-to-noise ratio at the sampling instant and is the optimum detector for a known signal in additive white Gaussian noise.

4. What are LDPC (Low-Density Parity-Check) Codes?

Model answer: Linear error-correcting codes defined by a sparse parity-check matrix. They can

approach the theoretical maximum capacity (the Shannon limit) and are decoded iteratively, giving excellent performance over noisy channels.

5. What is PCM (Pulse Code Modulation)?

Model answer: A method of representing an analogue signal digitally by sampling it at regular intervals and quantising each sample's amplitude into a binary code.

6. What is H-24?

Model answer: A notation meaning a service or facility (aerodrome, radio station, etc.) is available continuously, 24 hours a day.

7. What is Spread Spectrum?

Model answer: A technique in which the signal is deliberately transmitted over a bandwidth far wider than the minimum required, using a spreading code. It resists interference and jamming and lowers the probability of interception.

8. What is Nyquist Frequency?

Model answer: The minimum sampling rate needed to reconstruct a signal without aliasing: at least twice the highest frequency component, $f_s \geq 2 \cdot f_{xx}$.

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9. What are Turbo Codes / Modes?

Model answer: A high-performance class of error-correction codes using parallel concatenated convolutional encoders with interleaving and iterative decoding, achieving near-Shannon-limit performance in noisy channels.

10. How do Turbo Codes reduce ISI?

Model answer: Through iterative decoding and interleaving, which spreads bursts of errors apart in time, making the system more resilient to Inter-Symbol Interference (ISI).

11. What is Quantization Error?

Model answer: The difference between the true analogue sample value and the nearest discrete level assigned to it during quantisation; it appears as quantisation noise.

12. What is Electron Density?

Model answer: The number of free electrons per unit volume in the ionosphere. It sets the refractive index of each layer and therefore the Maximum Usable Frequency (MUF) for HF sky-wave propagation.

13. What is the ALOHA System?

Model answer: An early random-access protocol for shared radio/satellite channels: a station transmits whenever it has data, and if a collision occurs it waits a random time and retransmits.

14. What is PSK (Phase Shift Keying)?

Model answer: A digital modulation scheme that carries data by changing (keying) the phase of a constant-frequency, constant-amplitude carrier wave.

15. What is the International Emergency Frequency?

Model answer: 121.5 MHz (VHF) for civil aviation and 243.0 MHz (UHF) for military use; 406 MHz is used by modern ELTs for satellite alerting.

16. What is the Rayleigh Fading mechanism related to?

Model answer: Multipath propagation where there is no direct line-of-sight between transmitter and receiver, so the received signal is the sum of many reflected paths, causing rapid amplitude fluctuation (fading).

17. The Nyquist criterion is proportional to what?

Model answer: To the channel bandwidth B — the maximum symbol (signalling) rate without ISI is Rs = 2B symbols per second.

18. Why do pilots switch from HF to VHF?

Model answer: VHF gives higher-quality, static-free line-of-sight communication within range of a ground station. HF is reserved for long-range and oceanic areas beyond VHF coverage, but suffers fading and noise.

19. What is the significance when refractive index = 0?

Model answer: It marks the point at which the radio wave is completely refracted/reflected back towards Earth — effectively total internal reflection in the ionospheric layer, allowing sky-wave return.

20. What is the difference between 5 MHz and 15 MHz propagation?

Model answer: The lower frequency (5 MHz) works better at night and for shorter ranges, reflecting from lower ionospheric layers; the higher frequency (15 MHz) works better by day and for long-distance skips. As a rule: higher frequency by day, lower by night.

21. What is Radio Communication Failure (7600)?

Model answer: Transponder code 7600 is selected to alert ATC that the aircraft has lost radio communication capability.

22. If an aircraft reports a TCAS RA, what should ATC do?

Model answer: ATC must not attempt to modify the aircraft's flight path and shall not issue conflicting instructions until the pilot reports “Clear of Conflict.” The pilot is required to follow the Resolution Advisory in preference to any ATC clearance.

23. How is “Zulu” pronounced?

Model answer: ZOO-LOO. (Z = Zulu; used for UTC time.)

24. How is altimeter setting 1009 pronounced?

Model answer: “QNH ONE ZERO ZERO NINER.” Each digit is spoken separately.

25. How is RVR 1700 pronounced?

Model answer: “RVR ONE THOUSAND SEVEN HUNDRED” (metres) — commonly given as “RVR ONE SEVEN HUNDRED” in memory reports; state the value clearly in metres.

26. What is CPDLC (system & emergency use)?

Model answer: Controller-Pilot Data Link Communications — text-based messaging between controller and crew over a data link. In an emergency, pre-formatted MAYDAY or PAN-PAN messages can be sent, though voice remains primary for time-critical calls.

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27. What is the FDR system (PCM based)?

Model answer: Flight Data Recorder. Modern recorders use Pulse Code Modulation to digitise and store flight parameters in crash-protected solid-state memory.

28. What is the Spread Spectrum technique?

Model answer: Spreading the transmitted energy over a wide band using a pseudo-random code (Direct-Sequence or Frequency-Hopping). It provides anti-jam protection, low probability of intercept, and multiple access (CDMA).

29. What is the RCF (Radio Communication Failure) procedure?

Model answer: Squawk 7600. In VMC, continue in visual conditions and land at the nearest suitable aerodrome, reporting arrival to ATC. In IMC, continue per the last acknowledged clearance (route, level and estimate/EAT) as laid down in the rules, keep trying to re-establish contact, and observe light signals on approach.

30. How are MET reports published via AFTN (GG)?

Model answer: Meteorological messages are transmitted on the AFTN carrying the priority indicator GG (the priority reserved for meteorological traffic).

31. Explain Signal Propagation (technical).

Model answer: Radio energy reaches the receiver by three main modes: the ground/surface wave (dominant at LF/MF), the sky wave (HF, refracted by the ionosphere for beyond-horizon range), and the space or direct wave (VHF and above, essentially line-of-sight).

32. Explain HF propagation (detailed).

Model answer: HF sky waves are refracted by the ionospheric layers (D, E, F1, F2). Usable range depends on the MUF and LUF, the time of day, season and sunspot activity — higher frequencies by day, lower by night. It gives long-range/oceanic coverage but is subject to fading and a skip (dead) zone between ground-wave and first sky-wave return.

33. What is a Correlation Receiver in an AWGN channel?

Model answer: A receiver that correlates the incoming signal against stored reference waveforms and picks the largest output. It is mathematically equivalent to the matched filter and is the optimum detector in additive white Gaussian noise.

34. What is Duplex communication?

Model answer: A system allowing simultaneous two-way communication in both directions, like a telephone (full duplex), using separate transmit and receive frequencies.

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35. What are the ATS limitations (Aerodrome & CTR)?

Model answer: Air Traffic Services are bounded by the lateral/vertical limits of the Control Zone (CTR) and by the aerodrome's published hours of operation and available equipment/coverage.

36. What are Linear modes / linear modulation?

Model answer: Modulation in which the modulated output varies in direct (linear) proportion to the modulating signal — e.g. AM, DSB and SSB — as opposed to non-linear schemes such as FM and PM.

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37. Differentiate Differential PSK and Coherent PSK.

Model answer: Coherent PSK requires the receiver to recover and lock to the carrier's absolute phase reference. Differential PSK (DPSK) encodes data in the phase change between successive symbols, so no absolute phase reference is needed — simpler, at a small performance cost.

38. Differentiate Differential Entropy and Discrete Entropy.

Model answer: Discrete entropy $H = -\Sigma p \cdot \log_2 p$ measures the average information of a discrete source. Differential entropy is the analogous measure for a continuous random variable, defined by an integral over its probability density.

39. A Control Zone is above the surface of: Ground / Earth / Sea?

Model answer: A Control Zone (CTR) extends upward from the surface of the Earth, which includes both land and water.

⚡ Rapid-Fire Orals — Definitions, Units & Phraseology

25 Q&A

1. Abbreviation of VFR

Visual Flight Rules.

2. Meaning of AGL

Above Ground Level.

3. Unit of kilogram

The kilogram is the SI unit of mass.

4. Unit of length

The metre (SI unit of length).

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5. Once distress is over, the station shall transmit —

The distress-cancelling message: "… CANCEL DISTRESS" / "MAYDAY … DISTRESS TRAFFIC ENDED,

SEELONCE FEENEE."

6. Full form of PIC

Pilot-in-Command.

7. VHF frequency range

30 to 300 MHz (aviation voice band 118.0–136.975 MHz).

8. HF frequency range

3 to 30 MHz.

9. CPDLC full form

Controller-Pilot Data Link Communications.

10. Definition of Resistance

The opposition offered by a conductor to the flow of current, measured in ohms; R = V / I.

11. How to transmit FL200

"FLIGHT LEVEL TWO ZERO ZERO" (flight levels are spoken digit by digit).

12. How to transmit Squawk 2000

"SQUAWK TWO ZERO ZERO ZERO" (transponder codes are spoken digit by digit).

13. Frequency for International Emergency

121.5 MHz (VHF).

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14. Meaning of PAN-PAN (urgency)

A condition concerning the safety of the aircraft or a person on board but not requiring

immediate assistance.

15. How to pronounce the distress condition

“MAYDAY MAYDAY MAYDAY” — spoken three times.

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16. How to pronounce Heading 045

“HEADING ZERO FOUR FIFE.”

17. Squawk code for lost communication

7600.

18. Phonetic pronunciation for letter “B”

BRAVO.

19. Air-to-ground communication — which language is used?

English, or the language normally used by the ground station on the frequency.

20. Define Air-to-Ground communication

Two-way radio communication between aircraft stations and ground (aeronautical) stations.

21. What does WILCO mean?

“Understood and will comply.”

22. What is Bandwidth?

The range of frequencies a signal occupies or that a channel can pass, measured in hertz (Hz).

23. SELCAL — what is it?

Selective Calling: a system that alerts a specific aircraft's crew to an incoming call using a unique four-letter code, so they need not monitor continuously.

24. Airspace classification

ICAO Classes A to G — Class A the most restrictive controlled airspace, Class G uncontrolled.

25. Current, Watt & Voltage

Current (ampere) is the rate of flow of charge; Voltage (volt) is potential difference; Power (watt) = Voltage × Current.

118/121

🗣️ Standard R/T Phraseology Drills

5 items

Report visibility 900 m

“VISIBILITY NINER HUNDRED METRES.”

Report RVR 700 m

“RVR SEVEN HUNDRED METRES.”

Use of BREAK BREAK

Spoken to separate messages to different aircraft in a very busy environment: “BREAK BREAK.”

Read back

Repeat the clearance/instruction back to ATC exactly, so it can be verified (mandatory for level, heading, speed, runway, QNH, squawk, frequency).

Transmit frequency 118.000

“ONE ONE EIGHT DECIMAL ZERO” (trailing zeros after the decimal may be omitted; state the channel clearly).

🛩️ Practical Flight Papers (memory-based)

3 full papers

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How to use these

Each paper is a complete simulated flight the examiner talks you through — ground, departure, climb, cruise, and an emergency. Work through every step out loud, transmitting the correct call for each phase. Flight plan, weather and airfield details are shown on screen in the real exam.

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Practical Paper A — Guwahati to Imphal (memory, 25 Jun 2026)

Aircraft: ATR 72 · Call sign: IGO4518 (IndiGo 4518) · Route: VEGT → VEIM · IFR

1. Standard phraseology set at the start: visibility 900 m; RVR 700 m; use of "BREAK BREAK"; a read-back; and the frequency 118.000.

2. Ground: request pushback and start-up; during pushback another aircraft is taxiing behind you — hold/advise and take safe action; then request taxi and departure clearance.

3. Departure: holding short of RWY 20, ATC passes a previous aircraft's wind indicating a tailwind for RWY 20 — query/request the more favourable runway; read back the revised departure clearance; report ready for departure and accept the modified take-off clearance.

4. Initial climb: passing 2,600 ft Tower hands you to Approach; check in and request climb to cruising level; Approach clears "climb FL230, report passing FL150."

5. En-route: give a full standard position report (all ETAs provided); handle a TCAS Resolution Advisory ("TCAS RA" … "clear of conflict").

6. Emergency: cabin crew report unruly passengers possibly armed — transmit the MAYDAY call for a suspected hijack and squawk 7500 as appropriate.

Practical Paper B — Nagpur to Kolkata (memory, 17 Apr 2026 afternoon)

Flight: SEJ297 (SpiceJet 297) · IFR · FL310 · B738 · Route: VANP → VILK · Endurance 04:30, POB 145

1. Phraseology meanings asked: “Let me know that you have received and understood this

message” = ACKNOWLEDGE; “Authorised to proceed under the conditions specified” =

CLEARED; “True/Accurate”; “Establish communications with …” = CONTACT.

2. Ground: pushback and start-up; taxi.

3. ATC clearance: climb on runway heading, 2,600 ft, right turn, climb FL150.

4. Line-up: birds observed on the runway — take appropriate action before take-off; then take-

off (give the after-take-off call).

5. Climb: 20 NM from NNP passing FL80, report technical, request FL120; when technically clear

request further climb and set course; passing FL210 handle a TCAS RA.

6. Cruise: give a position report.

7. Emergency: standard engine fire indication — declare and divert to KKJ (Khajuraho); descend

with Khajuraho.

8. Approach: ILS unserviceable, runway in sight — carry out the visual approach actions; after

landing a tyre bursts — take the appropriate actions and advise ATC.

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Practical Paper C — Cochin to Mumbai (memory, 6 Apr 2026)

Aircraft: A320 · Call sign: VT-RSR · Route: VOCI → VABB · IFR · POB 159

1. Weather interpretation: RVR 1200, QNH 1000, weather TSRA (thunderstorm with rain),

visibility 3000 m — read and interpret.

2. Communication: pushback & start-up; ATC clearance & taxi.

3. In-flight: suspected bird strike after departure with all parameters normal — continue as per

flight plan with a normal call; request further climb from Cochin Approach; give a position report.

4. Abnormal: starboard engine vibration — return to Cochin; make a PAN-PAN call.

5. Emergency: starboard engine fire at 30 DME inbound Cochin — make a MAYDAY call.

Passing strategy

Drill Part 1 by section until you are consistently above 80% in each, then sit the whole set as one timed paper. For Part 2, rehearse the technical answers aloud and fly each practical paper end-to-end so the phraseology becomes automatic. The practical is where most candidates are failed — fluency, not memorisation, carries it.

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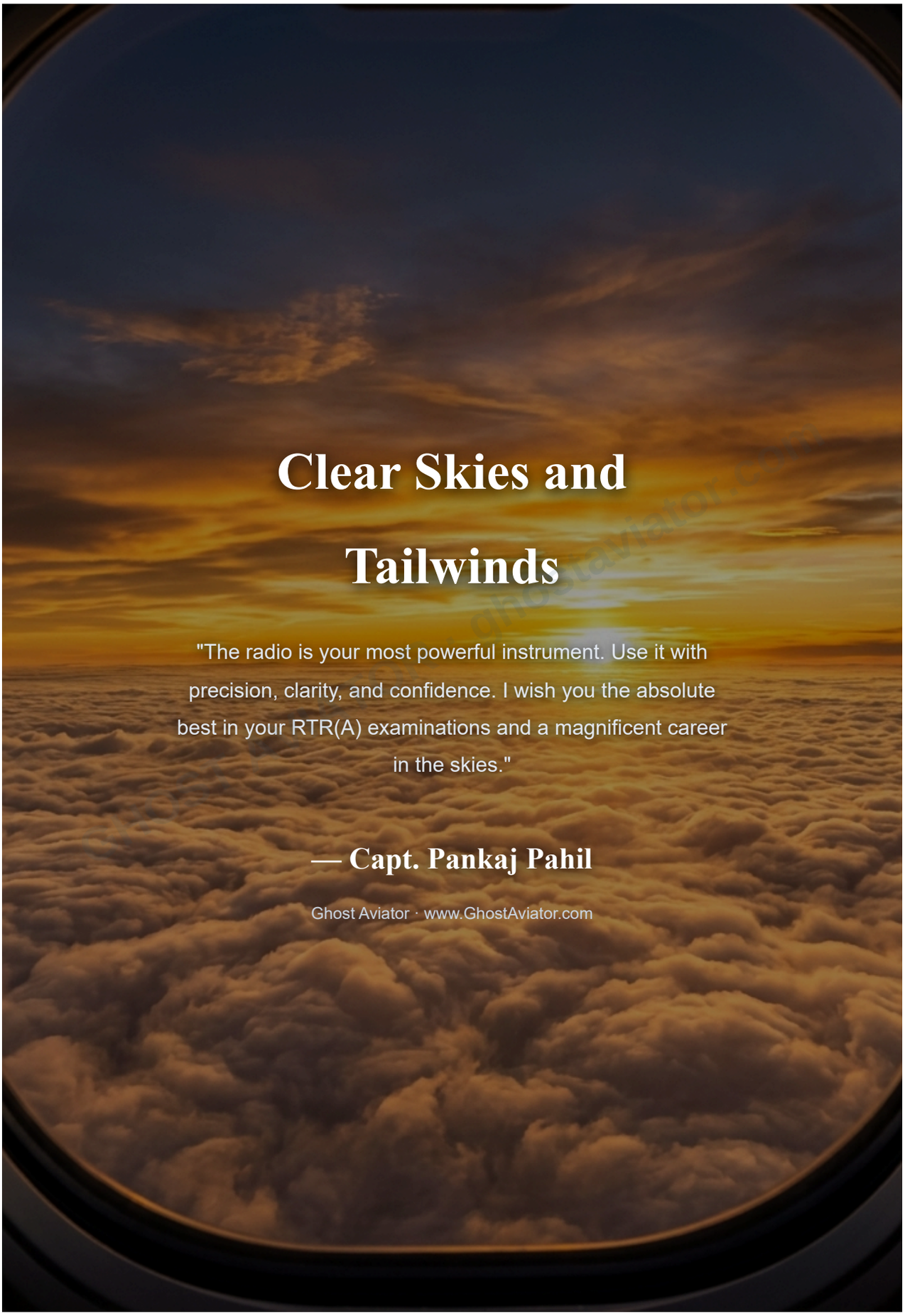
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— Capt. Pankaj Pahil
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